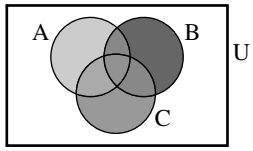
Chapter 1

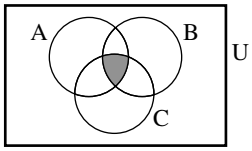
1–1. (a) $P(\overline{A \cup B \cup C}) = 1 - P(A \cup B \cup C) = 1 - 0.25 = 0.75$

(b) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.18$

1–2. (a)

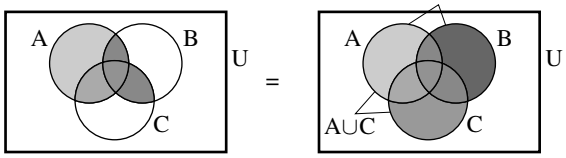


A∪(B∪C) = (A∪B)∪C

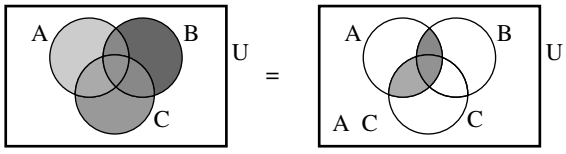


A∩(B∩C) = (A∩B)∩C

(b)

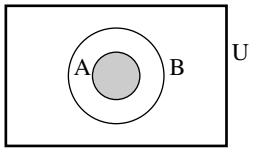


A∪(B∩C) = (A∪B)∩C(A∪C)



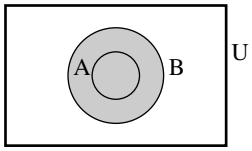
A∩(B∪C) = (A∩B)∪(A∩C)

(c)



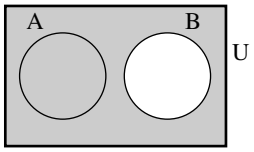
A∩B = A

(d)



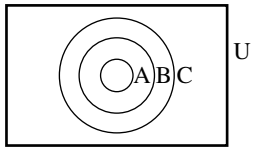
A∪B = B

(e)



A∩B = ∅ ⇒ A⊂$\overline{\text{B}}$

(f)



A⊂B and B⊂C ⇒ A⊂C

1–3. (a) $\overline{A} \cap B = \{5\}$,

(b) $\overline{A} \cup B = \{1, 3, 4, 5, 6, 7, 8, 9, 10\}$,

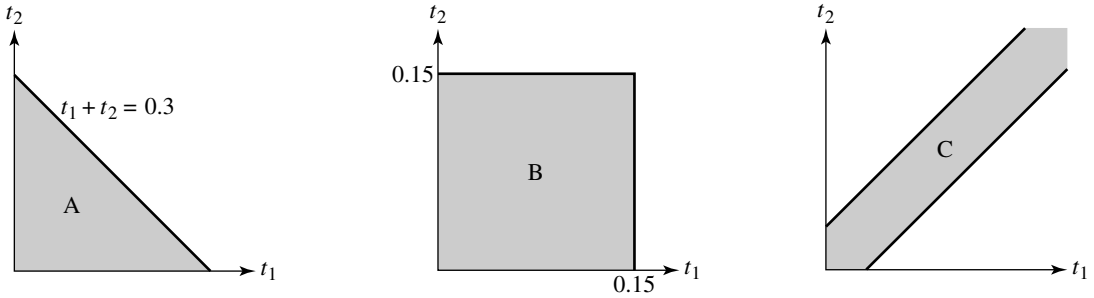
(c) $\overline{\overline{A} \cap \overline{B}} = \{2, 3, 4, 5\}$,

(d) $\overline{A \cap (B \cap C)} = U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$,

(e) $\overline{A \cap (B \cup C)} = \{1, 2, 5, 6, 7, 8, 9, 10\}$

1–4. $P(A) = 0.02,\ P(B) = 0.01,\ P(C) = 0.015$
$P(A \cap B) = 0.005,\ P(A \cap C) = 0.006$
$P(B \cap C) = 0.004,\ P(A \cap B \cap C) = 0.002$
$P(A \cup B \cup C) = 0.02 + 0.01 + 0.015 - 0.005 - 0.006 - 0.004 + 0.002 = 0.032$

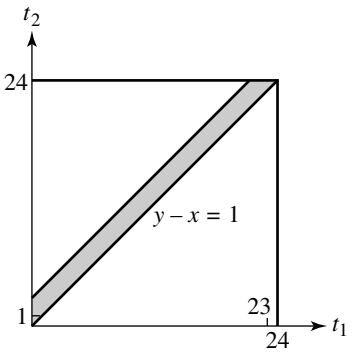
1–5. $S = \{(t_1, t_2)\colon\ t_1 \in R, t_2 \in R, t_1 \geq 0, t_2 \geq 0\}$



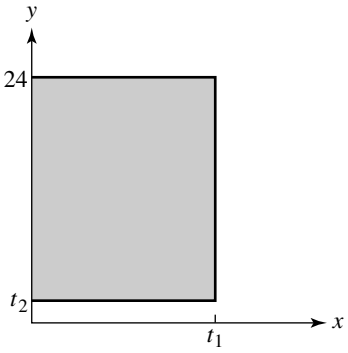
$A = \{(t_1, t_2)\colon\ t_1 \in R, t_2 \in R, 0 \leq t_1 \leq 0.3, 0 \leq t_2 \leq 0.3 - t_1\}$
$B = \{(t_1, t_2)\colon\ t_1 \in R, t_2 \in R, 0 \leq t_1 \leq 0.15, 0 \leq t_2 \leq 0.15\}$
$C = \{(t_1, t_2)\colon\ t_1 \in R, t_2 \in R, t_1 \geq 0, t_2 \geq 0, t_1 - 0.06 \leq t_2 \leq t_1 + 0.06\}$

1–6. (a) $S = \{(x, y)\colon\ x \in R, y \in R, 0 \leq x \leq y \leq 24\}$

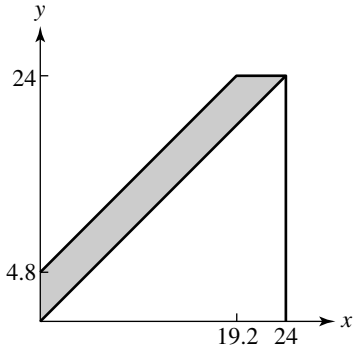
(b) i)



ii)



iii)



1–7. $S = \{NNNNN, NNNND, NNNDN, NNNDD, NNDNN,$
$NNDND, NNDD, NDNNN, NDNND, NDND,$
$NDD, DNNNN, DNNND, DNND, DND, DD\}$

1–8. $\{0,1\}^A = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a,b\}, \{a,c\}, \{a,d\}, \{b,c\}, \{b,d\},$
$\{c,d\}, \{a,b,c\}, \{a,b,d\}, \{a,c,d\}, \{b,c,d\}, \{a,b,c,d\}\}$

1–9. $N =$ Not Defective, $D =$ Defective

(a) $S = \{NNN, NND, NDN, NDD, DNN, DND, DDN, DDD\}$

(b) $S = \{NNNN, NNND, NNDN, NDNN, DNNN\}$

1–10. $p' =$ Lot Fraction Defective
$50 \cdot p' =$ Lot No. of Defectives

$$P(\text{Scrap Lot}|n = 10, N = 50, p') = 1 - \frac{\binom{50p'}{0}\binom{50(1-p')}{10}}{\binom{50}{10}}$$

If $p' = 0.1$, $P(\text{scrap lot}) \cong 0.689$.
She might wish to increase sample size.

1–11. $6 \cdot 5 = 30$ routes

1–12. $26^3 \cdot 10^3 = 17{,}576{,}000$ possible plates $\Rightarrow$ scheme feasible

1–13. $\binom{15}{6}\binom{8}{2}\binom{4}{1} = 560{,}560$ ways

1–14. $P(X \leq 2) = \sum_{k=0}^{2} \frac{\binom{20}{k}\binom{80}{4-k}}{\binom{100}{0.4}} \cong 0.97$

1–15. $P(\text{Accept}|p') = \sum_{k=0}^{1} \frac{\binom{300p'}{k}\binom{300(1-p')}{10-k}}{\binom{300}{10}}$

1–16. There are 5^{12} possibilities, so the probability of randomly selecting one is 5^{-12}.

1–17. $\binom{8}{2} = 28$ comparisons

1–18. $\binom{40}{2} = 780$ tests

1–19. $P_2^{40} = \dfrac{40!}{38!} = 1560$ tests

1–20. $\binom{10}{5} = 252$

1–21. $\binom{5}{1}\binom{5}{1} = 25$

$\binom{5}{2}\binom{5}{2} = 100$

1–22. $[1-(0.2)(0.1)(0.1)][1-(0.2)(0.1)](0.99) = 0.968$

1–23. $[1-(0.2)(0.1)(0.1)][1-(0.2)(0.1)](0.9) = 0.880$

1–24. $R_S = R_1\{1-[1-(1-(1-R_2)(1-R_4))(R_5)][1-R_3]\}$

1–25. $S =$ Siberia $\qquad U =$ Ural

$P(S) = 0.6, P(U) = 0.4, P(F|S) = P(\overline{F}|S) = 0.5$

$P(F|U) = 0.7, P(\overline{F}|U) = 0.3$

$$P(S|\overline{F}) = \frac{(0.6)\cdot(0.5)}{(0.6)(0.5)+(0.4)(0.3)} \doteq 0.714$$

1–26. $R_S = (0.995)(0.993)(0.994) = 0.9821$

1–27. A: 1st ball numbered 1
B: 2nd ball numbered 2

$$\begin{aligned} P(B) &= P(A)\cdot P(B|A) + P(\overline{A})\cdot P(B|\overline{A}) \\ &= \frac{1}{m}\cdot\frac{1}{m-1} + \frac{m-1}{m}\cdot\frac{1}{m} \\ &= \frac{m^2-m+1}{m^2(m-1)} \end{aligned}$$

1–28. $9\times 9 - 9 = 72$ possible numbers

$D_1 + D_2$ even: 32 possibilities

$P(D_1$ odd and D_2 odd$|D_1 + D_2$ even$) = \frac{20}{32}$.

1–29. A: over $6'$
M: male
F: female

$$P(M) = 0.6, P(F) = 0.4, P(A|M) = 0.2, P(A|F) = 0.01$$

$$\begin{aligned} P(F|A) &= \frac{P(F) \cdot P(A|F)}{P(F) \cdot P(A|F) + P(M) \cdot P(A|M)} \\ &= \frac{(0.04)(0.01)}{(0.4)(0.01) + (0.6)(0.2)} = \frac{0.004}{0.124} \\ &\cong 0.0323 \end{aligned}$$

1–30. A: defective
B_i: production on machine i

(a) $$\begin{aligned} P(A) &= P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3) \\ &\quad + P(B_4) \cdot P(A|B_4) \\ &= (0.15)(0.04) + (0.30)(0.03) + (0.20)(0.05) + (0.35)(0.02) \\ &= 0.032 \end{aligned}$$

(b) $$P(B_3|A) = \frac{(0.2)(0.05)}{0.032} = 0.3125$$

1–31. r = radius

$$P(\text{closer to center}) = \frac{\pi \left(\frac{r}{2}\right)^2}{\pi r^2} = \frac{1}{4}$$

1–32. $$\begin{aligned} P(A \cup B \cup C) &= P((A \cup B) \cup C) \quad \text{(associative law)} \\ &= P(A \cup B) + P(C) - P((A \cup B) \cap C) \\ &= P(A) + P(B) - P(A \cap B) + P(C) - P((A \cap C) \cup (B \cap C)) \\ &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) \\ &\quad - P(B \cap C) + P(A \cap B \cap C) \end{aligned}$$

1–33. For $k = 2$, $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$; Thm. 1–3.
Using induction we show that if true for $k - 1$, then true for k, i.e.,

If

$$P(A_2 \cup A_3 \cup \cdots \cup A_k) = \sum_{i=2}^{k} P(A_i) - \sum_{2 \le i < j \le k} P(A_i \cap A_j) + \sum_{2 \le i < j < r \le k} P(A_i \cap A_j \cap A_r) - \sum_{2 \le i < j < r < \ell \le k} P(A_i \cap A_j \cap A_r \cap A_\ell) + \cdots \qquad \text{(Eq. 1)}$$

Then

$$P(A_1 \cup A_2 \cup \cdots \cup A_k) = \sum_{i=1}^{k} P(A_i) - \sum_{1 \le i < j \le k} P(A_i \cap A_j) + \sum_{1 \le i < j < r \le k} P(A_i \cap A_j \cap A_r) - \sum_{1 \le i < j < r < \ell \le k} P(A_i \cap A_j \cap A_r \cap A_\ell) + \cdots \qquad \text{(Eq. 2)}$$

By hypothesis, and letting $A_1 \cap A_i$ replace A_i in Eq. 1,

$$P((A_1 \cap A_2) \cup (A_1 \cap A_3) \cup \cdots \cup (A_1 \cap A_k)) = \sum_{i=2}^{k} P(A_1 \cap A_i) - \sum_{2 \le i < j \le k} P(A_1 \cap A_i \cap A_j) + \sum_{2 \le i < j < r \le k} P(A_1 \cap A_i \cap A_j \cap A_r) - \sum_{2 \le i < j < r \le k} P(A_1 \cap A_i \cap A_j \cap A_r \cap A_\ell) + \cdots \qquad \text{(Eq. 3)}$$

By Thm. 1–3,

$$P(A_1 \cup (A_2 \cup A_3 \cup \cdots \cup A_k)) = P(A_1) + P(A_2 \cup A_3 \cup \cdots \cup A_k) - P((A_1 \cap A_2) \cup \cdots \cup (A_1 \cap A_k))$$

So from using Eq. 1 through 3,

$$\begin{aligned} & P(A_1 \cup A_2 \cup \cdots \cup A_k) \\ &= P(A_1) + \left[\sum_{i=2}^{k} P(A_i) - \sum_{2 \le i < j \le k} P(A_i \cap A_j) + \sum_{2 \le i < j < r \le k} P(A_i \cap A_j \cap A_r) - \cdots \right] \\ &\quad - \left[\sum_{i=2}^{k} P(A_1 \cap A_i) - \sum_{2 \le i < j \le k} P(A_1 \cap A_i \cap A_j) + \sum_{2 \le i < j < r \le k} P(A_1 \cap A_i \cap A_j \cap A_r) - \cdots \right] \\ &= \sum_{i=1}^{k} P(A_i) - \sum_{1 \le i < j \le k} P(A_i \cap A_j) + \sum_{1 \le i < j < r \le k} P(A_i \cap A_j \cap A_r) \\ &\quad + \cdots + (-1)^{k-1} \cdot P(A_1 \cap A_2 \cap \cdots \cap A_j) \end{aligned}$$

1–35. $P(\overline{B}) = \dfrac{(365)(364)\cdots(365-n+1)}{365^n}$

n	10	20	21	22	23	24	25	30	40	50	60
$P(B)$	0.117	0.411	0.444	0.476	0.507	0.538	0.569	0.706	0.891	0.970	0.994

1–36. $P(\text{win on 1st throw}) = \dfrac{6}{36} + \dfrac{2}{36} = \dfrac{8}{36}$

$$P(\text{win after 1st throw}) = P(\text{win on 4}) + P(\text{win on 5}) + P(\text{win on 6}) + P(\text{win on 8}) + P(\text{win on 9}) + P(\text{win on 10})$$

$$P(\text{win on 4}) = \frac{3}{36} \cdot \left[\frac{3}{36} + \left(\frac{27}{36}\right) \cdot \frac{3}{36} + \left(\frac{27}{36}\right)^2 \cdot \frac{3}{36} + \cdots\right] = \frac{1}{36}$$

$$P(\text{win on 5}) = \frac{4}{36} \cdot \left[\frac{4}{36} + \left(\frac{26}{36}\right)\left(\frac{4}{36}\right) + \left(\frac{26}{36}\right)^2 \cdot \left(\frac{4}{36}\right) + \cdots\right] = \frac{2}{45}$$

$$P(\text{win on 6}) = \frac{5}{36}\left[\frac{5}{36} + \left(\frac{25}{36}\right)\left(\frac{5}{36}\right) + \left(\frac{25}{36}\right)^2 \cdot \left(\frac{5}{36}\right) + \cdots\right] = \frac{25}{396}$$

$$P(\text{win on 8}) = P(\text{win on 6}) = \frac{25}{396}$$

$$P(\text{win on 9}) = P(\text{win on 5}) = \frac{2}{45} \qquad P(\text{win on 10}) = P(\text{win on 4}) = \frac{1}{36}$$

$$P(\text{win}) = \frac{8}{36} + \left[2 \cdot \frac{1}{36} + 2 \cdot \frac{2}{45} + 2 \cdot \frac{25}{396}\right] = 0.4929$$

1–37. $P_8^8 = 8! = 40{,}320$

1–38. Let B, C, D, E, X represent the events of arriving at points so labeled.

$$P(B) = P(C) = P(D) = P(E) = \frac{1}{4}$$

$$P(X|B) = \frac{1}{3}, \qquad P(X|C) = 1, \qquad P(X|D) = 1, \qquad P(X|E) = \frac{2}{5}$$

$$P(X) = P(B) \cdot P(X|B) + P(C) \cdot P(X|C) + P(D) \cdot P(X|D) + P(E) \cdot P(X|E)$$

$$= \left(\frac{1}{4} \cdot \frac{1}{3}\right) + \left(\frac{1}{4} \cdot 1\right) + \left(\frac{1}{4} \cdot 1\right) + \left(\frac{1}{4} \cdot \frac{2}{5}\right) = \frac{41}{60}$$

1–39. $$P(B_3|A) = \frac{P(B_3) \cdot P(A|B_3)}{\sum_{i=1}^{3} P(B_i) \cdot P(A|B_i)} = \left[\frac{(0.5)(0.3)}{(0.2)(0.2) + (0.3)(0.5) + (0.5)(0.3)}\right]$$
$$= 0.441$$

1–40. F: Structural Failure
D_S: Diagnosis as Structural Failure

$$P(F|D_S) = \frac{P(F) \cdot P(D_S|F)}{P(F) \cdot P(D_S|F) + P(\overline{F}) \cdot P(D_S|\overline{F})}$$
$$= \frac{(0.25)(0.9)}{(0.25)(0.9) + (0.75)(0.2)} = \frac{0.225}{0.225 + 0.150} = 0.6$$

Chapter 2

2–1. $R_X = \{0,1,2,3,4\}, P(X=0) = \dfrac{\binom{4}{0}\binom{48}{5}}{\binom{52}{5}}$

$$P(X=1) = \frac{\binom{4}{1}\binom{48}{4}}{\binom{52}{5}}, \qquad P(X=2) = \frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}},$$

$$P(X=3) = \frac{\binom{4}{3}\binom{48}{2}}{\binom{52}{5}}, \qquad P(X=4) = \frac{\binom{4}{4}\binom{48}{1}}{\binom{52}{5}}$$

2–2. $\mu = 0\cdot\frac{1}{6} + 1\cdot\frac{1}{6} + 2\cdot\frac{1}{3} + 3\cdot\frac{1}{12} + 4\cdot\frac{1}{6} + 5\cdot\frac{1}{12} = \frac{26}{12}$

$$\sigma^2 = \left[0^2\cdot\frac{1}{6} + 1^2\cdot\frac{1}{6} + 2^2\cdot\frac{1}{3} + 3^2\cdot\frac{1}{12} + 4^2\cdot\frac{1}{6} + 5^2\cdot\frac{1}{12}\right] - \left(\frac{26}{12}\right)^2 = \frac{83}{36}$$

2–3. $\displaystyle\int_0^\infty ce^{-x}\,dx = 1 \Rightarrow c = 1,$ so

$$f(x) = \begin{cases} e^{-x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\mu = \int_0^\infty xe^{-x}\,dx = -xe^{-x}|_0^\infty + \int_0^\infty e^{-x}\,dx = 1$$

$$\sigma^2 = \int_0^\infty x^2e^{-x}\,dx - 1^2 = \left[-x^2e^{-x}|_0^\infty + \int_0^\infty 2xe^{-x}\,dx\right] - 1$$

$$= \left[-2xe^{-x}|_0^\infty + \int_0^\infty 2e^{-x}\,dx\right] - 1$$

$$= 2 - 1 = 1$$

2–4. $F_T(t) = P_T(T \leq t) = 1 - e^{-ct}; t \geq 0$

$$\therefore f_T(t) = F_T'(t) = \begin{cases} ce^{-ct} & \text{if } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

2–5. (a) Yes

(b) No, since $G_X(\infty) \neq 1$ and $G_X(b) \not\geq G_X(a)$ if $b \geq a$

(c) Yes

2–6. (a) $f_X(x) = F'_X(x) = e^{-x}; 0 < x < \infty$
$= 0; x \leq 0$

(c) $h_X(x) = H'_X(x) = e^{x}; -\infty < x \leq 0$
$= 0; x > 0$

2–7. Both are since $p_X(x) \geq 0$, all x; and $\Sigma_{\text{all } x}\ p_X(x) = 1$

2–8. The probability mass function is

x	$p_X(x)$
-1	$\frac{1}{5}$
0	$\frac{1}{10}$
$+1$	$\frac{2}{5}$
$+2$	$\frac{3}{10}$
ow	0

$$E(X) = \left(-1 \cdot \frac{1}{5}\right) + \left(0 \cdot \frac{1}{10}\right) + \left(1 \cdot \frac{2}{5}\right) + \left(2 \cdot \frac{3}{10}\right) = \frac{4}{5}$$

$$V(X) = \left((-1)^2 \cdot \frac{1}{5}\right) + \left(0^2 \cdot \frac{1}{10}\right) + \left(1^2 \cdot \frac{2}{5}\right) + \left(2^2 \cdot \frac{3}{10}\right) - \left(\frac{4}{5}\right)^2 = \frac{29}{25}$$

$$F_X(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{5}, & -1 \le x < 0 \\ \frac{3}{10}, & 0 \le x < 1 \\ \frac{7}{10}, & 1 \le x < 2 \\ 1, & x \ge 2 \end{cases}$$

2–9. $P(X < 30) = P(X \le 29) = \sum_{x=0}^{29} \frac{e^{-20}(20)^x}{x!} = 0.978$

2–10. (a) $P_X(x) = F_X(x) - F_X(x-1)$

$$= \left[1 - \left(\frac{1}{2}\right)^{x+1}\right] - \left[1 - \left(\frac{1}{2}\right)^{x}\right]$$

$$= \left(\frac{1}{2}\right)^{x} - \left(\frac{1}{2}\right)^{x+1}$$

$$= \left(\tfrac{1}{2}\right)\left(\tfrac{1}{2}\right)^{x}; x = 0, 1, 2, \ldots$$

$$= 0; ow$$

(b) $P_X(0 < X \le 8) = F_X(8) - F_X(0) = 0.498$

(c) $P_X(X \text{ even}) = \frac{1}{2}\sum_{k=0}^{\infty}\left(\frac{1}{4}\right)^{k} = \frac{2}{3}$

2–11. (a) $\int_0^2 kx\,dx + \int_2^4 k(4-x)\,dx = 1 \Rightarrow k = \frac{1}{4}$

and $f_X(x) \ge 0$ for $k = \frac{1}{4}$

(b) $\mu = \displaystyle\int_0^2 \frac{1}{4}x^2\,dx + \int_2^4 \frac{1}{4}(4x - x^2)\,dx = 2$

$\sigma^2 = \displaystyle\int_0^2 \frac{1}{4}x^3\,dx + \int_2^4 \frac{1}{4}(4x^2 - x^3)\,dx - 2^2 \;=\; \frac{2}{3}$

(c) $F_X(x) = 0; \quad x < 0$

$= \displaystyle\int_0^x \frac{1}{4}t\,dt = \frac{x^2}{8}; \quad 0 \le x < 2$

$= \displaystyle\int_0^2 \frac{1}{4}x\,dx + \int_2^x \frac{1}{4}(4 - t)\,dt = -1 + x - \frac{x^2}{8}; \quad 2 \le x < 4$

$= 1; \quad x > 4$

2–12. (a) $k\left[\displaystyle\int_0^a x\,dx + \int_a^{2a} (2a - x)\,dx\right] = 1 \Rightarrow k = \frac{1}{a^2}$

(b) $F_X(x) = 0; x < 0$

$= \displaystyle\int_0^x kt\,dt = k\left(\frac{x^2}{2}\right); 0 \le x < a$

$= \displaystyle\int_0^a kx\,dx + \int_a^x k(2a - t)\,dt = k\left(\frac{a^2}{2}\right) + k\left[2a(x - a) + \frac{a^2 - x^2}{2}\right]$ for $a \le x \le 2a$

$= 1; x > 2a$

(c) $\mu = k\left[\displaystyle\int_0^a x^2\,dx + \int_a^{2a} (2ax - x^2)\,dx\right] = a$

$\sigma^2 = \dfrac{a^2}{6}$

2–13. From Chebyshev's inequality $1 - \frac{1}{k^2} = 0.75 \Rightarrow k = 2$ with $\mu = 2$, $\sigma = \sqrt{2}$, and the interval is $[14 - 2\sqrt{2}, 14 + 2\sqrt{2}]$.

2–14. (a) $\displaystyle\int_{-1}^0 kt^2\,dt = 1 \Rightarrow k = 3$

(b) $\mu = \int_{-1}^{0} 3t^3\,dt = 3\left[\left.\frac{t^4}{4}\right|_{-1}^{0}\right] = -\frac{3}{4}$

$\sigma^2 = \int_{-1}^{0} 3t^4\,dt - \left(-\frac{3}{4}\right)^2 = 3\left(\left.\frac{t^5}{5}\right|_{-1}^{0}\right) - \frac{9}{16} = \frac{3}{80}$

(c) $F_T(t) = 0; \quad t < -1$

$= \int_{-1}^{t} 3u^2\,du = t^3 + 1; \quad -1 \le t \le 0$

$= 1; \quad t > 0$

2–15. (a) $k\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8}\right) = 1 \Rightarrow k = \frac{8}{7}$

(b) $\mu = \frac{8}{7}\left[1 \cdot \left(\frac{1}{2}\right) + 2 \cdot \left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{2}\right)^3\right] = \frac{11}{7}$

$\sigma^2 = \frac{8}{7}\left[1^2 \cdot \left(\frac{1}{2}\right) + 2^2 \cdot \left(\frac{1}{2}\right)^2 + 3^2 \cdot \left(\frac{1}{2}\right)^3\right] - \left(\frac{11}{7}\right)^2 = \frac{26}{49}$

(c) $F_X(x) = 0; \quad x < 1$

$= \frac{8}{14}; \quad 1 \le x < 2$

$= \frac{12}{14}; \quad 2 \le x < 3$

$= 1; \quad x \ge 3$

2–16. $\sum_{n=0}^{\infty} kr^n = 1 => k = 1 - r$

2–17. Using Chebychev's inequality, $1 - \frac{1}{k^2} = 0.99 \Rightarrow k = 10$,
$\mu = 2$, $\sigma = \sqrt{0.4}$, so the interval is $[2 - 10(0.6324)]$; $2 + 10(0.6324)]$. The letters should be mailed 8.3 days before delivery date required.

2–18. (a) $\mu_A = 1000(0.2) + 1050(0.3) + 1100(0.1) + 1150(0.3) + 1200(0.05) + 1350(0.05)$
$= 1097.5$
$\mu_B = 1135$

(b) Assume independence so $\mu_{A|B=1130} = \mu_A = 1097.5$

(c) With independence, $P(A = k \text{ and } B = k) = P(A = k) \cdot P(B = k)$. So

$$\sum_k P(A = k)P(B = k) = (0.1)(0.2) + (0.3)(0.1) + (0.1)(0.3) + (0.3)(0.3) + (0.05)(0.1) + (0.05)(0.1) = 0.18$$

2–19. $p(x_i) = \dfrac{x_i - 1}{36}; \quad x_i = 2, 3, \ldots, 6$

$= \dfrac{13 - x_i}{36}; \quad x_i = 7, 8, \ldots, 12$

so

x_i	2	3	4	5	6	7	8	9	10	11	12
$p(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

2–20. $\mu = 7$, $\sigma^2 = \dfrac{105}{18}$

2–21. (a) $F_X(x) = 0; \quad x < 0$

$= x^2/9; \quad 0 \le x < 3$

$= 1; \quad x \ge 3$

(b) $\mu = 2$, $\sigma^2 = \dfrac{1}{2}$

(c) $\mu_3' = \dfrac{54}{5}$

(d) $m = \dfrac{3}{\sqrt{2}}$

2–22. $\mu = 0$, $\sigma^2 = 25$, $\sigma = 5$
$P[|X - \mu| \ge k\sigma] = P[|X| \ge 5k] = 0$ if $k > 1$ and $= 1$, $0 < k \le 1$.
From Chebychev's inequality, the upper bound is $\frac{1}{k^2}$.

2–23. $F(x) = 0; x < 0$

$$F(x) = \int_0^x \left(\frac{u}{t^2}\right) e^{-(u^2/2t^2)}\, du = \int_0^{x^2/2t^2} e^{-v}\, dv$$

$= 1 - e^{-x^2/2t^2}; x \ge 0$

2–24. $F(x) = \displaystyle\int_0^x \frac{1}{\sigma\pi} \frac{du}{\{1 + ((u-\mu)^2/\sigma^2)\}}; -\infty < x \leq \infty$

Let $t = \dfrac{u-\mu}{\sigma}$, $dt = \dfrac{1}{\sigma} du$ and

$$F(x) = \int_0^{\frac{x-u}{\sigma}} \frac{1}{\pi} \cdot \frac{dt}{1+t^2} = \frac{1}{\pi} \tan^{-1}\left(\frac{x-u}{\sigma}\right); -\infty < x < \infty$$

2–25. $\displaystyle\int_0^{\pi/2} k \sin y \, dy = 1 \Rightarrow k[-\cos y|_0^{\pi/2}] = 1 \Rightarrow k = 1$

$$\mu = \int_0^{\pi/2} y \sin y \, dy = \sin\left(\frac{\pi}{2}\right) = 1$$

2–26. Assume X continuous

$$\begin{aligned}
\mu_k &= \int_{-\infty}^{\infty} (x-\mu)^k f_X(x)\, dx = \int_{-\infty}^{\infty} \left[\sum_{j=0}^{k} \binom{k}{j} (-\mu^j) x^{k-j}\right] f_X(x)\, dx \\
&= \sum_{j=0}^{k} (-1)^j \binom{k}{j} \mu^j \int_{-\infty}^{\infty} x^{k-j} f_X(x)\, dx \\
&= \sum_{j=0}^{k} (-1)^j \binom{k}{j} \mu^j \mu'_{k-j}
\end{aligned}$$

Chapter 3

3–1. (a)

y	$p_Y(y)$
0	0.6
20	0.3
80	0.1
ow	0

(b) $E(Y) = \sum_y y \cdot p_Y(y) = 14$

$V(Y) = \sum_y y^2 \cdot p_Y(y) - (14)^2 = 564$

3–2. Let Profit $= P = 10 + 2X$

$$P(P \le p) = P(10 + 2X \le p) = P\left(X \le \frac{p-10}{2}\right) = \int_0^{\frac{p-10}{2}} \frac{x}{18} dx$$

$$F_P(p) = \frac{x^2}{36}\Big|_0^{\frac{p-10}{2}} = \frac{1}{144}(p^2 - 20p + 100)$$

$$f_P(p) = \frac{1}{72}(p-10); 10 \le p \le 22$$

$$= 0; ow$$

3–3. (a) $P(T < 1) = 1 - e^{-1/4} = 0.221$

(b) $E[P] = 200 - 200P(T < 1) = 155.80$

3–4. (a)

x	$p_X(x)$	$y = 2000(12 - x)$	$p_Y(y)$
10	0.1	4000	0.1
11	0.3	2000	0.3
12	0.4	0	0.4
13	0.1	-2000	0.1
14	0.1	-4000	0.1
ow	0	ow	0

(b) $E(X) = 10(0.1) + 11(0.3) + 12(0.4) + 13(0.1) + 14(0.1) = 11.8$ days

$V(X) = 10^2(0.1) + 11^2(0.3) + 12^2(0.4) + 13^2(0.1) + 14^2(0.1) - (11.8)^2$

$= 1.16$ days^2

$$E(Y) = (4000)(0.1) + (2000)(0.3) + 0(0.4) + (-2000)(0.1) + (-4000)(0.1) = \$400$$

$$V(Y) = (4000)^2(0.1) + (2000)^2(0.3) + 0^2(0.4) + (-2000)^2(0.1) + (-4000)^2(0.1) - 400^2 = 4{,}640{,}000(\$^2)$$

3–5. $F_Z(z) = P(Z \le z) = P(X^2 \le z) = P(|X| \le \sqrt{z})$
$= P(0 \le X \le \sqrt{z}) = \int_0^{\sqrt{z}} 2xe^{-x^2}\,dx$

Let $u = x^2$, $du = 2x\,dx$, so

$$F_Z(z) = \int_0^z e^{-u}\,du = 1 - e^z$$

$$f_Z(z) = e^{-z}; z \ge 0$$
$$= 0; ow$$

3–6. (a) $E(D_i) = \sum_{d=0}^{9} d \cdot p_{D_i}(d) = \frac{1}{10}(1+2+3+4+5+6+7+8+9) = 4.5$

(b) $V(D_i) = \sum_{d=0}^{9} d^2 \cdot p_{D_i}(d) - (4.5)^2$
$= \frac{1}{10}[1^2+2^2+3^2+4^2+5^2+6^2+7^2+8^2+9^2] - (20.25) = 8.25$

(c)

d	y
0	4
1	3
2	2
3	1
4	0
5	0
6	1
7	2
8	3
9	4

y	$p_Y(y)$
0	0.2
1	0.2
2	0.2
3	0.2
4	0.2
ow	0

$$E(Y) = \frac{2}{10}(1+2+3+4) = 2$$

$$V(Y) = \frac{2}{10}(1^2+2^2+3^2+4^2) - 4 = 2$$

3–7. $R =$ Revenue/Gal

$R = 0.92; A < 0.7$

$= 0.98; A \geq 0.7$

$E(R) = 0.92\ P(A < 0.7) + 0.98\ P(A \geq 0.7)$

$= 0.92(0.7) + 0.98(0.3) = 93.8¢/\text{gal.}$

3–8. $M_X(t) = \displaystyle\int_\beta^\infty e^{tx} \frac{1}{\theta} e^{-\frac{1}{\theta}(x-\beta)}\, dx = \frac{1}{\theta} e^{\beta/\theta} \int_\beta^\infty e^{-x(\frac{1}{\theta}-t)}\, dx$

$= \dfrac{1}{\theta}\left(\dfrac{1}{\theta} - t\right)^{-1} e^{\beta t},\ \text{for } \frac{1}{\theta} - t > 0$

$M_X'(t) = \dfrac{1}{\theta}\left(\dfrac{1-\theta t}{\theta}\right)^{-2} e^{\beta t}\left[\dfrac{1-\theta t}{\theta}\beta + 1\right]$

$M_X''(t) = \dfrac{1}{\theta}\left(\dfrac{1-\theta t}{\theta}\right)^{-3} e^{\beta t}\left[\left(\dfrac{1-\theta t}{\theta}\right)^2 \beta^2 + \left(\dfrac{1-\theta t}{\theta}\right)\beta + 2 + \left(\dfrac{1-\theta t}{\theta}\right)\beta\right]$

$E(X) = M_X'(0) = \beta + \theta$

$V(X) = M_X''(0) - (\beta+\theta)^2 = \theta^2$

3–9. (a) $F_Y(y) = P(2X^2 \leq y) = P\left[-\sqrt{\dfrac{y}{2}} \leq X \leq +\sqrt{\dfrac{y}{2}}\right]$

$= P\left(0 \leq X \leq \sqrt{\dfrac{y}{2}}\right)$

$= \displaystyle\int_0^{\sqrt{\frac{y}{2}}} e^{-x}\, dx = -e^{-x}\Big|_0^{\sqrt{\frac{y}{2}}} = 1 - e^{-\sqrt{\frac{y}{2}}}$

$f_Y(y) = F_Y'(y) = \dfrac{1}{4}\left(\dfrac{y}{2}\right)^{-1/2} \cdot e^{-(y/2)^{1/2}}; y > 0$

$= 0; \text{otherwise}$

(b) $F_V(v) = P(X^{1/2} \leq v) = P(X \leq v^2)$

$$= \int_0^{v^2} e^{-x}\,dx = -e^{-x}|_0^{v^2} = 1 - e^{-v^2}$$

$$f_V(v) = F_V'(v) = 2ve^{-v^2}; v > 0$$

$$= 0; \text{otherwise}$$

(c) $F_U(u) = P(\ell\text{n}(X) \leq u) = P(X \leq e^u) = \int_0^{e^u} e^{-x}\,dx$

$$= 1 - e^{-e^u}$$

$$f_U(u) = F_U'(u) = e^{-(e^u - u)}; u > 0$$

$$= 0; \text{otherwise}$$

3–10. Note that as stated $Y > y_0 \Rightarrow$ signal read (not $Y > |y_0|$), so

$$P_Y(Y > y) = \int_{\tan^{-1}y}^{\pi/2} \frac{1}{2\pi}dx + \int_{\tan^{-1}y+\pi}^{3\pi/2} \frac{1}{2\pi}dx$$

$$F_Y(y) = 1 - P_Y(Y > y)$$

$$= \frac{1}{2} + \frac{1}{\pi}\tan^{-1}y; y \geq 0$$

$$f_Y(y) = \frac{1}{\pi}\left(\frac{1}{1+y^2}\right); -\infty < y < +\infty$$

Note the symmetry in $(\frac{1}{1+y^2})$.

3–11. Let $S = \text{Stock Level}$

$$L(X,S) = 0.5(X - S), X > S$$

$$= 0.25(S - X), X \leq S$$

$$E(L(X,S)) = \int_{10^6}^{S} 0.25(S - X)\cdot 10^6\,dx + \int_S^{2\cdot 10^6} 0.5(X - S)\cdot 10^{-6}\,dx$$

$$\frac{dE[L(X,S)]}{dS} = \frac{6}{8}10^{-6}S - \frac{5}{4} = 0 \Rightarrow S = \frac{5}{3}10^6$$

3–12. $\mu_G = E(G) = \int_0^4 g f_G(g)\, dg = \int_0^4 \frac{g^2}{8} dg = 8/3.$

$$E(G^2) = \int_0^4 \frac{g^3}{8} dg = 8.$$

$$\sigma_G^2 = V(G) = E(G^2) - (E(G))^2 = 8/9.$$

$$H(G) = (3 + 0.05G)^2$$

$$H(\mu_G) = (3 + 0.05\mu_G)^2$$

$$H'(\mu_G) = (0.1)(3 + 0.05\mu_G)$$

$$H''(\mu_G) = 0.005.$$

Thus,

$$\mu_A \approx H(\mu_G) + \frac{1}{2}H''(\mu_G)\sigma_G^2 = H(8/3) + \frac{1}{2}(0.005)\frac{8}{9} = 9.82$$

and

$$\sigma_A^2 \approx (H'(\mu_G))^2\sigma_G^2 = 0.082.$$

3–13. (a) $F_Y(y) = P(4 - X^2 \le y) = P(X^2 \ge 4 - y)$

$$= P(X \le -\sqrt{4-y}) + P(X \ge \sqrt{4-y})$$

$$= 0 + \int_{\sqrt{4-y}}^2 dx = 2 - \sqrt{4-y}$$

$$f_Y(y) = \frac{1}{2}(4-y)^{-1/2}; 0 \le y \le 3$$

$$= 0; \text{otherwise}$$

(b) $F_Y(y) = P(e^x \le y) = P(X \le \ell\text{n}(y))$

$$= \int_1^{\ell\text{n}(y)} dx = \ell\text{n}(y) - 1$$

$$f_Y(y) = \frac{1}{y}; e^1 \le y \le e^2$$

$$= 0; \text{otherwise}$$

3–14.
$$y = \frac{3}{(1+x)^2}$$
$$x = \sqrt{3}y^{-1/2} - 1$$
$$\frac{dx}{dy} = -\frac{\sqrt{3}}{2}y^{-3/2}$$
$$\left|\frac{dx}{dy}\right| = \frac{\sqrt{3}}{2}y^{-3/2}$$
$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$
$$= f_X(\sqrt{3}y^{-1/2} - 1)\frac{\sqrt{3}}{2}y^{-3/2}$$
$$= \exp[-\sqrt{3}y^{-1/2} + 1]\frac{\sqrt{3}}{2}y^{-3/2}; \quad 0 \le y \le 3$$

3–15. With equal probability $p_X(1) = \cdots = p_X(6) = \frac{1}{6}$,

$$M_X(t) = \sum_{x=1}^{6}\left(\frac{1}{6}\right)e^{tx}$$
$$E(X) = M_X'(0) = \frac{7}{2}$$
$$V(X) = M_X''(0) - [M_X'(0)]^2 = \frac{35}{12}$$

3–16. (a) Using the substitution $y = bx^2$, we obtain

$$\begin{aligned} 1 &= \int_0^\infty ax^2 e^{-bx^2}\,dx \\ &= \int_0^\infty a\,\frac{y}{b}\,e^{-y}\frac{dy}{2\sqrt{by}} \\ &= \frac{a}{2b^{3/2}}\int_0^\infty y^{(3/2)-1}e^{-y}\,dy \\ &= \frac{a}{2b^{3/2}}\,\Gamma(\frac{3}{2}) \;=\; \frac{a}{2b^{3/2}}\,\frac{1}{2}\,\Gamma(\frac{1}{2}) \;=\; \frac{a\sqrt{\pi}}{4b^{3/2}} \end{aligned}$$

Thus,

$$a \;=\; \frac{4b^{3/2}}{\sqrt{\pi}}.$$

(b) Similarly,

$$\mu_X \;=\; E(X) \;=\; \int_0^\infty ax^3 e^{-bx^2}\,dx \;=\; \frac{a}{2b^2}\,\Gamma(2) \;=\; \frac{2}{\sqrt{\pi b}}$$

and

$$E(X^2) \;=\; \frac{a}{2b^{5/2}}\,\Gamma\left(\frac{5}{2}\right) \;=\; \frac{3}{2b}.$$

These facts imply that

$$\sigma_X^2 \;=\; E(X^2) - [E(X)]^2 \;=\; \frac{3}{2b} - \frac{4}{\pi b} \;=\; \frac{3\pi - 8}{2\pi b}.$$

Now we have

$$\begin{aligned} H(x) &= 18x^2 \\ H(\mu_X) &= 18\mu_X^2 \;=\; 18 \cdot \frac{4}{\pi b} \;=\; \frac{72}{\pi b} \\ H'(\mu_X) &= 36\mu_X \;=\; 36 \cdot \frac{2}{\sqrt{\pi b}} \;=\; \frac{72}{\sqrt{\pi b}} \\ H''(\mu_X) &= 36 \end{aligned}$$

Then

$$\begin{aligned} \mu_Y &\doteq H(\mu_X) + \frac{1}{2}H''(\mu_X)\cdot\sigma_X^2 \\ &= \frac{72}{\pi b} + \frac{1}{2}\cdot 36 \cdot \frac{3\pi - 8}{2\pi b} \;=\; \frac{27}{b} \end{aligned}$$

and

$$\begin{aligned} \sigma_Y^2 &\doteq (H'(\mu_X))^2\sigma_X^2 \\ &= \left(\frac{72}{\sqrt{\pi b}}\right)^2\left(\frac{3\pi - 8}{2\pi b}\right) \;=\; \frac{2592(3\pi - 8)}{\pi^2 b^2} \end{aligned}$$

3–17. $E(Y) = 1, V(Y) = 1 \qquad H(Y) = \sqrt{Y^2 + 36}$

$$E(X) \cong H(\mu_Y) + \frac{1}{2}H''(\mu_Y)\sigma_Y^2$$

$$V(X) \cong [H'(\mu_Y)]^2\sigma_Y^2$$

$$H'(y) = y(y^2 + 36)^{-1/2}, \qquad H''(y) = 36(y^2 + 36)^{-3/2}$$

$$E(X) \cong (37)^{1/2} + \frac{1}{2}(36)(37)^{-3/2} \;\doteq\; 6.16$$

$$V(X) \cong (37)^{-1} = 0.027027$$

3–18. $E(P) = \int_0^1 (1+3r)6r(1-r)\,dr = \int_0^1 (6r+12r^2-18r^3)\,dr \;=\; \frac{5}{2}$

Since $H(r) = 1+3r$ is increasing,

$$f_P(p) = 6\left(\frac{p-1}{3}\right)\left(1-\frac{p-1}{3}\right)\left(\frac{1}{3}\right) = \frac{2}{9}(5p-p^2-4); 1<p<4$$
$$= 0; \text{otherwise}$$

3–19. $M_X(t) = \int_0^\infty e^{tx}\cdot 4xe^{-2x}\,dx$

$$= \int_0^\infty 4xe^{x(t-2)}\,dx = \left(1-\frac{t}{2}\right)^{-2}$$

$$E(X) = M_X'(0) = 1$$

$$V(X) = M_X''(0) - [M_X'(0)]^2 = \frac{1}{2}$$

3–20. $V = \dfrac{\pi\cdot X^2}{4}\cdot 1$

$$E(V) = \frac{\pi}{4}E(X^2) = \frac{\pi}{4}[V(X)+(E(X))^2]$$

$$= \frac{\pi}{4}\cdot[25\cdot 10^{-6}+2^2] = 3.14162$$

3–21. $M_Y(t) = E(e^{tY}) = E(e^{t(aX+b)})$

$$= e^{tb}E(e^{(at)X})$$

$$= e^{tb}M_X(at)$$

3–22. (a) $\int_0^1 k(1-x)^{a-1}x^{b-1}\,dx = 1.$

Let $\Gamma(p) = \int_0^\infty u^{p-1}e^{-u}\,du$ define the gamma function. Integrating by parts and applying L'Hôspital's Rule, it can be shown that $\Gamma(p+1) = p\cdot\Gamma(p)$.

Make a variable change $u = v^2$ so that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Working with $\Gamma(p)$, let $u = v^2$ so $du = 2v\,dv$ and

$$\Gamma(p) = 2\int_0^\infty (v^2)^{p-1}e^{-v^2}v\,dv \;=\; 2\int_0^\infty v^{2p-1}e^{-v^2}\,dv.$$

$$\begin{aligned}\text{Then } \Gamma(a)\cdot\Gamma(b) &= 4\int_0^\infty s^{2a-1}e^{-s^2}\,ds\int_0^\infty t^{2b-1}e^{-t^2}\,dt\\ &= 4\int_0^\infty\int_0^\infty s^{2a-1}t^{2b-1}e^{-(s^2+t^2)}\,ds\,dt\end{aligned}$$

Let $s = \rho\cos\theta$, $t = \rho\sin\theta$, so the Jacobian $= \rho$

$$\begin{aligned}\Gamma(a)\cdot\Gamma(b) &= 4\int_0^\infty\int_0^{\pi/2}(\rho\cos\theta)^{2a-1}(\rho\sin\theta)^{2b-1}e^{-\rho^2}\rho\,d\theta\,d\rho\\ &= 4\int_0^\infty \rho^{2a+2b-1}e^{-\rho^2}\,d\rho\int_0^{\pi/2}(\cos\theta)^{2a-1}(\sin\theta)^{2b-1}\,d\theta\end{aligned}$$

Substitute $\rho^2 = y$ in the first integral and $\sin^2\theta = x$ in the second.

$$\begin{aligned}\Gamma(a)\Gamma(b) &= \int_0^\infty y^{a+b-1}e^{-y}\,dy\int_0^1 x^{b-1}(1-x)^{a-1}\,dx\\ &= \Gamma(a+b)\int_0^1 x^{b-1}(1-x)^{a-1}\,dx\end{aligned}$$

so $\displaystyle\int_0^1 x^{b-1}(1-x)^{a-1}\,dx = \frac{\Gamma(a)\cdot\Gamma(b)}{\Gamma(a+b)} \Rightarrow k = \frac{\Gamma(a+b)}{\Gamma(a)\cdot\Gamma(b)}$

(b) $E(X^k) = \dfrac{\Gamma(a+b)}{\Gamma(a)\cdot\Gamma(b)} \displaystyle\int_0^1 x^{b+k-1}(1-x)^{a-1}$

$= \dfrac{\Gamma(a+b)}{\Gamma(a)\cdot\Gamma(b)} \displaystyle\int_0^1 x^{(b+k)-1}(1-x)^{a-1}\,dx$

$= \dfrac{\Gamma(a+b)}{\Gamma(a)\cdot\Gamma(b)} \cdot \dfrac{\Gamma(a)\Gamma(b+k)}{\Gamma(a+b+k)}.$

Since $\Gamma(1) = 1$ we obtain $\Gamma(p+1) = p!$ for p a positive integer.

Then $E(X) = \dfrac{(a+b-1)!}{(a-1)!(b-1)!} = \dfrac{b}{a+b}$

$E(X^2) = \dfrac{[(a+b)-1]!}{(a-1)!(b-1)!} \cdot \dfrac{(a-1)!(b+1)!}{(a+b+1)!} = \dfrac{b(b+1)}{(a+b+1)(a+b)}$

so $V(X) = \dfrac{ab}{(a+b)^2(a+b+1)}$

3–23. $M_X(t) = \dfrac{1}{2} + \dfrac{1}{4}e^t + \dfrac{1}{8}e^{2t} + \dfrac{1}{8}e^{3t}$

$M_X'(t) = \dfrac{1}{4}e^t + \dfrac{1}{4}e^{2t} + \dfrac{3}{8}e^{3t}$

$M_X''(t) = \dfrac{1}{4}e^t + \dfrac{1}{2}e^{2t} + \dfrac{9}{8}e^{3t}$

(a) $E(X) = M_X'(0) = \dfrac{7}{8}$

$V(Y) = M_X''(0) - [M_X'(0)]^2 = \dfrac{15}{8} - \left(\dfrac{7}{8}\right)^2 = \dfrac{71}{64}$

(b)

x	y	y	$p_Y(y)$
0	4	0	$\frac{1}{8}$
1	1	1	$\frac{3}{8}$
2	0	4	$\frac{1}{2}$
3	1	*ow*	0
ow	0		

$$\begin{aligned} F_Y(y) &= 0; y < 0 \\ &= \frac{1}{8}; 0 \le y < 1 \\ &= \frac{1}{2}; 1 \le y < 4 \\ &= 1; y \ge 4 \end{aligned}$$

3–24. $\mu_3 = E(X - \mu_1')^3$

$= E(X^3) - 3\mu_1' \cdot E(X^2) + 3\mu_1' \cdot E(X) - (\mu_1')^3$

$= \mu_3' - 3\mu_1'\mu_2' + 2(\mu_1')^3.$

A r.v. is symmetric about a if $f(x + a) = f(-x + a)$ and we assume X continuous. Since $X - a$ and $a - X$ have the same p.d.f., $E(X - a) = E(a - X)$ or $E(X) - a = a - E(X)$, so $E(X) = a = \mu_1'$. Now,

$$\begin{aligned} E(X - \mu_1')^r = E(X - a)^r &= \int_{-\infty}^{\infty} (x - a)^r f(x)\, dx \\ &= \int_{-\infty}^{a} (x - a)^r f(x)\, dx + \int_{0}^{\infty} (x - a)^r f(x)\, dx \\ &= \int_{-\infty}^{0} y^r f(y + a)\, dy + \int_{0}^{\infty} y^r f(y + a)\, dy \\ &= -\int_{-\infty}^{0} (1 - y)^r f(-y + a)\, dy + \int_{0}^{\infty} y^r f(y + a)\, dy \\ &= \int_{0}^{\infty} (-y)^r f(y + a)\, dy + \int_{0}^{\infty} y^r f(y + a)\, dy \\ &= (-1)^r \int_{0}^{\infty} y^r f(y + a)\, dy + \int_{0}^{\infty} y^r f(y + a)\, dy \\ &= 0 \text{ for odd } r. \end{aligned}$$

Thus $E(X - \mu_1')^3 = 0$.

3–25. If $\int_{-\infty}^{\infty} |x|^r f(x)\,dx = k < \infty$, then $\int_{-\infty}^{\infty} |x|^n f(x)\,dx < \infty$, where $0 \le n < r$.

Proof: If $|x| \le 1$, then $|x|^n \le 1$ and if $|x| > 1$, then $|x|^n \le |x|^r$.

Then

$$\int_{-\infty}^{\infty} |x|^n f(x)\,dx = \int_{|x|\le 1} |x|^n \cdot f(x)\,dx + \int_{|x|>1} |x|^n \cdot f(x)\,dx$$
$$\le \int_{|x|\le 1} 1 f(x)\,dx + \int_{|x|>1} |x|^r f(x)\,dx \le 1 + k < \infty$$

3–26. $$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right) \Rightarrow \psi_X(t) = \mu t + \frac{\sigma^2 t^2}{2}$$

$$d\psi_X(t)/dt|_{t=0} = [\mu + \sigma^2 t]_{t=0} = \mu$$
$$d^2\psi_X(t)\,dt^2|_{t=0} = \sigma^2|_{t=0} = \sigma^2$$
$$d^r\psi_X(t)/dt^r|_{t=0} = 0; r \ge 3$$

3–27. From Table XV, using the first column with scaling: $u_1 = 0.10480$, $u_2 = 0.22368$, $u_3 = 0.24130$, $u_4 = 0.42167$, $u_5 = 0.37570$, $u_6 = 0.77921, \ldots, u_{20} = 0.07056$.

$$F_X(x) = 0; \quad x < 0$$
$$= 0.5; \quad 0 \le x < 1$$
$$= 0.75; \quad 1 \le x < 2$$
$$= 0.875; \quad 2 \le x < 3$$
$$= 1; \quad x \ge 3$$

So, $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$, $x_5 = 0$, $x_6 = 2, \ldots, x_{20} = 0$

3–28. The c.d.f. is

$$F_T(t) \;=\; 1 - e^{-t/4}$$

Thus, we set

$$t \;=\; -4\ell\mathrm{n}(1 - F_T(t)) \;=\; -4\ell\mathrm{n}(1-u)$$

From Table XV, using the second column with scaling: $u_1 = 0.15011$, $u_2 = 0.46573$, $u_3 = 0.48360, \ldots, u_{10} = 0.36257$

So,

$$\begin{aligned} t_1 &= -4\ell\mathrm{n}(0.84989) = 0.6506 \\ t_2 &= -4\ell\mathrm{n}(0.53427) = 2.5074 \\ &\vdots \\ t_{10} &= -4\ell\mathrm{n}(0.63143) = 1.8391 \end{aligned}$$

Chapter 4

4–1. (a)

x	0	1	2	3	4	5
$p_X(x)$	27/50	11/50	6/50	3/50	2/50	1/50

y	0	1	2	3	4
$p_Y(y)$	20/50	15/50	10/50	4/50	1/50

(b)

y	0	1	2	3	4
$p_{Y\|0}(y)$	11/27	8/27	4/27	3/27	1/27

(c)

x	0	1	2	3	4	5
$p_{X\|0}(x)$	11/20	4/20	2/20	1/20	1/20	1/20

4–2. (a)

y	1	2	3	4	5	6	7	8	9
$p_Y(y)$	44/100	26/100	12/100	8/100	4/100	2/100	2/100	1/100	1/100

x	1	2	3	4	5	6
$p_X(x)$	26/100	21/100	17/100	15/100	11/100	10/100

(b)

	y	1	2	3	4	5	6	7	8	9
$x=1$	$p_{Y\|1}(y)$	10/26	6/26	3/26	2/26	1/26	1/26	1/26	1/26	1/26
$x=2$	$p_{Y\|2}(y)$	8/21	5/21	3/21	2/21	1/21	1/21	1/21	0	0
$x=3$	$p_{Y\|3}(y)$	8/17	5/17	2/17	1/17	1/17	0	0	0	0
$x=4$	$p_{Y\|4}(y)$	7/15	4/15	2/15	1/15	1/15	0	0	0	0
$x=5$	$p_{Y\|5}(y)$	6/11	3/11	1/11	1/11	0	0	0	0	0
$x=6$	$p_{Y\|6}(y)$	5/10	3/10	1/10	1/10	0	0	0	0	0

4–3. (a) $\displaystyle\int_0^{10}\int_0^{100} \frac{k}{1000}\,dx_1\,dx_2 = 1 \Rightarrow k = 1$

(b) $\displaystyle f_{X_1}(x_1) = \int_0^{10} \frac{1}{1000}\,dx_2 = \frac{1}{100}; \quad 0 \le x_1 \le 100$

$= 0; \quad$ otherwise

$\displaystyle f_{X_2}(x_2) = \int_0^{100} \frac{1}{1000}\,dx_1 = \frac{1}{1000}x_1\Big|_0^{100} = \frac{1}{10}; \quad 0 \le x_2 \le 10$

$= 0; \quad$ otherwise

(c) $F_{X_1,X_2}(x_1,x_2) = 0; x_1 \le 0, x_2 \le 0$

$= 0; x_1 \le 0, x_2 > 0$

$= 0; x_1 > 0, x_2 \le 0$

$= \displaystyle\int_0^{x_1}\int_0^{x_2} \frac{1}{1000}\,dt_2\,dt_1 = \frac{x_1 x_2}{1000}; 0 < x_1 < 100, 0 < x_2 < 10$

$= \displaystyle\int_0^{x_1}\int_0^{10} \frac{dt_2\,dt_1}{1000} = \frac{x_1}{100}; 0 < x_1 < 100, x_2 \ge 10$

$= \displaystyle\int_0^{x_2}\int_0^{100} \frac{dt_1\,dt_2}{1000} = \frac{x_2}{10}; x_1 \ge 100, 0 < x_2 < 10$

$= 1; x_1 \ge 100, x_2 \ge 10.$

4–4. (a) $k\displaystyle\int_2^4\int_0^2 (6 - x_1 - x_2)\,dx_1\,dx_2 = 1 \Rightarrow k = \frac{1}{8}$

(b) $P(X_1 < 1, X_2 < 3) = \displaystyle\int_2^3\int_0^1 \frac{1}{8}(6 - x_1 - x_2)\,dx_1\,dx_2 = \frac{3}{8}$

(c) $P(X_1 + X_2 \le 4) = \dfrac{1}{8}\displaystyle\int_2^4\int_0^{4-x_2} (6 - x_1 - x_2)\,dx_1\,dx_2 = \frac{2}{3}$

(d) $f_{X_1}(x_1) = \dfrac{1}{8}\displaystyle\int_2^4 (6 - x_1 - x_2)\,dx_2 = \frac{1}{8}(6 - 2x_1); 0 \le x_1 \le 2$

$P(X_1 < 1.5) = \displaystyle\int_0^{3/2} \frac{1}{8}(6 - 2x_1)\,dx_1 \doteq 0.844$

(e) $f_{X_1}(x_1)$ see (d)

$f_{X_2}(x_2) = \displaystyle\int_0^2 \frac{1}{8}(6 - x_1 - x_2)\,dx_1 = \frac{1}{8}(10 - 2x_2); 2 \le x_2 \le 4$

4–5. (a) $P\left(W \le \dfrac{2}{3}, Y \le \dfrac{1}{2}\right) = \displaystyle\int_0^{2/3}\int_0^1\int_0^{1/2}\int_0^1 16wxyz\,dz\,dy\,dx\,dw = \frac{1}{9}$

(b) $P\left(X \le \dfrac{1}{2}, Z \le \dfrac{1}{4}\right) = \displaystyle\int_0^1\int_0^{1/2}\int_0^1\int_0^{1/4} 16wxyz\,dz\,dy\,dx\,dw = \frac{1}{64}$

(c) $\displaystyle\int_0^1\int_0^1\int_0^1 16wxyz\,dz\,dy\,dx = 2w; 0 \le w \le 1$

$= 0;$ otherwise

Note: x, y, z and w are independent

4–6. From Problem 4–4

$$f_X(x) = \frac{1}{8}(6-2x); 0 \le x \le 2$$

$$f_Y(y) = \frac{1}{8}(10-2y); 2 \le y \le 4$$

$$f_{X|y}(y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{6-x-y}{10-2y}; 0 \le x \le 2, 2 \le y \le 4$$

$$f_{Y|x}(x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{6-x-y}{6-2x}; 2 \le y \le 4, 0 \le x \le 2$$

4–7. $E(Y|X=3) = \sum_{y=1}^{9} y \cdot p_{Y|3}(y) = \frac{33}{17}$

4–8. (a)

x_1	51	52	53	54	55
$p_{X_1}(x_1)$	0.28	0.28	0.22	0.09	0.13
x_2	51	52	53	54	55
$p_{X_2}(x_2)$	0.18	0.15	0.35	0.12	0.20

(b)

x_2	51	52	53	54	55	otherwise
$P_{X_2\|51}(x_2)$	$\frac{6}{28}$	$\frac{7}{28}$	$\frac{5}{28}$	$\frac{5}{28}$	$\frac{5}{28}$	0
$P_{X_2\|52}(x_2)$	$\frac{5}{28}$	$\frac{5}{28}$	$\frac{10}{28}$	$\frac{2}{28}$	$\frac{6}{28}$	0
$P_{X_2\|53}(x_2)$	$\frac{5}{22}$	$\frac{1}{22}$	$\frac{10}{22}$	$\frac{1}{22}$	$\frac{5}{22}$	0
$P_{X_2\|54}(x_2)$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{5}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	0
$P_{X_2\|55}(x_2)$	$\frac{1}{13}$	$\frac{1}{13}$	$\frac{5}{13}$	$\frac{3}{13}$	$\frac{3}{13}$	0

$$E(X_2|x_1=51) = \sum_{x_2=51}^{55} x_2 \cdot p_{X_2|51}(x_2) = 52.857$$

$$E(X_2|x_1=52) = \sum_{x_2=51}^{55} x_2 \cdot p_{X_2|52}(x_2) = 52.964$$

$$E(X_2|x_1=53) = \sum_{x_2=51}^{55} x_2 \cdot p_{X_2|53}(x_2) = 53.0$$

$$E(X_2|x_1 = 54) = \sum_{x_2=51}^{55} x_2 \cdot p_{X_2|54}(x_2) = 53.0$$

$$E(X_2|x_1 = 55) = \sum_{x_2=51}^{55} x_2 \cdot p_{X_2|55}(x_2) = 53.461$$

4–9. $$f_{X_1}(x_1) = \int_0^1 6x_1^2 x_2 \, dx_2 = 3x_1^2 x_2^2|_{x_2=0}^1 = 3x_1^2; 0 \le x_1 \le 1$$

$$f_{X_2}(x_2) = \int_0^1 6x_1^2 x_2 \, dx_1 = 2x_1^3 x_2|_{x_1=0}^1 = 2x_2; 0 \le x_2 \le 1$$

$$f_{X_1|x_2}(x_1) = \frac{f_{X_1,X_2}(x_1,x_2)}{f_{X_2}(x_2)} = \frac{6x_1^2 x_2}{2x_2} = 3x_1^2; 0 \le x_1 \le 1$$

$$f_{X_2|x_1}(x_2) = \frac{f_{X_1,X_2}(x_1,x_2)}{f_{X_1}(x_1)} = 2x_2; 0 \le x_2 \le 1$$

$$E(X_1|x_2) = \int_0^1 x_1 f_{X_1|x_2}(x_1) \, dx_1 = \frac{3}{4}$$

$$E(X_2|x_1) = \int_0^1 x_2 f_{X_2|x_1}(x_2) \, dx_2 = \frac{2}{3}$$

4–10. (a) $f_{X_1}(x_1) = 2x_1 e^{-x_1^2}; \quad x_1 \ge 0$

$f_{X_2}(x_2) = 2x_2 e^{-x_2^2}; \quad x_2 \ge 0$

(b) $$f_{X_1|x_2}(x_1) = \frac{f_{X_1,X_2}(x_1,x_2)}{f_{X_2}(x_2)} = 2x_1 e^{-x_1^2}; x_1 \ge 0$$

$f_{X_2|x_1}(x_2) = 2x_2 e^{-x_2^2}; x_2 \ge 0$

Since $f_{X_1|x_2}(x_1) = f_{X_1}(x_1)$ and $f_{X_2|x_1}(x_2) = f_{X_2}(x_2)$, X_1 and X_2 are independent

(c) $$E(X_1|x_2) = E(X_1) = \int_0^\infty x_1(2x_1 e^{-x_1^2}) \, dx_1 = \int_0^\infty 2x_1^2 e^{-x_1^2} \, dx_1$$

Let $u = x_1$, $du = dx_1$, $dv = 2x_1 e^{-x_1^2}$, $v = -e^{-x_1^2}$, and integrate by parts, so

$$E(X_1) = -x_1 e^{-x_1^2}|_0^\infty + \int_0^\infty e^{-x_1^2} \, dx_1 = 0 + \frac{\pi}{4}$$

Similarly, $E(X_2) = \pi/4$.

4–11. $z = xy,\ t = x \Rightarrow x = t,\ y = \dfrac{z}{t}$

$$\begin{vmatrix} \partial x/\partial t & \partial x/\partial z \\ \partial y/\partial t & \partial y/\partial z \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -\dfrac{z}{t^2} & \dfrac{1}{t} \end{vmatrix} = \frac{1}{t}$$

so $p(z,t) = g(t)h(z/t)|1/t|$

Thus $\ell(z) = \int_{-\infty}^{\infty} g(t)h(z/t)|1/t|\,dt$

4–12. $a = s_1 \cdot s_2,\ t = s_1 \quad \text{so} \quad s_2 = \dfrac{a}{t},\ s_1 = t$

Jacobian is $\frac{1}{t}$

And $p(a,t) = (2t)(\frac{1}{8})(\frac{a}{t}) \cdot \frac{1}{t} = \frac{a}{4t}$

$$\therefore \ell(a) = \int_{a/4}^{1} \frac{a}{4t} dt = \frac{a}{4}\,\ell\mathrm{n}(t)\,\Big|_{a/4}^{1}$$

$$= -\frac{a}{4}\,\ell\mathrm{n}\,\frac{a}{4}; 0 < a < 4$$

4–13. $f(x,y) = g(x) \cdot h(y)$

$Z = \dfrac{X}{Y} \quad U = Y \quad y = u \quad x = uz$

$$\begin{vmatrix} \partial x/\partial u & \partial x/\partial z \\ \partial y/\partial u & \partial y/\partial z \end{vmatrix} = \begin{vmatrix} z & u \\ 1 & 0 \end{vmatrix} = u$$

$p(z,u) = g(uz) \cdot h(u)|u|$

$$\ell(z) = \int_{-\infty}^{\infty} g(uz)h(u)|u|\,du$$

4–14. $f(v,i) = 3e^{-3i} \cdot e^{-v}$

$r = \dfrac{v}{i} \quad u = i \quad v = ru \quad i = u$

The Jacobian is u, thus

$$p(r,u) = (e^{-ru})(3e^{-3u})(u) = 3ue^{-u(3+r)}; u \geq 0$$

$$\therefore \ell(r) = 3\int_0^{\infty} ue^{-u(3+r)}\,du = \frac{3}{(3+r)^2}; r \geq 0$$

4–15. $Y = X_1 + X_2 + X_3 + X_4$

$E(Y) = 80, V(Y) = 4(9) = 36$

4–16. If X_1 and X_2 are independent, then

$$p_{X_1,X_2}(x_1, x_2) = p_{X_1}(x_1)p_{X_2}(x_2) \quad \text{for all } x_1, x_2.$$

Thus,

$$p_{X_2|x_1}(x_2) = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_1}(x_1)} = \frac{p_{X_1}(x_1)p_{X_2}(x_2)}{p_{X_1}(x_1)} = p_{X_2}(x_2) \quad \text{for all } x_1, x_2.$$

Similarly,

$$p_{X_1|x_2}(x_1) = p_{X_1}(x_1).$$

Now prove the converse. Assume that

$$p_{X_2|x_1}(x_2) = p_{X_2}(x_2) \quad \text{and} \quad p_{X_1|x_2}(x_1) = p_{X_1}(x_1) \quad \text{for all } x_1, x_2.$$

Then

$$p_{X_1,X_2}(x_1, x_2) = p_{X_2|x_1}(x_2)p_{X_1}(x_1) = p_{X_2}(x_2)p_{X_1}(x_1) \quad \text{for all } x_1, x_2.$$

4–17. $X_2 = A + BX_1 \Rightarrow E(X_2) = A + B \cdot E(X_1)$

$V(X_2) = B^2 \cdot V(X_1), E(X_1X_2) = E(X_1(A + BX_1))$

$$= AE(X_1) + BE(X_1^2)$$

So

$$\rho^2 = \frac{[E(X_1X_2) - E(X_1)E(X_2)]^2}{V(X_1) \cdot V(X_2)}$$

$$= \frac{[AE(X_1) + BE(X_1^2) - E(X_1)(A + B \cdot E(X_1))]^2}{V(X_1) \cdot B^2V(X_1)}$$

$$= \frac{[BE(X_1^2) - B(E(X_1))^2]^2}{B^2V(X_1)^2} = \frac{B^2V(X_1)^2}{B^2V(X_1)^2} = 1$$

$\rho = -1$ if $B < 0$ and $\rho = 1$ if $B > 0$.

4–18. $M_{X_2}(t) = E(e^{tX_2}) = E(e^{t(A+BX_1)}) = E(e^{At} \cdot e^{BtX_1})$
$= e^{At} \cdot E(e^{BtX_1}) = e^{At} \cdot M_{X_1}(Bt)$

4–19. $f_{X_1}(x_1) = \int_{x_1}^{1} 2\,dx_2 = 2(1 - x_1); 0 \le x_1 \le 1$

$f_{X_2}(x_2) = \int_{0}^{x_2} 2\,dx_1 = 2x_2; 0 \le x_2 \le 1$

$E(X_1) = 2\int_0^1 x_1(1 - x_1)\,dx_1 = \frac{1}{3}$

$E(X_2) = 2\int_0^1 x_2^2\,dx_2 = \frac{2}{3}$

$E(X_1^2) = 2\int_0^1 x_1^2(1 - x_1)\,dx_1 = 2\left[\frac{x_1^3}{3} - \frac{x_1^4}{4}\right]_0^1 = \frac{1}{6}$

$E(X_2^2) = 2\int_0^1 x_2^3\,dx_2 = \frac{1}{2}$

$V(X_1) = E(X_1^2) - (E(X_1))^2 = \frac{1}{18}$

$V(X_2) = E(X_2^2) - (E(X_2))^2 = \frac{1}{18}$

$E(X_1X_2) = \int_0^1 \int_0^{1/2} 2x_1x_2\,dx_1\,dx_2 = 2\int_0^1 x_2\left[\frac{x_1^2}{2}\right]_0^{x_2} dx_2$

$= \int_0^1 x_2^3\,dx_2 = \frac{1}{4}$

$\rho = \frac{E(X_1X_2) - E(X_1)E(X_2)}{\sqrt{V(X_1)V(X_2)}}$

$= \frac{\frac{1}{4} - \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)}{\sqrt{\left(\frac{1}{18}\right)\left(\frac{1}{18}\right)}} = \frac{1}{2}$

4–20. $E(U) = A + BE(X_1), E(V) = C + DE(X_2)$

$V(U) = B^2 \cdot V(X_1), V(V) = D^2V(X_2)$

$E(UV) = E[(A + BX_1)(C + DX_2)]$

$= AC + AD \cdot E(X_2) + BC \cdot E(X_1) + BD \cdot E(X_1X_2)$

$$\rho_{U,V} = \frac{E(UV) - E(U) \cdot E(V)}{\sqrt{V(U) \cdot V(V)}}$$

$$= \frac{AC + AD \cdot E(X_2) + BC \cdot E(X_1) + BD \cdot E(X_1X_2) - [(A + BE(X_1))(C + DE(X_2))]}{\sqrt{B^2V(X_1)D^2V(X_2)}}$$

$$= \frac{BD[E(X_1X_2) - E(X_1)E(X_2)]}{|BD|\sqrt{V(X_1)V(X_2)}} = \frac{BD}{|BD|}\rho_{X_1,X_2}$$

4–21. X and Y are not independent. Note

(a) $P_{X,Y}(5, 4) = 0 \neq p_X(5) \cdot p_Y(4) = \frac{1}{50^2}$.

(b) $E(X) = 0.9$ $\quad E(Y) = 1.02$

$E(X^2) = 2.38$ $\quad E(Y^2) = 2.14$

$V(X) = 1.57$ $\quad V(Y) = 1.0996$

$E(XY) = 0.74$

$$\rho = \frac{E(XY) - E(X)E(Y)}{\sqrt{V(X)V(Y)}} = \frac{0.74 - (0.9)(1.02)}{\sqrt{(1.57)(1.0996)}} = -0.135$$

4–22. (a) A sale will take place if $Y \geq X$

(b) $\iint\limits_{x \leq y} f(x, y)\, dx\, dy$

(c) $E(Y|Y \geq x) = \iint\limits_{x \leq y} yf(x, y)\, dx\, dy \Big/ \iint\limits_{x \leq y} f(x, y)\, dx\, dy$

4–23. (a) $f_X(x) = \int_0^{\sqrt{1-x^2}} \frac{2}{\pi} dy = \frac{2}{\pi} y \Big|_0^{\sqrt{1-x^2}} = \frac{2}{\pi}\sqrt{1 - x^2}; -1 < x < +1$

$$f_Y(y) = \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \frac{2}{\pi} dx = \frac{4}{\pi}\sqrt{1 - y^2}; 0 < y < 1$$

(b) $f_{X|y}(x) = \dfrac{f_{X,Y}(x,y)}{f_Y(y)} = \dfrac{\dfrac{2}{\pi}}{\left(\dfrac{4}{\pi}\right)\sqrt{1-y^2}} = \dfrac{1}{2\sqrt{1-y^2}}; -\sqrt{1-y^2} < x < \sqrt{1-y^2}$

$$f_{Y|x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\dfrac{2}{\pi}}{\left(\dfrac{2}{\pi}\right)\sqrt{1-x^2}} = \frac{1}{\sqrt{1-x^2}}; 0 < y < \sqrt{1-x^2}$$

(c) $E(X|y) = \displaystyle\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \frac{x}{2\sqrt{1-y^2}} dx = \frac{1}{2\sqrt{1-y^2}} \left(\frac{x^2}{2} \Big|_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \right) = 0$

$$E(Y|x) = \int_0^{\sqrt{1-x^2}} \frac{y}{\sqrt{1-x^2}} dy = \frac{1}{\sqrt{1-x^2}} \left(\frac{y^2}{2} \Big|_0^{\sqrt{1-x^2}} \right) = \frac{\sqrt{1-x^2}}{2}$$

4–24. X and Y continuous

$$E(X|Y=y) = \int_{-\infty}^{\infty} x f_{X|y}(x)\, dx = \int_{-\infty}^{\infty} x \frac{f_{X,Y}(x,y)}{f_Y(y)} dx$$

$$= \int_{-\infty}^{\infty} \frac{f_X(x) f_Y(y)}{f_Y(y)} dx = \int_{-\infty}^{\infty} x f_X(x)\, dx = E(X)$$

Change X and Y, reversing roles to show $E(Y|X) = E(Y)$ if X and Y are independent.

4–25. $E(X|y) = \displaystyle\int_{-\infty}^{\infty} x f_{X|y}(x)\, dx = \int_{-\infty}^{\infty} x \frac{f_{X,Y}(x,y)}{f_Y(y)} dx$

$$E[E(X|Y)] = \int_{-\infty}^{\infty} E(X|y) \cdot f_Y(y)\, dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left[x \frac{f_{X,Y}(x,y)}{f_Y(y)} dx \right] f_Y(y)\, dy$$

$$= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dy \right] dx = \int_{-\infty}^{\infty} x f_X(x)\, dx = E(X).$$

Reverse roles of X and Y to show $E(E(Y|X)) = E(Y)$.

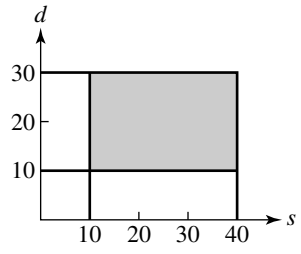
4–26. $w = s + d$, let $y = s - d$

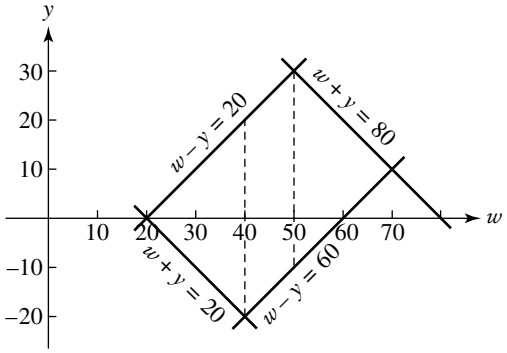
$$s = \frac{w+y}{2}, \quad d = \frac{w-y}{2}$$

$s = 10 \rightarrow w + y = 20$ $\qquad$ $d = 10 \rightarrow w - y = 20$

$s = 40 \rightarrow w + y = 80$ $\qquad$ $d = 30 \rightarrow w - y = 60$

As illustrated





The Jacobian $J = \begin{vmatrix} \frac{\partial s}{\partial w} & \frac{\partial s}{\partial y} \\ \frac{\partial d}{\partial w} & \frac{\partial d}{\partial y} \end{vmatrix} = \begin{vmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{vmatrix} = -\frac{1}{2}$

$$\therefore f_{W,Y}(w,y) = \frac{1}{30} \cdot \frac{1}{20} \cdot \frac{1}{2};\ 10 < \frac{w+y}{2} < 40,\ 10 < \frac{w-y}{2} < 30$$

$$f_W(w) = \int_{20-w}^{w-20} \frac{1}{1200} dy = \frac{w-20}{600}; 20 < w < 40$$

$$= \int_{w-60}^{w-20} \frac{1}{1200} dy = \frac{20}{600}; 40 < w \leq 50$$

$$= \int_{w-60}^{80-w} \frac{1}{1200} dy = \frac{70-w}{600}; 50 < w < 70$$

$$= 0; ow$$

4–27. (a) $\quad f_Y(y) = \int_0^1 (x+y)\, dx = y + \frac{1}{2}; 0 < y < 1$

$$f_{X|y}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{x+y}{y+\frac{1}{2}}; 0 < x < 1, 0 < y < 1$$

$$E(X|Y=y) = \int_0^1 x \cdot \frac{(x+y)}{\left(y+\frac{1}{2}\right)} dx = \frac{1}{y+\frac{1}{2}} \left[\frac{x^3}{3} + \frac{x^2 y}{2}\right]_0^1$$

$$= \frac{2+3y}{3(1+2y)}; 0 < y < 1.$$

(b) $f_X(x) = \int_0^1 (x+y)\, dy = \left[xy + \frac{y^2}{2}\right]_{y=0}^1 = x + \frac{1}{2}; 0 < x < 1$

$$E(X) = \int_0^1 [x^2 + (x/2)]\, dx = [(x^3/3) + (x^2/4)]_0^1 = \frac{7}{12}$$

(c) $E(Y) = \int_0^1 [y^2 + (y/2)]\, dy = [(y^3/3) + (y^2/4)]_0^1 = \frac{7}{12}$

4–28. (a) $\int_0^\infty \int_0^\infty \frac{k(1+x+y)}{(1+x)^4(1+y)^4} dx\, dy = 1 \Rightarrow k = \frac{9}{2}$

(b) $f_X(x) = \int_0^\infty \frac{(9/2)(1+x+y)\, dy}{(1+x)^4(1+y)^4} = \frac{(9/2)}{(1+x)^4} \int_0^\infty \frac{dy}{(1+y)^4}$

$$+ \frac{(9/2)x}{(1+x)^4} \int_0^\infty \frac{dy}{(1+y)^4} + \frac{(9/2)}{(1+x)^4} \int_0^\infty \frac{y\, dy}{(1+y)^4}$$

$$= \frac{3(3+2x)}{4(1+x)^4}; x > 0$$

4–29. (a) $k \int_0^\infty \int_0^\infty (1+x+y)^{-n}\, dx\, dy = 1, \quad k \int_0^\infty \left[\frac{(1+x+y)^{-n+1}}{(-n+1)}\right]_0^\infty dy = 1$

so $\frac{k}{n-1} \int_0^\infty \frac{dy}{(1+y)^{n-1}} = 1 \Rightarrow \frac{k}{(n-1)(n-2)} = 1 \Rightarrow k = (n-1)(n-2)$

(b) $F_{X,Y}(x,y) = P(X \le x, Y \le y) = k \int_0^x \int_0^y (1+s+t)^{-n}\, dt\, ds$

$$= k \left[\frac{(1+s+y)^{-n+2}}{(-n+2)(-n+1)} - \frac{(1+s)^{-n+2}}{(-n+2)(-n+1)}\right]_{s=0}^x$$

$$= 1 - (1+x)^{2-n} - (1+y)^{2-n} + (1+x+y)^{2-n}; x > 0, y > 0$$

4–30. From Eq. 4–58

$$n \geq \frac{p(1-p)}{\epsilon^2 \cdot \alpha} = \frac{(0.5)(0.5)}{(0.05)^2(0.05)} = 2000$$

4–31. (a) $g(x,y) = (2xe^{-x^2})(2ye^{-y^2})$ so X and Y are independent

(b) X and Y are not independent. For given X, Y may only assume values greater than that value of X.

(c) See Prob. 4–29 with $k = (4-1)(4-2) = 6$ and $n = 4$.

$$f_X(x) = 2(1+x)^{-3}; x > 0$$
$$f_Y(y) = 2(1+y)^{-3}; y > 0$$

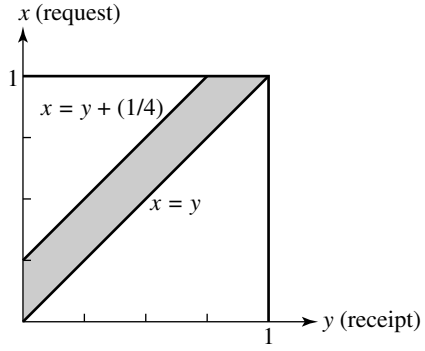
$f_{X,Y}(x,y) \neq f_X(x) \cdot f_Y(y)$ so the variables are not independent.

4–32. The probability is 2/3 since $P(X > Y > Z) = P(X < Y < Z) = \frac{1}{6}$

4–33. (a) $P\left(0 \leq X \leq \frac{1}{2}, 0 \leq Y \leq \frac{1}{2}\right) = \int_0^{1/2} \int_0^{1/2} 1\, dx\, dy = \frac{1}{4}$

(b) $P(X > Y) = \int_0^1 \int_0^x 1\, dy\, dx = \int_0^1 x\, dx = \frac{1}{2}.$

4–34. $P\left(0 \leq Y \leq 1, Y \leq X \leq Y + \frac{1}{4} \leq 1\right) = \frac{7}{32}$



4–35. (a) $F_Z(z) = P(Z \leq z) = P(a + bX \leq z) = P\left(X \leq \frac{z-a}{b}\right)$

$$= F_X\left(\frac{z-a}{b}\right)$$

$$f_Z(z) = f_X\left(\frac{z-a}{b}\right)\left|\frac{1}{b}\right|.$$

(b) $F_Z(z) = P\left(\frac{1}{X} \leq z\right) = P\left(X \geq \frac{1}{z}\right) = 1 - F_X\left(\frac{1}{z}\right)$

$f_Z(z) = f_X\left(\frac{1}{z}\right)\frac{1}{z^2}$

(c) $F_Z(z) = P(\ell\mathrm{n}(X) \leq Z) = P(e^{\ell n(x)} \leq e^z) = P(X \leq e^z) \;=\; F_X(e^z)$

$f_Z(z) = f_X(e^z)e^z$

(d) $F_Z(z) = P(e^X \leq z) = P(X \leq \ell\mathrm{n}\, z) = F_X(\ell\mathrm{n}\, z)$

$f_Z(z) = f_X(\ell\mathrm{n}(z))\left|\frac{1}{z}\right|$

The range space for Z is determined from the range space of X and the definition of the transformation.

Chapter 5

5–1. The probability mass function is

x	$p(x)$
0	$(1-p)^4$
1	$4p(1-p)^3$
2	$6p^2(1-p)^2$
3	$4p^3(1-p)$
4	p^4
otherwise	0

5–2.

$$\begin{aligned} P(X \geq 5) &= \sum_{x=5}^{6} \binom{6}{x} (0.95)^x (0.05)^{6-x} \\ &= 6(0.95)^5(0.05) + (0.95)^6 \\ &= 0.9672 \end{aligned}$$

5–3. Assume independence and let W represent the number of orders received.

$$\begin{aligned} P(W \geq 4) &= \sum_{w=4}^{12} \binom{12}{w} (0.5)^w (0.5)^{12-w} \\ &= (0.5)^{12} \sum_{w=4}^{12} \binom{12}{w} \\ &= 1 - (0.5)^{12} \sum_{w=0}^{3} \binom{12}{w} \;=\; 0.9270 \end{aligned}$$

5–4. Assume customer decisions are independent.

$$P(X \geq 10) \;=\; 1 - P(X \leq 9) \;=\; 1 - \sum_{x=0}^{9} \binom{20}{x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{20-x} \;=\; 0.0918$$

5–5.

$$P(X > 2) \;=\; 1 - P(X \leq 2) \;=\; 1 - \sum_{x=0}^{2} \binom{50}{x} (0.02)^x (0.98)^{50-x} \;=\; 0.0784.$$

5–6.

$$\begin{aligned} M_X(t) &= E[e^{tX}] = \sum_{x=0}^{n} e^{tx}\binom{n}{x}p^x(1-p)^{n-x} \\ &= (pe^t+q)^n, \quad \text{where } q=1-p \end{aligned}$$

$$E[X] = M_X'(0) = [n(pe^t+q)^{n-1}pe^t]_{t=0} = np$$

$$\begin{aligned} E[X^2] &= M_X''(0) = np[e^t(n-1)(pe^t+q)^{n-2}(pe^t)+(pe^t+q)^{n-1}e^t]_{t=0} \\ &= (np)^2 - np^2 + np \end{aligned}$$

$$V(X) = E[X^2]-(E[X])^2 = n^2p^2 - np^2 + np - n^2p^2 = np(1-p) = npq$$

5–7.

$$\begin{aligned} P(\hat{p}\le 0.03) &= P\left(\frac{X}{100}\le 0.03\right) = P(X\le 3) \\ &= \sum_{x=0}^{3}\binom{100}{3}(0.01)^x(0.99)^{100-x} = 0.9816 \end{aligned}$$

5–8.

$$\begin{aligned} P(\hat{p} > p+\sqrt{pq/n}) &= P\left(\hat{p} > 0.07+\sqrt{(0.07)(0.93)/200}\right) \\ &= P(X > 200(0.088)) \\ &= P(X > 17.6) \\ &= 1-P(X\le 17.6) \\ &= 1-\sum_{x=0}^{17}\binom{200}{x}(0.07)^x(0.93)^{200-x} \\ &= 0.1649 \end{aligned}$$

By two standard deviations,

$$\begin{aligned} P(\hat{p} > p + 2\sqrt{pq/n}) &= P(\hat{p} > 0.106) \\ &= P(X > 21.2) \\ &= 1 - P(X \leq 21) \\ &= 1 - \sum_{x=0}^{21} \binom{200}{x} (0.07)^x (0.93)^{200-x} \\ &= 0.0242 \end{aligned}$$

By three standard deviations,

$$\begin{aligned} P(\hat{p} > p + 3\sqrt{pq/n}) &= P(X > 24.8) \\ &= 1 - P(X \leq 24) \\ &= 1 - \sum_{x=0}^{24} \binom{200}{x} (0.07)^x (0.93)^{200-x} \\ &= 0.0036 \end{aligned}$$

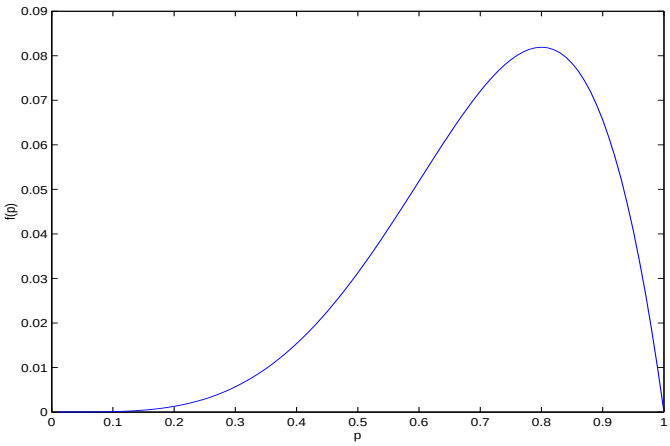
5–9. $P(X = 5) = (0.95)^4(0.05) = 0.0407$

5–10. A: Successful on first three calls, B: Unsuccessful on fourth call

$$P(B|A) = P(B) = 0.90 \quad \text{if } A \text{ and } B \text{ are independent}$$

5–11.

$$\begin{aligned} P(X = 5) &= p^4(1-p) = f(p) \\ \frac{df(p)}{dp} &= 5p^4 - 4p^3 = 0 \Rightarrow p = \frac{4}{5} \end{aligned}$$



5–12. (a) X = trials required up to and including the first sale

$$\begin{aligned} C(X) &= 1000X + 3000(X-1) \\ &= 4000X - 3000 \end{aligned}$$

$$E[C(X)] = 4000E[X] - 3000 = 4000\left(\frac{1}{0.1}\right) - 3000 = 37000$$

(b) Since \$37000 > \$15000, the trips should not be undertaken.

(c)

$$\begin{aligned} P(C(X) > 100000) &= P(4000X - 3000 > 100000) \\ &= P\left(X > \frac{103000}{4000}\right) \doteq P(X > 25.75) \\ &= 1 - P(X \leq 25) \\ &= 1 - \sum_{x=1}^{25} (0.90)^{x-1}(0.10) \end{aligned}$$

5–13.

$$M_X(t) = \frac{pe^t}{1 - qe^t}, \quad \text{where } q = 1 - p$$

$$\begin{aligned} E[X] &= M_X'(0) = \left[\frac{(1-qe^t)(pe^t) + (pe^t)(qe^t)}{(1-qe^t)^2}\right]_{t=0} \\ &= \frac{(1-q)p + pq}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p} \end{aligned}$$

$$\begin{aligned} E[X^2] &= M_X''(0) = \left[\frac{(1-qe^t)^2(pe^t) + 2pe^t(1-qe^t)(qe^t)}{(1-qe^t)^4}\right]_{t=0} \\ &= \frac{(1+q)p}{(1-q)^3} = \frac{1+q}{p^2} \end{aligned}$$

$$V(X) = \frac{q}{p^2}$$

5–14.

$$\begin{aligned} P(X \leq 2) &= P(X=1) + P(X=2) = 0.8 + (0.2)(0.8) = 0.96 \\ P(X \leq 3) &= \sum_{x=1}^{3} (0.2)^{x-1}(0.8) = 0.992 \end{aligned}$$

5–15.

$$P(X = 36) = (0.95)^{35}(0.05) = 0.0083$$

5–16. (a)

$$P(X = 8) = \binom{7}{2} (0.1)^2(0.9)^5 = 0.0124$$

(b)

$$P(X > 8) = \sum_{x=9}^{\infty} \binom{x-1}{2} (0.1)^2(0.9)^{x-3}$$

5–17.

$$P(X = 4) = \binom{3}{1} (0.8)^2(0.2)^2 = 0.0768$$

$$P(X < 4) = P(X=2) + P(X=3) = (0.8)^2 + \binom{2}{1} (0.8)^2(0.2) = 0.896$$

5–18. Suppose $X_1, X_2, \ldots, X_r$ are independent geometric random variables, each with parameter p. X_1 is the number of trials to first success, X_2 the number of trials from first to the second, etc. Let

$$X = X_1 + X_2 + \ldots + X_r$$

The moment generating function for the geometric is $\frac{pe^t}{1-qe^t}$, so

$$M_X(f) = \prod_{i=1}^{r} M_{X_i}(t) = \left[\frac{pe^t}{1-qe^t} \right]^r$$

$$E[X] = M_X'(t)|_{t=0} = \frac{r}{p}$$

We could also have obtained this result as follows.

$$E[X] = \sum_{i=1}^{r} E[X_i] = r\left(\frac{1}{p}\right) = \frac{r}{p}$$

Continuing,

$$V(X) = \left[M_X''(t)|_{t=0}\right] - \left(\frac{r}{p}\right)^2 = \frac{rq}{p^2}$$

We could also have obtained this result as follows.

$$V(X) = \sum_{i=1}^{r} V(X_i) = \frac{rq}{p^2}$$

5–19.

$$E[X] = \frac{r}{p} = \frac{5}{0.8} = 6.25, \quad V(X) = \frac{rq}{p^2} = \frac{(5)(0.2)}{(0.8)^2} = 1.5625$$

5–20. X = Mission number on which 4th hit occurs.

$$p(x) = \begin{cases} \binom{x-1}{3}(0.8)^4(0.2)^{x-4} & x = 4, 5, 6, \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$P(X \leq 7) = \sum_{x=4}^{7} \binom{x-1}{3}(0.8)^4(0.2)^{x-4}$$

5–21. $(X, Y, Z) \sim$ multinomial $(n = 3, p_1 = 0.4, p_2 = 0.3, p_3 = 0.3)$. The probability that one company receives all orders is

$$\begin{aligned} & P(3,0,0) + P(0,3,0) + P(0,0,3) \\ &= \frac{3!}{3!0!0!}(0.4)^3(0.3)^0(0.3)^0 + \frac{3!}{0!3!0!}(0.4)^0(0.3)^3(0.3)^0 + \frac{3!}{0!0!3!}(0.4)^0(0.3)^0(0.3)^3 \\ &= 0.4^3 + 0.3^3 + 0.3^3 \doteq 0.118 \end{aligned}$$

5–22. (a) $(X_1, X_2, X_3, X_4) \sim$ multinomial $(n = 5, p_1 = p_2 = p_3 = p_4 = \frac{1}{4})$. Therefore, $P(5,0,0,0) + P(0,5,0,0) + P(0,0,5,0) + P(0,0,0,5)$ is the probability that one company gets all five. That is,

$$4\left[\frac{5!}{5!0!0!0!}\right]\left(\frac{1}{4}\right)^5\left(\frac{1}{4}\right)^0\left(\frac{1}{4}\right)^0\left(\frac{1}{4}\right)^0 = \frac{1}{256}$$

(b)

$$1 - \left[4(60)\left(\frac{1}{4}\right)^5\right] \doteq 0.7656$$

5–23.

$$P(Y_1 = 4, Y_2 = 1, Y_3 = 3, Y_4 = 2) = \frac{10!}{4!1!3!2!}(0.2)^4(0.2)^1(0.2)^3(0.4)^2 \doteq 0.005$$

5–24.

$$P(Y_1 = 0, Y_2 = 0, Y_3 = 0, Y_4 = 10) = \frac{10!}{0!0!0!10!}(0.2)^0(0.2)^0(0.2)^0(0.4)^{10}$$

$$P(Y_1 = 5, Y_2 = 0, Y_3 = 0, Y_4 = 5) = \frac{10!}{5!0!0!5!}(0.2)^5(0.2)^0(0.2)^0(0.4)^5$$

5–25. (a)

$$P(X \le 2) = \sum_{x=0}^{2} \frac{\binom{4}{x}\binom{21}{5-x}}{\binom{24}{5}} \doteq 0.98$$

(b)

$$P(X \le 2) = \sum_{x=0}^{2} \binom{5}{x}\left(\frac{4}{25}\right)^x\left(\frac{21}{25}\right)^{5-x} \doteq 0.97$$

5–26. The approximation improves as $\frac{n}{N}$ decreases. $n = 5, N = 100$ is a better condition than $n = 5, N = 25$.

5–27. we want the smallest n such that

$$P(X \geq 1) \;=\; 1 - \frac{\binom{7}{0}\binom{18}{n}}{\binom{25}{n}} \;\geq\; 0.95 \qquad \Leftrightarrow \qquad \frac{\binom{18}{n}}{\binom{25}{n}} \;\leq\; 0.05.$$

By trial and error, we find that $n = 8$ does the job.

We could instead use the binomial approximation; now we want n such that

$$0.05 \;\geq\; \binom{n}{0}\left(\frac{7}{25}\right)^0\left(\frac{18}{25}\right)^n \;=\; \left(\frac{18}{25}\right)^n.$$

We find that $n \doteq 9$.

5–28.

$$M_X(t) \;=\; E[e^{tX}] \;=\; \sum_{x=0}^{\infty} e^{tx}\frac{e^{-c}c^x}{x!} \;=\; e^{-c}\sum_{x=0}^{\infty}\frac{(ce^t)^x}{x!} \;=\; e^{-c}e^{ce^t} \;=\; e^{c(e^t-1)}$$

5–29.

$$P(X < 10) \;=\; P(X \leq 9) \;=\; \sum_{x=0}^{9}\frac{e^{-25}(25)^x}{x!} \;\doteq\; 0.0002$$

5–30.

$$\begin{aligned} P(X > 20) = P(X \geq 21) \;&=\; \sum_{x=21}^{\infty}\frac{e^{-10}(10)^x}{x!} \\ &=\; 1 - P(X \leq 20) = 1 - \sum_{x=0}^{20}\frac{e^{-10}(10)^x}{x!} \\ &=\; 0.002 \end{aligned}$$

5–31.

$$\begin{aligned} P(X > 5) \;&=\; P(X \geq 6) \\ &=\; 1 - P(X \leq 5) = 1 - \sum_{x=0}^{5}\frac{e^{-4}4^x}{x!} \\ &\doteq\; 0.2149 \end{aligned}$$

5–32. Mean count rate $= (1-p)c$. Therefore,

$$P(Y_t = y) = \frac{e^{-[(1-p)c]t}[(1-p)ct]^y}{y!} \qquad y = 0, 1, 2, \ldots$$

5–33. Using a Poisson model,

$$\begin{aligned} P(X \le 3) &= \sum_{x=0}^{3} \frac{e^{-\lambda}\lambda^x}{x!} \qquad \lambda = 15000(0.002) = 30 \\ P(X \ge 5) &= \sum_{x=5}^{\infty} \frac{e^{-30}(30)^x}{x!} = 1 - \sum_{x=0}^{4} \frac{e^{-30}(30)^x}{x!} \end{aligned}$$

5–34. Y = Number of requests.

(a)
$$P(Y > 3) = 1 - P(Y \le 3) = 1 - \sum_{y=0}^{3} \frac{e^{-2}2^y}{y!}$$

(b)
$$E[Y] = c = 2$$

(c)
$$P(Y \le y) \ge 0.9 \quad \text{so } y = 4 \text{ and } P(Y \le 4) = 0.9473$$

(d) X = Number serviced.

y	x	$p(x)$	$xp(x)$
0	0	e^{-2}	0
1	1	$2e^{-2}$	$2e^{-2}$
2	2	$2e^{-2}$	$4e^{-2}$
3 or more	3	$1 - 5e^{-2}$	$3 - 15e^{-2}$

$$E[X] = 1.78$$

(e) Let M = number of crews going to central stores. Then $M = Y - X$

$$E[M] = E[Y] - E[X] = 2 - 1.78 = 0.22$$

5–35. Using a Poisson model,

$$P(X < 3) = P(X \le 2) = \sum_{x=0}^{2} \frac{e^{-2.5}(2.5)^x}{x!} \doteq 0.544$$

5–36. Let Y = No. Boarding
Let X = No. Recorded

Y	0	1	2	3	4	5	6	7	8	9	≥ 10
X	0	1	2	3	4	5	6	7	8	9	10

$$\begin{aligned} p_X(x) &= \frac{e^{-c}c^x}{x!}, \qquad x = 0, 1, 2, \ldots, 9 \\ &= \sum_{i=10}^{\infty} \frac{e^{-c}c^i}{i!} = 1 - \sum_{i=0}^{9} \frac{e^{-c}c^i}{i!}, \qquad x = 10 \end{aligned}$$

5–37. (a) Let X denote the number of errors on 50 pages. Then

$$X \sim \text{Binomial}(5, \frac{50}{200}) = \text{Binomial}(5, 1/4).$$

This implies that

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \binom{5}{0} (1/4)^0 (3/4)^5 = 0.763.$$

(b) Now $X \sim \text{Binomial}(5, \frac{n}{200})$, where n is the number of pages sampled.

We want the smallest n such that

$$\sum_{i=3}^{5} \binom{5}{i} \left(\frac{n}{200}\right)^i \left(\frac{200-n}{200}\right)^{5-i} \geq 0.90$$

By trial and error, we find that $n = 151$ does the trick.

We could also have done this problem using a Poisson approximation. For (a), we would use $\lambda = 0.025$ errors / page with 50 pages. Then $c = 50(0.025) = 1.25$, and we would eventually obtain $P(X \geq 1) = 1 - \frac{e^{-1.25}(1.25)^0}{0!} \doteq 0.7135$, which is a bit off of our exact answer. For (b), we would take $c = n(0.025)$, eventually yielding $n = 160$ after trial and error.

5–38.

$$P(X = 0) = \frac{e^{-c}c^0}{0!} \quad \text{with } c = 10000(0.0001) = 1,$$

$$P(X = 0) = e^{-1} = 0.3679$$

and

$$P(X \geq 2) = 1 - P(X \leq 1) = 0.265$$

5–39. X $\sim$ Poisson with $\alpha = 10(0.1) = 0.10$

$$P(X \geq 2) = 1 - P(X \leq 1) = 0.0047$$

5–40. Kendall and Stuart state: "the liability of individuals to accident varies." That is, the individuals who compose a population have different degrees of accident proneness.

5–41. Use Table XV and scaling by 10^{-5}.

(a) From Col. 3 of Table XV,

Realization 1	Realization 2
$u_1 = 0.01536 < 0.5 \Rightarrow x_1 = 1$	$u_1 = 0.63661 > 0.5 \Rightarrow x_1 = 0$
$u_2 = 0.25595 < 0.5 \Rightarrow x_2 = 1$	$u_2 = 0.53342 > 0.5 \Rightarrow x_2 = 0$
$u_3 = 0.22527 < 0.5 \Rightarrow x_3 = 1$	$u_3 = 0.88231 > 0.5 \Rightarrow x_3 = 0$
$u_4 = 0.06243 < 0.5 \Rightarrow x_4 = 1$	$u_4 = 0.48235 < 0.5 \Rightarrow x_4 = 1$
$u_5 = 0.81837 > 0.5 \Rightarrow x_5 = 0$	$u_5 = 0.52636 > 0.5 \Rightarrow x_5 = 0$
$u_6 = 0.11008 < 0.5 \Rightarrow x_6 = 1$	$u_6 = 0.87529 > 0.5 \Rightarrow x_6 = 0$
$u_7 = 0.56420 > 0.5 \Rightarrow x_7 = 0$	$u_7 = 0.71048 > 0.5 \Rightarrow x_7 = 0$
$u_8 = 0.05463 < 0.5 \Rightarrow x_8 = 1$	$u_8 = 0.51821 > 0.5 \Rightarrow x_8 = 0$
$x = 6$	$x = 1$

Continue to get three more realizations.

(b) Use Col. 4 of Table XV ($p = 0.4$).
Realization 1

$$u_1 = 0.02011 \leq 0.4 \Rightarrow x = 1$$

Realization 2

$$\begin{aligned} u_1 &= 0.85393 > 0.4 \\ u_2 &= 0.97265 > 0.4 \\ u_3 &= 0.61680 > 0.4 \\ u_4 &= 0.16656 < 0.4 \Rightarrow x = 4 \end{aligned}$$

Realization 3

$$\begin{aligned} u_1 &= 0.42751 > 0.4 \\ u_2 &= 0.69994 > 0.4 \\ u_3 &= 0.07972 < 0.4 \Rightarrow x = 3 \end{aligned}$$

Continue to get seven more realizations of X.

(c) $\lambda t = c = 0.15$, $e^{-0.15} = 0.8607$. Using Col. 6 of Table XV,

Realization	u_i	product	$< e^{-0.15}$?	x
#1	$u_1 = 0.91646$	0.91646	No	
	$u_2 = 0.89198$	0.81746	Yes	$x = 1$
#2	$u_1 = 0.64809$	0.64809	Yes	$x = 0$
#3	$u_1 = 0.16376$	0.16376	Yes	$x = 0$
#4	$u_1 = 0.91782$	0.91782	No	
	$u_2 = 0.53498$	0.49102	Yes	$x = 1$
#5	$u_1 = 0.31016$	0.31016	Yes	$x = 0$

5–42. $X \sim$ Geometric with $p = 1/6$.

$y = x^{1/3}$

Using Col. 5 of Table XV, we obtain the following realizations.

1

$$\begin{aligned} u_1 &= 0.81647 > 1/6 \\ u_2 &= 0.30995 > 1/6 \\ u_3 &= 0.76393 > 1/6 \\ u_4 &= 0.07856 < 1/6 \Rightarrow x = 4, y = 1.587 \end{aligned}$$

2

$$u_1 = 0.06121 < 1/6 \Rightarrow x = 4, y = 1$$

Continue to get additional realizations.

Chapter 6

6–1.

$$f_X(x) \;=\; \frac{1}{4}; \quad 0 < x < 4$$

$$P\left(\frac{1}{2} < X < \frac{7}{4}\right) \;=\; \int_{1/2}^{7/4} \frac{dx}{4} \;=\; \frac{5}{16}, \quad P\left(\frac{9}{4} < X < \frac{27}{8}\right) \;=\; \int_{9/4}^{27/8} \frac{dx}{4} \;=\; \frac{9}{32}$$

6–2.

$$f_X(x) \;=\; \frac{4}{34}; \quad \frac{143}{4} < x < \frac{177}{4}$$

$$P(X < 40) \;=\; \int_{143/4}^{40} \frac{4}{34}\, dx \;=\; \frac{1}{2}$$

$$P(40 < X < 42) \;=\; \int_{40}^{42} \frac{4}{34}\, dx \;=\; \frac{4}{17}$$

6–3.

$$f_X(x) \;=\; \frac{1}{2}, \quad 0 \le x \le 2$$

$$F_Y(y) \;=\; P(Y \le y) \;=\; P\left(X \le \frac{y-5}{2}\right) \;=\; \int_0^{\frac{y-5}{2}} \frac{dx}{2} \;=\; \frac{y-5}{4}$$

So

$$f_Y(y) \;=\; \frac{1}{4}, \quad 5 < y < 9$$

6–4. The p.d.f. of profit, X, is

$$f_X(x) \;=\; \frac{1}{2000}; \quad 0 < x < 2000$$

$$Y \;=\; \text{Brokers Fees} \;=\; 50 + 0.06X$$

$$\begin{aligned} F_Y(y) \;&=\; P(50 + 0.06X \le y) \\ &=\; P\left(X \le \frac{y-50}{0.06}\right) \\ &=\; \int_0^{\frac{y-50}{0.06}} \frac{dx}{2000} \;=\; \frac{y-50}{120} \end{aligned}$$

$$f_Y(y) \;=\; \frac{1}{120}; \quad 50 < y < 170$$

6–5.

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \int_\alpha^\beta e^{tx}\,\frac{dx}{\beta-\alpha} \\ &= \frac{e^{tx}}{(\beta-\alpha)t}\Big|_\alpha^\beta \\ &= \frac{e^{t\beta}-e^{t\alpha}}{(\beta-\alpha)t} \end{aligned}$$

Using L'Hôspital's rule when necessary, we obtain

$$\begin{aligned} E(X) &= M_X'(0) = \frac{1}{\beta-\alpha}\left[t^{-1}\beta e^{t\beta} - t^{-2}e^{t\beta} - t^{-1}\alpha e^{t\alpha} + t^{-2}e^{t\alpha}\right]_{t=0} \\ &= \frac{1}{\beta-\alpha}\left[\beta^2 e^{t\beta} - \beta^2 e^{t\beta}/2 - \alpha^2 e^{t\alpha} + \alpha^2 e^{t\alpha}/2\right]_{t=0} \\ &= \frac{\beta+\alpha}{2} \end{aligned}$$

and

$$\begin{aligned} E(X^2) &= M_X''(0) \\ &= \frac{1}{\beta-\alpha}\left[t^{-1}\beta^2 e^{t\beta} - \beta e^{t\beta}t^{-2} + 2e^{t\beta}t^{-3} - t^{-2}\beta e^{t\beta}\right. \\ &\qquad \left. -t^{-1}\alpha^2 e^{t\alpha} + \alpha e^{t\alpha}t^{-2} + t^{-2}\alpha e^{t\alpha} - 2e^{t\alpha}t^{-3}\right]_{t=0} \\ &= \frac{1}{\beta-\alpha}\left[\frac{\beta^3-\alpha^3}{3}\right] \end{aligned}$$

$$V(X) = E(X^2) - [E(X)]^2 = \frac{(\beta-\alpha)^2}{12}$$

6–6.

$$E(X) = \frac{\beta+\alpha}{2} = 0 \Rightarrow \beta+\alpha = 0$$

$$V(X) = \frac{(\beta-\alpha)^2}{12} = 1 \Rightarrow \beta^2 - 2\alpha\beta + \alpha^2 = 12$$

$$\Rightarrow \alpha = -\sqrt{3}, \quad \beta = +\sqrt{3}$$

6–7. The CDF for Y is

y	$F_Y(y)$
$y < 1$	0
$1 \le y < 2$	0.3
$2 \le y < 3$	0.5
$3 \le y < 4$	0.9
$y > 4$	1

Generate realizations of $u_i \sim$ Uniform[0,1] random numbers as described in Section 6–6; use these in the inverse as $y_i = F_Y^{-1}(u_i)$, $i = 1, 2, \ldots$. For example, if $u_1 = 0.623$, then $y_1 = F_Y^{-1}(0.623) = 3$.

6–8.

$$\begin{aligned} f_X(x) &= \frac{1}{4}; \quad 0 < x < 4 \\ &= 0; \quad \text{otherwise} \end{aligned}$$

The roots of $y^2 + 4xy + (x+1) = 0$ are real if $16x^2 - 4(x+1) \ge 0$, or if

$$(x - \frac{1}{8})^2 - \frac{17}{64} \ge 0$$

or where $x \le \frac{1}{8}(1 - \sqrt{17})$ or $x \ge \frac{1}{8}(1 + \sqrt{17})$

$$P\left(X \le \frac{1}{8}(1 - \sqrt{17})\right) = 0$$

$$P\left(X \ge \frac{1}{8}(1 + \sqrt{17})\right) = \int_{\frac{1}{8}(1+\sqrt{17})}^{4} \frac{dx}{4} = \frac{1}{32}(31 - \sqrt{17})$$

6–9.

$$M_X(t) = \int_0^\infty e^{tx} \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty e^{x(t-\lambda)}\, dx,$$

which converges if $t < \lambda$. Thus for $t < \lambda$,

$$M_X(t) = \frac{\lambda}{t - \lambda} \left[e^x (t - \lambda)\right]_{x=0}^{\infty} = \frac{\lambda}{\lambda - t} = \frac{1}{1 - \lambda/t}, \; t < \lambda$$

$$E(X) \;=\; M_X'(0) \;=\; \left[\lambda(\lambda - t)^{-2}\right]_{t=0} \;=\; \frac{1}{\lambda}$$

$$E(X^2) \;=\; M_X''(0) \;=\; \left[2\lambda(\lambda - t)^{-3}\right]_{t=0} \;=\; \frac{2}{\lambda^2}$$

$$V(X) \;=\; E(X^2) - [E(X)]^2 \;=\; \frac{2}{\lambda^2} - \frac{1}{\lambda^2} \;=\; \frac{1}{\lambda^2}$$

6–10. X denotes life length,

$$E(X) \;=\; \frac{1}{\lambda} \;=\; 3 \;\Rightarrow\; \lambda \;=\; \frac{1}{3}$$

P^* denotes profit

$$\begin{aligned} P^* &= 1000, \quad X > 1 \\ &= 750, \quad X \leq 1 \end{aligned}$$

$$\begin{aligned} E(P^*) &= 1000P(X > 1) + 750P(X \leq 1) \\ &= 1000e^{-1/3} + 750(1 - e^{-1/3}) \;=\; \$929.13 \end{aligned}$$

6–11.

$$P(X < \frac{1}{2}) \;=\; \int_0^{1/2} \frac{1}{3} e^{-x/3} dx \;=\; 1 - e^{(-1/3)(1/2)} \;=\; 0.154$$

or 15.4% experience failure in the first six months.

6–12. P^* = Profit, T = Life Length

$$\begin{aligned} P^* &= rY - dY, \quad T > Y \\ &= rT - dY, \quad T \leq Y \end{aligned}$$

$$\begin{aligned} E(P^*) &= (rY - dY)P(T \geq Y) - dY\,P(T \leq Y) + r\int_0^Y t\theta e^{-\theta t}\, dt \\ &= r(\theta^{-1} - \theta^{-1}e^{-\theta Y}) - dY \end{aligned}$$

$$\frac{dE(P^k)}{dY} \;=\; re^{-\theta Y} - d \;=\; 0 \;\Rightarrow\; Y \;=\; -\theta^{-1}\ell\mathrm{n}(\frac{d}{r})$$

For Y to be positive, $0 < \frac{d}{r} < 1$.

6–13. X = Life Length,

$$E(X) = \frac{1}{\lambda} = 3 \Rightarrow \lambda = \frac{1}{3}$$

$$P(X < 1) \;=\; 1 - e^{-1/3} \;=\; 0.283$$

28.3% of policies result in a claim.

6–14. No.

$$P(X \le 2) \;=\; 1 - e^{-2\lambda}$$

$$P(X \le 3) \;=\; 1 - e^{-3\lambda}$$

$$1 - e^{-2\lambda} \;=\; \frac{2}{3}(1 - e^{-3\lambda}) \;\Rightarrow\; 1 \;=\; 3e^{-2\lambda} - 2e^{-3\lambda}$$

Only $\lambda = 0$ satisfies this condition; but we must have $\lambda > 0$, so there is no value of λ for which

$$P(X \le 2) \;=\; \frac{2}{3}P(X \le 3)$$

6–15.

$$\begin{aligned} C_I &= C, \quad X > 15; \\ &= C + Z, \quad X \le 15 \\ C_{II} &= 3C, \quad X > 15 \\ &= 3C + Z, \quad X \le 15 \end{aligned}$$

$$\begin{aligned} E(C_I) &= CP(X > 15) + (C + Z)P(X \le 15) \\ &= Ce^{-15/25} + (C + Z)(1 - e^{-15/25}) \;\doteq\; C + 0.4512Z \\ E(C_{II}) &= 3CP(X > 15) + (3C + Z)P(X \le 15) \\ &= 3Ce^{-15/25} + (3C + Z)(1 - e^{-15/35}) \;\doteq\; 3C + 0.3426Z \end{aligned}$$

$$E(C_{II}) - E(C_I) = 2C - 0.1026Z$$

which favors process I if $C > 0.0513Z$.

6–16.

$$P(X > x+s|X > x) \;=\; P(X > s) \;=\; P(X > 10000) \;=\; e^{-10000/20000} \;=\; 0.6064$$

and $P(X < 30000|X > 20000) \;=\; 0.3936$.

6–17.

$$\Gamma(p) \;=\; \int_0^\infty x^{p-1}e^{-x}dx$$

Let $x = y^2 \Rightarrow \frac{dx}{dy} = 2y$. So $\Gamma(p) \;=\; \int_0^\infty y^{2p-1}e^{-y^2}dy$ and $\Gamma(\frac{1}{2}) \;=\; 2\int_0^\infty e^{-y^2}dy$.

$$\begin{aligned}\left(\Gamma(\frac{1}{2})\right)^2 &= 4\int_0^\infty e^{-y^2}\,dy \cdot \int_0^\infty e^{-x^2}\,dx \\ &= 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dx\,dy\end{aligned}$$

Let $x = \rho\cos(\theta)$ and $y = \rho\sin(\theta)$. So

$$\begin{aligned}\left(\Gamma(\frac{1}{2})\right)^2 &= 4\int_0^{\pi/2} \int_0^\infty \rho e^{-\rho^2}\,d\rho\,d\theta \\ &= 2\int_0^{\pi/2} d\theta \;=\; \pi\end{aligned}$$

So $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

6–18. Integrate by parts with $u = x^{n-1}, dv = e^{-x}dx$

$$\begin{aligned}\Gamma(n) &= \int_0^\infty x^{n-1}e^{-x}dx \;=\; \left[-e^{-x}x^{n-1}\right]_0^\infty + (n-1)\int_0^\infty x^{n-2}e^{-x}dx \\ &= 0 + (n-1)\int_0^\infty x^{n-2}e^{-x}dx \;=\; (n-1)\cdot\Gamma(n-1)\end{aligned}$$

Repeatedly using the approach above, we get

$$\begin{aligned}\Gamma(n) &= (n-1)\cdot\Gamma(n-1) \;=\; (n-1)(n-2)\cdot\Gamma(n-2) \\ &= \cdots \;=\; (n-1)(n-2)\cdots\Gamma(1).\end{aligned}$$

Since $\Gamma(1) = \int_0^\infty e^{-x}dx = 1$, we have $\Gamma(n) = (n-1)!$

6–19.

$$Y = X_1 + \cdots + X_{10}$$

$$g(x_i) \;=\; \begin{cases} 7e^{-7x_i} & \text{if } x_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} P(Y>1) \;&=\; \int_1^\infty \frac{7}{\Gamma(10)}(7y)^9 e^{-7y}dy \\ &=\; \sum_{k=0}^{9} e^{-7\cdot 1}\frac{7^k}{k!} \;=\; 0.8305 \end{aligned}$$

6–20.

$$\lambda \;=\; 6;\; t=4 \;\Rightarrow\; \lambda t \;=\; 24$$

$$P(X \geq 24) \;=\; 1 - P(X \leq 23) \;=\; 1 - \sum_{x=0}^{23} \frac{e^{-24}24^x}{x!}$$

6–21.

$$\begin{aligned} M_X(t) \;&=\; E(e^{tX}) \;=\; \int_0^\infty e^{tx}\,\frac{\lambda}{\Gamma(r)}(\lambda x)^{r-1}e^{-\lambda x}dx \\ &=\; \frac{\lambda^r}{\Gamma(r)}\int_0^\infty x^{r-1}e^{-x(\lambda-t)}dx \end{aligned}$$

The integral converges if $(\lambda - t) > 0$ or $\lambda > t$. Let $u = x(\lambda - t)$, $\frac{dx}{du} = \frac{1}{(\lambda - t)}$. So

$$\begin{aligned} M_X(t) \;&=\; \frac{\lambda^r}{\Gamma(r)}\int_0^\infty \left(\frac{u}{\lambda - t}\right)^{r-1} e^{-u}(\lambda - t)^{-1}\,du \\ &=\; \left(\frac{\lambda}{\lambda - t}\right)^r \cdot \frac{1}{\Gamma(r)}\int_0^\infty u^{r-1}e^{-u}\,du \\ &=\; \left(\frac{\lambda}{\lambda - t}\right)^r \cdot \frac{1}{\Gamma(r)} \cdot \Gamma(r) \;=\; \left(\frac{\lambda}{\lambda - t}\right)^r \\ &=\; (1 - (t/\lambda))^{-r},\ \text{where } \lambda > t \end{aligned}$$

6–22.

$$\begin{aligned} P(Y>24) \;&=\; \int_{24}^\infty \frac{0.25}{\Gamma(4)}\,(0.25y)^3 e^{-0.25y}\,dy \\ &=\; \sum_{k=0}^{3} e^{-0.25\cdot 24}\frac{(0.25\cdot 24)^k}{k!} \;=\; 0.1512 \end{aligned}$$

6–23. $E(X) = r/\lambda = 40$, $V(X) = r/\lambda^2 = 400$

$\lambda = 0.1$, $r = 4$

$$\begin{aligned} P(X < 20) &= \int_0^{20} \frac{0.1}{\Gamma(4)}(0.1x)^3 e^{-0.1x}\,dx \\ &= 1 - \sum_{k=0}^{3} e^{-0.1\cdot 20}\frac{(0.1\cdot 20)^k}{k!} = 0.1429 \end{aligned}$$

$$P(X < 60) = 1 - \sum_{k=0}^{3} e^{-0.1\cdot 60}\frac{(0.1\cdot 60)^k}{k!} = 0.8488$$

6–24.

$$E(X) = \frac{\lambda^r}{\Gamma(r)}\int_0^{\infty} x(x-u)^{r-1}e^{-\lambda(x-u)}dx$$

Let $y = \lambda(x-u) \Rightarrow \frac{dx}{dy} = \frac{1}{\lambda}$

$$\begin{aligned} E(X) &= \frac{\lambda^{r-2}}{\Gamma(r)}\int_0^{\infty}(y+\lambda u)(y/\lambda)^{r-1}e^{-y}\,dy \\ &= \frac{1}{\lambda\Gamma(r)}\int_0^{\infty} y^r e^{-y}dy + \frac{\lambda u}{\lambda\Gamma(r)}\int_0^{\infty} y^{r-1}e^{-y}\,dy \\ &= \frac{\Gamma(r+1)}{\lambda\Gamma(r)} + \frac{u\Gamma(r)}{\Gamma(r)} \\ &= \frac{r}{\lambda} + u \end{aligned}$$

6–26.

$$f_X(x) = \frac{\Gamma(\lambda+r)}{\Gamma(\lambda)\Gamma(r)}x^{\lambda-1}(1-x)^{r-1}$$

$\lambda = r = 1$ gives

$$\begin{aligned} f_X(x) &= \frac{\Gamma(2)}{\Gamma(1)\Gamma(1)}x^0(1-x)^0 \\ &= \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

6–27. $\lambda = 2$, $r = 1$

$$\begin{aligned} f_X(x) &= \frac{\Gamma(3)}{\Gamma(2)\Gamma(1)} x(1-x)^0 \\ &= \begin{cases} 2x & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

$\lambda = 1$, $r = 2$

$$\begin{aligned} f_X(x) &= \frac{\Gamma(3)}{\Gamma(1)\Gamma(2)} x^0(1-x) \\ &= \begin{cases} 2(1-x) & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

6–29. See solution to 3-22.

6–30.

$$E(X) = \int_0^\infty x \frac{\beta}{\delta} \left(\frac{x-\gamma}{\delta} \right)^{\beta-1} e^{-(\frac{x-\gamma}{\delta})^\beta} dx$$

First let $y = \frac{x-\gamma}{\delta} \Rightarrow dx = \delta dy$

$$\begin{aligned} E(X) &= \int_0^\infty (\delta y + \gamma) \frac{\beta}{\delta} y^{\beta-1} e^{-y^\beta} \delta \, dy \\ &= \beta \int_0^\infty (\delta y + \gamma) y^{\beta-1} e^{-y^\beta} dy \end{aligned}$$

Let $u = y^\beta \Rightarrow dy = \beta^{-1} y^{-\beta+1} du$

$$\begin{aligned} E(X) &= \beta \int_0^\infty (\delta u^{1/\beta} + \gamma) u^{(1-1/\beta)} e^{-u} \beta^{-1} u^{(1/\beta-1)} \, du \\ &= \gamma \int_0^\infty e^{-u} \, du + \int_0^\infty \delta u^{1/\beta} e^{-u} \, du \\ &= \gamma + \delta \Gamma\left(1 + \frac{1}{\beta}\right) \end{aligned}$$

Using the same approach

$$E(X^2) = \int_0^\infty x^2 \frac{\beta}{\delta} \left(\frac{x-\gamma}{\delta} \right)^{\beta-1} e^{-(\frac{x-\gamma}{\delta})} \, dx$$

Let $y = \frac{x-\gamma}{\delta} \Rightarrow dx = \delta dy$

$$E(X^2) \;=\; \int_0^\infty (\delta y + \gamma)^2 \, \frac{\beta}{\delta} \, y^{\beta-1} \, e^{-y^\beta} \, \delta \, dy$$

$u = y^\beta \Rightarrow dy = \beta^{-1} y^{1-\beta}$

$$\begin{aligned} E(X^2) \;&=\; \int_0^\infty (\delta u^{1/\beta} + \gamma)^2 \, \frac{\beta}{\delta} \, u^{1-1/\beta} \, e^{-u} \delta \, \beta^{-1} \, u^{1/\beta - 1} \, du \\ &=\; \delta^2 \, \Gamma\left(1 + \frac{2}{\beta}\right) + 2\gamma\delta \, \Gamma\left(1 + \frac{1}{\beta}\right) + \gamma^2 \end{aligned}$$

So

$$V(X) \;=\; E(X^2) - [E(X)]^2 \;=\; \delta^2 \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \Gamma^2\left(1 + \frac{1}{\beta}\right) \right]$$

6–31.

$$F(x) \;=\; 1 - e^{-(\frac{x-\gamma}{\delta})^\beta}$$

$$F(1.5) \;=\; 1 - e^{-(\frac{1.5-1}{0.5})^2} \doteq 0.63$$

6–32.

$$F(x) \;=\; 1 - e^{-(\frac{x-0}{400})^{1/3}}$$

$$1 - F(600) \;=\; e^{-(\frac{600}{400})^{1/3}} \doteq 0.32$$

6–33.

$$F(x) \;=\; 1 - e^{-(\frac{x-0}{400})^{1/2}}$$

$$1 - F(800) \;=\; 1 - e^{-2^{1/2}} \doteq 0.24$$

6–34. The graphs are identical to those shown in Figure 6–8.

6–35.

$$F(x) \;=\; 1 - e^{-(\frac{x-0}{200})^{1/4}}$$

(a) $1 - F(1000) = 1 - e^{-5^{1/4}} \doteq 0.22$
(b) $0 + 200\Gamma(5) = 200 \cdot 24 = 4800 = E(X)$

6–36. P^* = Profit

$$\begin{aligned} P^* &= \$100; \quad x \geq 8760 \\ &= -\$50; \quad x < 8760 \end{aligned}$$

$$\begin{aligned} E(P^*) &= -50 \int_0^{8760} 20000^{-1} e^{-20000^{-1}x}\, dx + 100 \int_{8760}^{\infty} 20000^{-1} e^{-20000^{-1}x}\, dx \\ &= -50(1 - e^{-876/2000}) + 100 e^{-876/2000} \\ &= \$46.80/\text{set} \end{aligned}$$

6–37. $r/\lambda = 20$

$(r/\lambda^2)^{1/2} = 10$

$r = 4$, $\lambda = 0.2$

$$P(X \leq 15) \;=\; F(15) \;=\; 1 - \sum_{k=0}^{3} e^{-3} 3^k / k! \;=\; 0.3528$$

6–38. (a) Use Table XV, Col. 1 with scaling and Equation 6–35.

$u_1 = 0.10480 \quad x_1 = 10 + u_1(10) = 11.0480$
$u_2 = 0.22368 \quad x_2 = 10 + u_2(10) = 12.2368$
$u_3 = 0.24130 \quad x_3 = 10 + u_3(10) = 12.4130$
$\vdots \qquad\qquad \vdots$
$u_{10} = 0.85475 \quad x_{10} = 10 + u_{10}(10) = 18.5475$

(b) Use Table XV, Col. 2; $x_i = -50000\, \ell\text{n}(1 - u_i)$

$u_1 = 0.15011 \quad x_1 = -50000\, \ell\text{n}(0.84989) = 8132.42$
$u_2 = 0.46573 \quad x_2 = -50000\, \ell\text{n}(0.53427) = 31345.51$
$u_3 = 0.48360 \quad x_3 = -50000\, \ell\text{n}(0.51640) = 33043.68$
$u_4 = 0.93093 \quad x_4 = -50000\, \ell\text{n}(0.06907) = 133631.74$
$u_5 = 0.39975 \quad x_5 = -50000\, \ell\text{n}(0.60025) = 25520.45$

(c) $a = \sqrt{3} = 1.732$, $b = 4 - \ell\text{n}(4) + 3^{-1/2} = 3.191$. Now use Table XV, Col. 3.

Realization 1: $u_1 = 0.01563$, $u_2 = 0.25595$

$$y = 2\left(\frac{0.01563}{0.98437}\right)^{1.732} = 0.00153$$

$$3.191 - \ell\text{n}[(0.01563)^2 0.25595] = 12.871$$

$$y \le 12.87$$

$$x_1 = y/4 = 0.0003383$$

Realization 2: $u_1 = 0.22527$, $u_2 = 0.06243$

$$y = 2\left(\frac{0.22527}{0.77473}\right)^{1.732} = 0.2354$$

$$3.191 - \ell\text{n}[(0.22527)^2 0.06243] = 8.9456$$

$$y \le 8.9456$$

$$x_2 = y/4 = 0.05885$$

Continue for additional realizations.

Note: Since r is an integer, an alternate scheme which may be more efficient here is to let $x_i = x_{i1} + x_{i2}$, where x_{ij} is exponential with parameter $\lambda = 4$.

$$x_{ij} = -0.25\,\ell\text{n}(1 - u_{ij}),\ i = 1, 2, \ldots, 5,\ j = 1, 2$$

Realization 1: $u_1 = 0.01563$, $u_2 = 0.25595$

$$x_{11} = -0.25\,\ell\text{n}(0.98437) = 0.003938$$

$$x_{12} = -0.25\,\ell\text{n}(0.74405) = 0.073712$$

This yields $x_1 = 0.07785$. Continue for more realizations.

(d) Use Table XV, Col. 4 with scaling.

$u_1 = 0.02011$	$x_1 = 0 + 100[-\ell\text{n}(0.02011)]^2 = 390.654$
$u_2 = 0.85393$	$x_2 = 0 + 100[-\ell\text{n}(0.85393)]^2 = 15.791$
$\vdots$	$\vdots$
$u_{10} = 0.53988$	$x_{10} = 0 + 100[-\ell\text{n}(0.53988)]^2 = 61.641$

6–39. (a) Using Table XV, Col. 5, and $y = x^{0.3}$, we get

$$\begin{array}{lll} u_1 = 0.81647 & x_1 = -10\,\ell\text{n}(0.18353) = 16.954 & y_1 = 2.3376 \\ u_2 = 0.30995 & x_2 = -10\,\ell\text{n}(0.69005) = 3.7099 & y_2 = 1.4818 \\ \vdots & \vdots & \vdots \\ u_{10} = 0.53060 & x_{10} = -10\,\ell\text{n}(0.46940) = 7.5630 & y_{10} = 1.8348 \end{array}$$

(b) Using the gamma variates in Problem 6–38(c) and Table XV, Col. 3 entry #25,

$$y_1 = \frac{(0.000383)^{1/2}}{(0.28834)^{1/2}} = 0.03645$$

$$y_2 = \frac{(0.05885)^{1/2}}{(0.04839)^{1/2}} = 1.102797$$

$\vdots$

etc.

Chapter 7

7–1. (a) $P(0 \le Z \le 2) = \Phi(2) - \Phi(0) = 0.97725 - 0.5 = 0.47725$

(b) $P(-1 \le Z \le 1) = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1 = 0.68268$

(c) $P(Z \le 1.65) = \Phi(1.65) = 0.95053$

(d) $P(Z \ge -1.96) = \Phi(1.96) = 0.9750$

(e) $P(|Z| \ge 1.5) = 2[1 - \Phi(1.5)] = 0.1336$

(f) $P(-1.9 \le Z \le 2) = \Phi(2) - \Phi(-1.9) = \Phi(2) - [1 - \Phi(1.9)] = 0.94853$

(g) $P(Z \le 1.37) = 0.91465$

(h) $P(|Z| \le 2.57) = 2\Phi(2.57) - 1 = 0.98984$

7–2. $X \sim N(10, 9)$.

(a) $P(X \le 8) = \Phi\left(\dfrac{8-10}{3}\right) = \Phi\left(-\dfrac{2}{3}\right) = 0.2525$

(b) $P(X \ge 12) = 1 - \Phi\left(\dfrac{12-10}{3}\right) = 1 - \Phi\left(\dfrac{2}{3}\right) = 0.2525$

(c) $P(2 \le X \le 10) = \Phi\left(\dfrac{10-10}{3}\right) - \Phi\left(\dfrac{2-10}{3}\right) = 0.5 - \Phi(-2.67) = 0.496$

7–3. From Table II of the Appendix

(a) $c = 1.56$

(b) $c = 1.96$

(c) $c = 2.57$

(d) $c = -1.645$

7–4. $P(Z \ge Z_\alpha) = \alpha \;\Rightarrow\; \Phi(Z_\alpha) = 1 - \alpha$.

(a) $Z_{0.025} = 1.96$

(b) $Z_{0.005} = 2.57$

(c) $Z_{0.05} = 1.645$

(d) $Z_{0.0014} = 2.99$

7–5. $X \sim N(80, 100)$.

(a) $P(X \leq 100) = \Phi\left(\dfrac{100-80}{10}\right) = \Phi(2) = 0.97725$

(b) $P(X \leq 80) = \Phi\left(\dfrac{80-80}{10}\right) = 0.5$

(c) $P(75 \leq X \leq 100) = \Phi\left(\dfrac{100-80}{10}\right) - \Phi\left(\dfrac{75-80}{10}\right) = \Phi(2) - \Phi(-0.5) = 0.97725 - 0.30854 = 0.66869$

(d) $P(X \geq 75) = 1 - \Phi\left(\dfrac{75-80}{10}\right) = 1 - \Phi(-0.5) = \Phi(0.5) = 0.69146$

(e) $P(|X - 80| \leq 19.6) = \Phi(1.96) - \Phi(-1.96) = 0.95$

7–6. (a) $P(X > 680) = 1 - \Phi\left(\dfrac{680-600}{60}\right) = 1 - \Phi(1.33) = 0.09176$

(b) $P(X \leq 550) = \Phi\left(\dfrac{550-600}{60}\right) = \Phi(-5/6) = 1 - \Phi(5/6) = 0.20327$

7–7. $P(X > 500) = 1 - \Phi\left(\dfrac{500-485}{30}\right) = 1 - \Phi(0.5) = 0.30854$, i.e., 30.854%

7–8. (a) $P(X \geq 28.5) = 1 - \Phi\left(\dfrac{28.5-30}{1.1}\right) = 1 - \Phi(-1.36) = \Phi(1.36) = 0.91308$

(b) $P(X \leq 31) = \Phi\left(\dfrac{31-30}{1.1}\right) = 0.819$

(c) $P(|X - 30| > 2) = 1 - \left[\Phi\left(\dfrac{2}{1.1}\right) - \Phi\left(-\dfrac{2}{1.1}\right)\right] = 1 - [0.96485 - 0.03515] = 0.0703$

7–9. $X \sim N(2500, 5625)$. Then $P(X < \ell) = 0.05$ implies that

$$P\left(Z < \frac{\ell - 2500}{75}\right) = 0.05$$

or

$$\frac{\ell - 2500}{75} = -1.645.$$

Thus, $\ell = 2376.63$ is the lower specification limit.

7–10.

$$\begin{aligned}
M_X(t) &= E(e^{tX}) = \frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{tx}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx \\
&= \frac{\sigma}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{t(y\sigma+\mu)}\, e^{-y^2/2}\, dy \quad (\text{letting } y=(x-\mu)/\sigma) \\
&= \frac{e^{\mu t}}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(y^2-2\sigma t y)/2}\, dy \\
&= \frac{e^{\mu t}}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(y^2-2\sigma t y+\sigma^2t^2-\sigma^2t^2)/2}\, dy \\
&= \frac{e^{\mu t}}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(y-\sigma t)^2/2} e^{\sigma^2t^2/2}\, dy \\
&= e^{\mu t+(1/2)\sigma^2t^2}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-w^2/2}\, dw \quad (\text{letting } w=y-\sigma t) \\
&= e^{\mu t+(1/2)\sigma^2t^2},
\end{aligned}$$

since the integral term equals 1.

7–11.

$$\begin{aligned}
F_Y(y) &= P(aX+b\le y) = P\left(X\le \frac{y-b}{a}\right) \\
&= \Phi\left(\frac{\frac{y-b}{a}-\mu}{\sigma}\right) \\
&= \Phi\left(\frac{y-b-a\mu}{a\sigma}\right) = \Phi\left(\frac{y-(a\mu+b)}{a\sigma}\right)
\end{aligned}$$

This implies that $Y \sim N(a\mu+b, a^2\sigma^2)$.

7–12. $X \sim N(12, (0.02)^2)$.

(a) $P(X > 12.05) = 1 - \Phi\left(\dfrac{12.05-12}{0.02}\right) = 1-\Phi(2.5) = 0.00621$

(b)

$$
\begin{aligned}
&P(X > c) = 0.9 \\
&\Rightarrow \quad 1 - \Phi\left(\frac{c-12}{0.02}\right) = 0.9 \\
&\Rightarrow \quad \Phi\left(\frac{c-12}{0.02}\right) = 0.1 \\
&\Rightarrow \quad \frac{c-12}{0.02} = -1.28 \\
&\Rightarrow \quad c = 12 - 0.0256 = 11.97
\end{aligned}
$$

(c)

$$
\begin{aligned}
P(11.95 \le X \le 12.05) &= \Phi\left(\frac{12.05-12}{0.02}\right) - \Phi\left(\frac{11.95-12}{0.02}\right) \\
&= \Phi(2.5) - \Phi(-2.5) = 0.9876
\end{aligned}
$$

7–13. $X \sim N(\mu, (0.1)^2)$.

(a) Take $\mu = 7.0$. Then

$$
\begin{aligned}
&P(X > 7.2) + P(X < 6.8) \\
&= 1 - \Phi\left(\frac{7.2-7}{0.1}\right) + \Phi\left(\frac{6.8-7}{0.1}\right) \\
&= 1 - \Phi(2) + \Phi(-2) \\
&= 1 - 0.97725 + 0.02275 \;=\; 0.0455
\end{aligned}
$$

(b)

$$
\begin{aligned}
&1 - \Phi\left(\frac{7.2-7.05}{0.1}\right) + \Phi\left(\frac{6.8-7.05}{0.1}\right) \\
&= 1 - \Phi(1.5) + \Phi(-2.5) \\
&= 1 - 0.93319 + 0.00621 \;=\; 0.07302
\end{aligned}
$$

(c)

$$
\begin{aligned}
&\Phi\left(\frac{7.2-7.25}{0.1}\right) - \Phi\left(\frac{6.8-7.25}{0.1}\right) \\
&= \Phi(-0.5) - \Phi(-4.5) \\
&\doteq 1 - \Phi(0.5) \;=\; 0.3085
\end{aligned}
$$

(d)

$$\Phi\left(\frac{7.2-6.75}{0.1}\right) - \Phi\left(\frac{6.8-6.75}{0.1}\right) \doteq 1 - \Phi(0.5) = 0.3085$$

7–14. $X \sim N(50, 25)$, $Y \sim N(45, 6.25)$.

If $Y \geq X$, i.e., if $Y - X \geq 0$, a transaction will occur.

Let $W = Y - X \sim N(-5, 31.25)$.

$$P(W > 0) = P\left(Z \geq \frac{0+5}{5.59}\right) = 1 - \Phi(0.89) = 0.1867.$$

7–15. \$9.00 = revenue / capacitor, k = manufacturing cost for process A, $2k$ = manufacturing cost for process B. The profits are

$$P_A^* = \begin{cases} 9-k & \text{if } 1000 \leq X \leq 5000 \\ 9-k-3 & \text{otherwise} \end{cases}$$

$$P_B^* = \begin{cases} 9-2k & \text{if } 1000 \leq X \leq 5000 \\ 9-2k-3 & \text{otherwise} \end{cases}$$

Therefore,

$$\begin{aligned} E(P_A^*) &= (9-k)P(1000 \leq X \leq 5000) + (6-k)[1 - P(1000 \leq X \leq 5000)] \\ &= (9-k)0.9544 + (6-k)0.0456 = 8.8632 - k \\ E(P_B^*) &= (9-2k)P(1000 \leq X \leq 5000) + (6-2k)[1 - P(1000 \leq X \leq 5000)] \\ &= (9-2k)(1) + (6-k)(0) = 9-2k \end{aligned}$$

Since $E(P_A^*) < E(P_B^*)$ when $k < 0.1368$, use process B; When $k \geq 0.1368$, use process A.

7–16. The profit P is

$$P = \begin{cases} C & \text{if } 6 \leq X \leq 8 \\ -R_1 & \text{if } X < 6 \\ -R_2 & \text{if } X > 8 \end{cases}$$

$$\begin{aligned} E(P) &= CP(6 \le X \le 8) - R_1 P(X < 6) - R_2 P(X > 8) \\ &= C[\Phi(8-\mu) - \Phi(6-\mu)] - R_1\Phi(6-\mu) - R_2[1 - \Phi(8-\mu)] \\ &= (C+R_2)\Phi(8-\mu) - (C+R_1)\Phi(6-\mu) - R_2 \end{aligned}$$

Then

$$\frac{dE(P)}{d\mu} \;=\; -(C+R_2)\Phi(8-\mu) + (C+R_1)\Phi(6-\mu) \;=\; 0,$$

or

$$\frac{C+R_2}{C+R_1} \;=\; \frac{\Phi(6-\mu)}{\Phi(8-\mu)} \;=\; e^{14-2\mu}$$

Thus,

$$\mu \;=\; 7 - \frac{1}{2}\,\ell\mathrm{n}\left(\frac{C+R_2}{C+R_1}\right).$$

7–17. If $R_1 = R_2 = R$, then $\mu = 7 - 0.5\,\ell\mathrm{n}(1) = 7$, which is the midpoint of the interval [6,8].

7–18. $\mu = 7 - \dfrac{1}{2}\,\ell\mathrm{n}\left(\dfrac{12}{10}\right) = 6.909.$

7–19. $X \sim N(70, 16)$.

(a) We have

$$\begin{aligned} P(62 \le X \le 72) &= \Phi\left(\frac{72-70}{4}\right) - \Phi\left(\frac{62-70}{4}\right) \\ &= \Phi(0.5) - \Phi(-2) \\ &= 0.69146 - 0.02275 \;=\; 0.66871. \end{aligned}$$

(b) $c = 1.96\sigma = 7.84.$

(c) $(9)(0.66871) = 6.018.$

7-20. $E(Y) = E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E(X_i) = n\mu$, so

$$E(Z_n) = \frac{E(Y) - n\mu}{\sqrt{\sigma^2 n}} = 0$$

$$V(Y) = V\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} V(X_i) = n\sigma^2$$

$$V(Z_n) = \frac{V(Y)}{n\sigma^2} = 1$$

7–21. $E(\bar{X}) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{n\mu}{n} = \mu$

$$V(\bar{X}) = V\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} V(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

7–22. $X_1 \sim N(1.25, 0.0009)$ and $X_2 \sim N(1.2, 0.0016)$.

$Y = X_1 - X_2$, $E(Y) = 0.05$, $V(Y) = 0.0025$.

$Y \sim N(0.05, 0.0025)$

$$P(Y < 0) = \Phi\left(\frac{0 - 0.05}{0.05}\right) = \Phi(-1) = 1 - \Phi(1) = 0.15866.$$

7–23. $X_i \sim N(2, 0.04)$, $i = 1, 2, 3$, and $Y = X_1 + X_2 + X_3 \sim N(6, 0.12)$.

Then

$$\begin{aligned} P(5.7 < Y < 6.3) &= \Phi\left(\frac{6.3 - 6.0}{0.3464}\right) - \Phi\left(\frac{5.7 - 6.0}{0.3464}\right) \\ &= \Phi(0.866) - \Phi(-0.866) = 0.6156. \end{aligned}$$

7–24. $E(Y) = E(X_1) + 2E(X_2) + E(X_3) + E(X_4) = 4 + 2(4) + 2 + 3 = 17.$

With independence,

$$V(Y) \;=\; V(X_1) + 2^2 V(X_2) + V(X_3) + V(X_4) \;=\; 3 + 4(4) + 4 + 2 \;=\; 25.$$

$$\begin{aligned} P(15 \le Y \le 20) &= \Phi\left(\frac{20-17}{5}\right) - \Phi\left(\frac{15-17}{5}\right) \\ &= \Phi(0.6) - \Phi(-0.4) \\ &= 0.72575 - 0.34458 \;=\; 0.38117. \end{aligned}$$

7–25. $E(X_i) = 0$, $V(X_i) = 1/12$.

$$Y \;=\; \sum_{i=1}^{50} X_i, \quad E(Y) \;=\; 0, \quad V(Y) \;=\; 50/12$$

$$P(Y > 5) \;=\; 1 - \Phi\left(\frac{5-0}{\sqrt{50/12}}\right) \;=\; 1 - \Phi(2.45) \;=\; 0.00714.$$

7–26. $E(X_i) = 1$, $V(X_i) = 0.0001$, $i = 1, 2, \ldots, 100$.

$$Y \;=\; \sum_{i=1}^{100} X_i.$$

Assuming that the X_i's are independent, we use the central limit theorem to approximate the distribution of $Y \sim N(1000, 0.01)$. Then

$$P(Y > 102) \;=\; P\left(Z > \frac{102-100}{0.1}\right) \;=\; 1 - \Phi(20) \;\doteq\; 0.$$

7–27. $\bar{X} \sim N(11.9, 0.0025)$ and $n = 9$. Thus, $\mu = 11.9$, $\sigma^2/n = \sigma^2/9 = 0.0025$, so $\sigma^2 = 0.0225$. All of this implies that $X \sim N(11.9, 0.0225)$. Then

$$P(11.8 < X < 12.2) \;=\; \Phi(2) - \Phi(-0.67) \;=\; 0.7258,$$

so that there are 27.4% defective.

If $\mu = 12$, then

$$P(11.8 < X < 12.2) \;=\; \Phi(1.33) - \Phi(-1.33) \;=\; 0.8164,$$

or 18.4% defective. This is the optimal value of the mean.

7–28. $Y = \sum_{i=1}^{n} E(X_i)$, where X_i is the travel time between pair i.

$$\begin{aligned} E(Y) &= \sum_{i=1}^{n} E(X_i) = 30 \\ V(Y) &= \sum_{i=1}^{n} V(X_i) \\ &= (0.4)^2 + (0.6)^2 + (0.3)^2 + (1.2)^2 + (0.9)^2 + (0.4)^2 + (0.4)^2 = 3.18. \end{aligned}$$

Thus,

$$P(Y \le 32) = \Phi\left(\frac{32-30}{\sqrt{3.18}}\right) = \Phi(1.12) = 0.86864.$$

7–29. $p = 0.08$, $n = 200$, $np = 16$, $\sqrt{npq} = 3.84$.

(a) $P(X \le 16) = \Phi\left(\frac{16.5-16}{3.84}\right) = \Phi(0.13) = 0.55172.$

(b) $\Phi\left(\frac{15.5-16}{3.84}\right) - \Phi\left(\frac{14.5-16}{3.84}\right) = \Phi(-0.13) - \Phi(-0.391) = 0.1.$

(c) $\Phi\left(\frac{20.5-16}{3.84}\right) - \Phi\left(\frac{11.5-16}{3.84}\right) = \Phi(1.17) - \Phi(-1.17) = 0.758.$

(d) $\Phi\left(\frac{14.5-16}{3.84}\right) - \Phi\left(\frac{13.5-16}{3.84}\right) = \Phi(-0.391) - \Phi(-0.651) = 0.09.$

7–30. $P(0.05 \le \hat{p} \le 0.15) = 0.95$ implies that

$$P\left(\frac{0.05-0.10}{\sqrt{0.09/n}} \le Z \le \frac{0.15-0.10}{\sqrt{0.09/n}}\right) = \Phi\left(\frac{0.05}{0.3/\sqrt{n}}\right) - \Phi\left(\frac{-0.05}{0.3/\sqrt{n}}\right) = 0.95$$

$$\Rightarrow 2\,\Phi\left(\frac{0.05}{0.3/\sqrt{n}}\right) = 1.95$$

$$\Rightarrow \Phi(0.167\sqrt{n}) = 0.9750$$

$$\Rightarrow 0.167\sqrt{n} = 1.96$$

$$\Rightarrow n = 139$$

7–31. $Z_1 = \sqrt{-2\,\ell\mathrm{n}(u_1)} \cdot \cos(2\pi u_2)$, $Z_2 = \sqrt{-2\,\ell\mathrm{n}(u_1)} \cdot \sin(2\pi u_2)$. Note that the sine and cosine calculations are carried out in radians.

Obtain uniforms from Col. 2.

u_1	u_2	z_1	z_2
0.15011	0.46573	−1.902	0.416
0.48360	0.93093	1.093	−0.507
0.39975	0.06907	1.229	0.569

These results give

$$\begin{aligned} x_1 &= 100 + 2(-1.902) = 96.196 \\ x_2 &= 100 + 2(0.416) = 100.832 \\ x_3 &= 100 + 2(1.093) = 102.186 \\ x_4 &= 100 + 2(-0.507) = 98.986 \\ x_5 &= 100 + 2(1.229) = 102.458 \\ x_6 &= 100 + 2(0.569) = 101.138 \end{aligned}$$

7–32. Calculate Z_1 and Z_2 as in Problem 7–31, obtaining uniforms from Col. 4.

u_1	u_2	z_1	z_2
0.02011	0.08539	2.402	1.429
0.97265	0.61680	−0.175	−0.158
0.16656	0.42751	−1.700	0.833

These results give realizations of X_1.

x_1	$3x_1$
$10 + 1.732(2.402) = 14.161$	42.483
$10 + 1.732(1.429) = 12.475$	37.425
$10 + 1.732(-0.175) = 9.697$	29.091
$10 + 1.732(-0.158) = 9.726$	29.179
$10 + 1.732(-1.700) = 7.056$	21.167
$10 + 1.732(0.833) = 11.443$	34.328

Meanwhile, we use Col. 5 to calculate realizations of X_2.

x_2	$-2x_2$
$20(0.81647) = 16.329$	-32.658
$20(0.30995) = 6.199$	-12.398
$20(0.76393) = 15.279$	-30.558
$20(0.07856) = 1.571$	-3.142
$20(0.06121) = 1.224$	-2.448
$20(0.27756) = 5.551$	-11.102

Finally, we get the six realizations of Y,

$$\begin{aligned} y_1 &= 9.825 \\ y_2 &= 25.027 \\ y_3 &= -1.467 \\ y_4 &= 26.039 \\ y_5 &= 18.719 \\ y_6 &= 23.226 \end{aligned}$$

7–33. Using the z_i realizations from Problem 7–31,

$$\begin{aligned} (-1.092)^2 &= 3.618 \\ (0.416)^2 &= 0.173 \\ (1.093)^2 &= 1.195 \\ (-0.507)^2 &= 0.257 \\ (1.229)^2 &= 1.510 \end{aligned}$$

7–34. Let $z_1, z_2, \ldots, z_n$ be realizations of $N(0,1)$ r.v.'s.

$y_i = \mu_Y + \sigma z_i$, $i = 1, 2, \ldots, n$.

$x_i = e^{y_i}$, $i = 1, 2, \ldots, n$.

7–35. Generate a pair z_1, z_2 of $N(0,1)$ r.v.'s.

Let $x_1 = \mu_1 + \sigma_1 z_1$ and $x_2 = \mu_2 + \sigma_2 z_2$. Thus, $X_i \sim N(\mu_i, \sigma_i^2)$, $i = 1, 2$.

Let $y_1 = \sqrt{x_1}/x_2^2$.

Repeat this procedure for as many realizations as desired.

7–36. This is a normal distribution truncated on the right, with p.d.f.

$$\begin{aligned} f(x) &= \frac{1}{\Phi(\frac{r-\mu}{\sigma})\,\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \quad \text{if } -\infty < x \le r \\ &= 0 \quad \text{if } x > r \end{aligned}$$

For our problem, $r = 2600$, $\mu = 2500$, and $\sigma = 50$. Now, after a bit of calculus,

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x)\,dx \\ &= \mu - \frac{\sigma}{\Phi(\frac{r-\mu}{\sigma})\sqrt{2\pi}} \exp\left[-\frac{(r-\mu)^2}{2\sigma^2}\right] \\ &= 2500 - \frac{50}{0.9772\sqrt{2\pi}} \exp\left[-\frac{(2600-2500)^2}{2(50)^2}\right] \\ &= 2497.24 \end{aligned}$$

7–37. $E(X) = e^{62.5}$, $V(X) = e^{125}(e^{25} - 1)$, median$(X) = e^{50}$, mode$(X) = e^{25}$

7–38. W is lognormal with parameters 17.06 and 7.0692, or $\ell\text{n}(W) \sim N(17.06, 7.0692)$.

Thus, $P(L \le W \le R) = 0.90$ implies $P(\ell\text{n}(L) \le \ell\text{n}(W) \le \ell\text{n}(R)) = 0.90$, or

$$\Phi\left(\frac{\ell\text{n}(R) - 17.06}{\sqrt{7.0692}}\right) - \Phi\left(\frac{\ell\text{n}(L) - 17.06}{\sqrt{7.0692}}\right) = 0.90.$$

Assuming that the interval $[\ell\text{n}(L), \ell\text{n}(R)]$ is symmetric about 17.06, we obtain $\ell\text{n}(L) = 17.06 - c$ and $\ell\text{n}(R) = 17.06 + c$, so that

$$\Phi\left(\frac{c}{2.6588}\right) - \Phi\left(\frac{-c}{2.6588}\right) = 0.90.$$

This means that $\dfrac{c}{2.6588} = 1.645$, or $c = 4.374$.

$\ell\text{n}(L) = 12.69$, or $L = 324486.8$ and
$\ell\text{n}(R) = 21.43$, or $R = 2027359410$

7–39. $Y \sim N(\mu, \sigma^2)$, $Y = \ell\text{n}(X)$, or $X = e^Y$. The function e^y is strictly increasing in y; thus, from Theorem 3–1,

$$f(x) \;=\; \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{(\ell\text{n}(x) - \mu)^2}{2\sigma^2}\right]; \quad x \geq 0.$$

7–41. $X_1 \sim N(2000, 2500)$, $X_2 \sim N(0.10, 0.01)$, $\rho = 0.87$.

$$\begin{aligned} E(X_1|x_2) &= \mu_1 + \rho(\sigma_1/\sigma_2)(x_2 - \mu_2) \\ &= 2000 + (0.87)(50/0.1)(0.098 - 0.10) \\ &= 1999.13 \end{aligned}$$

$$V(X_1|x_2) \;=\; \sigma_1^2(1 - \rho^2) \;=\; 2500(1 - 0.7569) \;=\; 607.75$$

$$\begin{aligned} P(X_1 > 1950|x_2 = 0.098) &= P\left(Z > \frac{1950 - 1999.13}{\sqrt{607.75}}\right) \\ &= 1 - \Phi(-1.993) \;=\; 0.9769 \end{aligned}$$

7–42. $X_1 \sim N(75, 25)$, $X_2 \sim N(83, 16)$, $\rho = 0.8$.

$$\begin{aligned} E(X_2|x_1) &= \mu_2 + \rho(\sigma_2/\sigma_1)(x_1 - \mu_1) \\ &= 83 + (0.8)(4/5)(80 - 75) \\ &= 86.2 \end{aligned}$$

$$V(X_2|x_1) \;=\; \sigma_2^2(1 - \rho^2) \;=\; 16(1 - 0.64) \;=\; 5.76$$

$$P(X_2 > 80|x_1 = 80) \;=\; P\left(Z > \frac{80 - 86.2}{\sqrt{5.76}}\right) \;=\; 0.9951$$

7–43. (a) $f(x_1, x_2) = k$ implies that

$$k = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{ \frac{-1}{2(1-\rho^2)} \left[\left(\frac{x_1-\mu_1}{\sigma_1} \right)^2 - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \left(\frac{x_2-\mu_2}{\sigma_2} \right)^2 \right] \right\}$$

For a selected value of k, the quantity in brackets assumes a value, say c; thus,

$$\left(\frac{x_1-\mu_1}{\sigma_1} \right)^2 - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \left(\frac{x_2-\mu_2}{\sigma_2} \right)^2 - c = 0,$$

which is a quadratic in $x_1 - \mu_1$ and $x_2 - \mu_2$. If we write the general second-degree equation as

$$Ay_1^2 + By_1y_2 + Cy_2^2 + Dy_1 + Ey_2 + F = 0,$$

we can determine the nature of the curve from the second-order terms. In particular, if $B^2 - 4AC < 0$, the curve is an ellipse. In any case,

$$B^2 - 4AC = \left(\frac{2\rho}{\sigma_1\sigma_2} \right)^2 - \frac{4}{\sigma_1^2\sigma_2^2} = \frac{4(\rho^2-1)}{\sigma_1^2\sigma_2^2} < 0,$$

the last inequality a result of the fact that $\rho^2 < 1$ (for $\rho \neq 0$). Thus, we have an ellipse.

(b) Let $\sigma_1^2 = \sigma_2^2 = \sigma^2$ and $\rho = 0$. Then the equation of the curve becomes

$$\left(\frac{x_1-\mu_1}{\sigma_1} \right)^2 + \left(\frac{x_2-\mu_2}{\sigma_2} \right)^2 - c = 0,$$

which is a circle with center (μ_1, μ_2) and radius $\sigma\sqrt{c}$.

7–44.

$$\begin{aligned} F(r) &= P(R \le r) \\ &= P\left(\sqrt{X_1^2 + X_2^2} \le r \right) \\ &= P(X_1^2 + X_2^2 \le r^2) \\ &= \int\int_A \frac{1}{2\pi\sigma^2} \exp\left[-\frac{(t_1^2 + t_2^2)}{2\sigma^2} \right] dt_1\, dt_2, \end{aligned}$$

where $A = \{(x_1, x_2) : x_1^2 + x_2^2 \le r^2\}$.

Let $x_1 = \rho\cos(\theta)$ and $x_2 = \rho\sin(\theta)$. Then

$$\begin{aligned} P(R \le r) &= \int_0^r \int_0^{2\pi} \frac{\rho}{2\pi\sigma^2} \exp(-\rho^2/2\sigma^2)\, d\theta\, d\rho \\ &= 1 - \exp(-r^2/2\sigma^2) \end{aligned}$$

Thus,

$$f(r) = (r/\sigma^2)\exp(-r^2/2\sigma^2); \quad r > 0.$$

7–45. Using the fact that $\sum_{i=1}^n X_i^2$ has a χ_n^2 distribution, we obtain

$$f(r) = \frac{r^{n-1}e^{-r^2/2}}{2^{(n-2)/2}\Gamma(n/2)}; \quad r \ge 0$$

7–46. Let $Y_1 = X_1/X_2$ and $Y_2 = X_2$, with $X_2 \neq 0$. Then the Jacobian is

$$|J| = \begin{vmatrix} y_2 & y_1 \\ 0 & 1 \end{vmatrix} = |y_2|$$

So we have

$$f(y_1, y_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}|y_2| \exp\left\{\frac{-1}{2(1-\rho^2)}\left[\left(\frac{y_1 y_2}{\sigma_1}\right)^2 - \frac{2\rho y_1 y_2^2}{\sigma_1\sigma_2} + \left(\frac{y_2}{\sigma_2}\right)^2\right]\right\}$$

So the marginal is

$$\begin{aligned} f_{Y_1}(y_1) &= \int_{-\infty}^{\infty} f(y_1, y_2)\, dy_2 \\ &= \frac{\sqrt{1-\rho^2}}{\pi\sigma_1\sigma_2}\left[\left(\frac{y_1}{\sigma_1} - \frac{\rho}{\sigma_2}\right)^2 + \frac{1-\rho^2}{\sigma_2^2}\right]^{-1}; \quad -\infty < y_1 < \infty \end{aligned}$$

When $\rho = 0$ and $\sigma_1^2 = \sigma_2^2 = \sigma^2$, the distribution becomes

$$f_{Y_1}(y_1) = \frac{1}{\pi(1+y_1^2)}; \quad -\infty < y_1 < \infty,$$

also known as the *Cauchy* distribution.

7–47. The CDF is

$$\begin{aligned} F_Y(y) &= P(Y \le y) = P(Z^2 \le y) \\ &= P(-\sqrt{y} \le Z \le \sqrt{y}) \\ &= 2\int_0^{\sqrt{y}} \frac{1}{2\pi} e^{-z^2/2}\, dz \end{aligned}$$

Take $z = \sqrt{u}$ so that $dz = (2\sqrt{u})^{-1}\, du$. Then

$$F_Y(y) = \int_0^y \frac{1}{2\pi} u^{(1/2)-1} e^{-u/2}\, du$$

and so, by Leibniz' rule,

$$f(y) = \frac{1}{2\pi}\, y^{-1/2} e^{-y/2}, \quad y > 0.$$

7–48. The CDF is

$$\begin{aligned} F_Y(y) &= P(Y \le y) = P\left(\sum_{i=1}^n X_i^2 \le y\right) \\ &= \int\int\cdots\int_A (2\pi)^{-n/2} \exp\left[-\frac{1}{2}\sum_{i=1}^n x_i^2\right] dx_1\, dx_2 \cdots dx_n, \end{aligned}$$

where

$$A = \left\{(x_1, x_2, \ldots, x_n) : \sum_{i=1}^n x_i^2 \le y\right\}.$$

Transform to polar coordinates:

$$\begin{aligned} x_1 &= y^{1/2}\cos(\theta_1) \\ x_2 &= y^{1/2}\sin(\theta_1)\cos(\theta_2) \\ x_3 &= y^{1/2}\sin(\theta_1)\sin(\theta_2)\cos(\theta_3) \\ &\vdots \\ x_{n-1} &= y^{1/2}\sin(\theta_1)\sin(\theta_2)\cdots\sin(\theta_{n-2})\cos(\theta_{n-1}) \\ x_n &= y^{1/2}\sin(\theta_1)\sin(\theta_2)\cdots\sin(\theta_{n-2})\sin(\theta_{n-1}) \end{aligned}$$

The Jacobian of this transformation is

$$\begin{vmatrix} \frac{\partial x_1}{\partial y} & \frac{\partial x_1}{\partial \theta_1} & \cdots & \frac{\partial x_1}{\partial \theta_{n-1}} \\ \frac{\partial x_2}{\partial y} & \frac{\partial x_2}{\partial \theta_1} & \cdots & \frac{\partial x_2}{\partial \theta_{n-1}} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial x_n}{\partial y} & \frac{\partial x_n}{\partial \theta_1} & \cdots & \frac{\partial x_n}{\partial \theta_{n-1}} \end{vmatrix}$$

or, after a little algebra,

$$\begin{vmatrix} \frac{\cos(\theta_1)}{2\sqrt{y}} & -\sqrt{y}\sin(\theta_1) & 0 & \cdots & 0 \\ \frac{\sin(\theta_1)\cos(\theta_2)}{2\sqrt{y}} & \sqrt{y}\cos(\theta_1)\cos(\theta_2) & -\sqrt{y}\cos(\theta_1)\sin(\theta_2) & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\sin(\theta_1)\cdots\sin(\theta_{n-1})}{2\sqrt{y}} & \sqrt{y}\cos(\theta_1)\sin(\theta_2)\cdots\sin(\theta_{n-1}) & \cdots & \cdots & \sqrt{y}\sin(\theta_1)\sin(\theta_2)\cdots\cos(\theta_{n-1}) \end{vmatrix}$$

In other words, $J = (1/2)y^{(n/2)-1}|\Delta_n|$, where Δ_n is an $n \times n$ matrix obtained by taking out $(2\sqrt{y})^{-1}$ from the first column and $\sqrt{y}$ from the last $n-1$ columns. Expanding this determinant with respect to the last column, we have

$$\begin{aligned} |\Delta_n| &= \sin(\theta_1)\sin(\theta_2)\cdots\sin(\theta_{n-2})|\Delta_{n-1}| \\ &= \sin^{n-2}(\theta_1)\sin^{n-3}(\theta_2)\cdots\sin(\theta_{n-2}) \end{aligned}$$

This transformation gives variables whose limits are much easier. In the region covered by A, we have $0 \le \theta_i \le \pi$ for $i = 1, 2, \ldots, n-2$, and $0 < \theta_{n-1} < 2\pi$. Thus,

$$\begin{aligned} & P\left(\sum_{i=1}^{n} X_i^2 \le y^*\right) \\ &= \int\int\cdots\int_A \frac{1}{(2\pi)^{n/2}}\,\frac{1}{2}\,y^{(n/2)-1}e^{-y/2}|\Delta_n|\,dy\,d\theta_{n-1}\cdots d\theta_1 \\ &= \frac{1}{2(2\pi)^{n/2}}\int_0^{y^*} y^{(n/2)-1}e^{-y/2}\,dy\int_0^{2\pi} d\theta_{n-1}\int_0^{\pi}\sin(\theta_{n-2})\,d\theta_{n-2}\cdots\int_0^{\pi}\sin^{n-2}(\theta_1)\,d\theta_1 \\ &= K\int_0^{y^*} y^{(n/2)-1}e^{-y/2}\,dy \;\equiv\; F(y^*) \end{aligned}$$

Thus,

$$f(y) \;=\; F'(y) \;=\; Ky^{(n/2)-1}e^{-y/2}, \quad y \ge 0.$$

To evaluate K, use $K\int_0^\infty f(y)\,dy = 1$. This finally gives

$$f(y) \;=\; \frac{1}{2^{n/2}\Gamma(n/2)} y^{(n/2)-1} e^{-y/2}, \quad y \geq 0.$$

7–49. For $x \geq 0$,

$$\begin{aligned} F(x) &= P(|X| \leq x) \;=\; P(-x \leq X \leq x) \\ &= \int_{-x}^{x} \frac{1}{\sqrt{2\pi}}\, e^{-t^2/2}\, dt \\ &= 2\int_{0}^{x} \frac{1}{\sqrt{2\pi}}\, e^{-t^2/2}\, dt, \end{aligned}$$

so that

$$\begin{aligned} f(x) &= \frac{2}{\sqrt{2\pi}}\, e^{-x^2/2}, \quad x > 0 \\ &= 0, \quad \text{otherwise} \end{aligned}$$

Chapter 8

8–1. $\bar{x} = 131.30$, $s^2 = 113.85$, $s = 10.67$.

8–2. Descriptive Statistics for Y

Mean	=	34.767
Variance	=	1.828
Standard Dev	=	1.352
Skewness	=	0.420
Kurtosis	=	2.765
Minimum	=	32.100
Maximum	=	37.900

$n = 64$

Lower Limit	Cell Count	
32.1000	1	X
32.4625	4	XXXX
32.8250	3	XXX
33.1875	2	XX
33.5500	7	XXXXXXX
33.9125	6	XXXXXX
34.2750	9	XXXXXXXXX
34.6375	7	XXXXXXX
35.0000	7	XXXXXXX
35.3625	5	XXXXX
35.7250	2	XX
36.0875	2	XX
36.4500	4	XXXX
36.8125	1	X
37.1750	1	X
37.5375	3	XXX

8–3. Descriptive Statistics for X

Mean	=	89.476
Variance	=	17.287
Standard Dev	=	4.158
Skewness	=	0.251
Kurtosis	=	1.988
Minimum	=	82.600
Maximum	=	98.000

$n = 90$

Lower Limit	Cell Count	
82.60	4	XXXX
83.37	6	XXXXXX
84.14	4	XXXX
84.91	6	XXXXXX
85.68	7	XXXXXXX
86.45	3	XXX
87.22	8	XXXXXXXX
87.99	4	XXXX
88.76	4	XXXX
89.53	7	XXXXXXX
90.30	6	XXXXXX
91.07	4	XXXX
91.84	3	XXX
92.61	4	XXXX
93.38	4	XXXX
94.15	5	XXXXX
94.92	4	XXXX
95.69	3	XXX
96.46	1	X
97.23	3	XXX

8–4.

Number of Defects	Frequency	Relative Freq
1	1	0.0067
2	14	0.0933
3	11	0.0733
4	21	0.1400
5	10	0.0667
6	18	0.1200
7	15	0.1000
8	14	0.0933
9	9	0.0600
10	15	0.1000
11	4	0.0267
12	4	0.0267
13	6	0.0400
14	5	0.0333
15	1	0.0067
16	1	0.0067
17	1	0.0067
	150	1.0000

$\bar{x} = 6.9334$, $s^2 = 12.5056$, $R = 16$, $\tilde{x} = 6.5$, MO $= 4$. The data appear to follow a Poisson distribution, though s^2 seems to be somewhat greater than $\bar{x}$.

8–5. $\bar{x} = 131.30$, $s^2 = 113.85$, $s = 10.67$.

8–6.

Class Interval	Frequency	Relative Freq
$32 \leq X < 33$	6	0.094
$33 \leq X < 34$	11	0.172
$34 \leq X < 35$	22	0.344
$35 \leq X < 36$	14	0.219
$36 \leq X < 37$	6	0.094
$37 \leq X < 38$	5	0.077
	64	1.000

$\bar{x} = 34.7672$, $s^2 = 1.828$, $\tilde{x} = (34.6 + 34.7)/2 = 34.65$. The data appear to follow a normal distribution.

8–7.

Class Interval	Frequency	Relative Freq
$82 \leq X < 84$	6	0.067
$84 \leq X < 86$	14	0.156
$86 \leq X < 88$	18	0.200
$88 \leq X < 90$	11	0.122
$90 \leq X < 92$	14	0.156
$92 \leq X < 94$	8	0.088
$94 \leq X < 96$	12	0.133
$96 \leq X < 98$	6	0.067
$98 \leq X < 100$	1	0.011

$\bar{x} = 89.4755$, $s^2 = 17.2870$. The data appear to follow a either a gamma or a Weibull distribution.

8–8. (a) Descriptive Statistics for Time

Mean	=	14.355
Variance	=	356.577
Standard Dev	=	18.883
Skewness	=	1.809
Kurtosis	=	5.785
Minimum	=	0.190
Maximum	=	72.890

$n = 19$

Lower Limit	Cell Count	
0.1900	13	XXXXXXXXXXXXX
10.5757	1	X
20.9614	0	
31.3471	4	XXXX
41.7329	0	
52.1186	0	
62.5043	1	X

(b) $\bar{x} = 14.355$, $s^2 = 356.577$, $s = 18.88$, $\tilde{x} = 6.5$.

8–9. $\bar{x} = 126.875$, $s^2 = 660.12$, $s = 25.693$

8–10. (a,b)

83	4
84	3
85	3
86	7 7
87	7 5 8 6 9 4
88	5 6 3 2 3 5 3 6 7 4 9
89	8 2 0 9 8 6 3 8 3 7
90	8 3 1 9 4 1 4 6 4 3 5 0 7
91	5 1 0 0 8 2 8 6 1 1 6 2 0
92	7 3 7 6 7 2 2 2
93	3 2 4 3 0 7
94	7 2 2 4
95	6
96	1
97	
98	8
99	
100	3

(c) $\bar{x} = 90.6425$, $s^2 = 7.837$, $s = 2.799$

(d) $\tilde{x}$ = median = 90.45. There are several modes, e.g., 91.0, 919.1, 92.7.

8–12. (a)

		Frequency
32	5 6 9 8 1 7	6
33	1 6 6 8 4 6 8 1 6 5 6	11
34	2 5 3 7 7 2 7 6 9 7 1 6 0 1 6 7 6 5 6 1 7 3	22
35	6 1 0 4 1 3 2 0 1 4 9 8 5 7	14
36	2 8 8 4 6 8	6
37	9 8 1 6 3	5

(b) $\bar{x} = 34.7672$, $s^2 = 1.828$

(c)

		Frequency
32	1 5 6 7 8 9	6
33	1 1 4 5 6 6 6 6 6 8 8	11
34	0 1 1 1 2 2 3 3 5 5 6 6 6 6 6 7 7 7 7 7 7 9	22
35	0 0 1 1 1 2 3 4 4 5 6 7 8 9	14
36	2 4 6 8 8 8	6
37	1 3 6 8 9	5

(d) $\tilde{x} = 34.65$

8–13. (a)

		Frequency
82	6 9	2
83	0 1 6 7	4
84	0 1 1 1 2 5 6 9	8
85	0 1 1 1 4 4	6
86	1 1 1 4 4 4 4 6 7 7	10
87	3 3 3 3 5 6 6 7	8
88	2 2 3 6 8	5
89	1 1 4 6 6 7	6
90	0 0 1 1 3 4 5 6 6 6	10
91	1 2 4 7	4
92	1 4 4	3
93	1 1 2 2 7	5
94	1 1 1 3 3 4 6 7	8
95	1 2 3 6	4
96	1 3 4 8	4
97	3 8	2
98	0	1

(b) $\bar{x} = 89.25$, $Q1 = 86.1$, $Q3 = 93.1$.

(c) $IQR = Q3 - Q1 = 7.0$

8–14. min = 82.6, $Q1 = 86.1$, $\bar{x} = 89.25$, $Q3 = 93.1$, max = 98.1

8–15. min = 32.1, $Q1 = 33.8$, $\bar{x} = 34.65$, $Q3 = 35.45$, max = 37.9

8–16. min = 1, $Q1 = 4$, $\bar{x} = 7$, $Q3 = 10$, max = 17

8–18. The descriptive measures developed in this chapter are for numerical data only. The mode, however, does have some meaning. For these data, the mode is the letter e.

8–19. (a)

$$\begin{aligned}\sum_{i=1}^{n}(X_i - \bar{X}) &= \sum_{i=1}^{n} X_i - \sum_{i=1}^{n} \bar{X} = \sum_{i=1}^{n} X_i - n\bar{X} \\ &= \sum_{i=1}^{n} X_i - \sum_{i=1}^{n} X_i = 0\end{aligned}$$

(b)

$$\begin{aligned}\sum_{i=1}^{n}(X_i - \bar{X})^2 &= \sum_{i=1}^{n}(X_i^2 + \bar{X}^2 - 2X_i\bar{X}) \\ &= \sum_{i=1}^{n} X_i^2 + n\bar{X}^2 - 2\bar{X}\sum_{i=1}^{n} X_i \\ &= \sum_{i=1}^{n} X_i^2 + n\bar{X}^2 - 2n\bar{X}^2 \\ &= \sum_{i=1}^{n} X_i^2 - n\bar{X}^2\end{aligned}$$

8–20. $\bar{x} = 1.1933$, $s^2 = 0.000266$, $s = 0.016329$, $\tilde{x}$ = median $= (1.19 + 1.20)/2 = 1.195$, mode = 1.21.

8–21. $\bar{x} = 74.0020$, $s = 0.0026$, $s^2 = 6.875 \times 10^{-6}$

8–22. $\bar{x} = 62.75$, $s = 2.12$, $s^2 = 4.5$

8–23. (a) Sample average will be reduced by 63.

(b) Sample mean and standard deviation will be 100 units larger; the sample variance will be 10000 units larger.

8–24. $\bar{y} = a + b\bar{x}$, $s_y = bs_x$

8–25. $a = \bar{x}$

8–26. (a) 89.336

(b) 89.237

8–27. There is no guarantee that LN is an integer. For example, if we want a 10% trimmed mean with 23 observations, then we would have to trim 2.3 observations from each end. Since we cannot do this, some other procedure must be used. A reasonable alternative is to calculate the trimmed mean with two observations trimmed from each end, then to repeat this procedure with three observations trimmed from each end, and finally to interpolate between the two different values of the trimmed mean.

8–29. (a) $\bar{x} = 120.22481$, $s^2 = 5.66001$

(b) median = 120, mode = 121

8–30. (a) $\bar{x} = -0.20472$, $s^2 = 3.96119$

(b) median = mode = 0

8–31. For 8–29, $cv = 2.379/120.225 = 0.01979$.
For 8–30, $cv = 1.990/(-0.205) = -9.722$.

8–32. $\bar{x} \approx 51.124$, $s^2 \approx 586.603$, $\tilde{x} \approx 48.208$, mode ≈ 36.334

8–33. $\bar{x} \approx 22.407$, $s^2 \approx 208.246$, $\tilde{x} \approx 22.813$, mode ≈ 23.64

8–34. $\bar{x} \approx 847.885$, $s^2 \approx 15987.81$, $s \approx 126.44$, $\tilde{x} \approx 858.98$, mode ≈ 1050

Chapter 9

9–1. Since

$$f(x_i) \;=\; \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x_i-\mu)^2}{2\sigma^2}\right],$$

we have

$$\begin{aligned} f(x_1, x_2, \ldots, x_5) &= \prod_{i=1}^{5} f(x_i) \\ &= \prod_{i=1}^{5} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x_i-\mu)^2}{2\sigma^2}\right] \\ &= \left(\frac{1}{\sigma^2 2\pi}\right)^{5/2} \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{5}(x_i-\mu)^2\right] \end{aligned}$$

9–2. Since

$$f(x_i) \;=\; \lambda e^{-\lambda x_i},$$

we have

$$\begin{aligned} f(x_1, x_2, \ldots, x_n) &= \prod_{i=1}^{n} f(x_i) \\ &= \prod_{i=1}^{n} \lambda e^{-\lambda x_i} \\ &= \lambda^n \exp\left[-\lambda \sum_{i=1}^{n} x_i\right] \end{aligned}$$

9–3. Since $f(x_i) = 1$, we have

$$f(x_1, x_2, x_3, x_4) \;=\; \prod_{i=1}^{4} f(x_i) \;=\; 1$$

9–4. The joint probability function for X_1 and X_2 is

$$\begin{aligned}
p_{X_1,X_2}(0,0) &= \frac{\binom{N-M}{0}\binom{M}{2}}{\binom{N}{2}} \\
p_{X_1,X_2}(0,1) &= \frac{\binom{N-M}{1}\binom{M}{1}}{2\binom{N}{2}} \\
p_{X_1,X_2}(1,0) &= \frac{\binom{N-M}{1}\binom{M}{1}}{2\binom{N}{2}} \\
p_{X_1,X_2}(1,1) &= \frac{\binom{N-M}{2}\binom{M}{0}}{\binom{N}{2}}
\end{aligned}$$

Of course,

$$\begin{aligned}
p_{X_1}(x_1) &= \sum_{x_2=0}^{1} p_{X_1,X_2}(x_1,x_2) \qquad \text{and} \\
p_{X_2}(x_2) &= \sum_{x_1=0}^{1} p_{X_1,X_2}(x_1,x_2)
\end{aligned}$$

So $p_{X_1}(0) = M/N$, $p_{X_1}(1) = 1 - (M/N)$, $p_{X_2}(0) = M/N$, $p_{X_2}(1) = 1 - (M/N)$.

Thus, X_1 and X_2 are not independent since

$$p_{X_1,X_2}(0,0) \neq p_{X_1}(0)p_{X_2}(0)$$

9–5. $N(\mu, \sigma^2/n) = N(5, 0.00125)$

9–6. $\sigma/\sqrt{n} = 0.1/\sqrt{8} = 0.0353$

9–7. Use estimated standard error $S/\sqrt{n}$.

9–8. $N(-5, 0.22)$

9–9. The standard error of $\bar{X}_1 - \bar{X}_2$ is

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \;=\; \sqrt{\frac{(1.5)^2}{25} + \frac{(2.0)^2}{30}} \;=\; 0.473$$

9–10. $Y = \bar{X}_1 - \bar{X}_2$ is a linear combination of the 55 variables X_{ij}, $i = 1, j = 1, 2, \ldots, 25$, $i = 2, j = 1, 2, \ldots, 30$. As such, we would expect Y to be very nearly normal with mean $\mu_Y = -0.5$ and variance $(0.473)^2 = 0.223$.

9–11. $N(0, 1)$

9–12. $N(\hat{p}, \hat{p}(1-\hat{p})/n)$

9–13. $se(\hat{p}) = \sqrt{p(1-p)/n}$, $\widehat{se}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$,

9–14.

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \int_0^\infty e^{tx} \frac{1}{2^{n/2}\Gamma(n/2)} x^{(n/2)-1} e^{-x/2}\, dx \\ &= \frac{1}{2^{n/2}\Gamma(n/2)} \int_0^\infty x^{(n/2)-1} e^{-x[(1/2)-t]}\, dx \end{aligned}$$

This integral converges if $1/2 > t$.

Let $u = x[(1/2) - t]$. Then $dx = [(1/2) - t]^{-1} du$. Thus,

$$\begin{aligned} M_X(t) &= \frac{1}{2^{n/2}\Gamma(n/2)} \int_0^\infty \frac{u^{(n/2)-1}}{[(1/2)-t]^{(n/2)-1}} e^{-u} \frac{1}{[(1/2)-t]}\, du \\ &= \frac{1}{2^{n/2}\Gamma(n/2)[(1-2t)/2]^{n/2}} \int_0^\infty u^{(n/2)-1} e^{-u}\, du \\ &= \frac{1}{(1-2t)^{n/2}}, \quad t < 1/2, \end{aligned}$$

since $\Gamma(n/2) = \displaystyle\int_0^\infty u^{(n/2)-1} e^{-u}\, du$.

9–15. First of all,

$$\begin{aligned} M'_X(t) &= n(1-2t)^{-(n/2)-1} \\ M''_X(t) &= n(n+2)(1-2t)^{-(n/2)-2} \end{aligned}$$

Then

$$\begin{aligned} E(X) &= M'_X(0) = n \\ E(X^2) &= M''_X(0) = n(n+2) \\ V(X) &= E(X^2) - [E(X)]^2 = 2n \end{aligned}$$

9–16. Let $T = Z/\sqrt{\chi^2_n/n} = Z\sqrt{n/\chi^2_n}$. Now

$E(T) = E(Z)E(\sqrt{n/\chi^2_n}) = 0$, because $E(Z) = 0$.

$V(T) = E(T^2)$, because $E(T) = 0$. Thus,

$V(T) = E[Z^2(n/\chi^2_n)] = E(Z^2)E(n/\chi^2_n)$.

Note that $E(Z^2) = V(Z) = 1$, so that

$$\begin{aligned} V(T) &= E(n/\chi^2_n) \\ &= \int_0^\infty \frac{(n/s)}{2^{n/2}\Gamma(n/2)} s^{(n/2)-1}\, e^{-s/2}\, ds \\ &= \frac{n}{2^{n/2}\Gamma(n/2)} \int_0^\infty s^{(n/2)-2}\, e^{-s/2}\, ds \\ &= \frac{n}{2^{(n/2)-1}\Gamma(n/2)} \int_0^\infty (2u)^{(n/2)-2}\, e^{-u}\, du \\ &= \frac{n\Gamma(\frac{n}{2}-1)}{2\Gamma(\frac{n}{2})}, \quad \text{if } n > 2 \\ &= \frac{n\Gamma(\frac{n}{2}-1)}{2(\frac{n}{2}-1)\Gamma(\frac{n}{2}-1)}, \quad \text{if } n > 2 \\ &= \frac{n}{n-2}, \quad \text{if } n > 2 \end{aligned}$$

9–17. $E(F_{m,n}) = E[(\chi^2_m/m)/(\chi^2_n/n)] = E(\chi^2_m/m)E(n/\chi^2_n)$.

$E(\chi^2_m/m) = (1/m)E(\chi^2_m) = 1$.

From Problem 9–16, we have $E(n/\chi^2_n) = n/(n-2)$.

Therefore, $E(F_{m,n}) = n/(n-2)$, if $n > 2$.

To find $V(F_{m,n})$, let $X \sim \chi^2_m$ and $Y \sim \chi^2_n$. Then

$E(F^2_{m,n}) = (n/m)^2 E(X^2)E(1/Y^2)$.

Since $E(X^2) = V(X) + [E(X)]^2$ and $X \sim \chi^2_m$, we have $E(X^2) = 2m + m^2$. Now

$$\begin{aligned} E(1/Y^2) &= \int_0^\infty \frac{(1/y^2)}{2^{n/2}\Gamma(n/2)} y^{(n/2)-1}\, e^{-y/2}\, dy \\ &= \frac{1}{2^{n/2}\Gamma(n/2)} \int_0^\infty y^{(n/2)-3}\, e^{-y/2}\, dy \\ &= \frac{1}{2^{n/2}\Gamma(n/2)} \int_0^\infty 2(2u)^{(n/2)-3}\, e^{-u}\, du \\ &= \frac{1}{(n-2)(n-4)}, \quad \text{if } n > 4 \end{aligned}$$

Thus,

$$\begin{aligned} V(F_{m,n}) &= (n/m)^2 E(X^2)E(1/Y^2) - \left(\frac{n}{n-2}\right)^2 \\ &= \frac{n^2(2m+m^2)}{m^2(n-2)(n-4)} - \frac{n^2}{(n-2)^2} = \frac{2n^2(m+n-2)}{m(n-2)^2(n-4)} \end{aligned}$$

9–18. $X_{(1)}$ is greater than t if and only if every observation is greater than t. Then

$$\begin{aligned} P(X_{(1)} > t) &= P(X_1 > t, X_2 > t, \ldots, X_n > t) \\ &= P(X_1 > t)P(X_2 > t)\cdots P(X_n > t) \\ &= P(X > t)P(X > t)\cdots P(X > t) \\ &= [1 - F(t)]^n \end{aligned}$$

So $F_{X(1)}(t) = 1 - P(X_{(1)} > t) = 1 - [1 - F(t)]^n$.

If X is continuous, then so is $X_{(1)}$; so

$$f_{X_{(1)}}(t) \;=\; F'_{X(1)}(t) \;=\; n[1 - F(t)]^{n-1} f(t)$$

Similarly,

$$\begin{aligned} F_{X(n)}(t) &= P(X_{(n)} \le t) \\ &= P(X_1 \le t, X_2 \le t, \ldots, X_n \le t) \\ &= P(X_1 \le t)P(X_2 \le t) \cdots P(X_n \le t) \\ &= P(X \le t)P(X \le t) \cdots P(X \le t) \\ &= [F(t)]^n \end{aligned}$$

Since $X_{(n)}$ is continuous,

$$f_{X_{(n)}}(t) \;=\; F'_{X(n)}(t) \;=\; n[F(t)]^{n-1} f(t)$$

9–19.

$$F(t) \;=\; \begin{cases} 0 & t < 0 \\ 1 - p & 0 \le t < 1 \\ 1 & t \ge 1 \end{cases}$$

Then

$$P(X_{(n)} = 1) \;=\; F_{X_{(n)}}(1) - F_{X_{(n)}}(0) \;=\; [F(1)]^n - [F(0)]^n \;=\; 1 - (1 - p)^n$$

$$P(X_{(1)} = 0) \;=\; 1 - [1 - F(0)]^n \;=\; 1 - [1 - (1 - p)]^n \;=\; 1 - p^n$$

9–20.

$$f_{X_{(1)}}(t) \;=\; n\left[1 - \Phi\left(\frac{t-\mu}{\sigma}\right)\right]^{n-1} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

$$f_{X_{(n)}}(t) \;=\; n\left[\Phi\left(\frac{t-\mu}{\sigma}\right)\right]^{n-1} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

9–21. $f(t) = \lambda e^{-\lambda t}, \quad t > 0$

$F(t) = 1 - e^{-\lambda t}$

$F_{X_{(1)}}(t) = 1 - [1 - F(t)]^n = 1 - [1 - (1 - e^{-\lambda t})]^n = 1 - e^{-n\lambda t}$

$f_{X_{(1)}}(t) = n\lambda e^{-n\lambda t}, \quad t > 0$

$F_{X_{(n)}}(t) = [F(t)]^n = (1 - e^{-\lambda t})^n$

$f_{X_{(n)}}(t) = n(1 - e^{-\lambda t})^{n-1}\lambda e^{-\lambda t}, \quad t > 0$

9–22. $f_{X_{(n)}}(X_{(n)}) = n[F(X_{(n)})]^{n-1} f(X_{(n)})$

Treat $F(X_{(n)})$ as a random variable giving the fraction of objects in the population having values of $X \leq X_{(n)}$.

Let $Y = F(X_{(n)})$. Then $dy = f(X_{(n)})dx_{(n)}$, and thus $f(y) = ny^{n-1}$, $0 \leq y \leq 1$.

This gives

$$E(Y) \;=\; \int_0^1 ny^n \, dy \;=\; \frac{n}{n+1}.$$

Similarly, $f_{X_{(1)}}(X_{(1)}) = n[1 - F(X_{(1)})]^{n-1} f(X_{(1)})$

Treat $F(X_{(1)})$ as a random variable giving the fraction of objects in the population having values of $X \leq X_{(1)}$.

Let $Y = F(X_{(1)})$. Then $dy = f(X_{(1)})dx_{(1)}$, and thus $f(y) = n(1-y)^{n-1}$, $0 \leq y \leq 1$.

This gives

$$E(Y) \;=\; \int_0^1 ny(1-y)^{n-1} \, dy$$

The family of Beta distributions is defined by p.d.f.'s of the form

$$g(x) = \begin{cases} [\beta(r,s)]^{-1}x^{r-1}(1-x)^{s-1} & 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

where $\beta(r,s) = \Gamma(r)\Gamma(s)/\Gamma(r+s)$.

Thus,

$$\begin{aligned} E(Y) &= n\int_0^1 y(1-y)^{n-1}\,dy = n\beta(2,n) \\ &= \frac{n\Gamma(2)\Gamma(n)}{\Gamma(n+2)} = \frac{n!1!}{(n+1)!} = \frac{1}{n+1} \end{aligned}$$

9–23. (a) 2.73
(b) 11.34
(c) 34.17
(d) 20.48

9–24. (a) 2.228
(b) 0.687
(c) 1.813

9–25. (a) 1.63
(b) 2.85
(c) 0.241
(d) 0.588

Chapter 10

10–1. Both estimators are unbiased. Now, $V(\overline{X}_1) = \sigma^2/2n$ while $V(\overline{X}_2) = \sigma^2/n$. Since $V(\overline{X}_1) < V(\overline{X}_2)$, $\overline{X}_1$ is a more efficient estimator than $\overline{X}_2$.

10–2. $E(\hat{\theta}_1) = \mu$, $E(\hat{\theta}_2) = (1/2)E(2X_1 - X_6 + X_4) = (1/2)(2\mu - \mu + \mu) = \mu$. Both estimators are unbiased.

$$V(\hat{\theta}_1) = \sigma^2/7,$$

$$V(\hat{\theta}_2) = \left(\frac{1}{2}\right)^2 V(2X_1 - X_6 + X_4)$$

$$= \left(\frac{1}{4}\right)[4V(X_1) + V(X_6) + V(X_4)] = \left(\frac{1}{4}\right)6\sigma^2 = 3\sigma^2/2$$

$\hat{\theta}_1$ has a smaller variance than $\hat{\theta}_2$.

10–3. Since $\hat{\theta}_1$ is unbiased, $MSE(\hat{\theta}_1) = V(\hat{\theta}_1) = 10$.

$$MSE(\hat{\theta}_2) = V(\hat{\theta}_2) + (Bias)^2 = 4 + (\theta - \theta/2)^2 = 4 + \theta^2/4.$$

If $\theta < \sqrt{24} = 4.8990$, $\hat{\theta}_2$ is a better estimator of θ than $\hat{\theta}_1$, because it would have smaller MSE.

10–4. $MSE(\hat{\theta}_1) = V(\hat{\theta}_1) = 12$, $MSE(\hat{\theta}_2) = V(\hat{\theta}_2) = 10$,
$MSE(\hat{\theta}_3) = E(\hat{\theta}_3 - \theta)^2 = 6$. $\hat{\theta}_3$ is a better estimator because it has smaller MSE.

10–5. $E(S^2) = (1/24)E(10S_1^2 + 8S_2^2 + 6S_3^2) = (1/24)(10\sigma^2 + 8\sigma^2 + 6\sigma^2)$
$= (1/24)24\sigma^2 = \sigma^2$

10–6. Any linear estimator of μ is of the form $\hat{\theta} = \sum_{i=1}^n a_i X_i$ where a_i are constants. $\hat{\theta}$ is an unbiased estimator of μ only if $E(\hat{\theta}) = \mu$, which implies that $\sum_{i=1}^n a_i = 1$. Now $V(\hat{\theta}) = \sum_{i=1}^n a_i^2\sigma^2$. Thus we must choose the a_i to minimize $V(\hat{\theta})$ subject to the constraint $\sum a_i = 1$. Let λ be a Lagrange multiplier. Then

$$F(a_i, \lambda) = \sum_{i=1}^n a_i^2\sigma^2 - \lambda\left(\sum_{i=1}^n a_i - 1\right)$$

and $\partial F/\partial a_i = \partial F/\partial\lambda = 0$ gives

$$2a_i\sigma^2 - \lambda = 0; i = 1, 2, \ldots, a$$

$$\sum_{i=1}^n a_i = 1$$

The solution is $a_i = 1/n$. Thus $\hat{\theta} = \overline{X}$ is the best linear unbiased estimator of μ.

10–7. $$L(\alpha) = \prod_{i=1}^{n} \alpha^{X_i} e^{-\alpha}/X_i! = \alpha^{\Sigma X_i} e^{-n\alpha} \Big/ \prod_{i=1}^{n} X_i!$$

$$\ell\mathrm{n}\, L(\alpha) = \sum_{i=1}^{n} X_i\, \ell\mathrm{n}\, \alpha - n\alpha - \ell\mathrm{n} \left(\prod_{i=1}^{n} X_i! \right)$$

$$\frac{d\, \ell\mathrm{n}\, L(\alpha)}{d\alpha} = \sum_{i=1}^{n} X_i/\alpha - n = 0$$

$$\hat{\alpha} = \sum_{i=1}^{n} X_i/n = \overline{X}$$

10–8. For the Poisson distribution, $E(X) = \alpha = \mu_1'$. Also, $M_1' = \overline{X}$. Thus $\hat{\alpha} = \overline{X}$ is the moment estimator of α.

10–9. $$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda t_i} = \lambda^n e^{-\lambda \sum_{i=1}^{n} t_i}$$

$$\ell\mathrm{n}\, L(\lambda) = n\, \ell\mathrm{n}\, \lambda - \lambda \sum_{i=1}^{n} t_i$$

$$\frac{d\, \ell\mathrm{n}\, L(\lambda)}{d\lambda} = (n/\lambda) - \sum_{i=1}^{n} t_i = 0$$

$$\hat{\lambda} = n \Big/ \sum_{i=1}^{n} t_i = (\bar{t})^{-1}$$

10–10. $E(t) = 1/\lambda = \mu_1'$, and $M_1' = \bar{t}$. Thus $1/\lambda = \bar{t}$ or $\hat{\lambda} = (\bar{t})^{-1}$.

10–11. If X is a gamma random variable, then $E(X) = r/\lambda$ and $V(X) = r/\lambda^2$. Thus $E(X^2) = (r + r^2)\lambda^2$. Now $M_1' = \overline{X}$ and $M_2' = (1/n) \sum_{i=1}^{n} X_i^2$. Equating moments, we obtain

$$r/\lambda = \overline{X}, \quad (r + r^2)\lambda^2 = (1/n) \sum_{i=1}^{n} X_i^2$$

or,

$$\hat{\lambda} = \overline{X} \Big/ \left[(1/n) \sum_{i=1}^{n} X_i^2 - \overline{X}^2 \right]$$

$$\hat{r} = \overline{X}^2 \Big/ \left[(1/n) \sum_{i=1}^{n} X_i^2 - \overline{X}^2 \right]$$

10–12. $E(X) = 1/p$, $M_1' = \overline{X}$. Thus $1/p = \overline{X}$ or $\hat{p} = 1/\overline{X}$.

10–13. $L(p) = \prod_{i=1}^{n}(1-p)^{X_i-1}p = p^n(1-p)^{\Sigma X_i - n}$

$\ell\text{n}\, L(p) = n\,\ell\text{n}\, p + (\sum_{i=1}^{n} X_i - n)\ell\text{n}\,(1-p)$. From $d\,\ell\text{n}\, L(p)/d\rho = 0$, we obtain

$$(n/\hat{p}) - \left(\sum_{i=1}^{n} X_i - n\right) \Big/ (1-\hat{\rho}) = 0$$

$$\hat{p} = n \Big/ \sum_{i=1}^{n} X_i = 1/\overline{X}$$

10–14. $E(X) = p$, $M_1' = \overline{X}$. Thus $\hat{p} = \overline{X}$.

10–15. $E(X) = np$ (n is known), $M_1' = \overline{X}_N$ ($\overline{X}$ is based on a sample of N observations.) Thus $np = \overline{X}_N$ or $\hat{p} = \overline{X}_N/n$.

10–16. $E(X) = np$, $V(X) = np(1-p)$, $E(X^2) = np - np^2 + n^2p^2$
$M_1' = \overline{X}$, $M_2' = (1/N)\sum_{i=1}^{N} X_i^2$. Equating moments,

$$np = \overline{X}, np - np^2 + n^2p^2 = (1/N)\sum_{i=1}^{N} X_i^2$$

$$\hat{n} = \overline{X}^2 \Big/ \left[\overline{X} - (1/N)\sum_{i=1}^{N}(X_i - \overline{X})^2\right],\ \hat{p} = \overline{X}/\hat{N}$$

10–17. $$L(p) = \prod_{i=1}^{N}\binom{n}{X_i} p^{X_i}(1-p)^{n-X_i} = \left[\prod_{i=1}^{N}\binom{n}{X_i}\right] p^{\sum_{i=1}^{n} X_i}(1-p)^{nN-\sum_{i=1}^{N} X_i}$$

$$\ell\text{n}\, L(p) = \sum_{i=1}^{N} \ell\text{n}\binom{n}{X_i} + \left(\sum_{i=1}^{N} X_i\right)\ell\text{n}\, p + \left(nN - \sum_{i=1}^{N} X_i\right)\ell\text{n}(1-p)$$

$$\frac{d\,\ell\text{n}\, L(p)}{dp} = \sum_{i=1}^{N} X_i \Big/ p - \left(nN - \sum_{i=1}^{N} X_i\right) \Big/ (1-p) = 0$$

$$\hat{p} = \overline{X}/n$$

10–18. $L = \prod_{i=1}^{n} \left(\frac{\beta}{\delta}\right) \left(\frac{X_i - \gamma}{\delta}\right)^{\beta\tau} \exp\left[-\left(\frac{X_i - \gamma}{\delta}\right)^{\beta}\right]$

The system of partial derivatives $\partial L/\partial\delta = \partial L/\partial\beta = \partial L/\partial\gamma = 0$ yield simultaneous nonlinear equations that must be solved to produce the maximum likelihood estimators. In general, iterative methods must be used to find the maximum likelihood estimates. A number of special cases are of practical interest; for example, if $\gamma = 0$, the two-parameter Weibull distribution results. Both iterative and linear estimation techniques can be used for the two-parameter case.

10–19. Let X be a random variable and c be a constant. Then Chebychev's inequality is

$$P(|X - c| \geq \epsilon) \leq (1/\epsilon^2)E(X - c)^2$$

Thus,

$$P(|\hat{\theta} - \theta| \geq \epsilon) \leq (1/\epsilon^2)E(\hat{\theta} - \theta)^2$$

Now $E(\hat{\theta} - \theta)^2 = V(\hat{\theta}) + [E(\hat{\theta}) - \theta]^2$.
Then

$$P(|\hat{\theta} - \theta| \geq \epsilon) \leq (1/\epsilon^2)\{V(\hat{\theta}) + [E(\hat{\theta}) - \theta]^2\}$$

If $\hat{\theta}$ is unbiased then $E(\hat{\theta}) - \theta = 0$ and if $\lim_{n\to\infty} V(\hat{\theta}) = 0$ we see that $\lim_{n\to\infty} P(|\hat{\theta} - \theta| \geq \epsilon) \leq 0$, or $\lim_{n\to\infty} P(|\hat{\theta} - \theta| \geq \epsilon) = 0$.

10–20. $$E(\overline{X}) = E[a\overline{X}_1 + (1-a)\overline{X}_2] = aE(\overline{X}_1) + (1-a)E(\overline{X}_2)$$
$$= a\mu + (1-a)\mu = \mu$$
$$V(\overline{X}) = a^2V(\overline{X}_1) + (1-a)^2V(\overline{X}_2) = a^2(\sigma^2/n_1) + (1-a)^2(\sigma^2/n_2)$$
$$\frac{dV(\overline{X})}{da} = 2a\left(\frac{\sigma^2}{n_1}\right) - 2(1-a)\left(\frac{\sigma^2}{n_2}\right) = 0$$
$$a^* = \frac{\sigma^2/n_2}{\sigma^2/n_1 + \sigma^2/n_2} = \frac{n_1}{n_1 + n_2}$$

10–21. $$L(\gamma) = \prod_{i=1}^{n}(\gamma+1)X_i^{\gamma} = (\gamma+1)^n \prod_{i=1}^{n} X_i^{\gamma}$$
$$\ell\mathrm{n}\, L(\gamma) = n\,\ell\mathrm{n}(\gamma+1) + \sum_{i=1}^{n} \gamma\,\ell\mathrm{n}\, X_i$$

$$\frac{d\,\ell\mathrm{n}\,L(\gamma)}{d\gamma} = n \Big/ (\gamma+1) + \sum_{i=1}^{n} \ell\mathrm{n}\,X_i = 0$$

$$\hat{\gamma} = -1 - \left(n \Big/ \sum_{i=1}^{n} \ell\mathrm{n}\,X_i \right)$$

10–22. $$L(\gamma) = \prod_{i=1}^{n} \lambda e^{-\lambda(X_i - X_\ell)} = \lambda^n e^{-\lambda(\sum_{i=1}^{n} X_i - nX_\ell)}$$

$$\ell\mathrm{n}\,L(\lambda) = n\,\ell\mathrm{n}\,\lambda - \lambda \left(\sum_{i=1}^{n} X_i - nX_\ell \right)$$

$$\frac{d\,\ell\mathrm{n}\,L(\lambda)}{d\lambda} = n \Big/ \lambda - \left(\sum_{i=1}^{n} X_i - nX_\ell \right) = 0$$

10–23. Assume X_ℓ unknown, and we want to maximize $n\,\ell\mathrm{n}\,\lambda - \lambda \sum_{i=1}^{n}(X_i - X_\ell)$ with respect to X_ℓ, subject to $X_i \geq X_\ell$. Thus we want $\sum_{i=1}^{n}(X_i - X_\ell)$ to be a minimum, subject to $X_i \geq X_\ell$. Thus $\hat{X}_\ell = \min(X_1, X_2, \ldots, X_n) = X_{(1)}$.

10–24. $$E(G) = E\left[K \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2 \right] = K \left[\sum_{i=1}^{n-1} E(X_{i+1} - X_i)^2 \right]$$

$$= K \left[\sum_{i=1}^{n-1} E\left(X_{i+1}^2 - 2X_iX_{i+1} + X_i^2 \right) \right]$$

$$= K \sum_{i=1}^{n-1} [E(X_{i+1}^2) - 2E(X_iX_{i+1}) + E(X_i^2)]$$

$$= K[(n-1)(\sigma^2 + \mu^2) - 2(n-1)\mu^2 + (n-1)(\mu^2 + \sigma^2)]$$

$$= K[2(n-1)\sigma^2]$$

For $K[2(n-1)\sigma^2]$ to equal σ^2, $K = [2(n-1)]^{-1}$. Thus

$$G = \frac{1}{2(n-1)} \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2$$

is an unbiased estimator of σ^2.

10–25. $f(x_1, x_2, \ldots, x_n|\mu) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2}\Sigma(x_i - \mu)^2/\sigma^2\right)$,

$$f(\mu) = (2\pi\sigma_0^2)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu_0)^2/\sigma_0^2\right)$$

$$f(\mu|x_1, x_2, \ldots, x_n) = c^{-1/2}(2\pi)^{-1/2} \exp\left\{-\frac{c}{2}\left[\mu - \frac{1}{c}\left(\frac{n\overline{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}\right)\right]^2\right\}$$

$$\text{where } c = \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}$$

10–26. $f(x_1, x_2, \ldots, x_n|1/\sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2}\Sigma(x_i - \mu)^2/\sigma^2\right)$

$$f(1/\sigma^2) = \frac{1}{\Gamma(m+1)}(m\sigma_0^2)^{m+1}(1/\sigma^2)^m e^{-m\sigma_0^2/\sigma^2}$$

The posterior density for $1/\sigma^2$ is gamma with parameters $m + (n/2) + 1$ and $m\sigma_0^2 + \Sigma(x_i - \mu)^2$.

10–27. $f(x_1, x_2, \ldots, x_n|p_0) = p^n(1-p)^{\Sigma x_i - n}$,

$$f(p) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}p^{a-1}(1-p)^{b-1}$$

The posterior density for p is a beta distribution with parameters $a + n$ and $b + \Sigma x_i - n$.

10–28. $f(x_1, x_2, \ldots, x_n|p) = p^{\Sigma x_i}(1-p)^{n - \Sigma x_i}$,

$$f(p) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}p^{a-1}(1-p)^{b-1},$$

The posterior density for p is a beta distribution with parameters $a + \Sigma x_i$ and $b + n - \Sigma x_i$

10–29. $f(x_1, x_2, \ldots, x_n|\lambda) = \lambda^{\Sigma x_i} e^{-n\lambda}/\Pi x_i!$

$$f(\lambda) = \frac{1}{\Gamma(m+1)}\left(\frac{m+1}{\lambda_0}\right)^{m+1}\lambda^m e^{-(m+1)\lambda/\lambda_0}$$

The posterior density for λ is gamma with parameters $r = m + \Sigma x_i + 1$ and $\delta = n + (m+1)/\lambda_0$.

10–30. From Exercise 10–25 and using the relationship that the Bayes' estimator for μ using a squared-error loss function is given by $\hat{\mu} = \frac{1}{c}[\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}]$, we have

$$\hat{\mu} - \left[\frac{25}{40} + \frac{1}{8}\right]^{-1} \left[\frac{25(4.85)}{40} + \frac{4}{8}\right] = 4.708$$

10–31. $$\hat{\mu} = 1.05 - \frac{\frac{0.1}{2}}{\Phi\left(\frac{1.20 - 1.05}{\frac{0.1}{2}}\right) - \Phi\left(\frac{0.98 - 1.05}{\frac{0.1}{2}}\right)} (2\pi)^{-1/2}$$

$$\times \left[e^{-1/2(\frac{1.20-1.05}{0.1/2})} - e^{-1/2(\frac{0.98-1.05}{0.1/2})}\right]$$

$$= 1.05 - (0.0545)(0.399)(0.223 - 4.055) = \underline{0.967}$$

10–32. $\Sigma\chi_i = 6270$, $\hat{\lambda} = 0.000323$

10–33. $$\hat{\mu} = \left[\frac{25}{0.1} + \frac{1}{0.04}\right]^{-1} \left[\frac{25(10.05)}{0.1} + \frac{10}{0.04}\right] = 10.045$$

$$\text{weight} \sim N(10.045, 0.1)$$

$$P(\text{weight} < 9.95) = P\left(Z < \frac{9.95 - 10.045}{0.316}\right) = \Phi(-0.301) \approx 0.3783$$

10–34. From a previous exercise, the posterior is gamma with parameters $r = \Sigma x_i + 1$ and $\delta = n + (1/\lambda_0)$. Since $n = 10$ and $\Sigma x_i = 45$,

$$f(\lambda | x_1, \ldots, x_{10}) = \frac{1}{\Gamma(46)} (14)^{46} \lambda^{45} e^{-46\lambda}$$

The Bayes interval requires us to find L and U so that

$$\frac{(14)^{46}}{\Gamma(46)} \int_L^U \lambda^{45} e^{-46\lambda}\, d\lambda = 0.95$$

Since r is integer, tables of the Poisson distribution could be used to find L and U.

10–35. (a) $f(x_1|\theta) = \dfrac{2x}{\theta^2}$, $\quad f(\theta) = 1, 0 < \theta < 1, \quad f(x_1, \theta) = \dfrac{2x}{\theta^2}$, and

$$f(x_1) = \int_x^1 \frac{2x}{\theta^2} d\theta = -2x \frac{1}{\theta} \int_x^1 = 2 - 2x; 0 < x < 1$$

The posterior density is

$$f(\theta|x_1) = \frac{f(x_1,\theta)}{f(x_1)} = \frac{2x}{\theta^2(2-2x)}$$

(b) The estimator must minimize

$$\begin{aligned} Z &= \int \ell(\hat{\theta};\theta) f(\theta|x_1)\, d\theta = \int_0^1 \theta^2(\hat{\theta}-\theta)^2 \frac{2x}{\theta^2(2-2x)} d\theta \\ &= \frac{2x}{2-2x}\left[\hat{\theta}^2 - \hat{\theta} + \frac{1}{3}\right] \end{aligned}$$

From $\frac{dZ}{d\hat{\theta}} = 0$ we get $\hat{\theta} = \frac{1}{2}$

10–36. $f(x_1|p) = p^x(1-p)^{1-x}, f(x_1,p) = 6p^{x+1}(1-p)^{2-x},$ and

$$f(x_1) = \int_0^1 6p^{x+1}(1-p)^{2-x}\, dp = \frac{\Gamma(x+2)\Gamma(3-x)}{4}$$

The posterior density for p is

$$f(p|x_1) = \frac{24p^{x+1}(1-p)^{2-x}}{\Gamma(x+2)\Gamma(3-x)}$$

For a squared-error loss, the Bayes estimator is

$$\hat{p} = E(p|x_1) = \int_0^1 \frac{24p^{x+2}(1-p)^{2-x}}{\Gamma(x+2)\Gamma(3-x)} = \frac{\Gamma(x+3)}{5\Gamma(x+2)} = \frac{x+3}{5}$$

If $\ell(\hat{p};p) = 2(\hat{p}-p)^2$, the Bayes estimator must minimize

$$\begin{aligned} Z &= \frac{24}{\Gamma(x+2)\Gamma(3-x)} \int_0^1 2(\hat{p}-p)^2 p^{x+1}(1-p)^{2-x}\, dp \\ &= \frac{2}{\Gamma(x+2)}\left[\hat{p}^2\Gamma(x+2) - \frac{2\hat{p}\Gamma(x+3)}{5} + \frac{\Gamma(x+4)}{30}\right] \end{aligned}$$

From $dz/d\hat{p} = 0$, $\hat{p} = \frac{x+2}{5}$

10–37. For $\alpha_1 = \alpha_2 = \alpha/2$, $\alpha = 0.05$; $\overline{x} \pm 1.96(\frac{\sigma}{\sqrt{n}})$

For $\alpha_1 = 0.01$, $\alpha_2 = 0.04$; $\overline{x} - 1.751(\frac{\sigma}{\sqrt{n}}) \le \mu \le \overline{x} + 2.323(\frac{\sigma}{\sqrt{n}})$

$\alpha_1 = \alpha_2 = \alpha/2$ is shorter.

10–38. (a) $N(0,1)$

(b) $\overline{X} - Z_{\alpha/2}\left(\sqrt{\dfrac{\overline{X}}{n}}\right) \leq \lambda \leq \overline{X} + Z_{\alpha/2}\left(\sqrt{\dfrac{\overline{X}}{n}}\right)$

10–39. (a) $\overline{x} - Z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{x} + Z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}$

$74.03533 \leq \mu \leq 74.03666$

(b) $\overline{x} - Z_{\alpha}\dfrac{\sigma}{\sqrt{n}} \leq \mu$

$74.0356 \leq \mu$

10–40. (a) $1003.04 \leq \mu \leq 1024.96$

(b) $1004.80 \leq \mu$

10–41. (a) $3232.11 \leq \mu \leq 3267.89$

(b) $3226.49 \leq \mu \leq 3273.51$

The width of the confidence interval in (a) is 35.78, and the width of the interval in (b) is 47.01. The wider confidence interval in (b) reflects the higher confidence coefficient.

10–42. $n = (Z_{\alpha/2}\sigma/E)^2 = [(1.96)25/5]^2 = 96.04 \simeq 97$

10–43. For the total width to be 8, the half-width must be 4, therefore $n = (Z_{\alpha/2}\sigma/E)^2 = [(1.96)25/4]^2 = 150.06 \simeq 150$ or 151.

10–44. $n = (Z_{\alpha/2}\sigma/E)^2 = 1000(1.96/15)^2 = 17.07 \simeq 18.$

10–45. (a) $0.0723 \leq \mu_1 - \mu_2 \leq 0.3076$

(b) $0.0499 \leq \mu_1 - \mu_2 \leq 0.33$

(c) $\mu_1 - \mu_2 \leq 0.3076$

10–46. $3.553 \leq \mu_1 - \mu_2 \leq 8.447$

10–47. $-3.68 \leq \mu_1 - \mu_2 \leq -2.12$

10–48. (a) $2238.6 \leq \mu \leq 2275.4$

(b) $2242.63 \leq \mu$

(c) $2240.11 \leq \mu \leq 2275.39$

10–49. $183.0 \leq \mu \leq 256.6$

10–50. $4.05 - t_{0.10,24}(0.08/\sqrt{25}) \leq \mu \Rightarrow 4.029 \leq \mu$

10–51. 13

10–52. (a) $546.12 \leq \mu \leq 553.88$
(b) $546.82 \leq m$
(c) $\mu \leq 553.18$

10–53. $94.282 \leq \mu \leq 111.518$

10–54. (a) $7.65 \leq \mu_1 - \mu_2 \leq 12.346$
(b) $8.03 \leq \mu_1 - \mu_2$
(c) $\mu_1 - \mu_2 \leq 11.97$

10–55. $-0.839 \leq \mu_1 - \mu_2 \leq -0.679$

10–56. (a) $-0.561 \leq \mu_1 - \mu_2 \leq 1.561$
(b) $\mu_1 - \mu_2 \leq 1.384$
(c) $-0.384 \leq \mu_1 - \mu_2$

10–57. $0.355 \leq \mu_1 - \mu_2 \leq 0.455$

10–58. $-30.24 \leq \mu_1 - \mu_2 \leq -19.76$

10–59. From 10–48, $s = 34.51$, $n = 16$, and $(n-1)s^2 = 17864.1$
(a) $649.60 \leq \sigma^2 \leq 2853.69$
(b) $714.56 \leq \sigma^2$
(c) $\sigma^2 \leq 2460.62$

10–60. (a) $1606.18 \leq \sigma^2 \leq 26322.15$
(b) $1755.68 \leq \sigma^2$
(c) $\sigma^2 \leq 21376.78$

10–61. $0.0039 \leq \sigma^2 \leq 0.0157$

10–62. $\sigma^2 \leq 193.09$

10–63. $0.574 \leq \sigma_1^2/\sigma_2^2 \leq 3.614$

10–64. $s_1^2 = 0.29$, $s_2^2 = 0.34$, $s_1^2/s_2^2 = 1.208$, $n_1 = 12$, $n_2 = 18$

(a) $0.502 \leq \sigma_1^2/\sigma_2^2 \leq 2.924$

(b) $0.423 \leq \sigma_1^2/\sigma_2^2 \leq 3.468$

(c) $0.613 \leq \sigma_1^2/\sigma_2^2$

(d) $\sigma_1^2/\sigma_2^2 \leq 2.598$

10–65. $0.11 \leq \sigma_1^2/\sigma_2^2 \leq 0.86$

10–66. $0.089818 \leq p \leq 0.155939$

10–67. $n = 4057$

10–68. $p \leq 0.00348$

10–69. $n = (Z_{\alpha/2}/E)^2 p(1-p) = (2.575/0.01)^2 p(1-p) = 66306.25p(1-p)$. The most conservative choice of p is $p = 0.5$, giving $n = 16576.56$ or $n = 16577$ homeowners.

10–70. $0.0282410 \leq p_1 - p_2 \leq 0.0677590$

10–71. $-0.0244 \leq p_1 - p_2 \leq 0.0024$

10–72. $-8.50 \leq \mu_1 - \mu_2 \leq 1.94$

10–73. $-2038 \leq \mu_1 - \mu_2 \leq 3774.8$

10–74. Since $\overline{X}$ and S^2 are independent, we can construct confidence intervals for μ and σ^2 such that we are 90 percent confident that both intervals provide correct conclusions by constructing a $100(0.90)^{1/2}$ percent confidence interval for each parameter. That is, we need a 95 percent confidence interval on μ and σ^2. Thus, $3.938 \leq \mu \leq 4.057$ and $0.0049 \leq \sigma^2 \leq 0.0157$ provides the desired simultaneous confidence intervals.

10–75. Assume that all three variances are equal. A 95 percent simultaneous confidence interval on $\mu_1 - \mu_2$, $\mu_1 - \mu_3$, and $\mu_2 - \mu_3$ will require that the individual intervals use $\alpha/3 = 0.05/3 = 0.0167$.

For $\mu_1 - \mu_2, s_p^2 = 1.97, t_{0.0167/2,18} \simeq 2.64; -3.1529 \leq \mu_1 - \mu_2 \leq 0.1529$

For $\mu_1 - \mu_3, s_p^2 = 1.76, t_{0.0167/2,23} \simeq 2.59; -1.9015 \leq \mu_1 - \mu_3 \leq 0.9015$

For $\mu_2 - \mu_3, s_p^2 = 1.24, t_{0.0167/2,23} \simeq 2.59; -0.1775 \leq \mu_2 - \mu_3 \leq 2.1775$

10–76. The posterior density for μ is truncated normal:

$$f(\mu|x_1,\ldots,x_{16}) = \sqrt{\frac{16}{2\pi 10}}\left[\Phi\left(\frac{12-8}{\sqrt{10/16}}\right) - \Phi\left(\frac{6-8}{\sqrt{10/16}}\right)\right]^{-1} e^{(-1/2)(\frac{\mu-8}{10/16})^2}$$

for $6 < \mu \leq 12$. From the normal tables, the 90% interval estimate for μ is centered at 8 and is from $8-(1.795)(1.054) = 6.108$ to 9.892. Since $6.108 < 9 < 9.892$, we have no evidence to reject H_0.

10–77. The posterior density for $1/\sigma^2$ is gamma w/parameters $r+(n/2)$ and $\lambda+\Sigma(x_i-\mu)^2$. For $r=3$, $\lambda=1$, $n=10$, $\mu=5$, $\Sigma(x_i-5)^2 = 4.92$, the Bayes estimate of $1/\sigma^2$ is $(1/\sigma^2) = \frac{3+5}{1+4.92} = 1.35$. The integral:

$$0.90 = \int_L^U \frac{1}{8}(5.92)^8(1/\sigma^2)^7 e^{-5.92/\sigma^2}$$

10–78. $$Z = \int_{-\infty}^{\infty}(\hat{\theta}-\theta)^2 f(\theta|x_1,x_2,\ldots,x_n)\,d\theta$$

$$= \hat{\theta}^2\int_{-\infty}^{\infty} f(\theta|x_1,x_2,\ldots,x_n)\,d\theta - 2\hat{\theta}\int_{-\infty}^{\infty}\theta f(\theta|x_1,x_2,\ldots,x_n)\,d\theta$$

$$+ \int_{-\infty}^{\infty}\theta^2 f(\theta|x_1,x_2,\ldots,x_n)\,d\theta$$

Let

$$E(\hat{\theta}) = \int_{-\infty}^{\infty}\theta f(\theta|x_1,x_2,\ldots,x_n)\,d\theta = \mu_\theta$$

$$E(\hat{\theta}^2) = \int_{-\infty}^{\infty}\theta^2 f(\theta|x_1,x_2,\ldots,x_n)\,d\theta = \tau_\theta$$

Then

$$Z = \hat{\theta}^2 - 2\hat{\theta}\mu_\theta + \tau_\theta$$

$$\frac{dZ}{d\hat{\theta}} = 2\hat{\theta} - 2\mu_\theta = 0 \quad \text{so} \quad \hat{\theta} = \mu_\theta.$$

Chapter 11

11–1. (a) H_0: $\mu \leq 160$
H_1: $\mu > 160$
$$Z_0 = \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{158 - 160}{3/2} = -1.333$$

The fiber is acceptable if $Z_0 > Z_{0.05} = 1.645$. Since $Z_0 = -1.33 < 1.645$, the fiber is not acceptable.

(b) $d = \frac{\mu - \mu_0}{\sigma} = \frac{165-160}{3} = 1.67$, if $n = 4$ then using the OC curves, we get $\beta \simeq 0.05$.

11–2. (a) H_0: $\mu = 90$
H_1: $\mu < 90$
$$Z_0 = \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{90.48 - 90}{\sqrt{5/5}} = 0.48$$

Since Z_0 is not less than $-Z_{0.05} = -1.645$, do not reject H_0. There is no evidence that mean yield is less than 90 percent.

(b) $n = (Z_\alpha + Z_\beta)^2\sigma^2/\delta^2 = (1.645 + 1.645)^2 5/(5)^2 = 2.16 \simeq 3$. Could also use the OC curves, with $d = (\mu_0 - \mu)/\sigma = (90 - 85)/\sqrt{5} = 2.24$ and $\beta = 0.05$, also giving $n = 3$.

11–3. (a) H_0: $\mu = 0.255$
H_1: $\mu \neq 0.255$
$$Z_0 = \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{0.2546 - 0.255}{0.0001/\sqrt{10}} = -12.65$$

Since $|Z_0| = 12.65 > Z_{0.025} = 1.96$, reject H_0.

(b) $d = \frac{|\mu - \mu_0|}{\sigma} = \frac{|0.2552-0.225|}{0.0001} = 2$, and using the OC curves with $\alpha = 0.05$ and $\beta = 0.10$ gives $n \simeq 3$. Could also use $n \simeq (Z_{\alpha/2} + Z_\beta)^2\sigma^2/\delta^2 = (1.96 + 1.28)^2(0.0001)^2/(0.0002)^2 = 2.62 = 3$.

11–4. (a) H_0: $\mu = 74.035$
H_1: $\mu \neq 74.035$
$$Z_0 = \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{74.036 - 74.035}{0.001/\sqrt{15}} = 3.87$$

Since $Z_0 > Z_{\alpha/2} = 2.575$, reject H_0.

(b) $n \cong \dfrac{(Z_{\alpha/2} + Z_\beta)^2\sigma^2}{\delta^2} = 0.712, n = 1$

11–5. (a) H_0: $\mu = 1.000$ $\quad Z_0 = \dfrac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} = \dfrac{1014 - 1000}{25/\sqrt{20}} = 2.50$

H_1: $\mu \neq 1.000$ $\quad |Z_0| = 2.50 > Z_{0.005} = 1.96$, reject H_0.

11–6. (a) H_0: $\mu = 3500$ $\quad Z_0 = \dfrac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} = \dfrac{3250 - 3500}{\sqrt{1000/12}} = -27.39$

H_1: $\mu \neq 3500$ $\quad |Z_0| = 27.39 > Z_{0.005} = 2.575$, reject H_0.

11–7. (a) H_0: $\mu_1 = \mu_2$ $\qquad Z_0 = \dfrac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = 1.349$

H_1: $\mu_1 \neq \mu_2$

Since $Z_0 < Z_{\alpha/2} = 1.96$, do not reject H_0.

(d) $d = 3.2$, $n = 10$, $\alpha = 0.05$, OC curves gives $\beta \approx 0$, therefore power ≈ 1.

11–8. μ_1 = New machine, μ_2 = Current machine

H_0: $\mu_1 - \mu_2 \leq 2$, H_1: $\mu_1 - \mu_2 > 2$

Use the t-distribution assuming equal variances: $t_0 = -5.45$, do not reject H_0.

11–9. H_0: $\mu_1 - \mu_2 = 0$, $\qquad Z_0 = \dfrac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = 2.656$

H_1: $\mu_1 - \mu_2 \neq 0$

Since $Z_0 > Z_{\alpha/2} = 1.645$, reject H_0.

11–10. H_0: $\mu_1 - \mu_2 = 0$, $\qquad Z_0 = \dfrac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = -6.325$

H_1: $\mu_1 - \mu_2 > 0$

Since $Z_0 < Z_\alpha = 1.645$, do not reject H_0.

11–11. H_0: $\mu_1 - \mu_2 = 0$, $\qquad Z_0 = \dfrac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_2^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = -7.25$

H_1: $\mu_1 - \mu_2 < 0$

Since $Z_0 < -Z_\alpha = -1.645$, reject H_0.

11–12. H_0: $\mu = 0$ $\qquad t_0 = \dfrac{\overline{x} - \mu_0}{s/\sqrt{n}} = \dfrac{-0.168 - 0}{8.5638/\sqrt{10}} = -0.062$

H_1: $\mu \neq 0$ $\qquad |t_0| = 0.062 < t_{0.025,9} = 2.2622$, do not reject H_0.

11–13. (a) $t_0 = 1.842$, do not reject H_0.

(b) $n = 8$ is not sufficient; $n = 10$.

11–14. H_0: $\mu = 9.5$ $\qquad t_0 = \dfrac{10.28 - 9.5}{2.55/\sqrt{6}} = 0.7492$

H_1: $\mu > 9.5$ $\qquad t_{0.05,5} = 2.015$, do not reject H_0.

11–15. $t_0 = 1.47$, do not reject at $\alpha = 0.05$ level of significance. It can be rejected at the $\alpha = 0.10$ level of significance.

11–16. (a) H_0: $\mu = 7.5$ $\quad t_0 = \dfrac{6.997 - 7.5}{1.279/\sqrt{18}} = -1.112$

H_1: $\mu < 7.5$ $\quad t_{0.05,7} = 1.895$, do not reject H_0,
the true scrap rate is not $< 7.5\%$.

(b) $n = 5$

(c) 0.95

11–17. $n = 3$

11–18. $d = \dfrac{|\delta|}{\sigma} = \dfrac{20}{10} = 2$, $n = 3$

11–19. (a) H_0: $\mu_1 = \mu_2$ $\quad t_0 = \dfrac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}} = \dfrac{25.617 - 21.7}{0.799\sqrt{1/6 + 1/6}} = 8.49$

H_1: $\mu_1 > \mu_2$ $\quad t_0 > t_{0.01,10} = 2.7638$

(b) H_0: $\mu_1 - \mu_2 = 5$ $\quad t_0 = \dfrac{\overline{x}_1 - \overline{x}_2 - 5}{s_p\sqrt{1/n_1 + 1/n_2}} = 2.35$, do not reject H_0.

H_1: $\mu_1 - \mu_2 > 5$

(c) Using $s_p = 0.799$ as an estimate of σ,
$d = (\mu_1 - \mu_2)/(2\sigma) = 5/2(0.799) = 3.13$, $n_1 = n_2 = 6$, $\alpha = 0.01$, OC curves give $\beta \approx 0$, so power $\simeq 1$.

(d) OC curves give $n = 5$.

11–20. $t_0 = -0.02$, do not reject H_0.

11–21. H_0: $\sigma_1^2 = \sigma_2^2$ $\quad s_1 = 9.4186$ $\quad s_1^2 = 88.71$

H_1: $\sigma_1^2 \neq \sigma_2^2$ $\quad s_2 = 10.0222$ $\quad s_2^2 = 100.44$

$\alpha = 0.05$

Reject H_0 if $F_0 > F_{0.025,9,9} = 3.18$.

$F_0 = \dfrac{88.71}{100.44} = 0.8832$ $\quad \therefore$ do not reject H_0: $\sigma_1^2 = \sigma_2^2$.

11–22. (a) H_0: $\sigma_1^2 = \sigma_2^2$ $\quad F_0 = s_1^2/s_2^2 = 101.17/94.73 = 1.07$

H_1: $\sigma_1^2 \neq \sigma_2^2$ $\quad$ do not reject H_0.

(b) H_0: $\mu_1 = \mu_2$ $\quad t_0 = \dfrac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}} = \dfrac{12.5 - 10.2}{9.886\sqrt{1/8 + 1/9}}$

H_1: $\mu_1 > \mu_2$ $\quad = -0.48$, do not reject.

11–23. (a) H_0: $\mu_1 = \mu_2$ $\quad s_p = \sqrt{\dfrac{1480 + 1425}{18}} = 12.704$

H_1: $\mu_1 \neq \mu_2$ $\quad t_0 = \dfrac{20.0 - 15.8}{12.704\sqrt{1/10 + 1/10}} = 0.74$, do not reject H_0.

Reject H_0 if $|t_0| < t_{0.005,9} = 3.250$.

(b) $d = 10/[2(12.7)] = 0.39$, Power $= 0.13$, $n^* = 19$

(c) $n_1 = n_2 = 75$

11–24. (a) H_0: $\mu_1 = \mu_2$ $\quad t_0 = \dfrac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}} = \dfrac{20.0 - 21.5}{1.40\sqrt{1/10 + 1/10}} = -2.40$

H_1: $\mu_1 \neq \mu_2$ $\quad$ reject H_0.

(b) Use $s_p = 1.40$ as an estimate of σ. Then $d = |\mu_1 - \mu_2|/2\sigma = 2/2(1.40) = 0.7$. If $\alpha = 0.05$ and $n_1 = n_2 = 10$, OC curves gives $\beta \simeq 0.5$. For $\beta \simeq 0.15$, we must have $n_1 = n_2 \simeq 30$.

(c) $F_0 = s_1^2/s_2^2 = 2.25/1.69 = 1.33$, do not reject H_0.

(d) $\lambda = \sigma_1/\sigma_2 = 2$, $\alpha = 0.05$, $n_1 = n_2 = 10$, OC curves give $\beta \simeq 0.50$.

11–25. H_0: $\mu_1 = \mu_2$ $\quad t_0 = \dfrac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}} = \dfrac{8.75 - 8.63}{0.57\sqrt{1/12 + 1/18}} = 0.56$

H_1: $\mu_1 \neq \mu_2$ $\quad$ do not reject H_0.

11–26. H_0: $\sigma^2 = 16$ $\quad$ If $\alpha = 0.05$, $\lambda = \sigma_1/\sigma_0 = 3/4$, and $\beta = 0.10$, then
H_1: $\sigma^2 < 16$ $\quad$ $n \simeq 55$. Thus $n = 10$ is not good.

For the sample of $n = 10$ given, $\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{9(14.69)}{16} = 8.26$. Since $\chi_{0.05,9}^2 = 3.325$, do not reject H_0.

11–27. (a) H_0: $\sigma = 0.00002$ $\quad \chi_0^2 = \dfrac{(n-1)s^2}{\sigma_0^2} = \dfrac{7(0.00005)^2}{(0.00002)^2} = 43.75$

H_1: $\sigma > 0.00002$ $\quad$ Since $\chi_0^2 > \chi_{0.01,7}^2 = 18.475$, reject H_0.
The claim is unjustified.

(b) A 99% one-sided lower confidence interval is $0.3078 \times 10^{-4} \leq \sigma^2$.

(c) $\lambda = \sigma_1/\sigma_2 = 2$, $\alpha = 0.01$, $n = 8$, OC curves give $\beta \simeq 0.30$.

(d) $\lambda = 2$, $\beta \simeq 0.05$, $\alpha = 0.01$, OC curves give $n = 17$.

11–28. H_0: $\sigma = 0.005$
H_1: $\sigma > 0.005$

If $\alpha = 0.01$, $\beta = 0.10$, $\lambda = 0.010/0.005 = 2$, then the OC curves give $n = 14$. Assuming $n = 14$, then $\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{13(0.007)^2}{(0.005)^2} = 25.48 < \chi^2_{0.01,13} = 27.688$, and we do not reject. The 95% one-sided upper confidence interval is $\sigma^2 \leq 0.155 \times 10^{-3}$.

11–29. (a) H_0: $\sigma^2 = 0.5$ $\quad \chi_0^2 = \dfrac{(n-1)s^2}{\sigma_0^2} = \dfrac{11(0.10388)}{0.5} = 2.28,$ reject H_0.
H_1: $\sigma^2 \neq 0.5$

(b) $\lambda = \sigma/\sigma_0 = 1/0.707 = 1.414$, $\beta \approx 0.58$

11–30. H_0: $\sigma_1^2 = \sigma_2^2$ $\quad F_0 = s_1^2/s_2^2 = 2.25 \times 10^{-4}/3.24 \times 10^{-4}$
H_1: $\sigma_1^2 \neq \sigma_2^2$ $\quad = 0.69,$ do not reject H_0.

Since $\sigma_1^2 = \sigma_2^2$, the test on means in Exercise 11–7 is appropriate. If $\lambda = \sigma_1/\sigma_2 = \sqrt{2.5} = 1.58$, then using $\alpha = 0.01$, $\beta = 0.10$, the OC curves give $n \simeq 75$.

11–31. H_0: $\sigma_1^2 = \sigma_2^2$ $\quad F_0 = s_1^2/s_2^2 = 0.9027/0.0294 = 30.69$
H_1: $\sigma_1^2 > \sigma_2^2$ $\quad F_0 > F_{0.01,8,10} = 5.06,$ so reject H_0.

If $\lambda = \sqrt{4} = 2$, $\alpha = 0.01$, and taking $n_1 \simeq n_2 = 10$ (say), we get $\beta \simeq 0.65$.

11–32. (b) H_0: $\mu_1 - \mu_2 = 0$, $\quad t_0 = \dfrac{(0.984 - 0.907)}{\sqrt{11.37(\frac{1}{25} + \frac{1}{30})}} = 0.0843,$ do not reject H_0.
H_1: $\mu_1 - \mu_2 \neq 0$

11–33. H_0: $\mu_d = 0$ $\quad t_0 = \dfrac{\overline{d} - 0}{s_d/\sqrt{n}} = \dfrac{5.0 - 0}{15.846/\sqrt{10}} = 0.998,$
H_1: $\mu_d \neq 0$ $\quad$ do not reject H_0.

11–34. Using $\mu_D = \mu_A - \mu_B$, $t_0 = -1.91$, do not reject H_0.

11–35. H_0: $\mu_d = 0$
H_1: $\mu_d \neq 0$

Reject H_0 if $|t_0| > t_{0.025,5} = 2.571$.

$t_0 = \dfrac{3 - 0}{1.41/\sqrt{6}} = 5.21$ $\quad \therefore$ reject H_0

11–36. $t_0 = 2.39$, reject H_0.

11–37. H_0: $p = 0.70$, H_1: $p \neq 0.70$; $Z_0 = \frac{(699-700)}{\sqrt{1000(0.7)(0.3)}} = 0.586$, do not reject H_0.

11–38. H_0: $p = 0.025$, H_1: $p \neq 0.025$; $Z_0 = \frac{(18-200)}{\sqrt{8000(0.975)(0.025)}} = -13.03$, reject H_0.

11–39. The "best" test will maximize the probability that H_0 is rejected, so we want to

$$\max Z_0 = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$
$$\text{subject to } n_1 + n_2 = N.$$

Since for a given sample, $\overline{X}_1 - \overline{X}_2$ is fixed, this is equivalent to

$$\min L = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$
$$\text{subject to } n_1 + n_2 = N.$$

Since $n_2 = n_1 - N$, we have

$$\min L = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{N - n_1}$$

and from $dL/dn_1 = 0$ we find

$$\frac{\sigma_1}{\sigma_2} = \frac{n_1}{n_2},$$

which says that the observations should be assigned to the populations in the same ratio as the standard deviations.

11–40. $z_0 = 6.26$, reject H_0.

11–41. H_0: $p_1 = p_2$, $\quad H_1$: $p_1 < p_2$; $Z_0 = \frac{(0.01-0.021)}{\sqrt{0.016(0.984)(\frac{1}{1000}+\frac{1}{1200})}} = -2.023$, do not reject H_0.

11–42. H_0: $p_1 = p_2$, $\quad H_1$: $p_1 \neq p_2$; $Z_0 = \frac{(0.042-0.064)}{\sqrt{0.053(0.947)(\frac{2}{500})}} = -1.55$, do not reject H_0.

11–43. Let $2\sigma^2 = \sigma_1^2/n_1 + \sigma_2^2/n_2$ be the specified sample variance. If we minimize $c_1 n_1 + c_2 n_2$ subject to the constraint $\sigma_1^2/n_1 + \sigma_2^2/n_2 = 2\sigma^2$, we obtain the solution

$$\frac{n_1}{n_2} = \sqrt{\frac{\sigma_1^2 c_2}{\sigma_2^2 c_1}}.$$

11–44. H_0: $\mu_1 = 2\mu_2$ $\qquad Z_0 = \dfrac{\overline{X}_1 - 2\overline{X}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{4\sigma_2^2}{n_2}}}$

H_1: $\mu_1 > 2\mu_2$

11–45. H_0: $\sigma^2 = \sigma_0^2$
H_1: $\sigma^2 \neq \sigma_0^2$

$$\beta = P\left(\chi^2_{1-\alpha/2,n-1} \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi^2_{\alpha/2,n-1} \middle| \sigma^2 = \sigma_1^2 \neq \sigma_0^2\right)$$
$$= P\left(\frac{\sigma_0^2}{\sigma_1^2}\chi^2_{1-\alpha/2,n-1} \leq \frac{(n-1)S^2}{\sigma_1^2} \leq \frac{\sigma_0^2}{\sigma_1^2}\chi^2_{\alpha/2,n-1}\right)$$

which can be evaluated using tables of χ^2.

11–46. H_0: $\sigma_1^2 = \sigma_2^2$
H_1: $\sigma_1^2 \neq \sigma_2^2$

$$\beta = P\left(F_{1-\alpha/2,u,v} \leq \frac{S_1^2}{S_2^2} \leq F_{\alpha/2,u,v} \middle| \frac{\sigma_1^2}{\sigma_2^2}\right)$$
$$= P\left(\frac{\sigma_2^2}{\sigma_1^2}F_{1-\alpha/2,u,v} \leq \frac{S_1^2}{S_2^2}\frac{\sigma_2^2}{\sigma_1^2} \leq \frac{\sigma_2^2}{\sigma_1^2}F_{\alpha/2,u,v}\right)$$

Since $(\sigma_2^2/\sigma_1^2)(S_1^2/S_2^2)$ follows an F-distribution, β may be evaluated by using tables of F.

11–47. (a) Assume the class intervals are defined as follows:

Class Interval	O_i	E_i	$(O_i - E_i)^2/E_i$
$-\infty < X < 11$	6	6.15	0.004
$11 \leq X < 16$	11	10.50	0.024
$16 \leq X < 21$	16	19.04	0.485
$21 \leq X < 26$	28	24.68	0.447
$26 \leq X < 31$	22	24.54	0.263
$31 \leq X < 36$	19	14.63	1.305
$36 \leq X < 41$	11	12.36	0.150
$41 \leq X$	4	5.10	0.237
			$\chi_0^2 = 2.915$

The expected frequencies are obtained by evaluating $n[\Phi(\frac{c_i-\overline{x}}{s}) - \Phi(\frac{c_{i-1}-\overline{x}}{s})]$ where c_i is the upper boundary of cell i. For our problem,

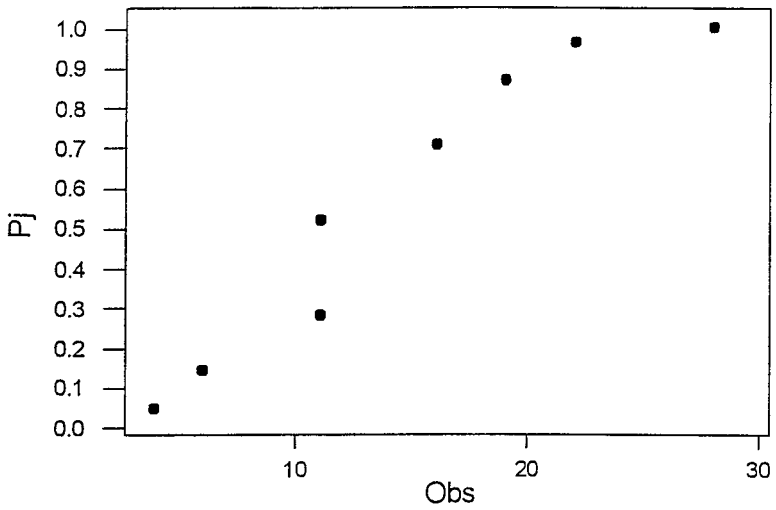
$$E_i = 117\left[\Phi\left(\frac{c_i - 25.61}{9.02}\right) - \Phi\left(\frac{c_{i-1} - 25.61}{9.02}\right)\right].$$

Since $\chi_0^2 = 2.915 < \chi_{0.05,5}^2 = 11.070$, do not reject H_0.

(b) To use normal probability paper for data expressed in a histogram, find the cumulative probability associated with each interval, and plot this against the upper boundary of each cell.

Cell Upper Bound	Observed Frequency	P_j
11	6	0.051
16	11	0.145
21	16	0.282
26	28	0.521
31	22	0.709
36	19	0.872
41	11	0.966
45	4	1.006

Normal probability plot.



11–48. Estimate $\hat{\lambda} = \overline{x} = 4.9775$. The expected frequencies are

Defects	0	1	2	3	4	5	6	7	8	9	10	11	12
O_i	4	13	34	56	70	70	58	42	25	15	9	3	1
E_i	2.76	13.72	34.15	56.66	70.50	70.50	70.18	58.22	41.40	25.76	14.25	3.21	1.33

Three cells have expected values less than 5, so they are combined with other cells to get:

Defects	0–1	2	3	4	5	6	7	8	9	10–12
O_i	17	34	56	70	70	58	42	25	15	13
E_i	16.48	34.15	56.66	70.50	70.50	70.18	58.22	41.40	25.76	18.79

$\chi_0^2 = 1.8846$, $\chi_{0.05,8}^2 = 15.51$, do not reject H_0, the data could follow a Poisson distribution.

11–49.

x	O_i	E_i	$(O_i - E_i)^2/E_i$
0	967	1000	1.089
1	1008	1000	0.064
2	975	1000	0.625
3	1022	1000	0.484
4	1003	1000	0.009
5	989	1000	0.121
6	1001	1000	0.001
7	981	1000	0.361
8	1043	1000	1.849
9	1011	1000	0.121

$\chi_0^2 = 4.724 < \chi_{0.05,9}^2 = 16.919$. Therefore, do not reject H_0.

11–50. (a) Assume that data given are the midpoints of the class intervals.

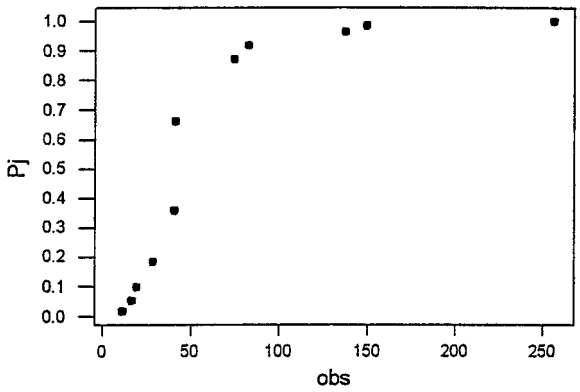
Class Interval	O_i	E_i	$(O_i - E_i)^2/E_i$
$X < 2.095$	0	1.79*	
$2.095 \le X < 2.105$	16	6.65	6.77
$2.105 \le X < 2.115$	28	22.18	1.53
$2.115 \le X < 2.125$	41	56.39	4.20
$2.125 \le X < 2.135$	74	108.92	11.19
$2.135 \le X < 2.145$	149	159.60	0.70
$2.145 \le X < 2.155$	256	178.02	34.16
$2.155 \le X < 2.165$	137	150.81	1.26
$2.165 \le X < 2.175$	82	96.56	2.20
$2.175 \le X < 2.185$	40	47.59	1.21
$2.185 \le X < 2.195$	19	18.09	0.04
$2.195 \le X < 2.205$	11	5.12	3.31
$2.205 \le X$	0	1.28*	

*Group into next cell

$\chi_0^2 = 66.57 > \chi_{0.05,8}^2 = 15.507$, reject H_0.

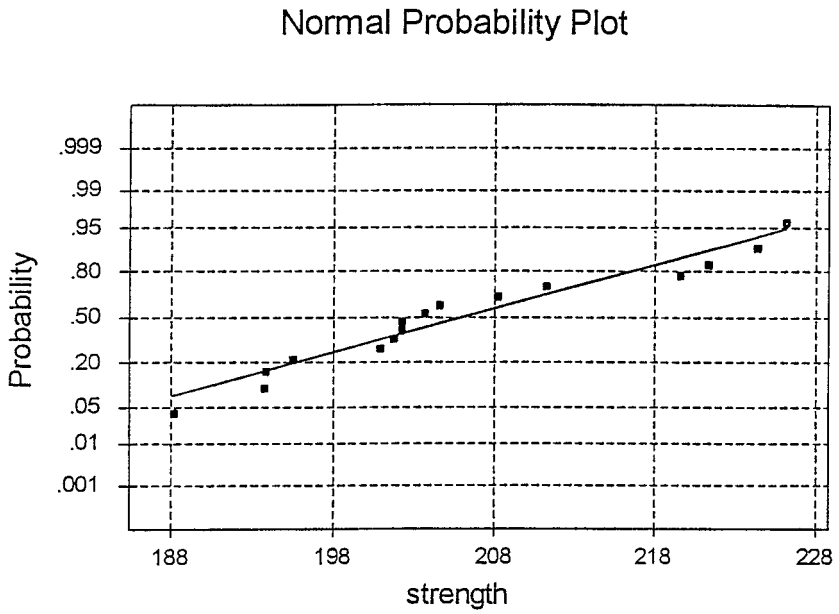
(b)

Upper Cell Bound	Observed Frequency	P_j
2.105	16	0.019
2.115	28	0.052
2.125	41	0.100
2.135	74	0.186
2.145	149	0.361
2.155	256	0.661
2.165	137	0.872
2.175	82	0.918
2.185	40	0.965
2.195	19	0.987
2.205	11	1.000



11–51.

$X_{(j)}$	$P_{(j)}$	$X_{(j)}$	$P_{(j)}$
188.12	0.0313	203.62	0.5313
193.71	0.0938	204.55	0.5938
193.73	0.1563	208.15	0.6563
195.45	0.2188	211.14	0.7188
200.81	0.2813	219.54	0.7813
201.63	0.3438	221.31	0.8438
202.20	0.4063	224.39	0.9063
202.21	0.4688	226.16	0.9688



11–52. $\chi_0^2 = 11.649 < \chi_{0.05,6}^2 = 12.592$. Do not reject.

11–53. $\chi_0^2 = 0.0331 < \chi_{0.05,1}^2 = 3.841$. Do not reject.

11–54. $\chi_0^2 = 25.554 < \chi_{0.05,9}^2 = 16.919$. Reject H_0.

11–55. $\chi_0^2 = 2.465 < \chi_{0.05,4}^2 = 9.488$. Do not reject.

11–56. $\chi_0^2 = 10.706 > \chi_{0.05,3}^2 = 7.81$. Reject H_0.

11–57. The observed and expected frequencies are

	IA	A	Total
L	216	245	461
	170.08	290.92	
M	226	409	635
	234.28	400.72	
H	114	297	411
	151.64	259.36	
Total	556	951	1507

$\chi_0^2 = 34.909$, reject H_0. Based on this data, physical activity is not independent of socioeconomic status.

11–58. $\chi_0^2 = 13.6289 < \chi_{0.05.8}^2 = 15.507$. Do not reject.

11–59. Expected Frequencies:

17	62	55	$\hat{\mu}_1 = 0.304$	$\hat{\mu}_1 = 0.127$
22	81	71	$\hat{\mu}_2 = 0.394$	$\hat{\mu}_2 = 0.465$
17	62	54	$\hat{\mu}_3 = 0.302$	$\hat{\mu}_3 = 0.408$

$$\chi_0^2 = \sum_{i=1}^{3}\sum_{j=1}^{3}\frac{(O_{ij}-E_{ij})^2}{E_{ij}} = 2.88 + 1.61 + 0.16 + 2.23 + 0.79 + 13.17 + 0 + 5.23 + 6 = 22.06$$

$\chi_{0.05,4}^2 = 9.488$

$\therefore$ reject H_0, pricing strategy and facility conditions are not independent

11–60. (a)

	Non Defective	Defective			
Machine 1	468	32	500	473.5	26.5
Machine 2	479	21	500	473.5	26.5
	947	53	1000		

$\Rightarrow$

$$\chi_0^2 = 0.064 + 1.141 + 0.013 + 1.41 = 2.897$$

$\chi_{0.05,1}^2 = 3.841$, do not reject H_0, the populations do not differ

(b) homogeneity

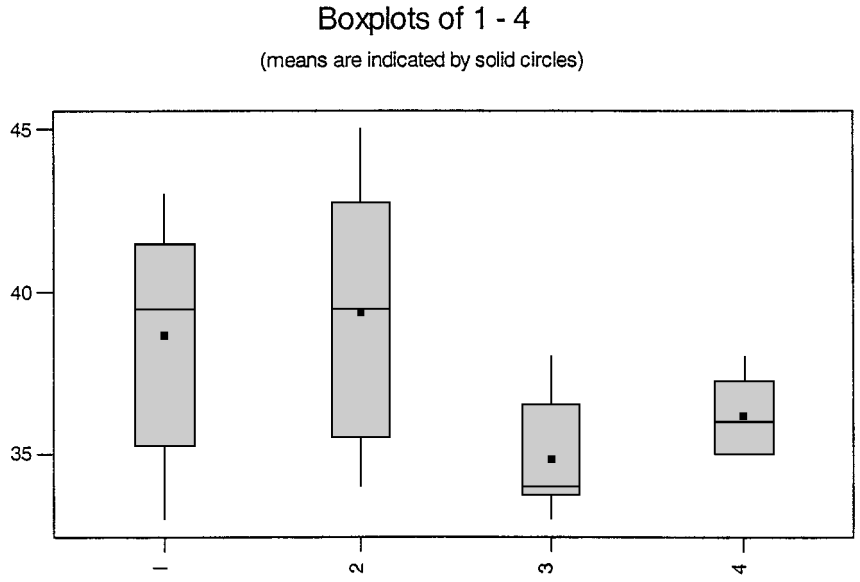
(c) yes

Chapter 12

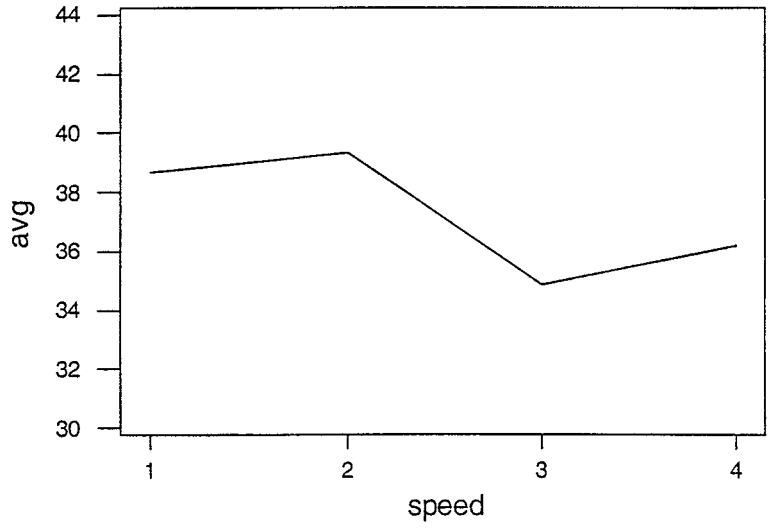
12–1. (a)

Analysis of Variance

Source	DF	SS	MS	F	P
Factor	3	80.17	26.72	3.17	0.047
Error	20	168.33	8.42		
Total	23	248.50			



(b)



(c) Tukey's pairwise comparisons

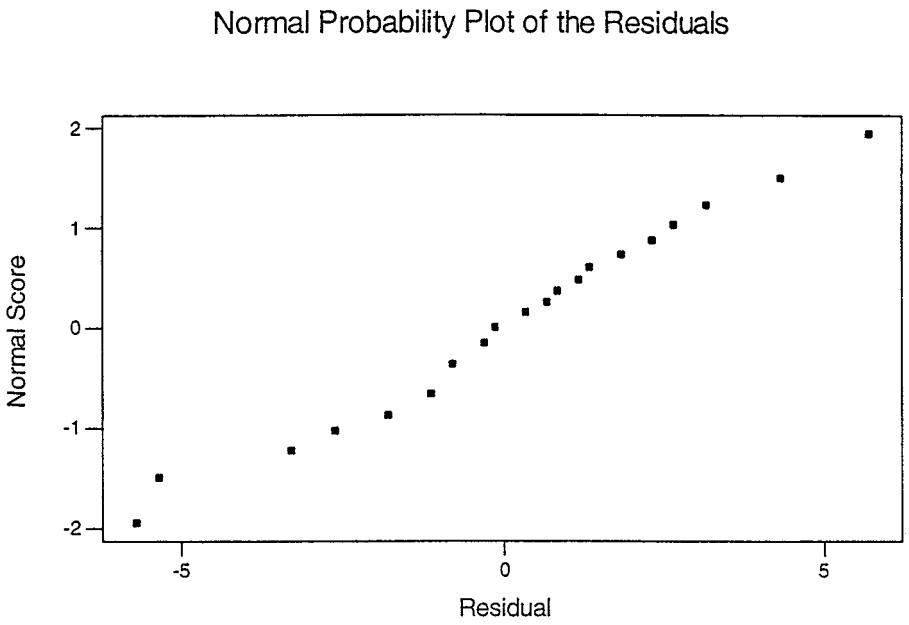
```
    Family error rate = 0.0500
Individual error rate = 0.0111
Critical value = 3.96
Intervals for (column level mean) - (row level mean)

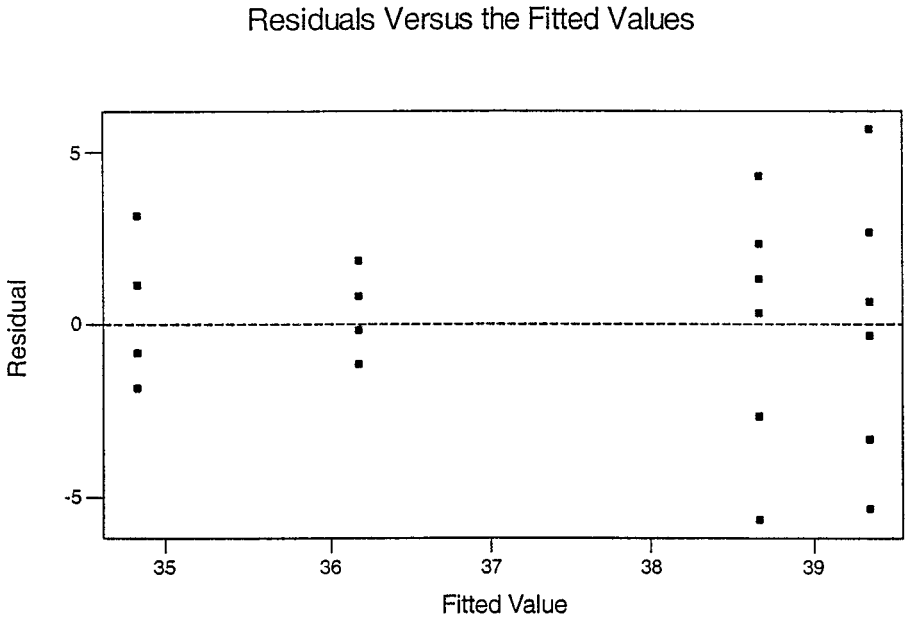
                1           2           3
2       -5.357
         4.024

3       -0.857      -0.190
         8.524       9.190

4       -2.190      -1.524      -6.024
         7.190       7.857       3.357
```

(d)

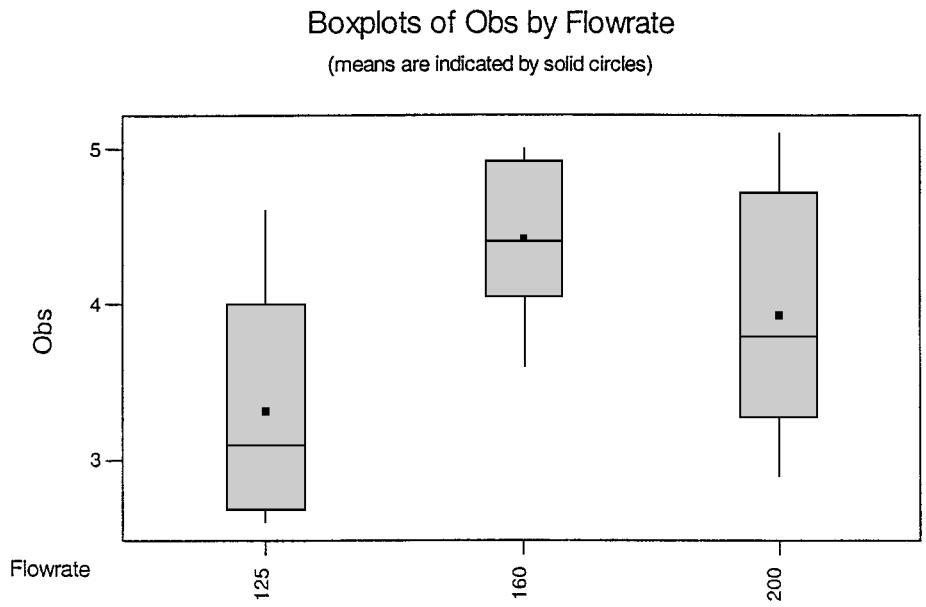




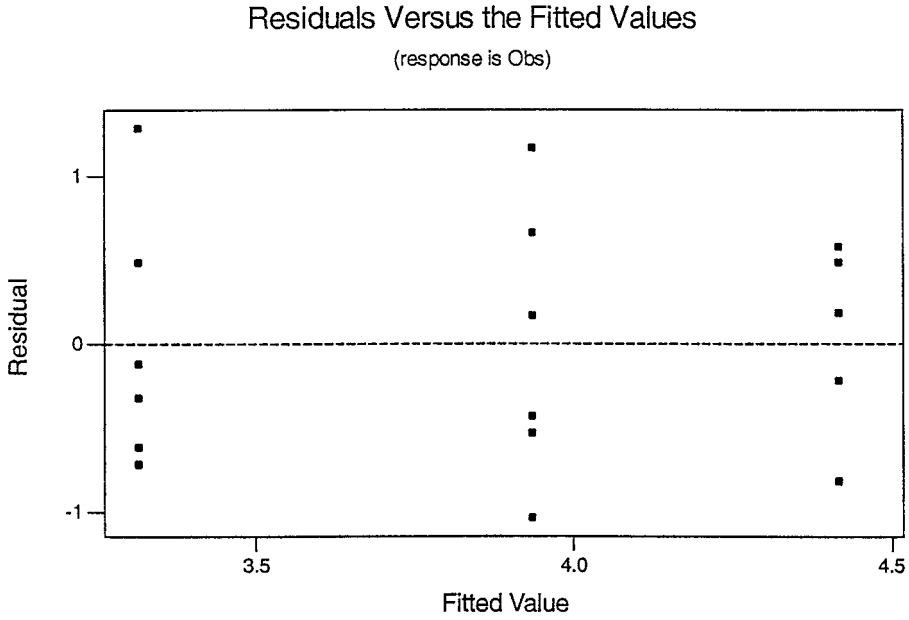
12–2. (a)

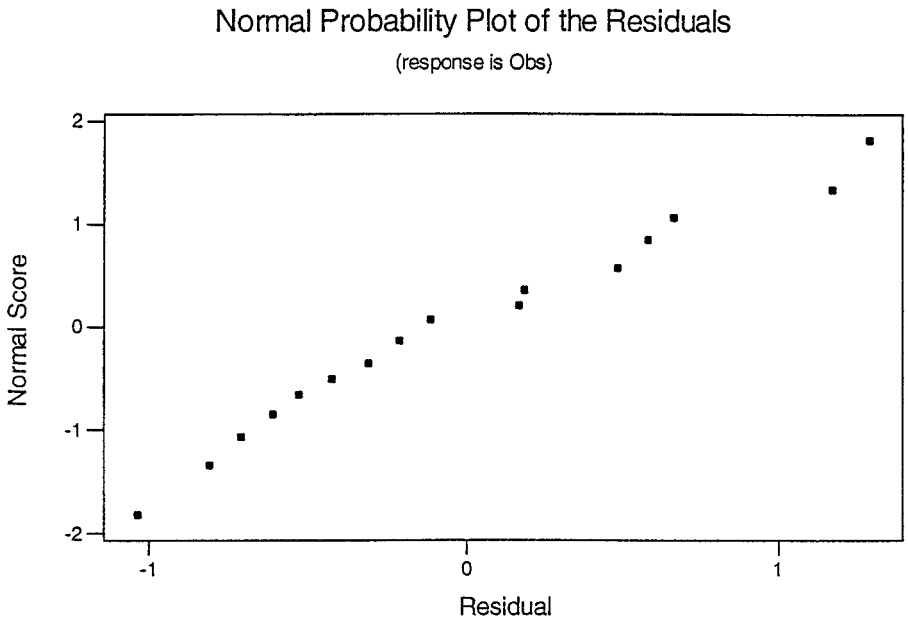
Analysis of Variance for Obs

Source	DF	SS	MS	F	P
Flowrate	2	3.648	1.824	3.59	0.053
Error	15	7.630	0.509		
Total	17	11.278			


Boxplots of Obs by Flowrate
(means are indicated by solid circles)
Obs
5
4
3
Flowrate
125
160
200

(b)


Residuals Versus the Fitted Values
(response is Obs)
Residual
1
0
-1
3.5
4.0
4.5
Fitted Value



12–3. (a)

```
Analysis of Variance for Strength
Source      DF        SS        MS        F        P
Technique    3    489740    163247    12.73    0.000
Error       12    153908     12826
Total       15    643648
```

(b) Tukey's pairwise comparisons

```
    Family error rate = 0.0500
Individual error rate = 0.0117
Critical value = 4.20
Intervals for (column level mean) - (row level mean)

                1          2          3
2         -423
            53

3         -201         -15
            275         460

4           67         252          30
            543         728         505
```

12–4. (a) Random effects

Analysis of Variance for Output

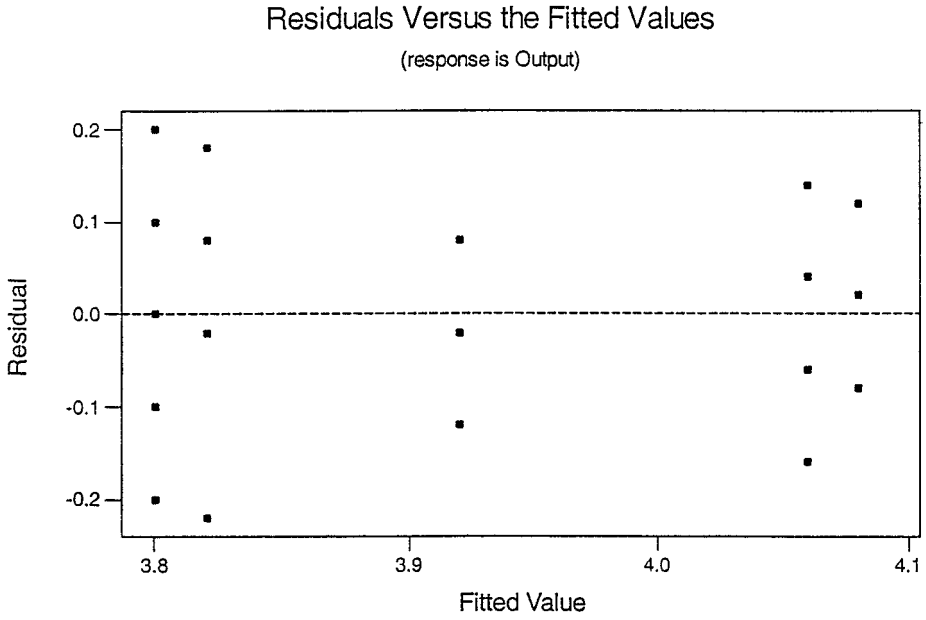
Source	DF	SS	MS	F	P
Loom	4	0.34160	0.08540	5.77	0.003
Error	20	0.29600	0.01480		
Total	24	0.63760			

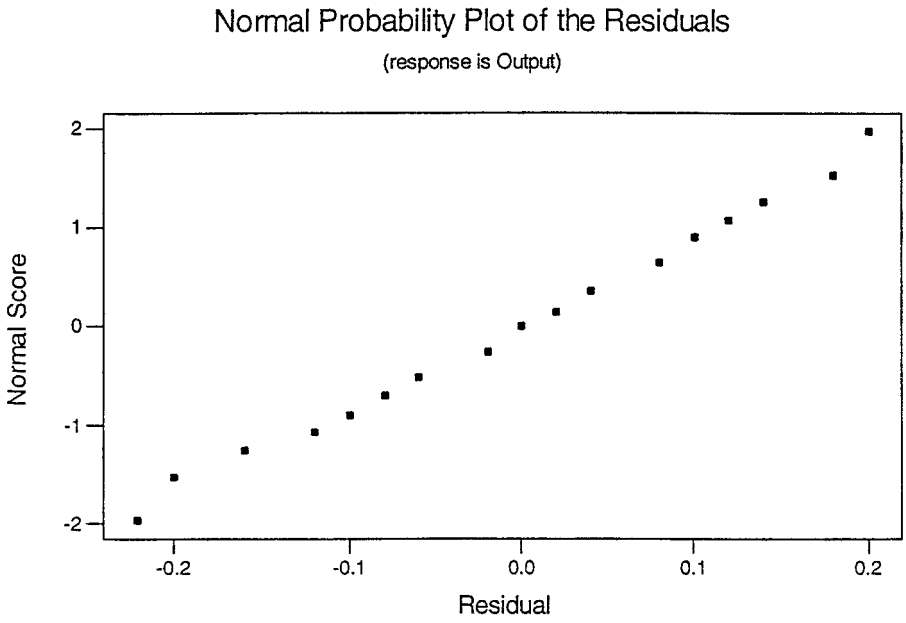
(b) $\sigma_\tau^2 = 0.01412$

(c) $\sigma^2 = 0.0148$

(d) 0.035

(e)





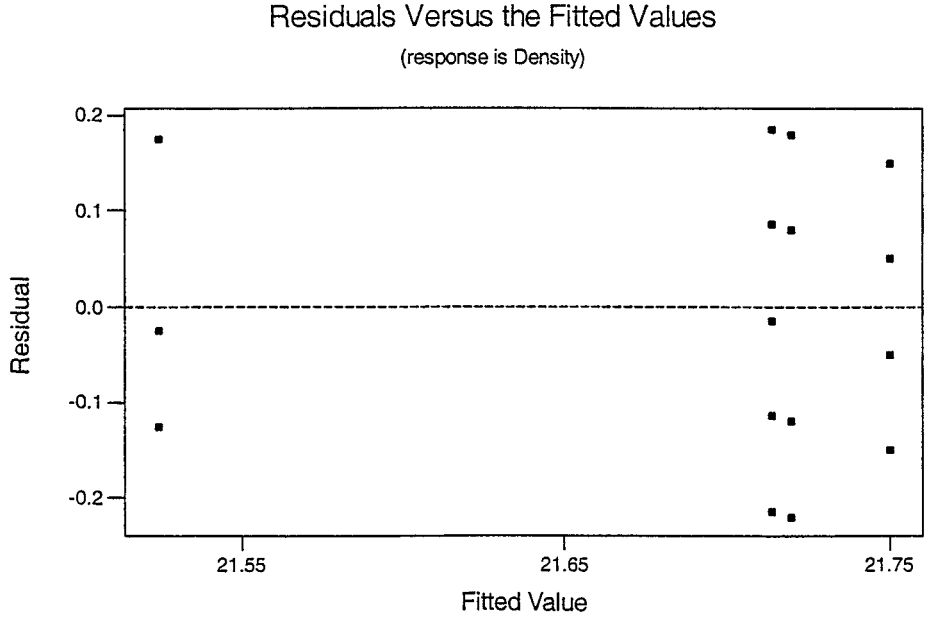
12–5. (a)

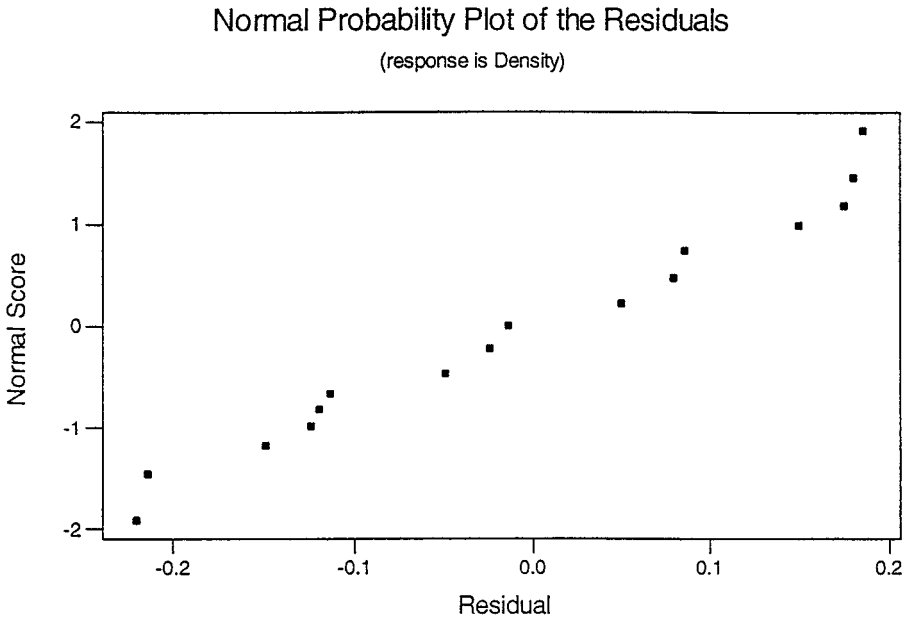
Analysis of Variance for Density

Source	DF	SS	SS	MS	F	P
Temp	3	0.13911	0.13911	0.04637	2.62	0.083
Error	18	0.31907	0.31907	0.01773		
Total	21	0.45818				

(b) $\hat{\mu} = 21.70$, $\hat{\tau}_1 = 0.023 \quad \hat{\tau}_2 = -0.166 \quad \hat{\tau}_3 = 0.029 \quad \hat{\tau}_4 = 0.059$

(c)





12–6. (a)

Analysis of Variance for Conductivity

Source	DF	SS	MS	F	P
Coating	4	1060.50	265.13	16.35	0.000
Error	15	243.25	16.22		
Total	19	1303.75			

(b) $\hat{\mu} = 139.25$, $\hat{\tau}_1 = 5.75$ $\hat{\tau}_2 = 6.00$ $\hat{\tau}_3 = -7.75$ $\hat{\tau}_4 = -10.00$ $\hat{\tau}_5 = 6.00$

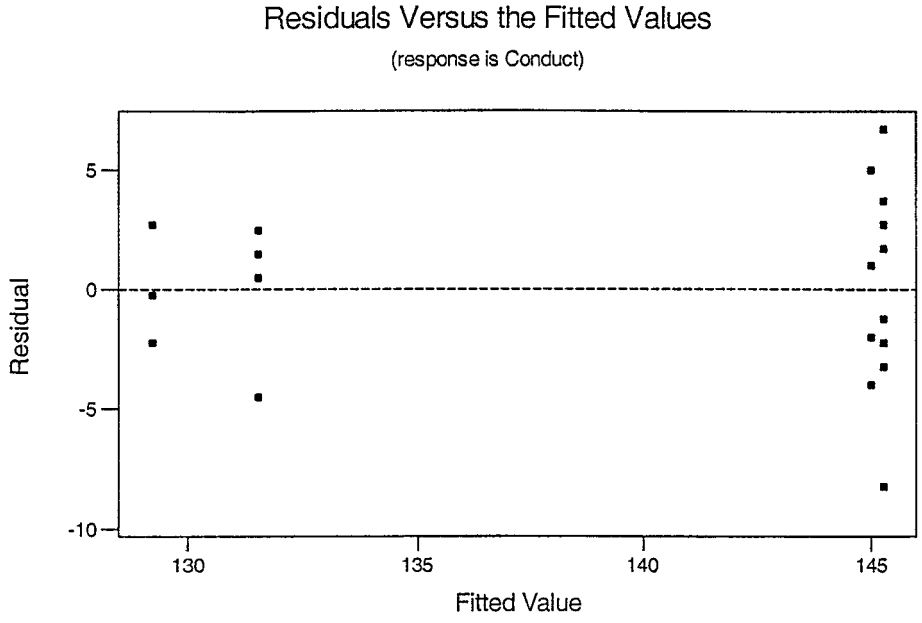
(c) $142.87 \leq \mu_1 \leq 147.13$, $7.363 \leq \mu_1 - \mu_2 \leq 24.137$

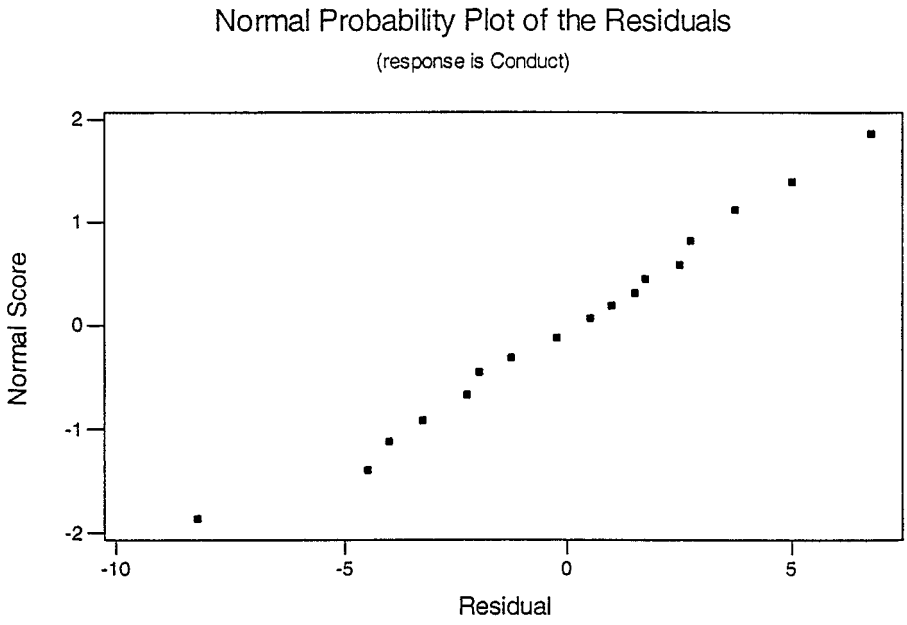
(d) Tukey's pairwise comparisons

```
    Family error rate = 0.0500
Individual error rate = 0.00747
Critical value = 4.37
Intervals for (column level mean) - (row level mean)
```

	1	2	3	4
2	-9.049 8.549			
3	4.701 22.299	4.951 22.549		
4	6.951 24.549	7.201 24.799	-6.549 11.049	
5	-9.049 8.549	-8.799 8.799	-22.549 -4.951	-24.799 -7.201

(e)





12–7. (a)

```
Analysis of Variance for Response Time
Source     DF        SS        MS        F         P
Circuit     2     260.9     130.5     4.01     0.046
Error      12     390.8      32.6
Total      14     651.7
```

(b)

```
Tukey's pairwise comparisons
    Family error rate = 0.0500
Individual error rate = 0.0206
Critical value = 3.77
Intervals for (column level mean) - (row level mean)

                1          2
2     -17.022
        2.222

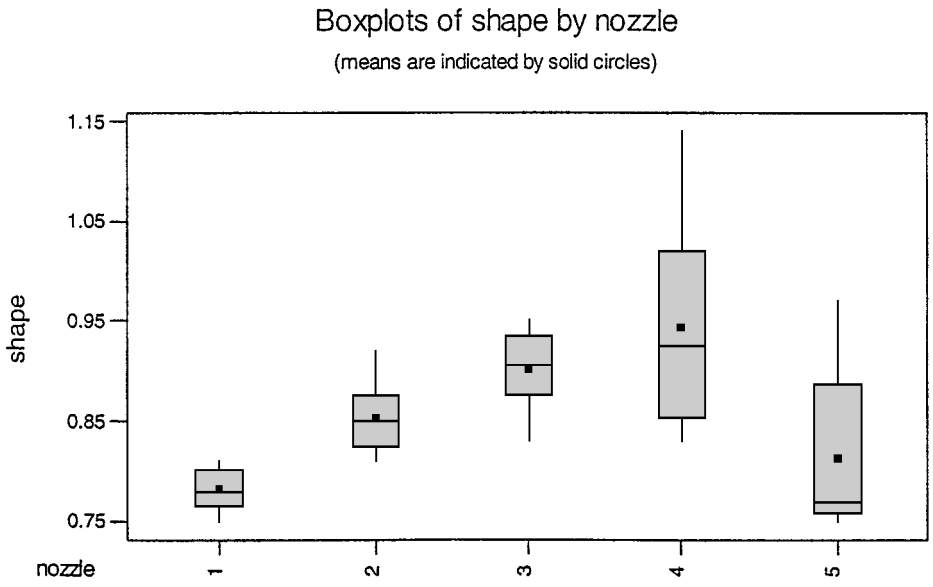
3      -7.222       0.178
       12.022      19.422
```

(c) $c_1 = y_{1\cdot} - 2y_{2\cdot} + y_{3\cdot}$, $SS_{c_1} = 14.4$
$c_2 = y_{1\cdot} - y_{3\cdot}$, $SS_{c_2} = 246.53$
Only c_2 is significant at $\alpha = 0.05$

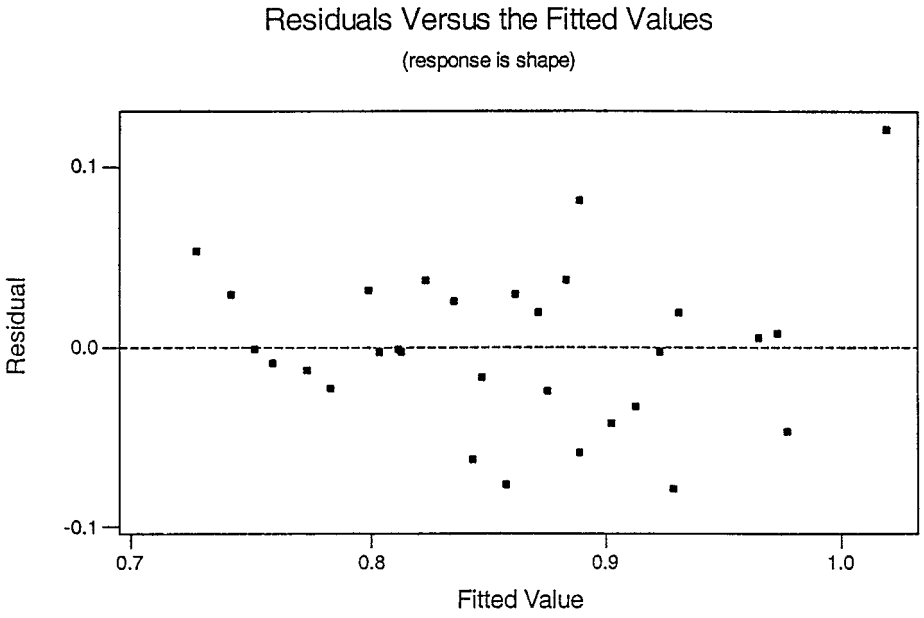
(d) 0.88

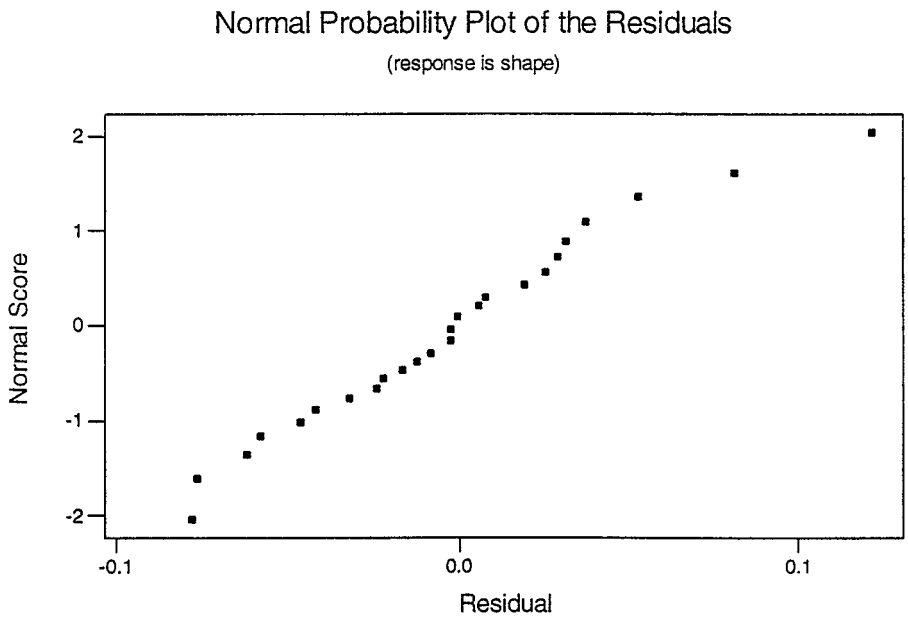
12–8. (a) Analysis of Variance for Shape

Source	DF	SS	MS	F	P
Nozzle	4	0.10218	0.02554	8.92	0.000
Efflux	5	0.06287	0.01257	4.39	0.007
Error	20	0.05730	0.00286		
Total	29	0.22235			



(c)



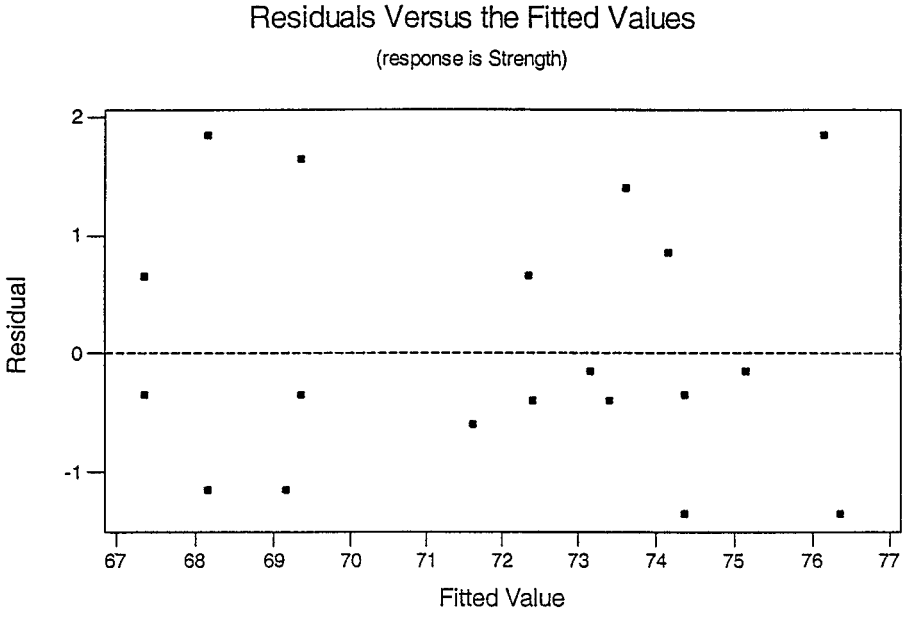


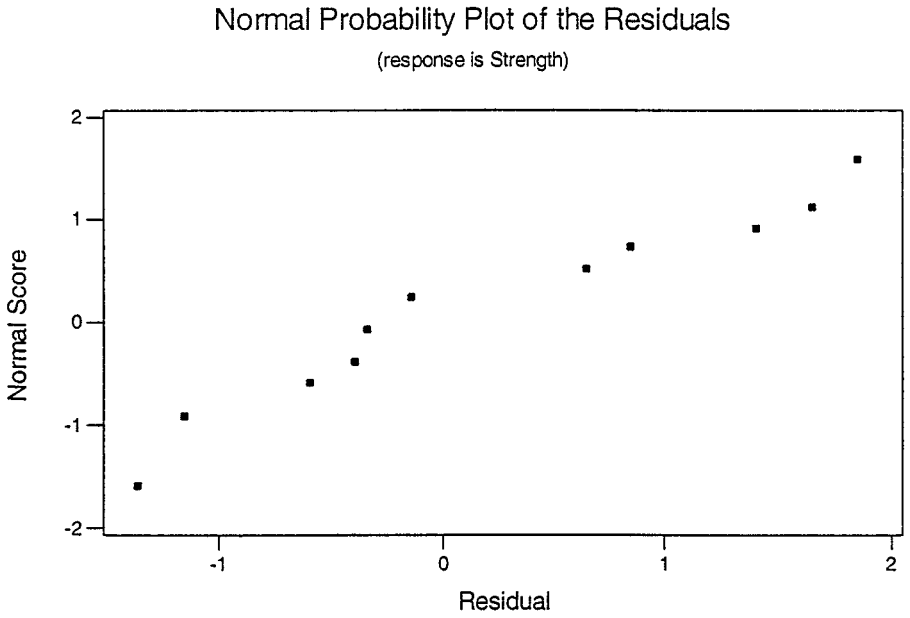
12–9. (a)

Analysis of Variance for Strength

Source	DF	SS	MS	F	P
Chemical	3	12.95	4.32	2.38	0.121
Bolt	4	157.00	39.25	21.61	0.000
Error	12	21.80	1.82		
Total	19	191.75			

(c)





12–10. $\mu = \frac{1}{4}\Sigma\mu_1 = \frac{220}{4} = 55$, $\tau_1 = 50 - 55 = -5$, $\tau_2 = 60 - 55 = 5$, $\tau_3 = 50 - 55 = -5$, $\tau_4 = 60 - 55 = 5$. Then $\Phi^2 = \frac{2}{a}\frac{n\Sigma\tau_i^2}{2\sigma^2} = \frac{2}{4}\frac{n(100)}{2(25)} = n$, $\Phi = \sqrt{n}$. $\beta < 0.1$, $\alpha = 0.05$, OC curves give $n = 7$.

12–11. $\mu = \dfrac{1}{5}\Sigma\mu_i = \dfrac{940}{5} = 188, \tau_i = \mu_i - 188, i = 1, 2, \ldots, 5$

$\tau_1 = -13, \tau_2 = 2, \tau_3 = -28, \tau_4 = 12, \tau_5 = 27$

$$\Phi^2 = \frac{n}{a}\frac{\Sigma\tau_i^2}{\sigma^2} = \frac{n}{5}\left(\frac{1830}{100}\right) = 3.66n, \Phi = 1.91\sqrt{n}$$

If $\beta \leq 0.05$, $\alpha = 0.01$, OC curves give $n = 3$. If $n = 3$, then $\beta \cong 0.03$.

12–12. The test statistic for the two-sample t-test (with $n_1 = n_2 = n$) is

$$t_0 = \frac{\overline{y}_{1\cdot} - \overline{y}_{2\cdot}}{S_p\sqrt{2/n}} - t_{2n-2}$$

$$t_0^2 = \frac{(\overline{y}_{1\cdot} - \overline{y}_{2\cdot})^2(n/2)}{S_p^2}$$

$$= \frac{\left(\dfrac{y_{1\cdot}}{n} + \dfrac{y_{2\cdot}}{n}\right)^2 \cdot \left(\dfrac{n}{2}\right)}{S_p^2} = \frac{\dfrac{y_{1\cdot}^2}{2n} + \dfrac{y_{2\cdot}^2}{2n} - \dfrac{y_{1\cdot}y_{2\cdot}}{n}}{S_p^2}$$

But since $y_{1\cdot}y_{2\cdot} = \frac{y_{1\cdot}^2}{2} + \frac{y_{2\cdot}^2}{2} - \frac{(y_{1\cdot}+y_{2\cdot})^2}{2}$, the last equation becomes

$$t_0^2 = \frac{\dfrac{y_{1\cdot}^2}{n} + \dfrac{y_{2\cdot}^2}{n} - \dfrac{(y_{1\cdot}+y_{2\cdot})^2}{2n}}{S_p^2} = \frac{SS_{\text{Treatments}}}{S_p^2}$$

Note that $S_p^2 = \frac{\sum_{i=1}^{2}\sum_{j=1}^{n}(y_{ij}-\overline{y}_{i\cdot})^2}{2n-2} = MS_E$. Therefore, $t_0^2 = \frac{SS_{\text{Treatments}}}{MS_E}$, and since the square of t_u is $F_{1,u}$ (in general), we see that the two tests are equivalent.

12–13. $$V\left(\sum_{i=1}^{a} c_i y_{i\cdot}\right) = \sum_{i=1}^{a} V(c_i y_{i\cdot}) = \sum_{i=1}^{a} c_i^2 V(y_{i\cdot})$$
$$= \sum_{i=1}^{a} c_i^2 V\left(\sum_{j=1}^{n_i} y_{ij}\right) = \sum_{i=1}^{a} c_i^2 \sum_{j=1}^{n_i} V(y)$$
$$= \sigma^2 \sum_{i=1}^{a} n_i c_i^2$$

12–14. For 4 treatments, a set of orthogonal contrasts is

$$3y_{1\cdot} - y_{2\cdot} - y_{3\cdot} - y_{4\cdot}$$
$$2y_{2\cdot} - y_{3\cdot} - y_{4\cdot}$$
$$y_{3\cdot} - y_{4\cdot}$$

Assuming equal n, the contrast sums of squares are

$$Q_1^2 = \frac{(3y_{1\cdot} - y_{2\cdot} - y_{3\cdot} - y_{4\cdot})^2}{12n}, \quad Q_2^2 = \frac{(2y_{2\cdot} - y_{3\cdot} - y_{4\cdot})^2}{6n}$$

$$Q_3^2 = \frac{(y_{3\cdot} - y_{4\cdot})^2}{2n}$$

Now

$$Q_1^2 + Q_2^2 + Q_3^2 = \frac{9\sum_{i=1}^{4} y_{i\cdot}^2 - 6\sum\sum_{i<j} y_{i\cdot}y_{j\cdot}}{12n}$$

and since

$$\sum\sum_{i<j} y_{i\cdot}y_{j\cdot} = \frac{1}{2}\left[y_{\cdot\cdot}^2 - \sum_{i=1}^{4} y_{i\cdot}^2\right],$$

$$Q_1^2 + Q_2^2 + Q_3^2 = \frac{12\sum_{i=1}^{4} y_{i\cdot}^2 - 3y_{\cdot\cdot}^2}{12n} = \frac{\sum_{i=1}^{4} y_{i\cdot}^2}{n} - \frac{y_{\cdot\cdot}^2}{4n}$$
$$= SS_{\text{Treatments}}$$

12–15. (a)
$$\begin{aligned} 15\hat{\mu} + 5\hat{\tau}_1 + 5\hat{\tau}_2 + 5\hat{\tau}_3 &= 307 \\ 5\hat{\mu} + 5\hat{\tau}_1 \qquad\qquad &= 104 \\ 5\hat{\mu} \qquad + 5\hat{\tau}_2 \qquad &= 111 \\ 5\hat{\mu} \qquad\qquad + 5\hat{\tau}_3 &= 92 \end{aligned}$$

If $\sum_{i=1}^{3} \hat{\tau}_i = 0$, then $\hat{\mu} = 20.47$, $\hat{\tau}_1 = 0.33$, $\hat{\tau}_2 = 1.73$, and $\hat{\tau}_3 = 2.07$.

$$\widehat{\tau_1 - \tau_2} = \hat{\tau}_1 - \hat{\tau}_2 = -1.40$$

(b) If $\hat{\tau}_3 = 0$, then $\hat{\mu} = 18.40$, $\hat{\tau}_1 = 2.40$, $\hat{\tau}_2 = 3.80$, and $\hat{\tau}_3 = 0$. These estimators differ from those found in part (a). However, note that

$$\widehat{\tau_1 - \tau_2} = \hat{\tau}_1 - \hat{\tau}_2 = 2.40 - 3.80 = -1.40$$

which agrees with part (a), because *contrasts* in the τ_i are uniquely estimated.

(c)

	Estimate Using	
Function	Part (a) Solution	Part (b) Solution
$\mu + \tau_1$	20.80	20.80
$2\tau_1 - \tau_2 - \tau_3$	1.00	1.00
$\mu + \tau_1 + \tau_2$	22.53	24.60

Chapter 13

13–1. Analysis of Variance for Wear

Source	DF	SS	MS	F	P
CS	2	0.0317805	0.0158903	15.94	0.000
DC	2	0.0271854	0.0135927	13.64	0.000
CS*DC	4	0.0006873	0.0001718	0.17	0.950
Error	18	0.0179413	0.0009967		
Total	26	0.0775945			

13–2. Analysis of Variance for Finish

Source	DF	SS	MS	F	P
Drying	2	27.4	13.7	0.01	0.986
Paint	1	355.6	355.6	0.38	0.601
Drying*Paint	2	1878.8	939.4	5.03	0.026
Error	12	2242.7	186.9		
Total	17	4504.4			

13–3. $-23.93 \leq \mu_1 - \mu_2 \leq 5.15$

13–4. Analysis of Variance for Strength

Source	DF	SS	MS	F	P
operator	2	2.250	1.125	0.29	0.759
machine	3	28.833	9.611	2.46	0.160
operator*machine	6	23.417	3.903	0.84	0.565
Error	12	56.000	4.667		
Total	23	110.500			

13–5. The results would be a mixed model. The test statistics would be:

Effect	F_0
Operator	0.241
Machine	2.46
Operator*Machine	0.84

There is no change in conclusions.

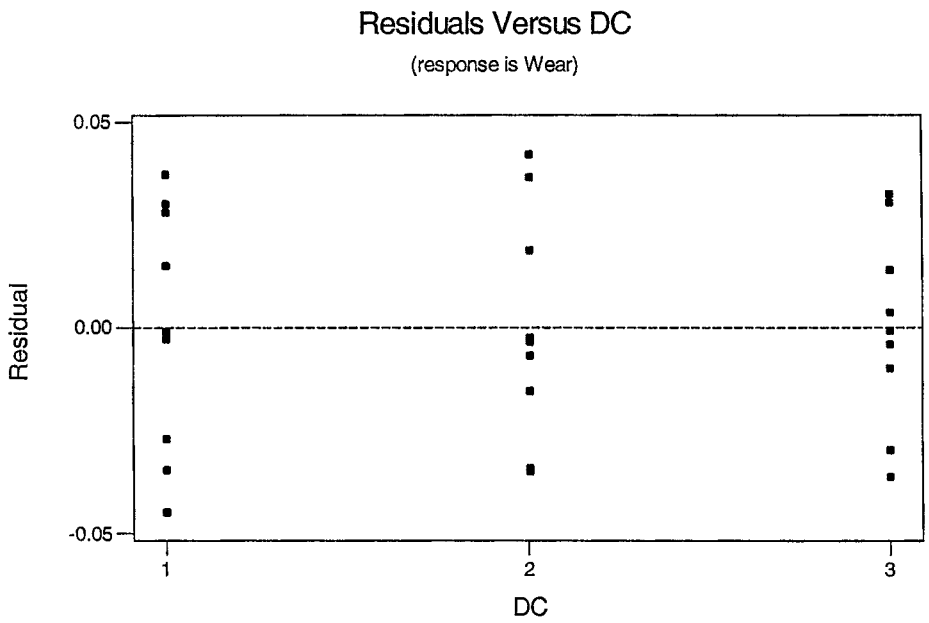
13–6. Analysis of Variance for time

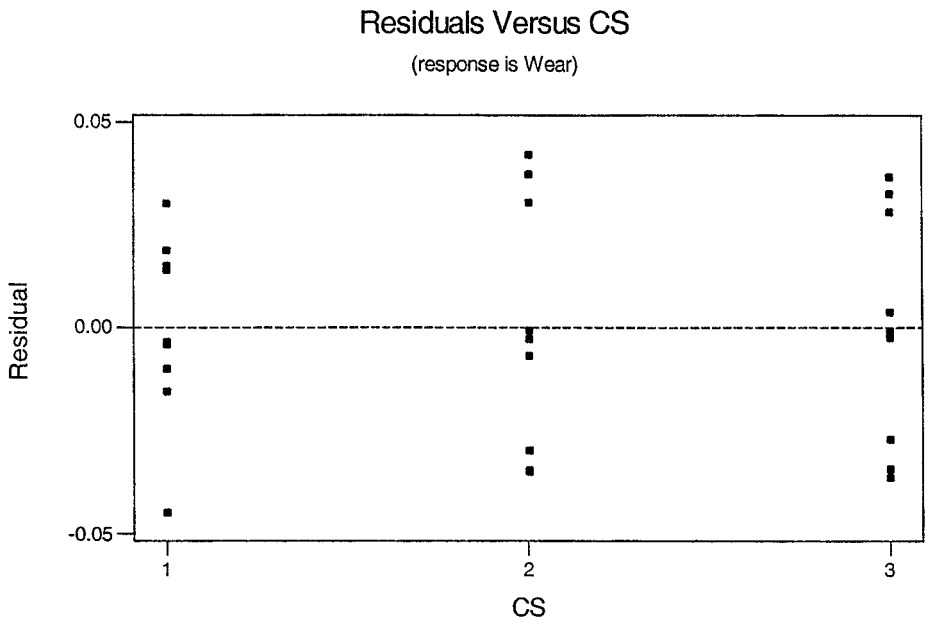
Source	DF	SS	MS	F	P
operator	2	0.01005	0.00503	0.07	0.937
engineer	1	0.04688	0.04688	0.62	0.512
operator*engineer	2	0.15005	0.07503	1.26	0.350
Error	6	0.35785	0.05964		
Total	11	0.56483			

13–7. Analysis of Variance for current

Source	DF	SS	MS	F	P
glass	1	14450.0	14450.0	273.79	0.000
phos	2	933.3	466.7	8.84	0.004
glass*phos	2	133.3	66.7	1.26	0.318
Error	12	633.3	52.8		
Total	17	16150.0			

13–8.





There does not appear to be a problem with constant variance across levels of either factor.

13–9.

Analysis of Variance for strength

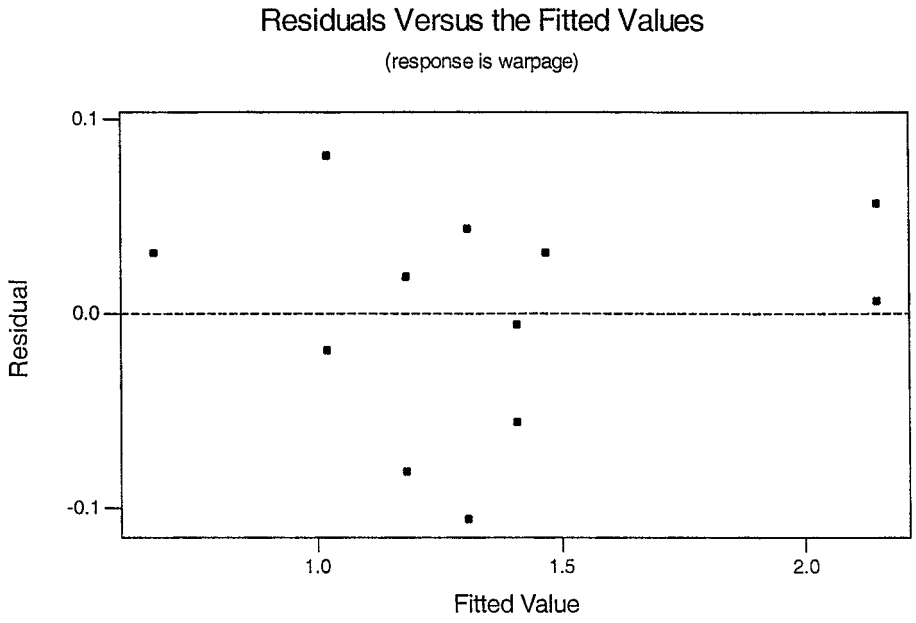
Source	DF	SS	MS	F	P
Conc	2	7.7639	3.8819	10.62	0.001
Freeness	2	19.3739	9.6869	26.50	0.000
Time	1	20.2500	20.2500	55.40	0.000
Conc*Freeness	4	6.0911	1.5228	4.17	0.015
Conc*Time	2	2.0817	1.0408	2.85	0.084
Freeness*Time	2	2.1950	1.0975	3.00	0.075
Conc*Freeness*Time	4	1.9733	0.4933	1.35	0.290
Error	18	6.5800	0.3656		
Total	35	66.3089			

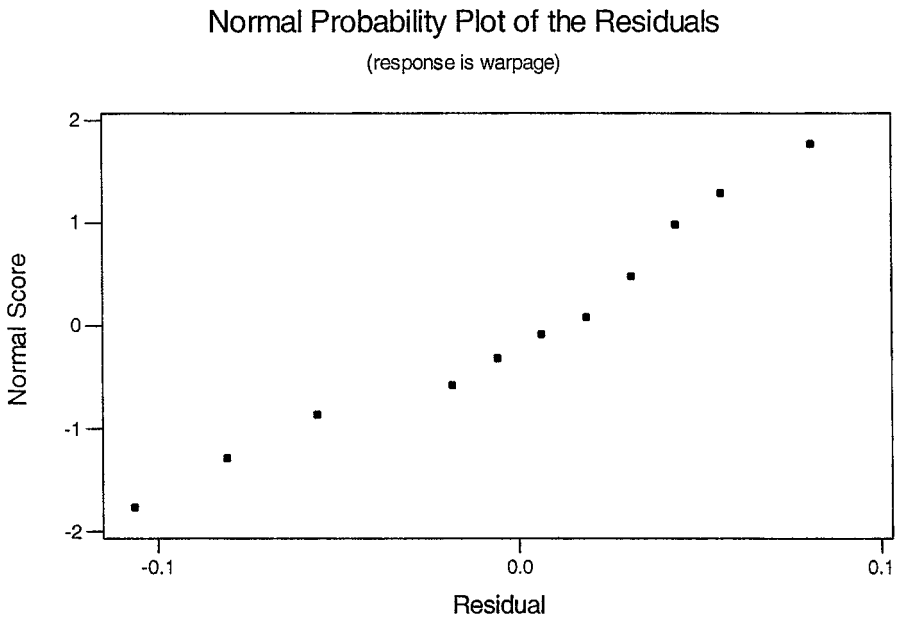
13–10. Estimated Effects and Coefficients for warpage (coded units)

Term	Effect	Coef	SE Coef	T	P
Constant		1.3250	0.01398	94.81	0.000
A	0.2250	0.1125	0.01398	8.05	0.000
B	-0.1625	-0.0813	0.01398	-5.81	0.000
C	-0.4500	-0.2250	0.01398	-16.10	0.000
A*B	-0.5125	-0.2563	0.01398	-18.34	0.000
A*C	0.0000	0.0000	0.01398	0.00	1.000
B*C	0.2875	0.1438	0.01398	10.29	0.000
A*B*C	0.0625	0.0313	0.01398	2.24	0.056

Based on the p-values, factors A, B, C, and the interactions AB and BC are significant at the 5% level of significance.

13–11. Using only the significant factors and interactions, the resulting residuals are as follows.

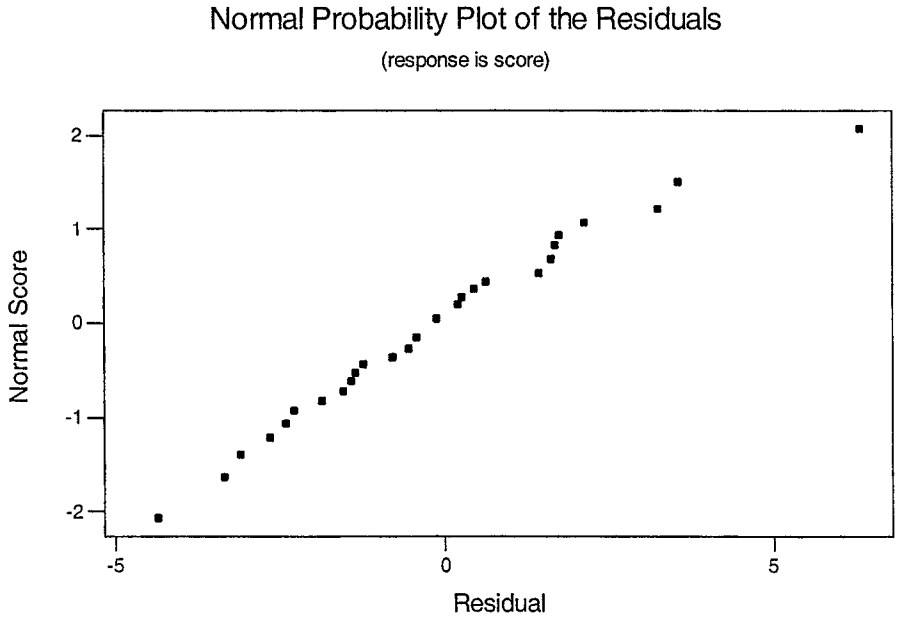


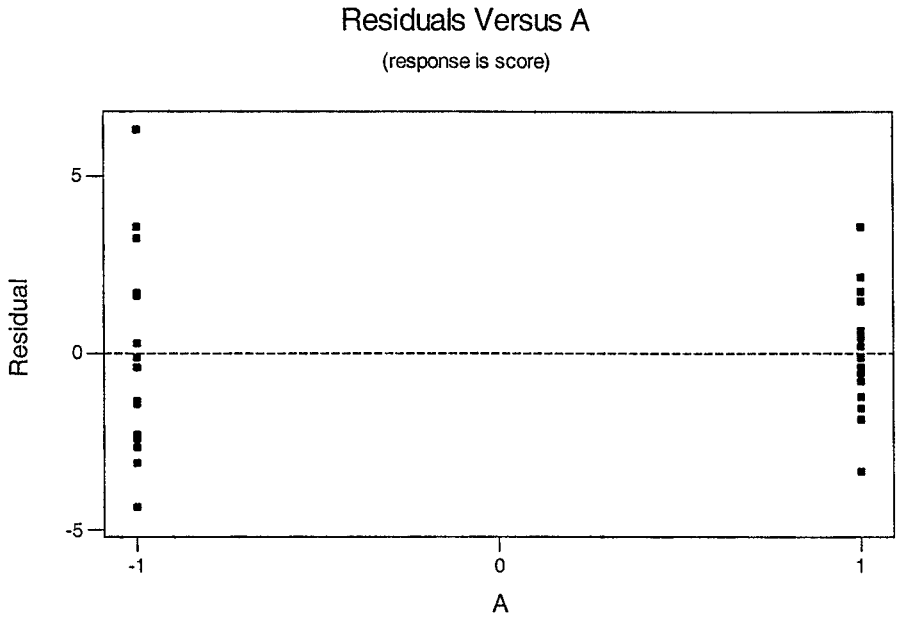


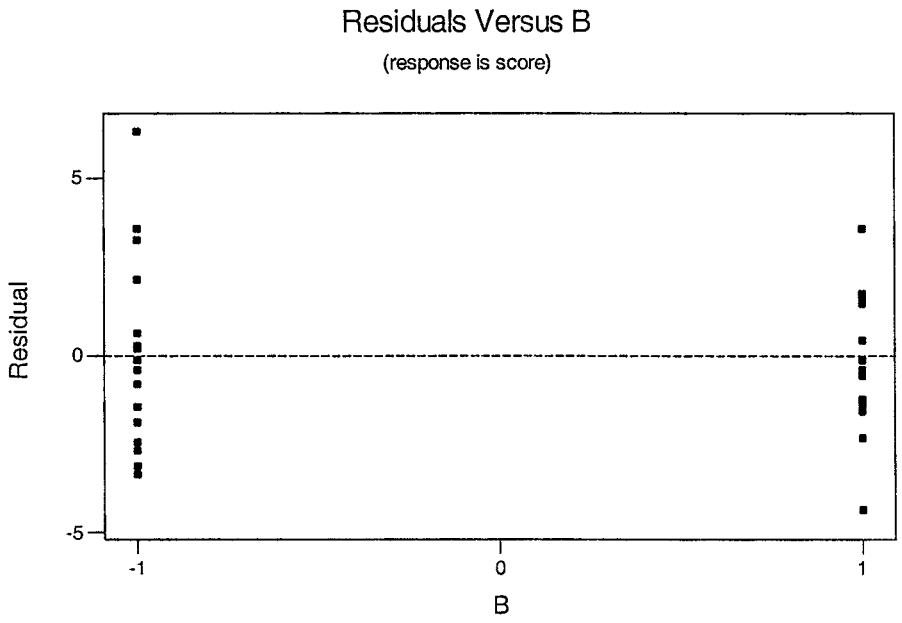
13–12. Estimated Effects and Coefficients for score (coded units)

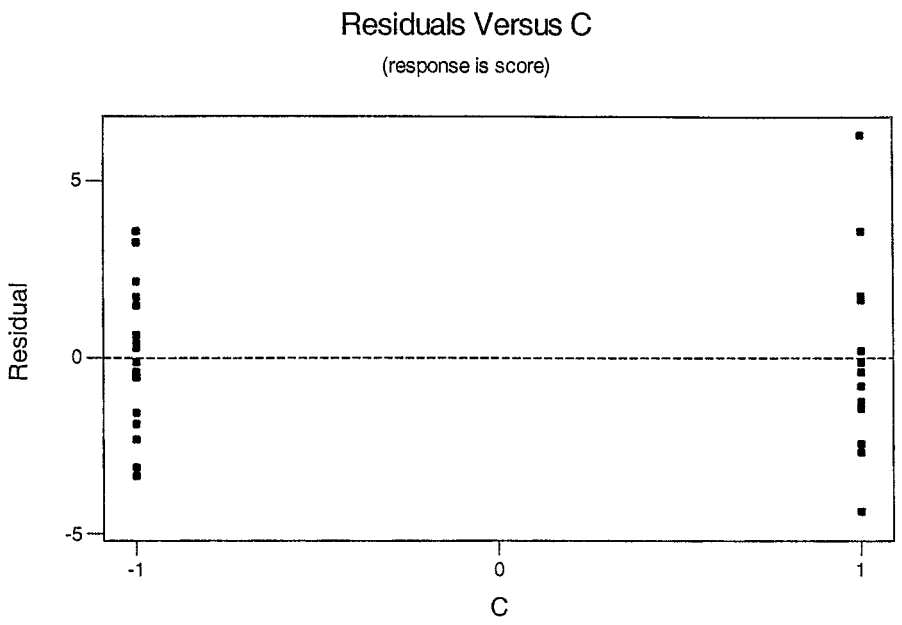
Term	Effect	Coef	SE Coef	T	P
Constant		182.781	0.4891	373.68	0.000
A	-9.063	-4.531	0.4891	-9.26	0.000
B	-1.312	-0.656	0.4891	-1.34	0.198
C	-2.687	-1.344	0.4891	-2.75	0.014
D	3.937	1.969	0.4891	4.02	0.001
A*B	4.062	2.031	0.4891	4.15	0.001
A*C	0.688	0.344	0.4891	0.70	0.492
A*D	-2.187	-1.094	0.4891	-2.24	0.040
B*C	-0.563	-0.281	0.4891	-0.57	0.573
B*D	-0.188	-0.094	0.4891	-0.19	0.850
C*D	1.688	0.844	0.4891	1.72	0.104
A*B*C	-5.187	-2.594	0.4891	-5.30	0.000
A*B*D	4.687	2.344	0.4891	4.79	0.000
A*C*D	-0.938	-0.469	0.4891	-0.96	0.352
B*C*D	-0.938	-0.469	0.4891	-0.96	0.352
A*B*C*D	2.437	1.219	0.4891	2.49	0.024

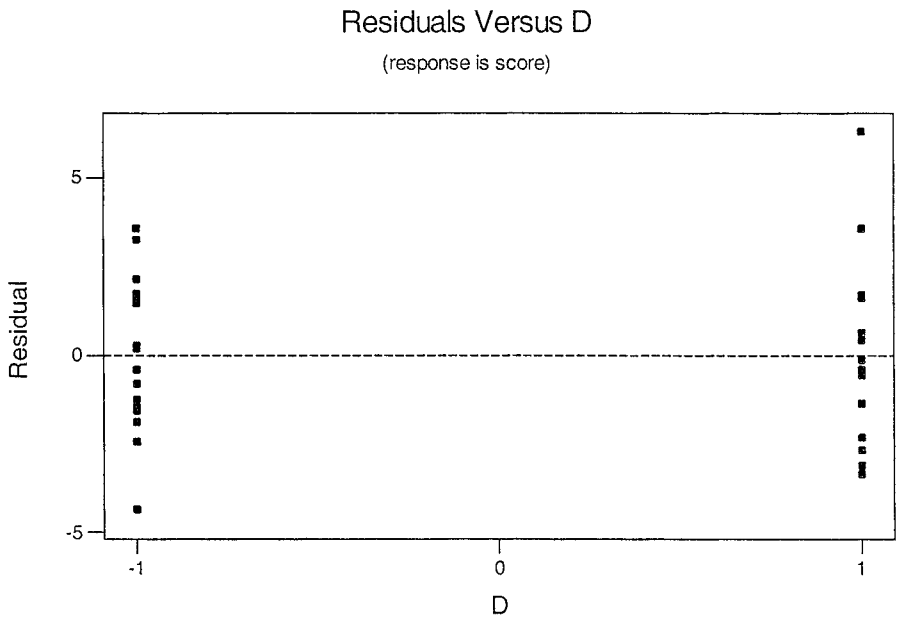
13–13.


Normal Probability Plot of the Residuals
(response is score)
Normal Score
2
1
0
-1
-2
-5
0
5
Residual


Residuals Versus A
(response is score)
Residual
5
0
-5
-1
0
1
A


Residuals Versus B
(response is score)
Residual
5
0
-5
-1
0
1
B


Residuals Versus C
(response is score)
Residual
5
0
-5
-1
0
1
C

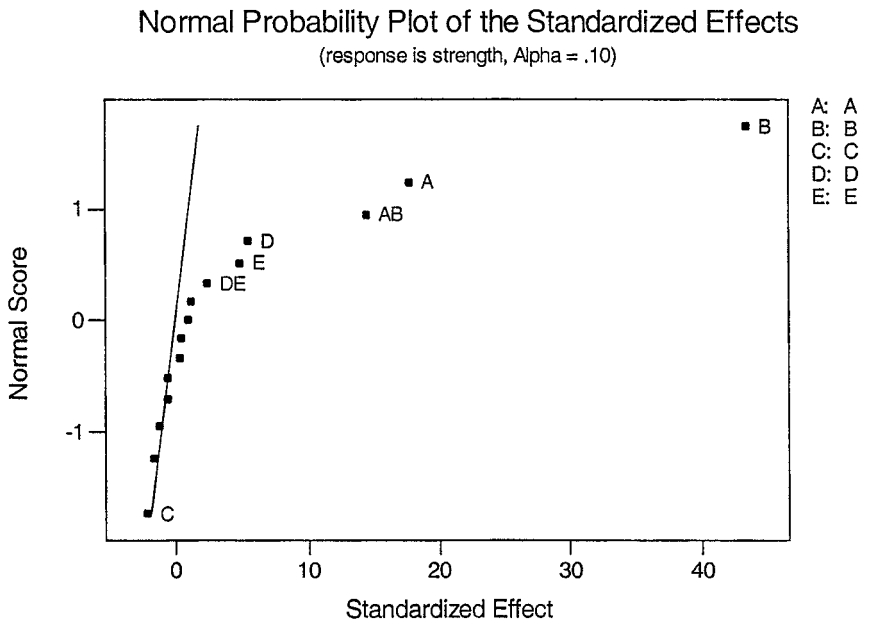


13–14. See solution for 13–12 for the standard errors.

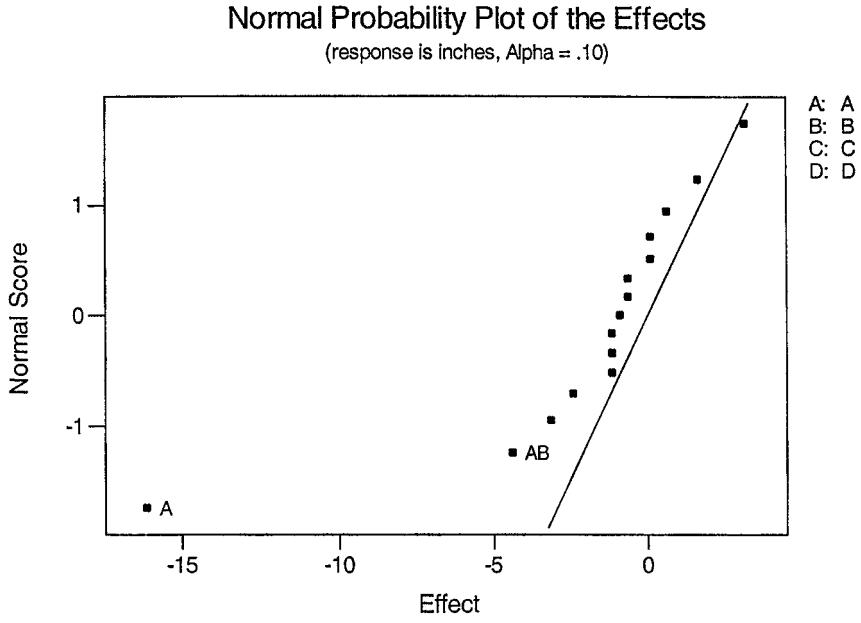
13–15. Estimated Effects and Coefficients for strength

Term	Effect	Coef	SE Coef	T	P
Constant		2872.06	40.47	70.97	0.000
A	1430.88	715.44	40.47	17.68	0.000
B	3506.62	1753.31	40.47	43.33	0.000
C	-168.37	-84.19	40.47	-2.08	0.054
D	443.37	221.69	40.47	5.48	0.000
E	394.13	197.06	40.47	4.87	0.000
A*B	1168.37	584.19	40.47	14.44	0.000
A*C	93.37	46.69	40.47	1.15	0.266
A*D	31.62	15.81	40.47	0.39	0.701
A*E	30.88	15.44	40.47	0.38	0.708
B*C	-130.87	-65.44	40.47	-1.62	0.125
B*D	-44.12	-22.06	40.47	-0.55	0.593
B*E	-43.37	-21.69	40.47	-0.54	0.599
C*D	80.88	40.44	40.47	1.00	0.333
C*E	-93.38	-46.69	40.47	-1.15	0.266
D*E	193.38	96.69	40.47	2.39	0.030

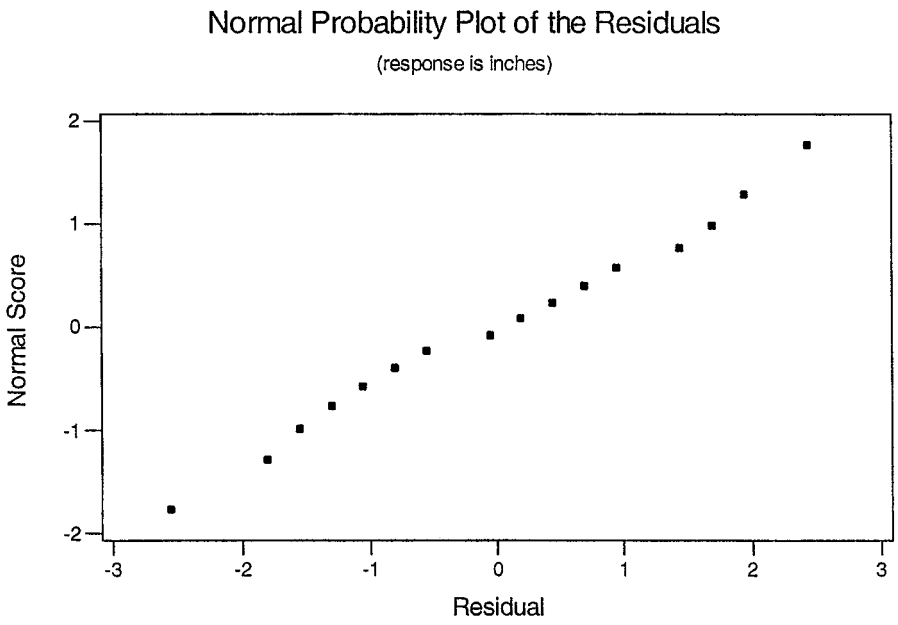
Main effects A, B, D, E and interactions AB and DE are significant.


Normal Probability Plot of the Standardized Effects
(response is strength, Alpha = .10)
Normal Score
Standardized Effect
A: A
B: B
C: C
D: D
E: E
B
A
AB
D
E
DE
C
1
0
-1
0
10
20
30
40

13–16. (a)


Normal Probability Plot of the Effects
(response is inches, Alpha = .10)
Normal Score
Effect
A: A
B: B
C: C
D: D
AB
A
1
0
-1
-15
-10
-5
0

(b)



(c) Estimated Effects and Coefficients for inches

Term	Effect	Coef	SE Coef	T	P
Constant		35.938	0.6355	56.55	0.000
A	-16.125	-8.062	0.6355	-12.69	0.000
B	3.125	1.562	0.6355	2.46	0.057
C	-1.125	-0.562	0.6355	-0.89	0.417
D	-1.125	-0.562	0.6355	-0.89	0.417
A*B	-4.375	-2.188	0.6355	-3.44	0.018
A*C	-0.625	-0.313	0.6355	-0.49	0.644
A*D	-3.125	-1.563	0.6355	-2.46	0.057
B*C	1.625	0.812	0.6355	1.28	0.257
B*D	0.125	0.063	0.6355	0.10	0.925
C*D	-0.625	-0.312	0.6355	-0.49	0.644

13–17.

Block 1	Block 2
(1)	a
ab	b
ac	c
bc	abc

13–18.

Block 1	Block 2
(1) *ad*	*a abc*
ab bd	*b bcd*
ac cd	*c acd*
bc abcd	*d abd*

13–19.

Block 1	Block 2	Block 3	Block 4
(1)	*a*	*c*	*d*
ab	*b*	*abc*	*abd*
bcd	*cd*	*bd*	*bc*
acd	*abcd*	*ad*	*ac*

13–20. AC and BDE confounded: $ABCDE$ generalized interaction

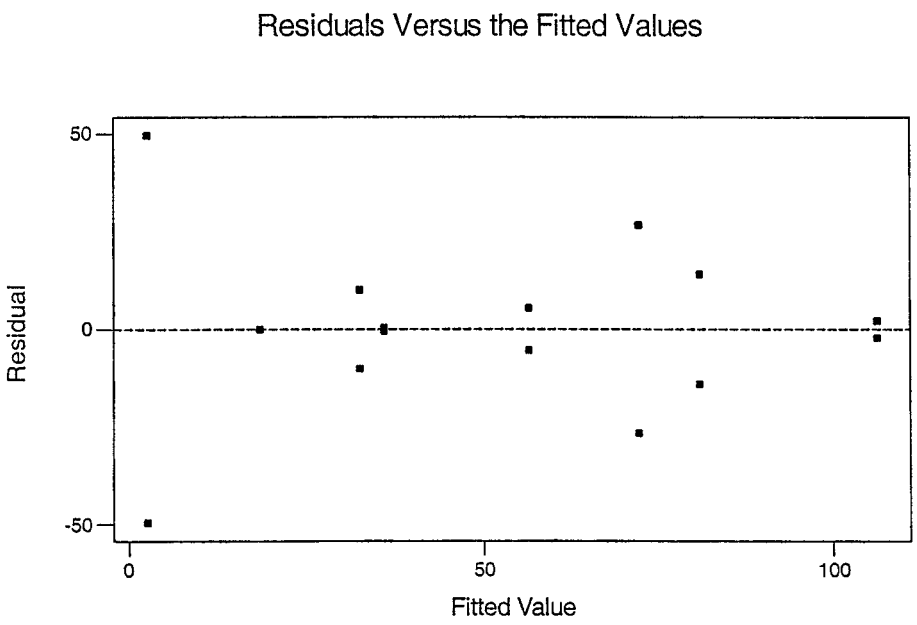
Block 1	Block 2	Block 3	Block 4
(1)	*a*	*b*	*ab*
ac	*c*	*abc*	*bc*
bd	*abd*	*d*	*ad*
abcd	*bcd*	*acd*	*cd*
be	*abe*	*e*	*ae*
abce	*bce*	*ace*	*ce*
de	*ade*	*bde*	*abde*
acde	*cde*	*abcde*	*bcde*

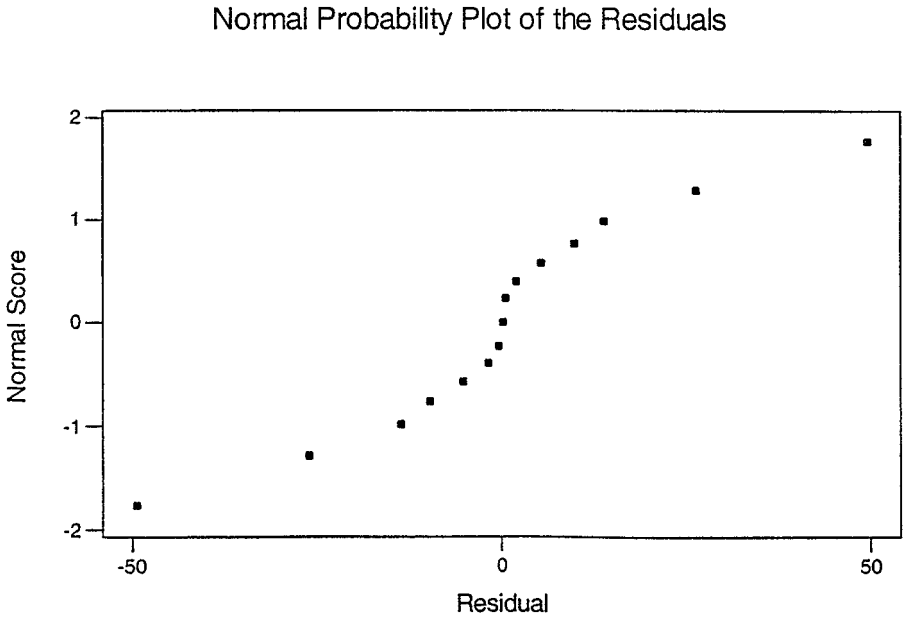
13–21. (a)

Estimated Effects and Coefficients for strength

Term	Effect	Coef	SE Coef	T	P
Constant		50.50	7.377	6.85	0.000
Block		-3.50	7.377	-0.47	0.648
A	-57.00	-28.50	7.377	-3.86	0.005
B	-13.25	-6.62	7.377	-0.90	0.395
C	26.25	13.12	7.377	1.78	0.113
A*B	7.25	3.62	7.377	0.49	0.636
A*C	-2.75	-1.38	7.377	-0.19	0.857
B*C	2.50	1.25	7.377	0.17	0.870

(b)





(c) & (d) This design is not as efficient as possible. If we were to confound a different interaction in each replicate this would provide some information on all interactions.

13–22. Please refer to the original reference for an analysis of the data from this experiment.

13–23. (a) $I = ABCD$. Alias Structure:

$$\begin{array}{lll} \ell_A = A + BCD & \ell_{AB} = AB + CD & \ell_{CE} = CE + ABD \\ \ell_B = B + ACD & \ell_{AC} = AC + BD & \ell_{AD} = AD + CB \\ \ell_C = C + ABD & \ell_{AE} = AE + BCD & \ell_{BD} = BD + AC \\ \ell_D = D + ABC & \ell_{BC} = BC + AD & \ell_{CD} = CD + AB \\ \ell_E = E + ABCD & \ell_{BE} = BE + ACD & \ell_{DE} = DE + ABC \end{array}$$

(b) design: $2^{5-1}D = ABC$

$$\begin{array}{lll} \ell_A = 0.238 & \ell_{AB} = -0.024 & \ell_{BD} = 0.042 \\ \ell_B = -0.16 & \ell_{AC} = 0.0042 & \ell_{CD} = -0.024 \\ \ell_C = -0.043 & \ell_{BC} = -0.026 & \ell_{BE} = 0.1575 \\ \ell_D = 0.0867 & \ell_{AD} = -0.026 & \ell_{CE} = -0.029 \\ \ell_E = -0.242 & \ell_{AE} = 0.059 & \ell_{DE} = 0.036 \end{array}$$

Conclusion: assuming 3 and 4 factor interactions insignificant, factors A & E are important; possible also B and BE.

13–24. (a) The generators used were $I = ACE$ and $I = BDE$.

(b) $I = ACE = BDE = ABCD$

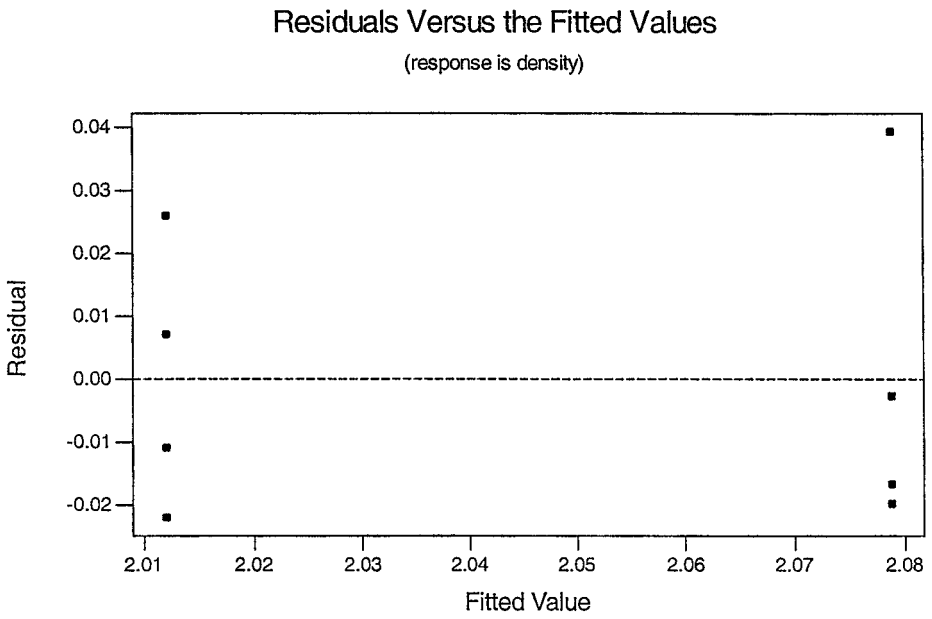
13–25. (a) $D = ABC$

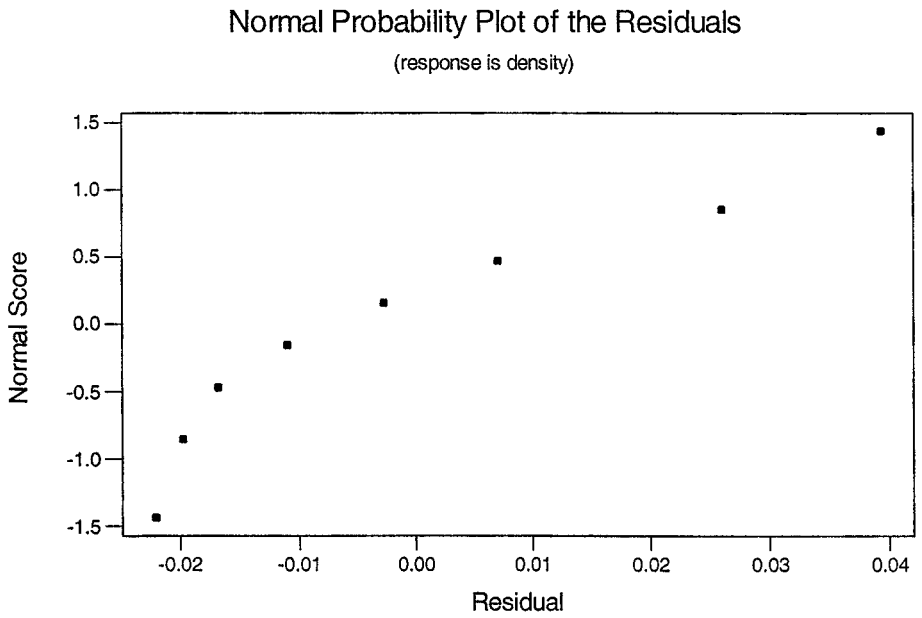
(b)

Term	Effect	Coef	SE Coef	T	P
Constant		2.04538	0.007423	275.55	0.000
A	0.06675	0.03338	0.007423	4.50	0.021
B	0.02625	0.01313	0.007423	1.77	0.175
C	0.02025	0.01012	0.007423	1.36	0.266
D	-0.00375	-0.00187	0.007423	-0.25	0.817

Only factor A appears to be significant.

(c)





13–26. 2^{3-1} with 2 replicates.

13–27. 2^{4-1} $I = ABCD$ Aliases:

$$\ell_A = A + BCD \qquad \ell_{AB} = AB + CD$$
$$\ell_B = B + ACD \qquad \ell_{AC} = AC + BD$$
$$\ell_C = C + ABD \qquad \ell_{AD} = AD + BC$$
$$\ell_D = D = ABC$$

	A	B	C	$D = ABC$	
(1)	$-$	$-$	$-$	$-$	190
a	$+$	$-$	$-$	$+$	174
b	$-$	$+$	$-$	$+$	181
ab	$+$	$+$	$-$	$-$	183
c	$-$	$-$	$+$	$+$	177
ac	$+$	$-$	$+$	$-$	181
bc	$-$	$+$	$+$	$-$	188
abc	$+$	$+$	$+$	$+$	173

$A = -6.25$ $\quad \ell_{AB} = -0.25$

$B = 0.75$ $\quad \ell_{AC} = 0.75$

$C = -2.25$ $\quad \ell_{AD} = 0.75$

$D = -9.25$

Sweetener (A) & Temperature (D) influence taste.

13–28. 2^{5-1} $\quad I = ABCDE$

A	B	C	D	$E = ABCD$	Treatment Combination	Strength
$-$	$-$	$-$	$-$	$+$	e	800
$+$	$-$	$-$	$-$	$-$	a	900
$-$	$+$	$-$	$-$	$-$	b	3400
$+$	$+$	$-$	$-$	$+$	abe	6200
$-$	$-$	$+$	$-$	$-$	c	600
$+$	$-$	$+$	$-$	$+$	ace	1200
$-$	$+$	$+$	$-$	$+$	bce	3006
$+$	$+$	$+$	$-$	$-$	abc	5300
$-$	$-$	$-$	$+$	$-$	d	1000
$+$	$-$	$-$	$+$	$+$	ade	1500
$-$	$+$	$-$	$+$	$+$	bde	4000
$+$	$+$	$-$	$+$	$-$	abd	6100
$-$	$-$	$+$	$+$	$+$	cde	1500
$+$	$-$	$+$	$+$	$-$	acd	1100
$-$	$+$	$+$	$+$	$-$	bcd	3300
$+$	$+$	$+$	$+$	$+$	$abcde$	6300

$\ell_A = 10{,}994$ $\quad \ell_{AB} = 9394$

$\ell_B = 29{,}006$

$\ell_C = -1594$

$\ell_D = 3394$

$\ell_E = 2806$

The estimates for A, B, and AB are large

13–29. 2^{5-2}

$$I = ABCD,\ I = ACE$$

A	B	C	$D = ABC$	$E = AC$	Treatment Combination	Strength
$-$	$-$	$-$	$-$	$+$	e	800
$+$	$-$	$-$	$+$	$-$	ade	1500
$-$	$+$	$-$	$+$	$+$	bde	4000
$+$	$+$	$-$	$-$	$-$	abe	6200
$-$	$-$	$+$	$+$	$-$	cde	1500
$+$	$-$	$+$	$-$	$+$	ace	1200
$-$	$+$	$+$	$-$	$-$	bce	3006
$+$	$+$	$+$	$+$	$+$	$abcde$	6300

$\ell_A = 5894$ $\quad \ell_C = -494$ $\quad \ell_{AB} = 8094$

$\ell_B = 14506$ $\quad \ell_D = 2094$

$\ell_E = 94$

13–30. 2^{6-3}_{III}

$$I = ABD = ACE = BCF = BCDE = ABEF = ACDF = DEF$$

$$\begin{aligned}
A &= BD = CE = ABCF = ABCDE = BEF = CDF = ADEF \\
B &= AD = CE = ABCF = CDE = AEF = ABCEF = BDEF \\
C &= ABCD = AE = BD = BDE = ABCEF = ADF = CDEF \\
D &= AB = ACDE = BCDF = BCE = ABDEF = ACF = EF \\
E &= ABDE = AC = BCEF = BCD = ABF = ACDEF = DF \\
F &= ABDF = ACEF = BC = BCDEF = ABE = ACD = DE
\end{aligned}$$

Chapter 14

14–1. (a) $\hat{y} = 10.5 - 0.00156$ yards

(b)

Predictor	Coef	SE Coef	T	P
Constant	10.451	2.514	4.16	0.000
yards	-0.001565	0.001090	-1.43	0.163

S = 3.414 R-Sq = 7.3% R-Sq(adj) = 3.8%

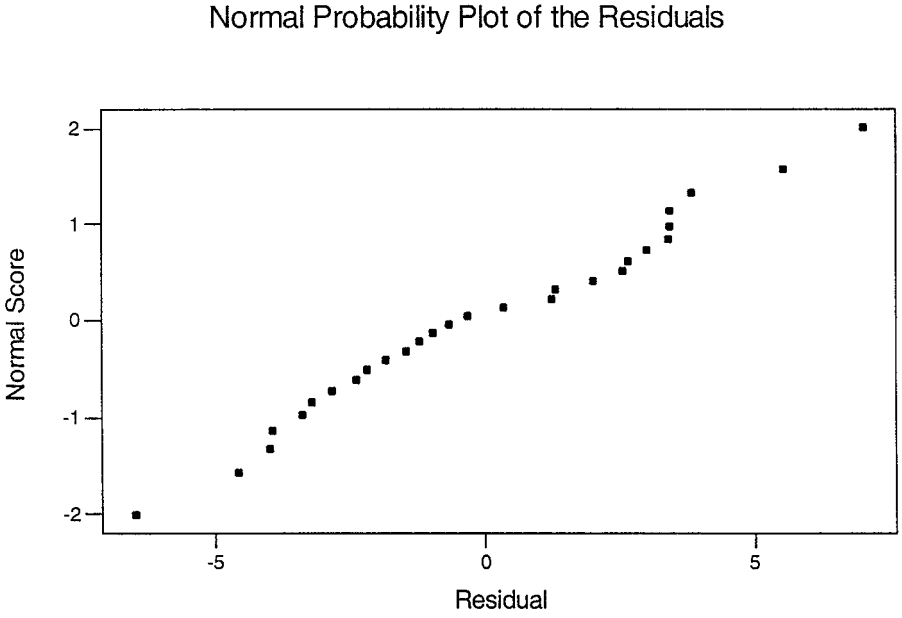
Analysis of Variance

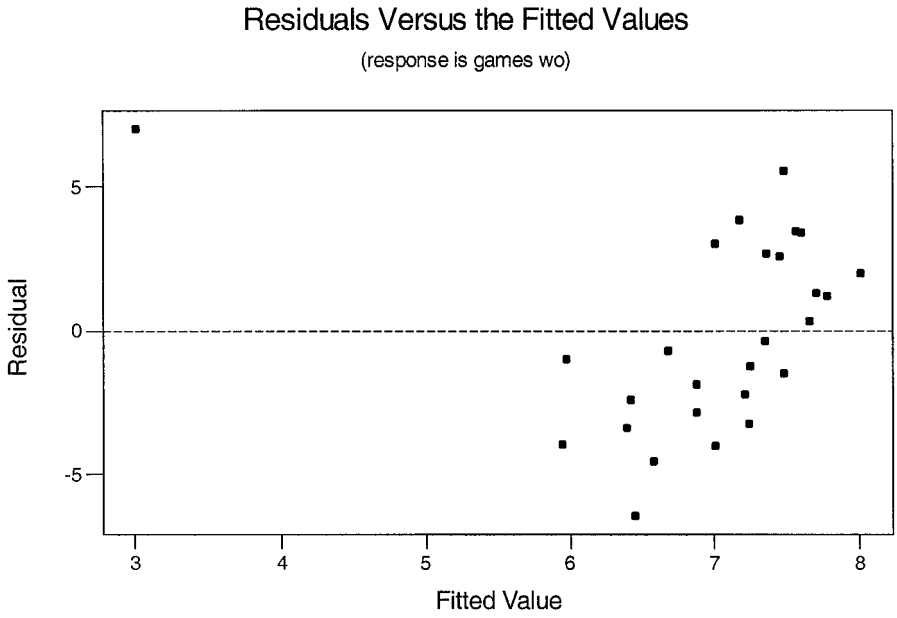
Source	DF	SS	MS	F	P
Regression	1	23.99	23.99	2.06	0.163
Residual Error	26	302.97	11.65		
Total	27	326.96			

(c) $-0.00380 \leq \beta_1 \leq 0.00068$

(d) $R^2 = 7.3\%$

(e)





Based on the residual plots there appears to be a severe outlier. This point should be investigated and if necessary, the point removed and the analysis rerun.

14–2. $\hat{y} = 10.5 - 0.00156(1800) = 7.63$ or approximately 8.
$6 \leq E(y|x_0 = 1800) \leq 9.27$

14–3. (a) $\hat{y} = 31.6 - 0.0410$ displacement

(b)

Predictor	Coef	SE Coef	T	P
Constant	31.648	1.812	17.46	0.000
displace	-0.040975	0.005406	-7.58	0.000

S = 1.976 R-Sq = 81.5% R-Sq(adj) = 80.1%$

Analysis of Variance

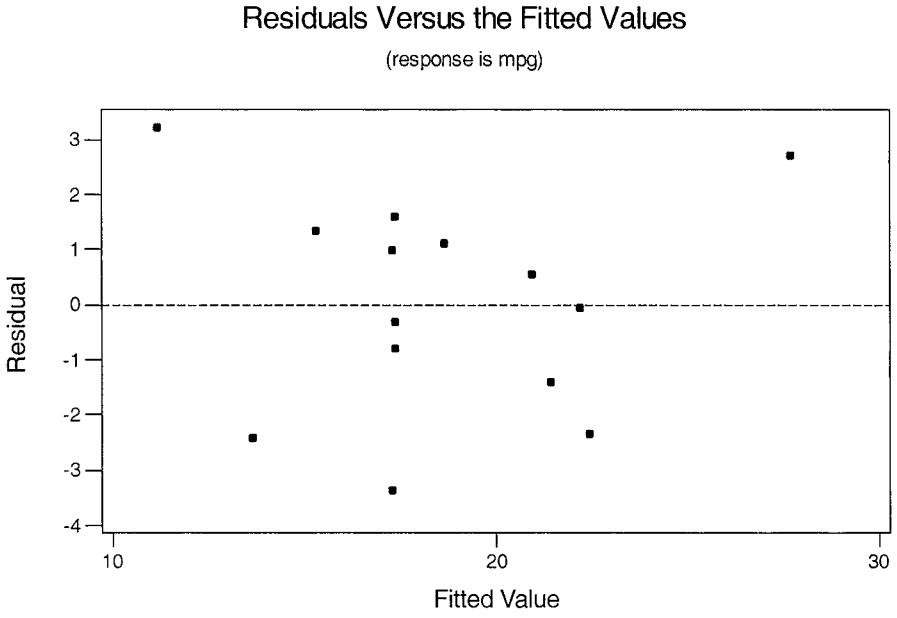
Source	DF	SS	MS	F	P
Regression	1	224.43	224.43	57.45	0.000
Residual Error	13	50.79	3.91		
Total	14	275.21			

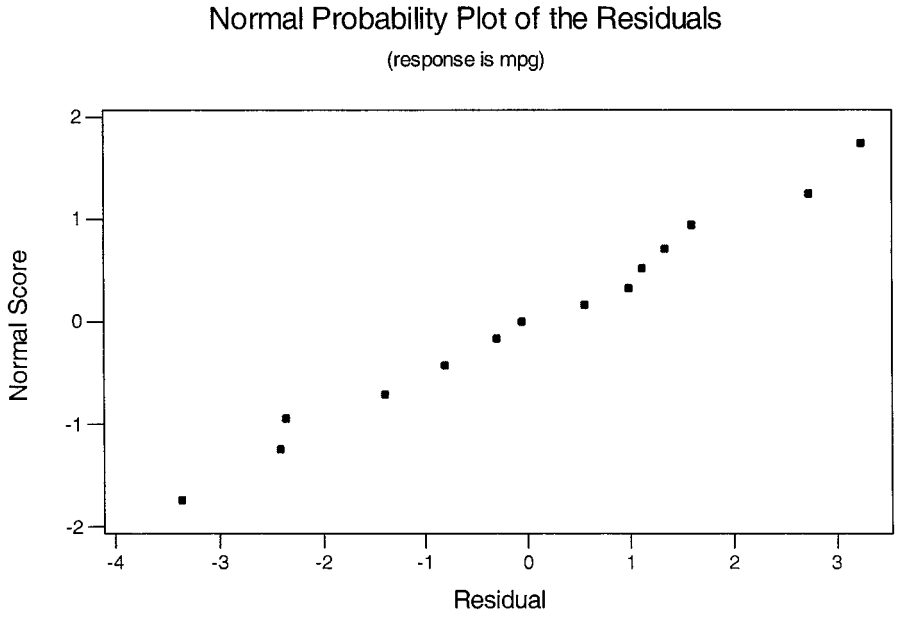
(c) $R^2 = 81.5\%$

(d) $\hat{y} = 20.381$, $19.374 \leq E(y|x_0 = 275) \leq 21.388$

14–4. $\hat{y} = 31.6 - 0.0410(275) = 20.381$, $10.37 \leq E(y|x_0 = 275) \leq 21.39$

14–5.





14–6. (a) $\hat{y} = 13.3 + 3.32$ taxes

(b)

Predictor	Coef	SE Coef	T	P
Constant	13.320	2.572	5.18	0.000
taxes	3.3244	0.3903	8.52	0.000

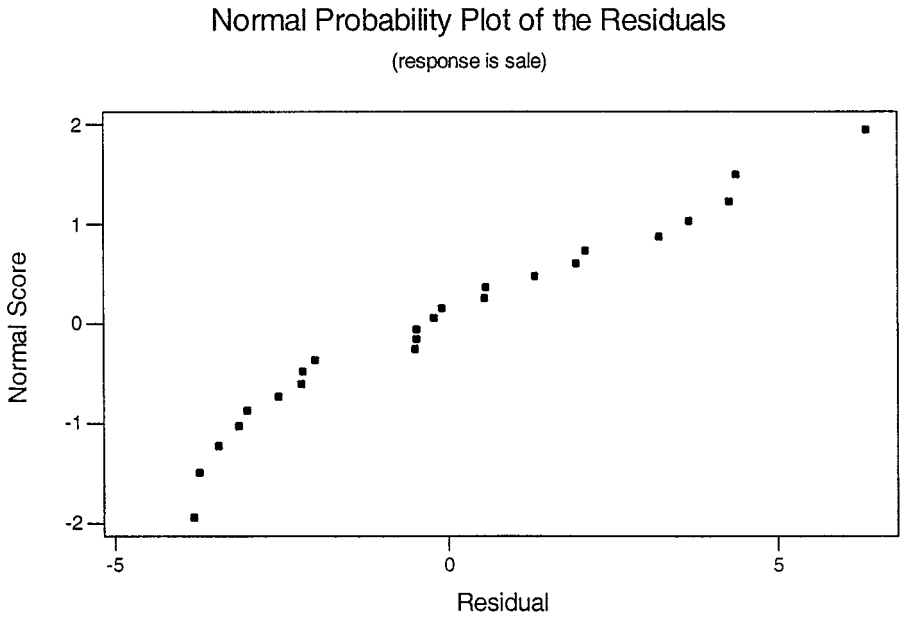
S = 2.961 R-Sq = 76.7% R-Sq(adj) = 75.7%

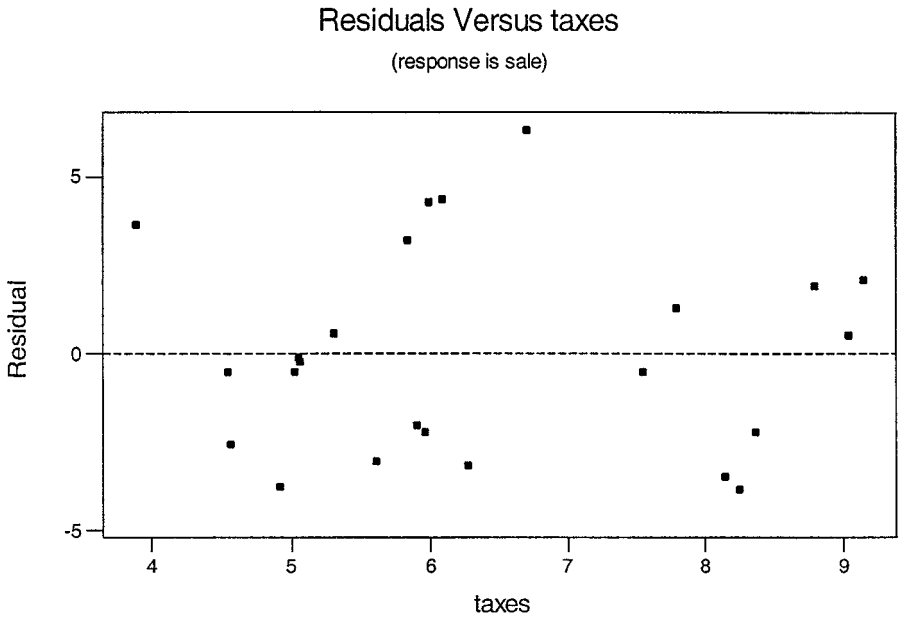
Analysis of Variance

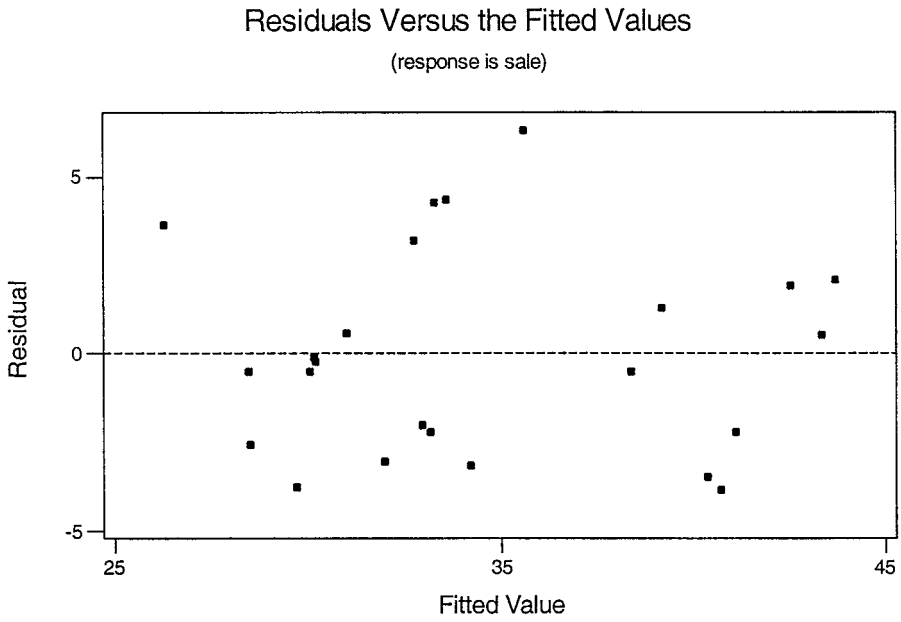
Source	DF	SS	MS	F	P
Regression	1	636.16	636.16	72.56	0.000
Residual Error	22	192.89	8.77		
Total	23	829.05			

(c) 76.7%

(d)




Residuals Versus taxes
(response is sale)
Residual
5
0
-5
4
5
6
7
8
9
taxes


Residuals Versus the Fitted Values
(response is sale)
Residual
5
0
-5
25
35
45
Fitted Value

14–7. (a) $\hat{y} = 93.3 + 15.6x$

(b)

Predictor	Coef	SE Coef	T	P
Constant	93.34	10.51	8.88	0.000
x	15.649	4.345	3.60	0.003

S = 11.63 R-Sq = 48.1% R-Sq(adj) = 44.4%

Analysis of Variance

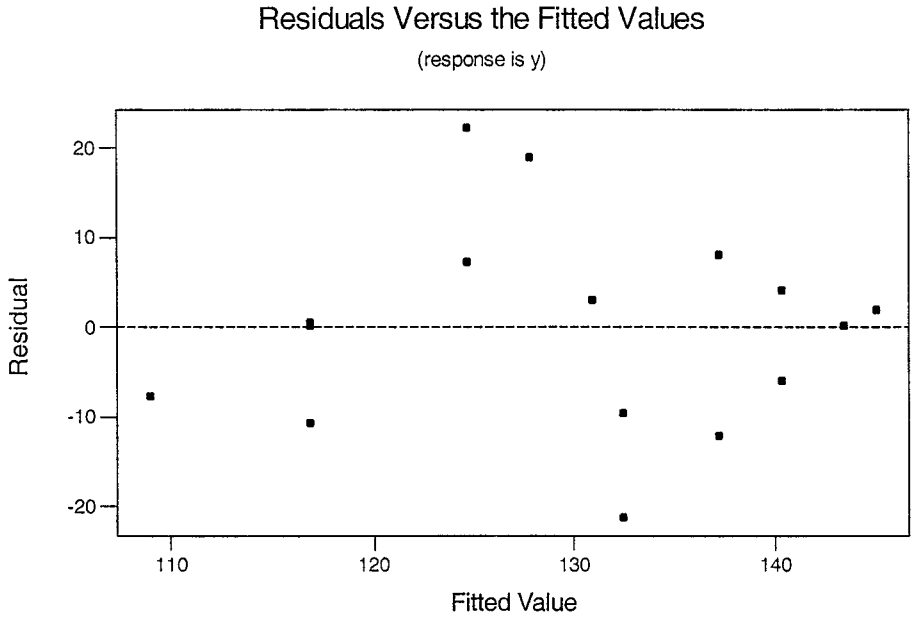
Source	DF	SS	MS	F	P
Regression	1	1755.8	1755.8	12.97	0.003
Residual Error	14	1895.0	135.4		
Lack of Fit	8	1378.6	172.3	2.00	0.207
Pure Error	6	516.4	86.1		
Total	15	3650.8			

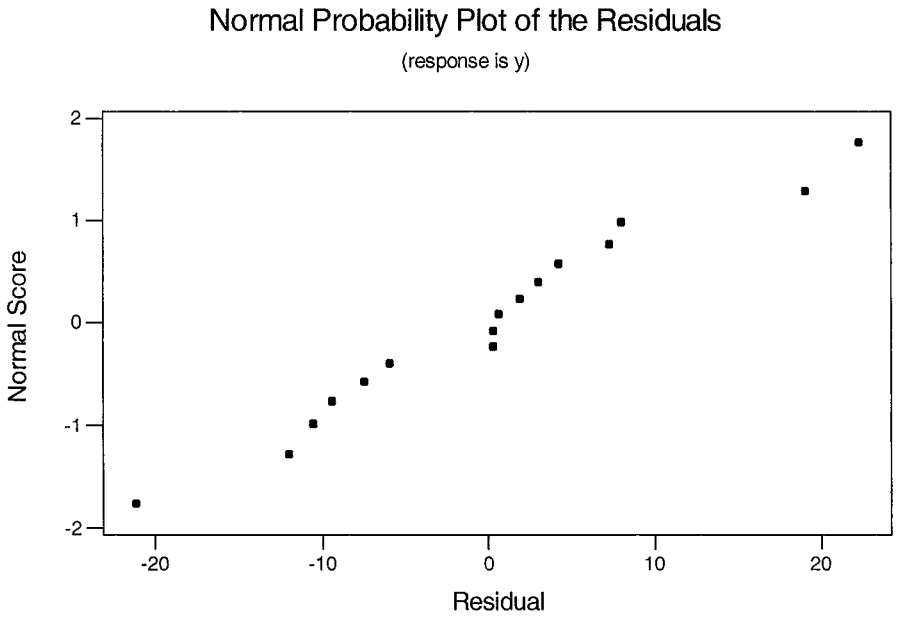
(c) $7.997 \leq \beta_1 \leq 23.299$

(d) $74.828 \leq \beta_0 \leq 111.852$

(e) $(126.012, 138.910)$

14–8.





14–9. (a) $\hat{y} = -6.34 + 9.21$ temp

(b)

Predictor	Coef	SE Coef	T	P
Constant	-6.336	1.668	-3.80	0.003
temp	9.20836	0.03377	272.64	0.000

S = 1.943 R-Sq = 100.0% R-Sq(adj) = 100.0%

Analysis of Variance

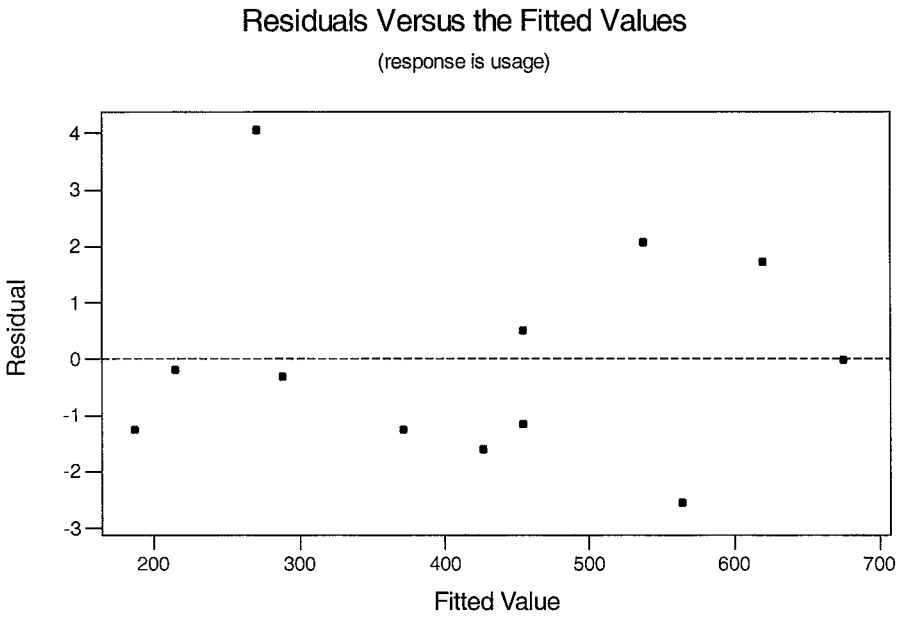
Source	DF	SS	MS	F	P
Regression	1	280583	280583	74334.36	0.000
Residual Error	10	38	4		
Total	11	280621			

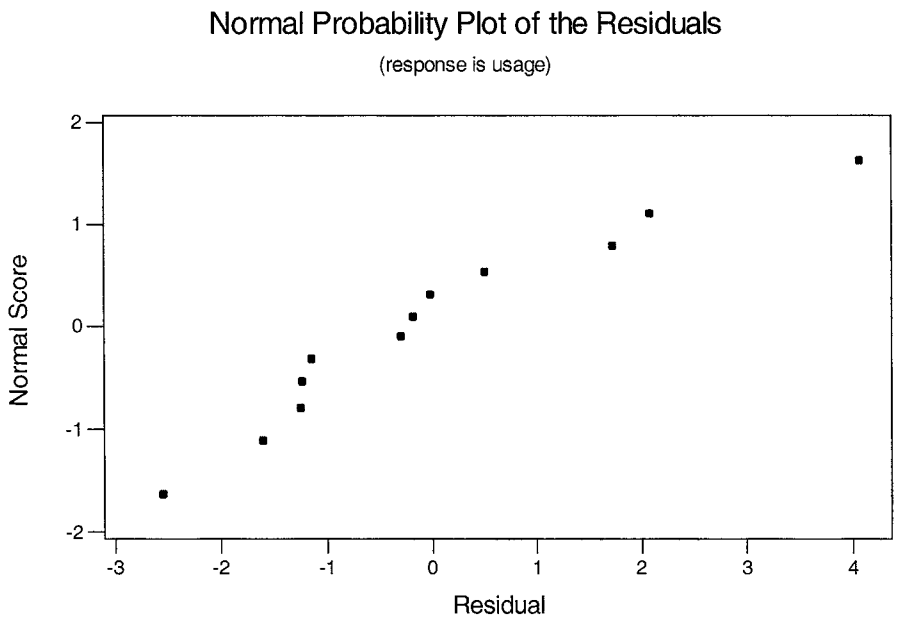
(c) $t_0 = \dfrac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\dfrac{MS_E}{S_{XX}}}} = \dfrac{9.20836 - 10}{\sqrt{\dfrac{4}{3309}}} = -23.41$; $t_{0.025,10} = 2.228$; Reject $\beta_1 = 0$

(d) $525.58 \leq E(y|X = 50) \leq 529.91$

(e) $521.22 \leq y_{x=58} \leq 534.28$

14–10.


Residuals Versus the Fitted Values
(response is usage)
Residual
Fitted Value
4
3
2
1
0
-1
-2
-3
200
300
400
500
600
700


Normal Probability Plot of the Residuals
(response is usage)
Normal Score
Residual
2
1
0
-1
-2
-3
-2
-1
0
1
2
3
4

14–11. (a) $\hat{y} = 77.7895 + 11.8634x$

(b)

Predictor	Coef	SE Coef	T	P
Constant	77.863	4.199	18.54	0.000
hydrocar	11.801	3.485	3.39	0.003

S = 3.597 R-Sq = 38.9% R-Sq(adj) = 35.5%

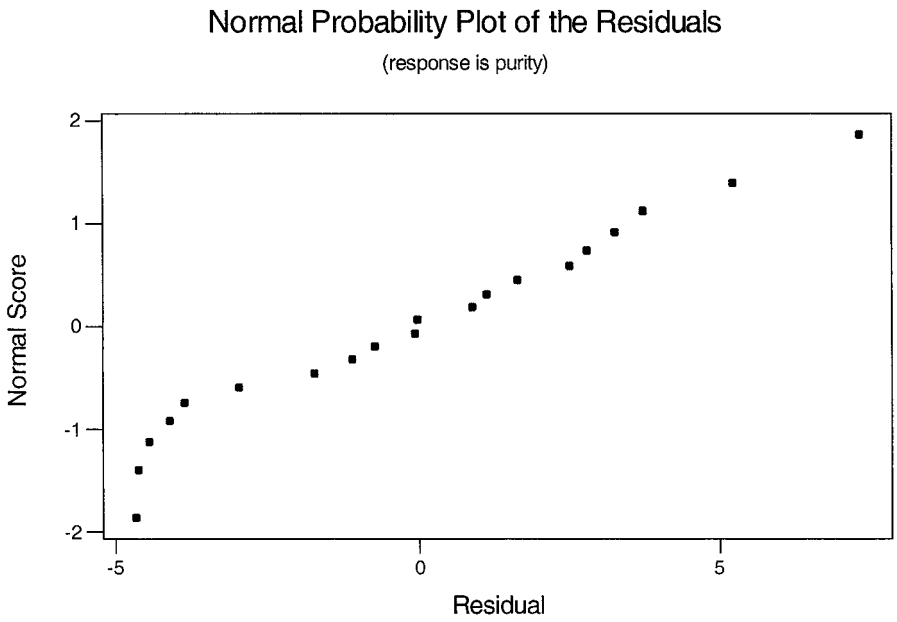
Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	148.31	148.31	11.47	0.003
Residual Error	18	232.83	12.94		

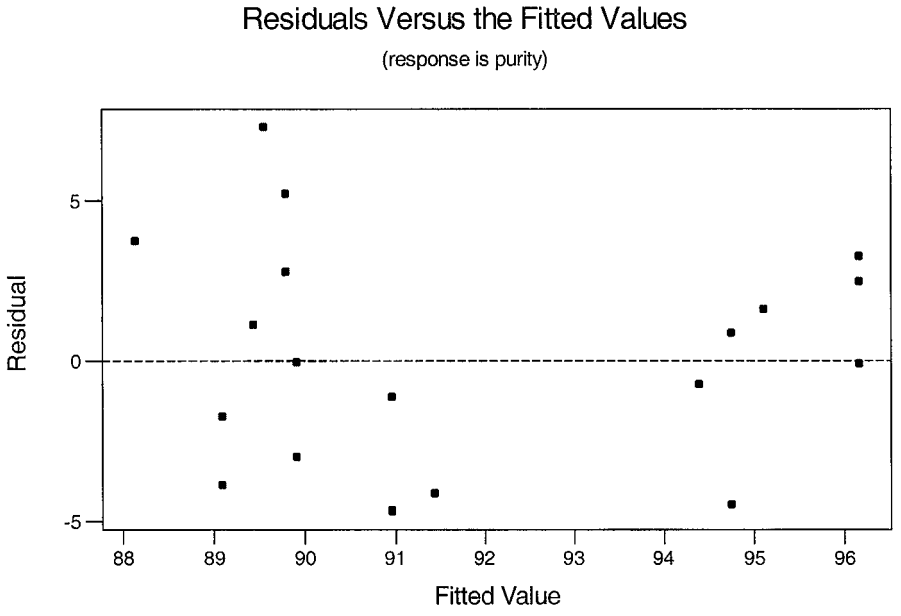
(c) 38.9%

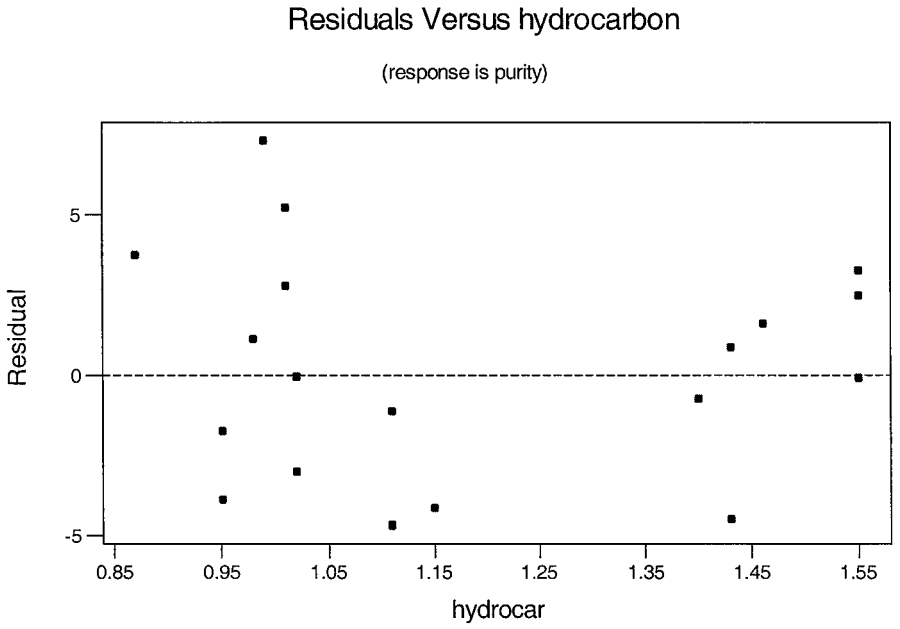
(d) $4.5661 \leq \beta \leq 19.1607$

14–12. (a)



(b)


Residuals Versus the Fitted Values
(response is purity)
Residual
5
0
-5
88
89
90
91
92
93
94
95
96
Fitted Value


Residuals Versus hydrocarbon
(response is purity)
Residual
5
0
-5
0.85
0.95
1.05
1.15
1.25
1.35
1.45
1.55
hydrocar

14–13. (a) $AvgSize = -1922.7 + 564.54$ Level

(b)

Predictor	Coef	SE Coef	T	P
Constant	-1922.7	530.9	-3.62	0.003
Level	564.54	32.74	17.24	0.000

S = 459.0 R-Sq = 95.5% R-Sq(adj) = 95.2%

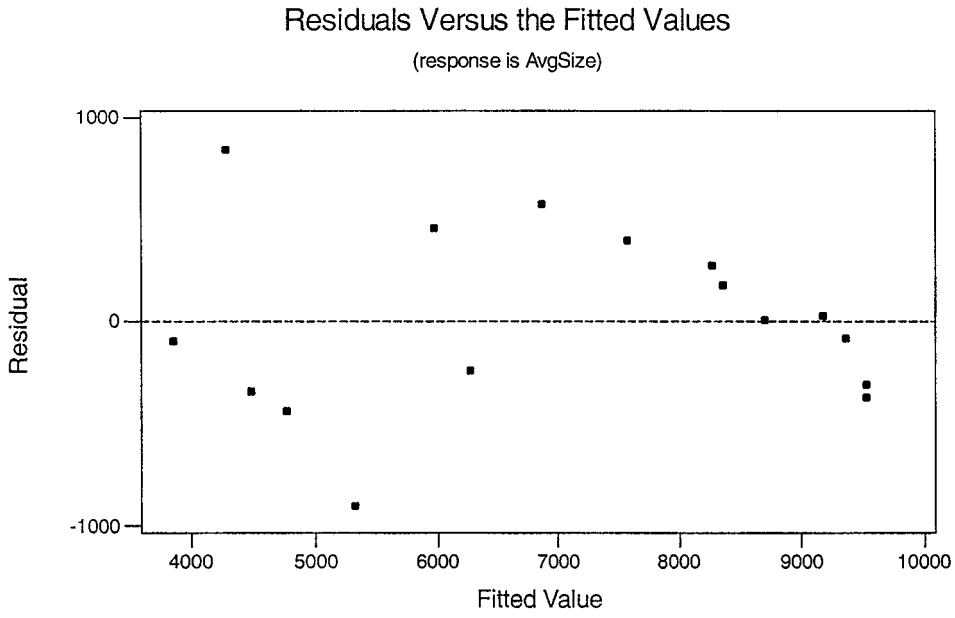
Analysis of Variance

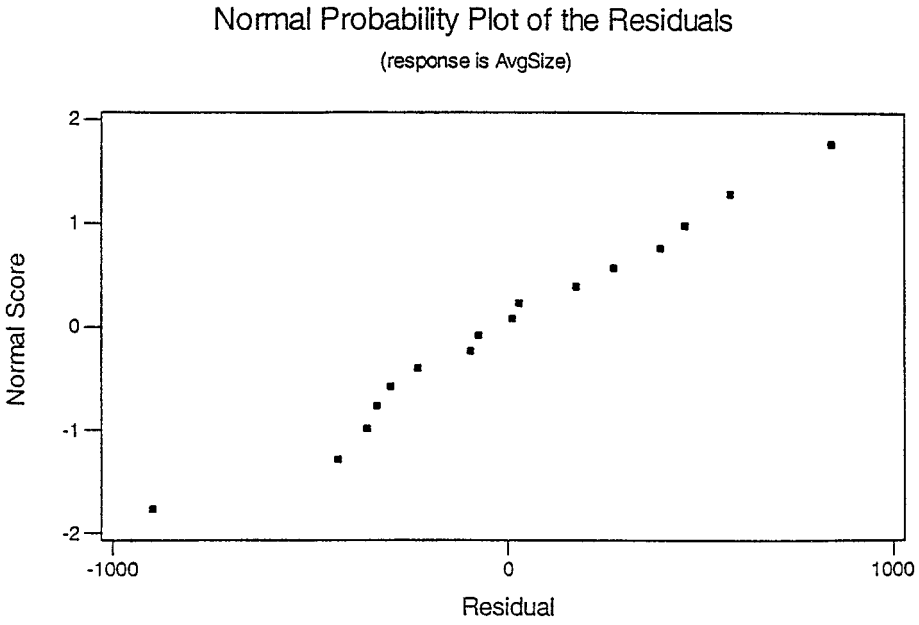
Source	DF	SS	MS	F	P
Regression	1	62660784	62660784	297.38	0.000
Residual Error	14	2949914	210708		
Total	15	65610698			

(c) $(0.0015, 0.0019)$

(d) 95.5%

(e)





14–14. x = OR Grade, y = Statistics Grade

(a) $\hat{y} = -0.0280 + 0.9910x$

(b) $r = 0.9033$

(c) $t_0 = \dfrac{r\sqrt{n-2}}{r\sqrt{1-r^2}} = \dfrac{0.9033\sqrt{18}}{\sqrt{1-0.8160}} = 8.93$, reject H_0.

(d) $Z_0 = (\text{arctanh}(0.9033) - \text{arctanh}(0.5))\sqrt{17} = 3.88$, reject H_0.

(e) $0.7676 \leq \rho \leq 0.9615$

14–15. x = weight, y = BP

(a) $\hat{y} = 69.1044 + 0.4194x$

(b) $r = 0.7735$

(c) $t_0 = \dfrac{r\sqrt{n-3}}{\sqrt{1-r^2}} = \dfrac{0.7735\sqrt{23}}{\sqrt{1-0.5983}} = 5.85$, reject H_0.

(d) $Z_0 = (\text{arctanh}(0.7735) - \text{arctanh}(0.6))\sqrt{23} = 1.61$, do not reject

(e) $0.5513 \leq \rho \leq 0.8932$

14–16. Note that $SS_R = \hat{\beta}_1 S_{xy} = \hat{\beta}_1^2 S_{xx}$, and

$$V(\hat{\beta}_1) = E(\hat{\beta}_1^2) - [E(\hat{\beta}_1)]^2 = \frac{\sigma^2}{S_{xx}}$$

$$E(\hat{\beta}_1^2) = \beta_1^2 + \frac{\sigma^2}{S_{xx}}$$

Therefore

$$E(SS_R) = E(\hat{\beta}_1^2) S_{xx} = \sigma^2 + \beta_1^2 S_{xx}$$

$$E(MS_R) = E\left(\frac{SS_R}{1}\right) = \sigma^2 + \beta_1^2 S_{xx}$$

14–17. $E(\hat{\beta}_1) = E\left(\dfrac{S_{xy}}{S_{xx}}\right) = \dfrac{1}{S_{xx}} E(S_{xy})$

$$= \frac{1}{S_{xx}} E \sum_{i=1}^{n} (x_{i1} - \overline{x}_1) y_i$$

$$= \frac{1}{S_{xx}} \sum_{i=1}^{n} (x_{i1} - \overline{x}_1) E(y_i)$$

$$= \frac{1}{S_{xx}} \sum_{i=1}^{n} (x_{i1} - \overline{x}_1)(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})$$

$$= \beta_1 + \beta_2 \frac{\displaystyle\sum_{i=1}^{n} x_{i2}(x_{i1} - \overline{x}_1)}{S_{xx}}$$

In general, $\hat{\beta}_1$ is a biased estimator of β_1.

14–18. $V(\hat{\beta}_1) = \sigma^2 / S_{xx}$, which is minimized if we can make S_{xx} is as large as possible. Since $\sum_{i=1}^{n} (x_i - \overline{x})^2 = S_{xx}$, place the x's as far from $\overline{x}$ as possible. If n is even, put $n/2$ trials at each end of the region of interest. If n is odd, put 1 trial at the center and $(n-1)/2$ trials at each end. These designs should be used only when you are positive that the relationship between y and x is linear.

14–19. $$L = \sum_{i=1}^{n} w_i(y_i - \beta_0 - \beta_1 x_i)^2$$

$$\frac{\partial L}{\partial \beta_0} = -2\sum_{i=1}^{n} w_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

$$\frac{\partial L}{\partial \beta_1} = -2\sum_{i=1}^{n} w_i x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

Simplification of these two equations gives the normal equations for weighted least squares.

14–20. If $y = (\beta_0 + \beta_1 x + \epsilon)^{-1}$, then $1/y = \beta_0 + \beta_1 x + \epsilon$ is a straight-line regression model. The scatter diagram of $y^* = 1/y$ versus x is linear.

x	10	15	18	12	9	8	11	6
y^*	5.88	7.69	11.11	6.67	5.00	4.76	5.56	4.17

14–21. The no-intercept model would be the form $y = \beta_1 x + \epsilon$. This model is likely not a good one for this problem, because there is no data near the point $x = 0$, $y = 0$, and it is probably unwise to extrapolate the linear relationship back to the origin. The intercept is often just a parameter that improves the fit of the model to the data in the region where the data were collected.

14–22. $b = 14$ $\quad \Sigma x_i = 65.262$ $\quad \Sigma x_i^2 = 385.194$ $\quad \overline{x} = 4.662$

$\Sigma y_i = 208$ $\quad \Sigma y_i^2 = 3490$ $\quad \overline{y} = 14.857$ $\quad \Sigma x_i y_i = 1148.08$

$S_{xx} = 80.989$ $\quad S_{xy} = 178.473$ $\quad S_{yy} = 599.714$

$$\hat{\beta}_1 = 2.204$$

$$\hat{\beta}_0 = 4.582$$

$$\hat{y} = 4.582 + 2.204x$$

$$r = \frac{S_{xy}}{(S_{xx}S_{yy})^{1/2}} = 0.9919, \quad R^2 = 0.9839$$

H_0: $\rho = 0$ $\quad t_0 = 27.08 > t_{0.05,12} = 1.782.$ $\quad \therefore$ reject H_0

H_1: $\rho \neq 0$

A strong correlation does not imply a cause and effect relationship.

Chapter 15

15–1. (a) $\hat{y} = 7.30 + 0.0183x_1 - 0.399x_4$

(b)

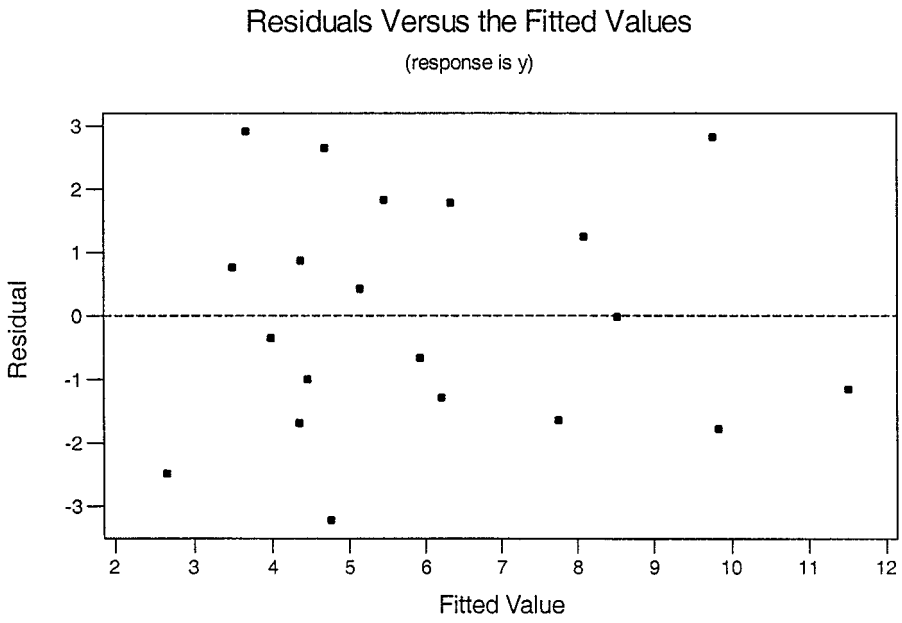
Predictor	Coef	SE Coef	T	P
Constant	7.304	5.179	1.41	0.176
x1	0.018299	0.004972	3.68	0.002
x4	-0.3986	0.1912	-2.08	0.053

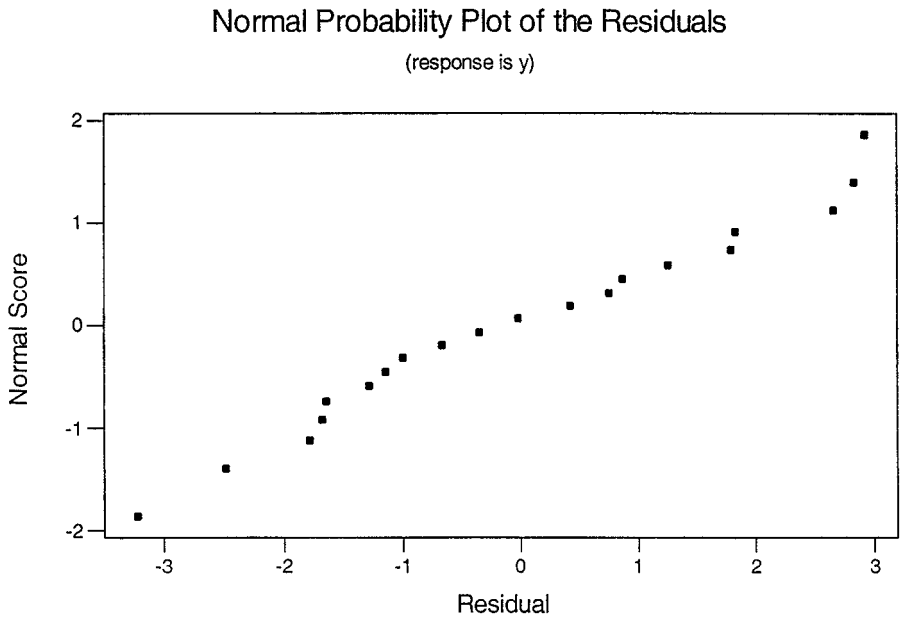
S = 1.922 R-Sq = 64.1% R-Sq(adj) = 59.9%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	112.263	56.131	15.19	0.000
Residual Error	17	62.824	3.696		
Total	19	175.086			

(c)





(d) The MS_E has improved with the x_1x_4 model, but the R^2 and adjusted R^2 have decreased.

15–2. (a) $\hat{y} = -27.9 + 0.0136x_1 + 30.7x_2 - 0.0670x_3$

(b)

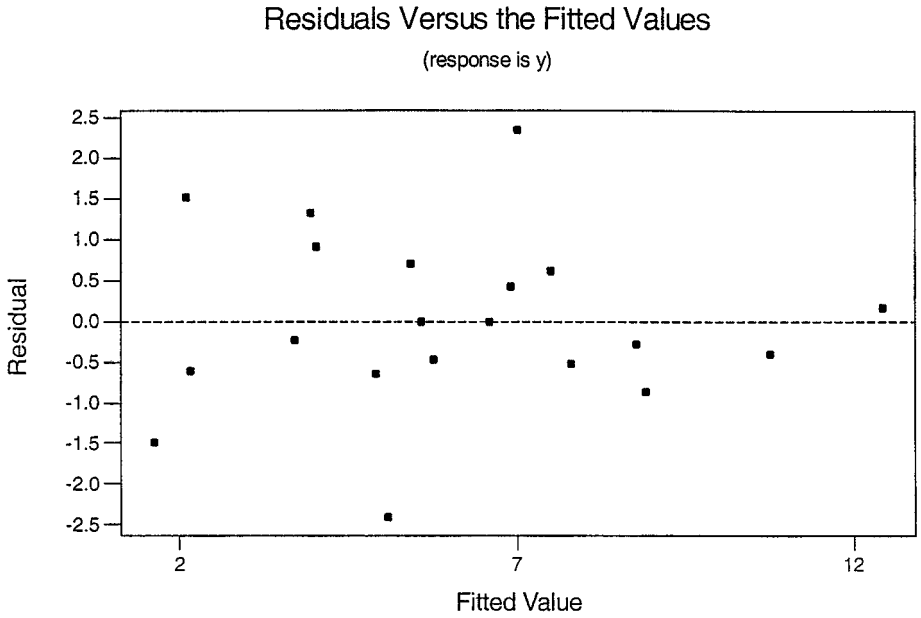
Predictor	Coef	SE Coef	T	P
Constant	-27.892	6.035	-4.62	0.000
x1	0.013597	0.003247	4.19	0.001
x2	30.685	6.513	4.71	0.000
x3	-0.06701	0.01916	-3.50	0.003

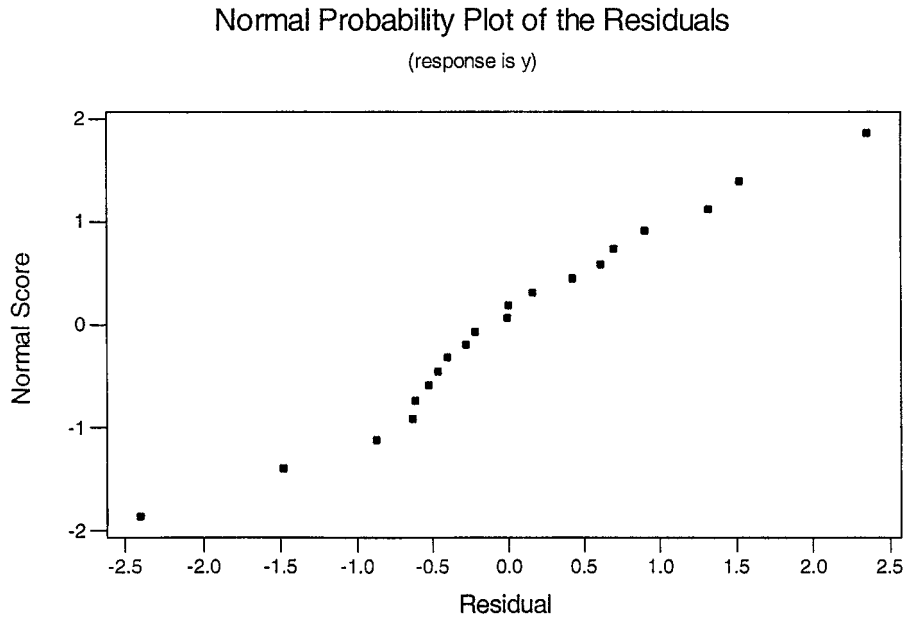
S = 1.167 R-Sq = 87.6% R-Sq(adj) = 85.2%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	3	153.305	51.102	37.54	0.000
Residual Error	16	21.782	1.361		
Total	19	175.086			

(c)



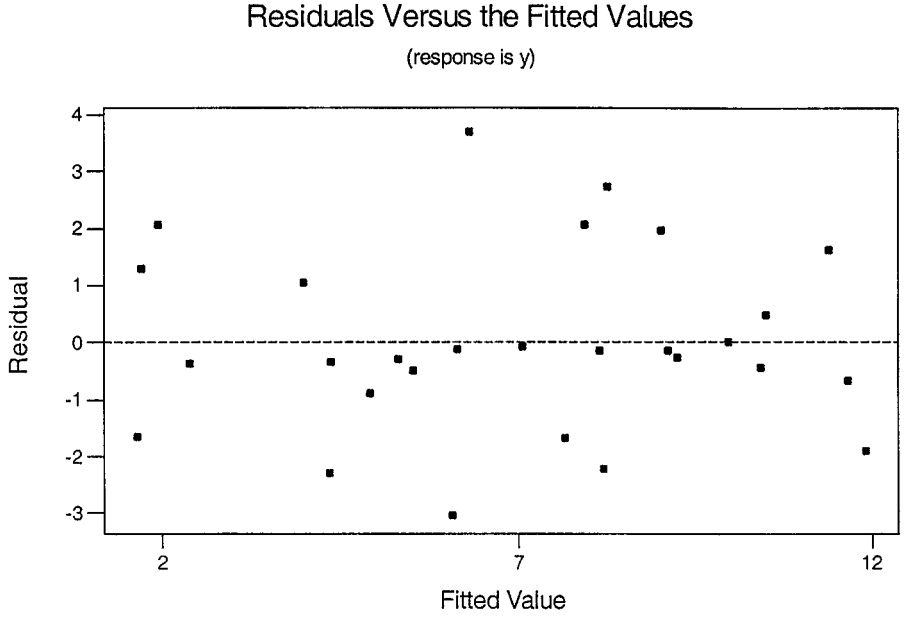


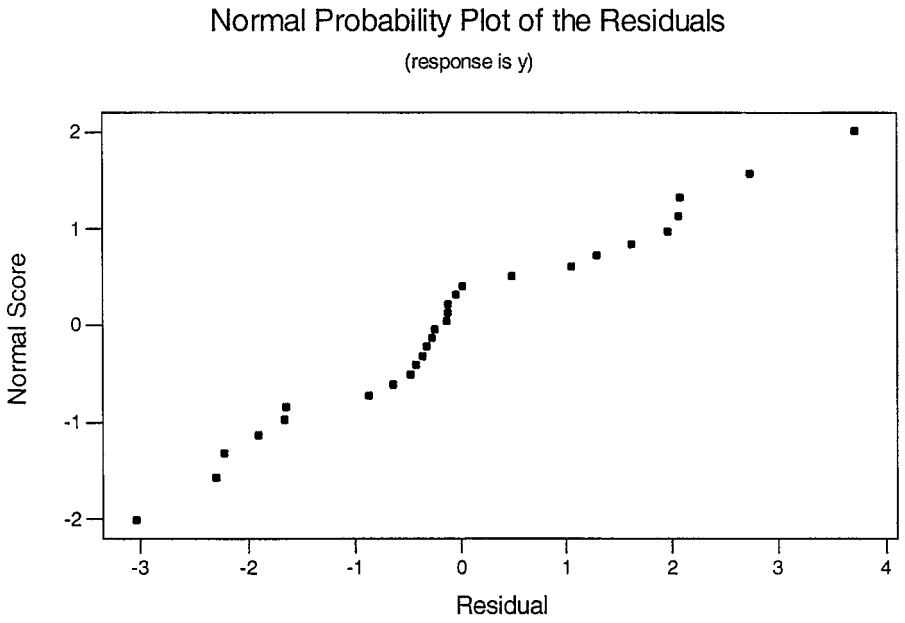
15–3. $(-0.8024, 0.0044)$

15–4. $(-0.1076, -0.0264)$

15–5. (a) $\hat{y} = -1.81 + 0.00360x_2 + 0.194x_7 - 0.00482x_8$

(b)





(c)

Predictor	Coef	SE Coef	T	P
Constant	-1.808	7.901	-0.23	0.821
x2	0.0035981	0.0006950	5.18	0.000
x7	0.19396	0.08823	2.20	0.038
x8	-0.004815	0.001277	-3.77	0.001

S = 1.706 R-Sq = 78.6% R-Sq(adj) = 76.0%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	3	257.094	85.698	29.44	0.000
Residual Error	24	69.870	2.911		
Total	27	326.964			

15–6. (a) $\hat{y} = 33.45 - 0.05435x_1 + 1.0782x_6$

(b)

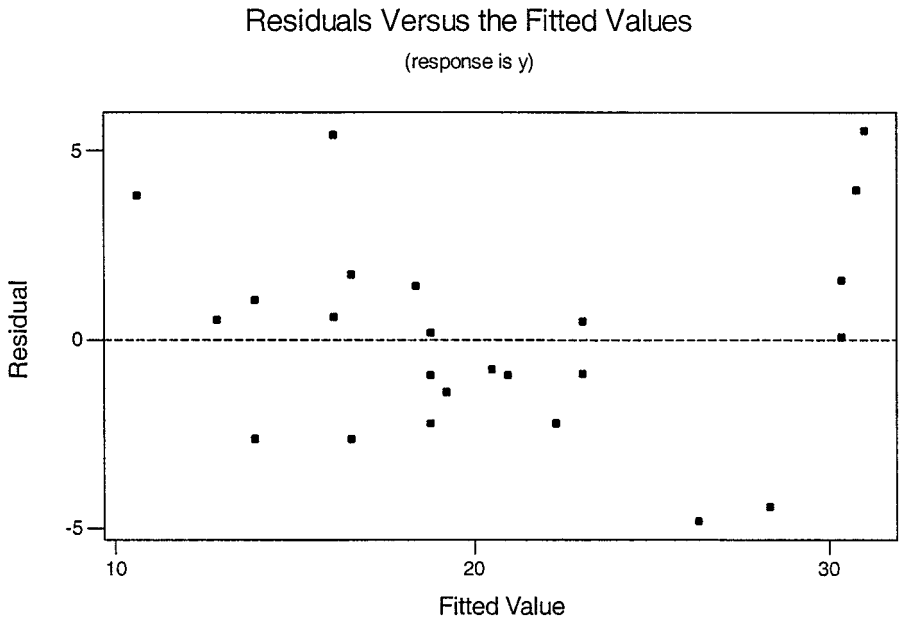
Predictor	Coef	SE Coef	T	P
Constant	33.449	1.576	21.22	0.000
x1	-0.054349	0.006329	-8.59	0.000
x6	1.0782	0.6997	1.54	0.138

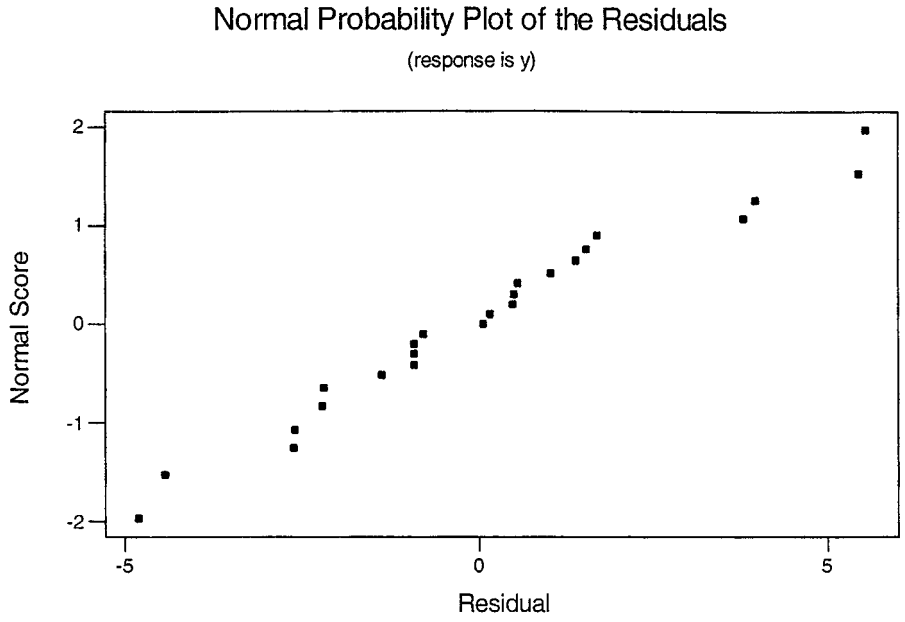
S = 2.834 R-Sq = 82.9% R-Sq(adj) = 81.3%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	856.24	428.12	53.32	0.000
Residual Error	22	176.66	8.03		
Total	24	1032.90			

(c)





(d) x_6 is not significant with x_1 included in the model.

15–7. (a) $\hat{y} = -103 + 0.605\,x_1 + 8.92\,x_2 + 1.44\,x_3 + 0.014\,x_4$

(b)

Predictor	Coef	SE Coef	T	P
Constant	-102.7	207.9	-0.49	0.636
x1	0.6054	0.3689	1.64	0.145
x2	8.924	5.301	1.68	0.136
x3	1.437	2.392	0.60	0.567
x4	0.0136	0.7338	0.02	0.986

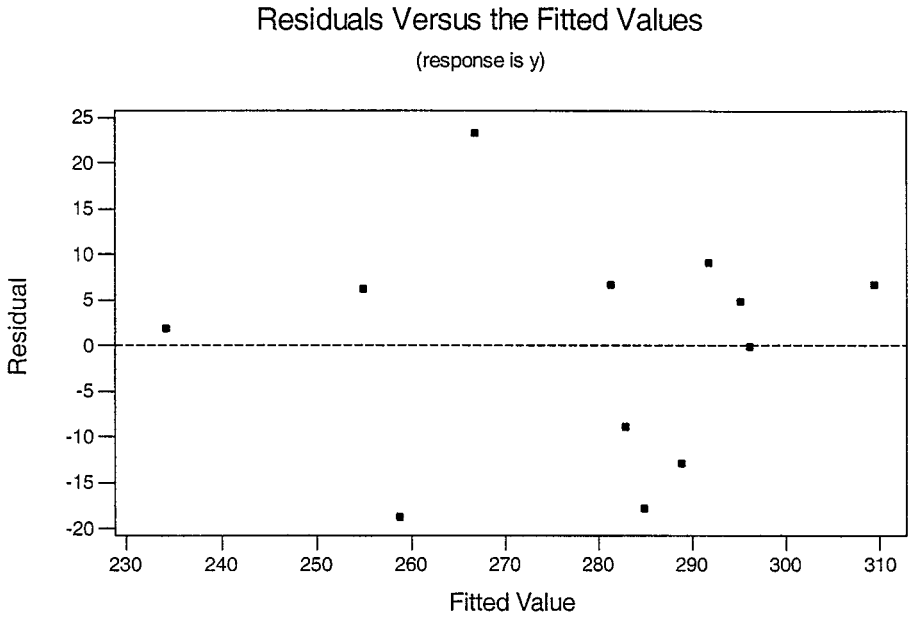
S = 15.58 R-Sq = 74.5% R-Sq(adj) = 59.9%

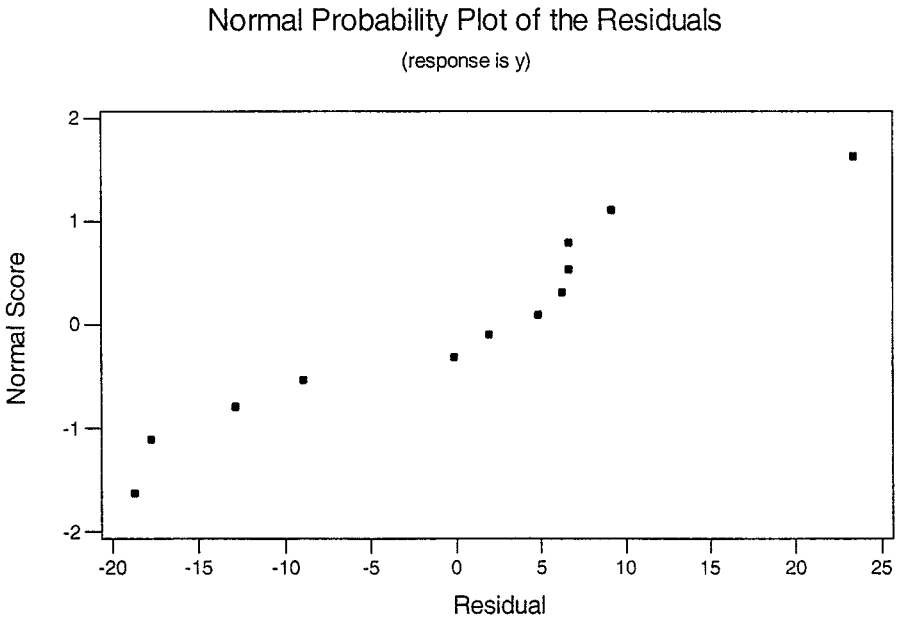
Analysis of Variance

Source	DF	SS	MS	F	P
Regression	4	4957.2	1239.3	5.11	0.030
Residual Error	7	1699.0	242.7		
Total	11	6656.3			

(c) H_0: $\beta_3 = 0, F_0 = 0.361$ (not significant)
H_0: $\beta_4 = 0, F_0 = 0.0004$ (not significant)

(d)





15–8. (a) $y = 62.4 + 1.55\,x_1 + 0.510\,x_2 + 0.102\,x_3 - 0.144\,x_4$

(b)

Predictor	Coef	SE Coef	T	P
Constant	62.41	70.07	0.89	0.399
x1	1.5511	0.7448	2.08	0.071
x2	0.5102	0.7238	0.70	0.501
x3	0.1019	0.7547	0.14	0.896
x4	-0.1441	0.7091	-0.20	0.844

S = 2.446 R-Sq = 98.2% R-Sq(adj) = 97.4%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	4	2667.90	666.97	111.48	0.000
Residual Error	8	47.86	5.98		
Total	12	2715.76			

(c) H_0: $\beta_4 = 0$, $F_0 = 0.04$ (not significant)

(d) The t statistics are given in part (b)

(e) H_0: $\beta_2 = \beta_3 = \beta_4 = 0$,

$$\begin{aligned} SS_R(\beta_2, \beta_3, \beta_4 | \beta_1, \beta_0) &= SS_R(\beta_1, \beta_2, \beta_3, \beta_4 | \beta_0) - SS_R(\beta_1 | \beta_0) \\ &= 2667.90 - 1450.08 \\ &= 1217.82 \end{aligned}$$

$F_0 = (1217.82/3)/5.98 = 67.88$; at least one of the variables is significant.

(f) $(-1.1588, 2.1792)$

15–9. (a) $y = -26219 + 189x - 0.331x^2$

(b)

Predictor	Coef	SE Coef	T	P
Constant	-26219	11911	-2.20	0.079
x	189.20	80.24	2.36	0.065
x2	-0.3312	0.1350	-2.45	0.058

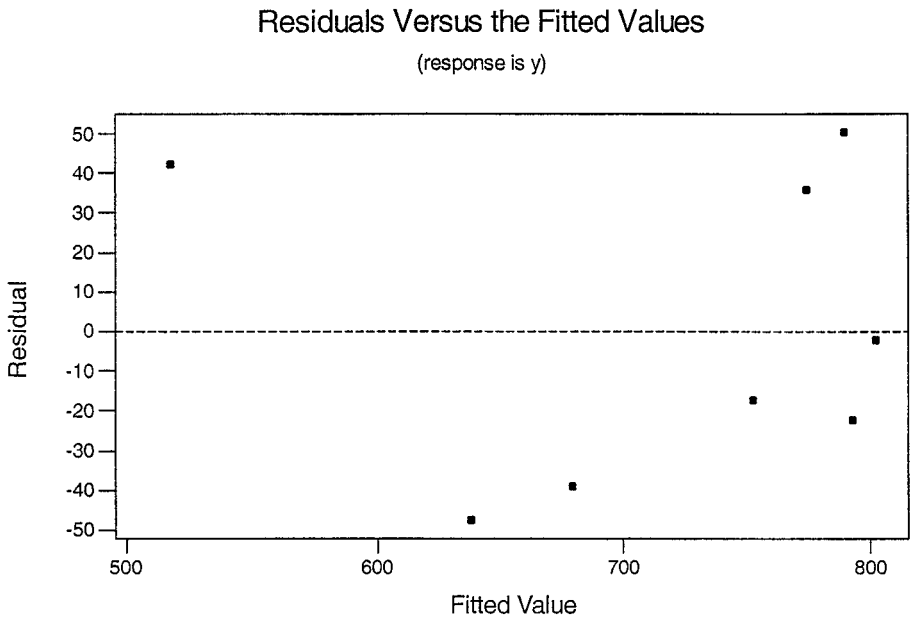
S = 45.20 R-Sq = 87.3% R-Sq(adj) = 82.2%

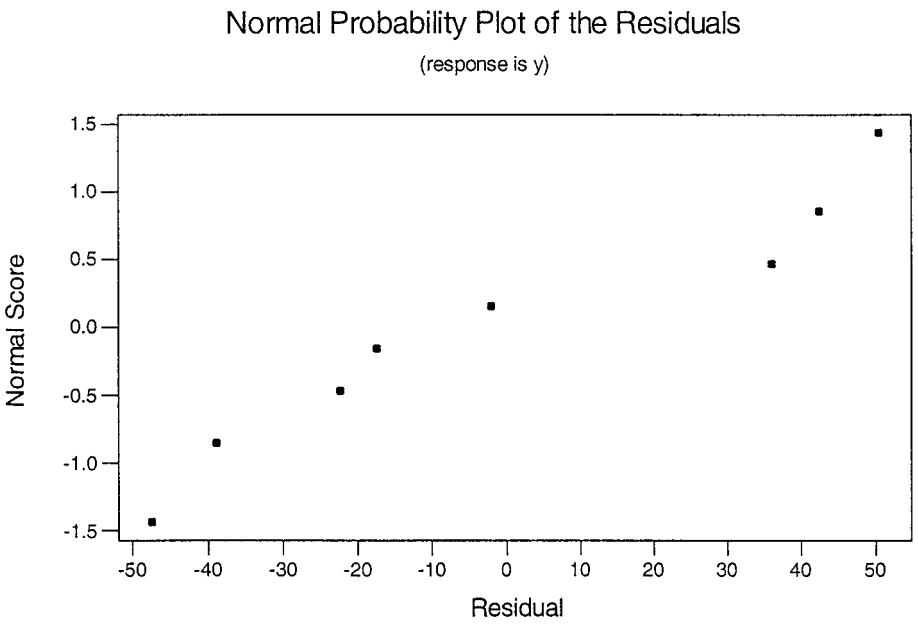
Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	70284	35142	17.20	0.006
Residual Error	5	10213	2043		
Total	7	80497			

(c) See the t-test results in part (b)

(d)





15–10. (a) $y = -4.33 + 4.89x - 2.59x^2$

(b)

Predictor	Coef	SE Coef	T	P
Constant	-4.3330	0.8253	-5.25	0.001
x	4.887	1.379	3.54	0.009
x2	-2.5855	0.4886	-5.29	0.001

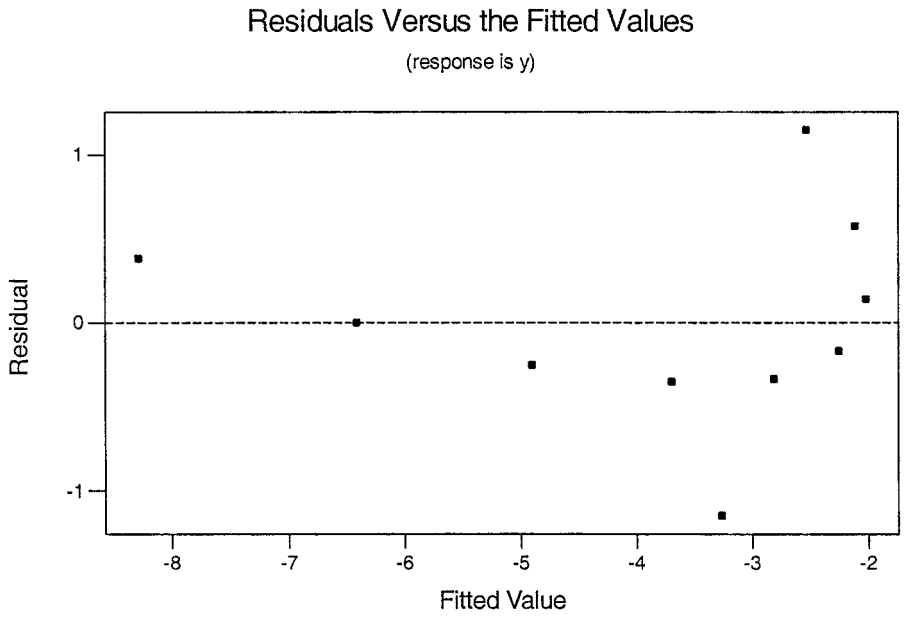
S = 0.7017 R-Sq = 91.9% R-Sq(adj) = 89.6%

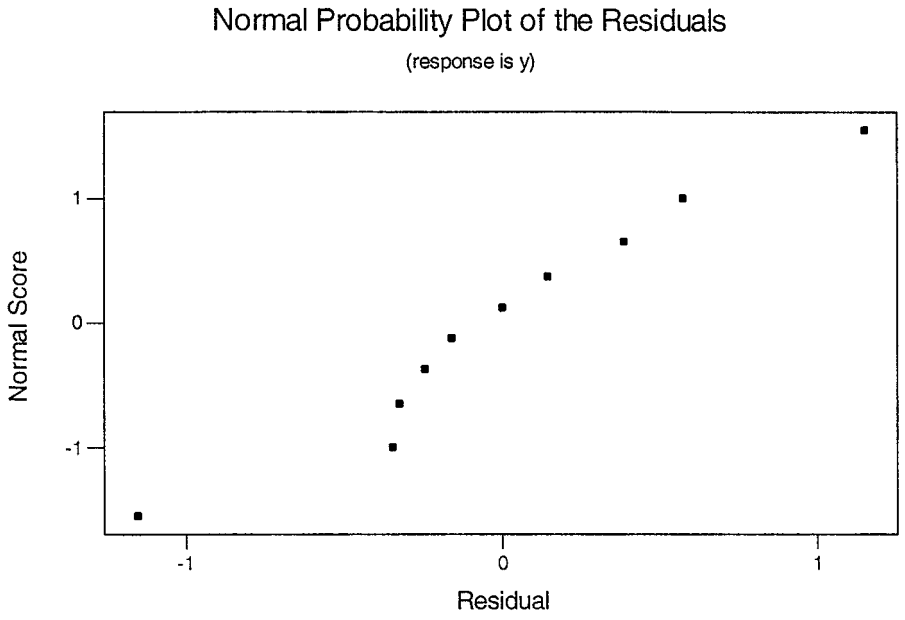
Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	39.274	19.637	39.89	0.000
Residual Error	7	3.446	0.492		
Total	9	42.720			

(c) see part (b)

(d)


Residuals Versus the Fitted Values
(response is y)
Residual
Fitted Value
1
0
-1
-8
-7
-6
-5
-4
-3
-2


Normal Probability Plot of the Residuals
(response is y)
Normal Score
Residual
1
0
-1
-1
0
1

15–11. $\hat{y} = 759.39 - 7.607x' - 0.331(x')^2$

15–13. (a) $y = -4.5 + 1.38x + 1.47x^2$

(b)

Predictor	Coef	SE Coef	T	P
Constant	-4.46	14.63	-0.30	0.768
x	1.384	5.497	0.25	0.807
x2	1.4670	0.4936	2.97	0.016

S = 1.657 R-Sq = 99.6% R-Sq(adj) = 99.5%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	5740.6	2870.3	1044.99	0.000
Residual Error	9	24.7	2.7		
Total	11	5765.3			

(c) see part (b)

15–15. The fitted model is $\hat{y} = 11.503 + 0.153x_1 - 0.094x_2 - 0.0306x_1x_2$
The t-statistic for H_0: $\beta_3 = 0$ is $t_0 = 1.79$. We conclude that the slopes are the same.

15–16. $y = \beta_0 + \beta_1 x_1 + \beta_2(x_1 - x^*)x_2 + \varepsilon$, where x_2 is an indicator variable with $x_2 = 0$ if $x_1 \leq x^*$ and $x_2 = 1$ if $x_1 > x^*$.

15–17. $y = \beta_0 + \beta_1 x_1 + \beta_2(x_1 - x^*)x_2 + \beta_3 + \varepsilon$, where x_2 and x_3 are indicator variables with $x_2 = x_3 = 0$ if $x_1 \leq x^*$ and $x_2 = x_3 = 1$ if $x_1 > x^*$. β_3 estimates the effect of the discontinuity.

15–18. The model is as in Exercise 15–16, except now X^* is unknown and must be estimated. This is a nonlinear regression problem. It could be solved by using one-dimensional or line search methods, which could be used to obtain the trial values of x^*.

15–19. $\hat{b}_1 = 0.594 \qquad \hat{b}_4 = -0.336$

15–20. $\hat{b}_1 = 0.441 \qquad \hat{b}_2 = 0.505 \qquad \hat{b}_3 = -0.315$

15–21. $VIF_1 = 1.2$, $VIF_4 = 1.2$

15–22. (a) All possible regressions from Minitab displaying the best two models for each combination of variables.

Vars	R-Sq	R-Sq(adj)	C-p	S	x 1	x 2	x 3	x 4	x 5	x 6	x 7	x 8	x 9
1	54.5	52.7	20.4	2.3929								X	
1	35.2	32.7	39.3	2.8548	X								
2	74.3	72.3	3.1	1.8324		X						X	
2	66.0	63.2	11.2	2.1097		X					X		
3	78.6	76.0	0.9	1.7062		X					X	X	
3	77.8	75.0	1.7	1.7410	X	X						X	
4	80.1	76.7	1.4	1.6812		X					X	X	X
4	79.5	75.9	2.0	1.7073	X	X						X	X
5	80.7	76.3	2.8	1.6941	X	X					X	X	X
5	80.7	76.3	2.9	1.6957		X		X			X	X	X
6	81.2	75.8	4.4	1.7118		X	X	X			X	X	X
6	81.1	75.6	4.5	1.7174	X	X		X			X	X	X
7	81.4	74.9	6.2	1.7442		X	X	X		X	X	X	X
7	81.3	74.8	6.2	1.7470	X	X		X		X	X	X	X
8	81.6	73.8	8.0	1.7814	X	X	X	X		X	X	X	X
8	81.4	73.6	8.2	1.7895		X	X	X	X	X	X	X	X
9	81.6	72.3	10.0	1.8302	X	X	X	X	X	X	X	X	X

(b) Stepwise regression from Minitab:

```
 Alpha-to-Enter: 0.15  Alpha-to-Remove: 0.15
Response is    y     on  9 predictors, with N =   28

    Step         1        2        3
Constant    21.788   14.713   -1.808
x8        -0.00703 -0.00681 -0.00482
T-Value      -5.58    -7.05    -3.77
P-Value      0.000    0.000    0.001

x2                  0.00311  0.00360
T-Value                4.40     5.18
P-Value                0.000    0.000

x7                              0.194
T-Value                          2.20
P-Value                         0.038
```

```
S             2.39     1.83     1.71
R-Sq         54.47    74.33    78.63
R-Sq(adj)    52.72    72.27    75.96
C-p           20.4      3.1      0.9
```

The stepwise procedure found variables x_2, x_7, and x_8 significant.

(c) Forward selection

```
Forward selection.  Alpha-to-Enter: 0.25
 Response is    y     on  9 predictors, with N =  28
    Step          1        2        3        4
Constant     21.788   14.713   -1.808   -1.822

x8         -0.00703 -0.00681 -0.00482 -0.00401
T-Value       -5.58    -7.05    -3.77    -2.87
P-Value       0.000    0.000    0.001    0.009

x2                   0.00311  0.00360  0.00382
T-Value                 4.40     5.18     5.42
P-Value                 0.000    0.000    0.000

x7                               0.194    0.217
T-Value                          2.20     2.45
P-Value                          0.038    0.023

x9                                       -0.0016
T-Value                                    -1.31
P-Value                                    0.202

S              2.39     1.83     1.71     1.68
R-Sq          54.47    74.33    78.63    80.12
R-Sq(adj)     52.72    72.27    75.96    76.66
C-p            20.4      3.1      0.9      1.4
```

Forward selection found variables x_2, x_7, x_8, and x_9 significant.

(d) Backward elimination

```
Backward elimination.  Alpha-to-Remove: 0.1
 Response is    y     on  9 predictors, with N =  28
    Step          1        2        3        4        5        6        7
Constant     -7.292   -7.294   -9.130   -7.695   -4.627   -1.822   -1.808

x1           0.0008   0.0008
T-Value        0.40     0.42
P-Value       0.690    0.681

x2          0.00363  0.00363  0.00363  0.00358  0.00371  0.00382  0.00360
T-Value        4.32     4.59     4.69     4.76     5.13     5.42     5.18
P-Value       0.000    0.000    0.000    0.000    0.000    0.000    0.000

x3             0.12     0.12     0.17     0.17
T-Value        0.47     0.49     0.75     0.77
P-Value       0.643    0.632    0.461    0.451

x4            0.032    0.032    0.037    0.035    0.026
T-Value        0.77     0.80     1.00     0.97     0.78
P-Value       0.453    0.431    0.329    0.342    0.445

x5            0.000
T-Value        0.00
P-Value       1.000

x6           0.0016   0.0016   0.0015
T-Value        0.49     0.51     0.48
P-Value       0.630    0.618    0.639

x7            0.154    0.154    0.189    0.193    0.235    0.217    0.194
T-Value        1.02     1.10     1.72     1.79     2.54     2.45     2.20
P-Value       0.324    0.284    0.102    0.088    0.019    0.023    0.038

x8          -0.0039  -0.0039  -0.0042  -0.0044  -0.0037  -0.0040  -0.0048
T-Value       -1.90    -1.95    -2.34    -2.50    -2.48    -2.87    -3.77
P-Value       0.074    0.066    0.030    0.021    0.021    0.009    0.001

x9          -0.0018  -0.0018  -0.0017  -0.0017  -0.0018  -0.0016
T-Value       -1.26    -1.30    -1.26    -1.28    -1.40    -1.31
P-Value       0.222    0.210    0.221    0.213    0.176    0.202

S              1.83     1.78     1.74     1.71     1.70     1.68     1.71
R-Sq          81.56    81.56    81.39    81.18    80.65    80.12    78.63
R-Sq(adj)     72.34    73.80    74.88    75.80    76.25    76.66    75.96
C-p            10.0      8.0      6.2      4.4      2.9      1.4      0.9
```

Backward elimination found variables x_2, x_7, and x_8 significant.

15–23. (a) All possible regressions from Minitab displaying the best two models for each combination of variables.

```
24 cases used; 1 case contains missing values.
```

Vars	R-Sq	R-Sq(adj)	C-p	S	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10	x11
1	80.9	80.1	5.6	2.9679	X										
1	77.1	76.1	10.7	3.2514			X								
2	82.6	81.0	5.2	2.8974	X					X					
2	82.5	80.9	5.4	2.9082	X			X							
3	84.3	82.0	5.0	2.8231					X			X		X	
3	84.0	81.5	5.5	2.8552	X		X			X					
4	85.0	81.9	6.0	2.8283			X		X			X		X	
4	84.9	81.7	6.2	2.8411	X	X	X			X					
5	86.7	83.0	5.8	2.7400	X		X		X			X		X	
5	85.8	81.8	7.1	2.8347	X		X	X	X		X				
6	88.6	84.5	5.3	2.6141	X		X		X		X	X		X	
6	87.6	83.2	6.6	2.7244	X		X		X			X		X	X
7	89.6	85.1	5.9	2.5649	X		X	X	X		X	X		X	
7	89.0	84.1	6.8	2.6465	X		X		X		X	X		X	X
8	90.2	84.9	7.2	2.5786	X		X	X	X		X	X		X	X
8	90.0	84.6	7.4	2.6068	X	X	X		X	X	X	X		X	
9	90.5	84.4	8.8	2.6285	X	X	X		X	X	X	X		X	X

(b) Stepwise regression

```
 Alpha-to-Enter: 0.15   Alpha-to-Remove: 0.15
 Response is    y     on 11 predictors, with N =   24
 N(cases with missing observations) =   1 N(all cases) =   25
    Step          1
Constant      34.43

x1          -0.0482
T-Value       -9.66
P-Value       0.000

S              2.97
R-Sq          80.93
R-Sq(adj)     80.06
C-p             5.6
```

The stepwise procedure found x_1 significant.

(c) Forward selection

```
Forward selection.  Alpha-to-Enter: 0.25
 Response is    y     on 11 predictors, with N =   24
 N(cases with missing observations) =   1 N(all cases) =   25
    Step          1        2        3
Constant      34.43    33.50    32.36

x1          -0.0482  -0.0544  -0.1034
T-Value       -9.66    -8.40    -2.65
P-Value       0.000    0.000    0.015

x6                      1.05     1.02
T-Value                 1.44     1.43
P-Value                0.164    0.169

x3                              0.070
T-Value                          1.28
P-Value                         0.217

S              2.97     2.90     2.86
R-Sq          80.93    82.65    83.95
R-Sq(adj)     80.06    80.99    81.55
C-p             5.6      5.2      5.5
```

The forward selection procedure found x_1, x_3, and x_6 significant.

(d) Backward elimination

```
Backward elimination.  Alpha-to-Remove: 0.1
 Response is    y     on 11 predictors, with N =   24
 N(cases with missing observations) =   1 N(all cases) =   25

    Step          1        2        3        4        5        6        7        8        9
Constant   -17.6442 -18.5202  -3.7497  -0.9652  -0.6957  -4.4555  -1.9112  -8.9148   0.3409

x1           -0.142   -0.142   -0.139   -0.127   -0.102   -0.089   -0.061
T-Value       -2.57    -2.68    -2.66    -2.54    -2.18    -2.11    -1.50
P-Value       0.024    0.019    0.019    0.023    0.044    0.050    0.152

x2           -0.076   -0.075   -0.096   -0.092
T-Value       -0.93    -0.97    -1.30    -1.26
P-Value       0.369    0.348    0.215    0.227

x3            0.231    0.230    0.240    0.224    0.140    0.146    0.099    0.035
T-Value        2.34     2.44     2.58     2.48     2.26     2.44     1.79     0.96
P-Value       0.037    0.030    0.022    0.026    0.038    0.026    0.090    0.348

x4              2.4      2.4
T-Value        0.87     0.90
P-Value       0.402    0.383

x5              6.8      6.7      6.8      6.4      6.1      6.6      3.0      3.6      2.9
T-Value        2.06     2.34     2.40     2.31     2.15     2.47     1.83     2.23     2.02
P-Value       0.062    0.036    0.031    0.036    0.047    0.024    0.083    0.038    0.057

x6             1.11     1.14     1.42     1.37     0.66
T-Value        0.88     0.99     1.30     1.26     0.70
P-Value       0.398    0.338    0.215    0.226    0.494

x7             -4.2     -4.1     -3.4     -3.8     -3.9     -3.6
T-Value       -1.47    -1.67    -1.46    -1.72    -1.74    -1.67
P-Value       0.167    0.118    0.165    0.106    0.100    0.114

x8             0.28     0.28     0.30     0.29     0.28     0.31     0.26    0.324    0.246
T-Value        2.11     2.20     2.40     2.31     2.26     2.54     2.12     2.72     2.81
P-Value       0.056    0.046    0.031    0.035    0.038    0.021    0.048    0.014    0.011

x9           -0.02
T-Value      -0.05
P-Value      0.959

x10        -0.0119  -0.0121  -0.0126  -0.0121  -0.0115  -0.0133   0.0113  -0.0142  -0.0099
T-Value      -1.82    -2.09    -2.21    -2.15    -2.01    -2.65    -2.21    -2.90    -5.00
P-Value      0.093    0.057    0.044    0.049    0.061    0.017    0.040    0.009    0.000

x11            -2.4     -2.5     -2.3
T-Value       -0.89    -0.93    -0.87
P-Value       0.393    0.370    0.400

S              2.75     2.65     2.63     2.61     2.65     2.61     2.74     2.83     2.82
R-Sq          91.04    91.04    90.48    89.97    88.90    88.57    86.70    85.04    84.31
R-Sq(adj)     82.83    84.15    84.36    84.62    84.05    84.53    83.00    81.89    81.96
C-p            12.0     10.0      8.8      7.4      6.9      5.3      5.8      6.0      5.0
```

Backward elimination found x_5, x_8, and x_{10} significant.

15–24. (a) All possible regressions from Minitab displaying the best two models for each combination of variables.

Vars	R-Sq	R-Sq(adj)	C-p	S	x1	x2	x3	x4
1	67.5	64.5	138.7	8.9639				X
1	66.6	63.6	142.5	9.0771		X		
2	97.9	97.4	2.7	2.4063	X	X		
2	97.2	96.7	5.5	2.7343	X			X
3	98.2	97.6	3.0	2.3087	X	X		X
3	98.2	97.6	3.0	2.3121	X	X	X	
4	98.2	97.4	5.0	2.4460	X	X	X	X

(b) Stepwise regression

Alpha-to-Enter: 0.15 Alpha-to-Remove: 0.15
Response is y on 4 predictors, with N = 13

Step	1	2	3	4
Constant	117.57	103.10	71.65	52.58
x4	-0.738	-0.614	-0.237	
T-Value	-4.77	-12.62	-1.37	
P-Value	0.001	0.000	0.205	
x1		1.44	1.45	1.47
T-Value		10.40	12.41	12.10
P-Value		0.000	0.000	0.000
x2			0.416	0.662
T-Value			2.24	14.44
P-Value			0.052	0.000
S	8.96	2.73	2.31	2.41
R-Sq	67.45	97.25	98.23	97.87
R-Sq(adj)	64.50	96.70	97.64	97.44
C-p	138.7	5.5	3.0	2.7

Stepwise procedure found x_1 and x_2 significant.

(c) Forward selection

```
Forward selection.   Alpha-to-Enter: 0.25
 Response is    y      on  4 predictors, with N =   13

    Step           1        2        3
Constant      117.57   103.10    71.65

x4            -0.738   -0.614   -0.237
T-Value        -4.77   -12.62    -1.37
P-Value        0.001    0.000    0.205

x1                      1.44     1.45
T-Value                10.40    12.41
P-Value                0.000    0.000

x2                              0.42
T-Value                         2.24
P-Value                        0.052

S              8.96     2.73     2.31
R-Sq          67.45    97.25    98.23
R-Sq(adj)     64.50    96.70    97.64
C-p           138.7      5.5      3.0
```

Forward selection found x_1, x_2, and x_4 significant.

(d) Backward elimination

```
Backward elimination.   Alpha-to-Remove: 0.1
 Response is    y      on  4 predictors, with N =   13
    Step          1         2         3
Constant      62.41     71.65     52.58

x1             1.55      1.45      1.47
T-Value        2.08     12.41     12.10
P-Value       0.071     0.000     0.000

x2            0.510     0.416     0.662
T-Value        0.70      2.24     14.44
P-Value       0.501     0.052     0.000

x3             0.10
T-Value        0.14
P-Value       0.896

x4            -0.14     -0.24
T-Value       -0.20     -1.37
P-Value       0.844     0.205

S              2.45      2.31      2.41
R-Sq          98.24     98.23     97.87
R-Sq(adj)     97.36     97.64     97.44
C-p             5.0       3.0       2.7
```

Backward elimination found x_1 and x_2 significant.

15–25. $VIF_1 = 38.5$, $VIF_2 = 254.4$, $VIF_3 = 46.9$, $VIF_4 = 282.5$. The variance inflation factors indicate a problem with multicollinearity.

Chapter 16

16–1. H_0: $\tilde{\mu} = 7.0$ $\quad n = 10$
H_1: $\tilde{\mu} = 7.0$

$\alpha = 0.05$

CR: $R \le R^*_\alpha$ (Table X)

$R^+ = 8,\ R^- = 2 \ \Rightarrow\ R = 2$

Since $R > R^*_\alpha$, do not reject H_0

16–2. H_0: $\tilde{\mu} = 8.5$
H_1: $\tilde{\mu} \neq 8.5$

$\alpha = 0.05$

$R^+ = 8,\ R^- = 11,\ R = \min(8, 11) = 8,\ R^*_{0.05} = 5$

Since $R > R^*_{0.05}$, do not reject H_0

16–3. (a) H_0: $\tilde{\mu} = 3.5$ $\quad$ Critical Region: $R^- < R^*_d$
H_1: $\tilde{\mu} > 3.5$

(b) $n = 10$ $\quad \alpha = 0.05$ $\quad R^*_{0.05} = 1$
$R = 3$ $\quad$ Since $R^- > R^*_{0.05}$, do not reject H_0

(c) Probability of not rejecting H_0 ($\tilde{\mu} = 3.5$) when $\tilde{\mu} = 4.5$

$$p \ = \ P(X > 1) \ = \ \int_1^\infty \frac{1}{\beta} e^{-(1/\beta)x}\, dx \ = \ e^{-1/\beta}$$

16–4. $n = 10$ $\quad \sigma = 1$
H_0: $\mu = 0$
H_1: $\mu > 0$

(a) $\alpha = 0.025$

(b) $p = P(X > 0) = P(Z > -1) = 1 - \Phi(-1) = 0.8413$
$R^*_{0.05} = 1$
$\beta = 1 - \sum_{x=0}^{1} \binom{10}{x} (0.1587)^x (0.8413)^{10-x} = 0.487$

16–5. H_0: $\tilde{\mu}_d = 0$
H_1: $\tilde{\mu}_d \neq 0$

$\alpha = 0.05$

CR: $R < R^*_{0.05} = 1$
$R^- = 2,\ R^+ = 6 \ \Rightarrow\ R = 2$
Since R is not less than R^*, do not reject H_0

16–8. H_0: $\mu = 7$ $\quad R^+ = 50.5$
H_1: $\mu \neq 7$ $\quad R^- = 4.5$
$R = \min(50.5, 4.5) = 4.5$
$R_{0.05} = 8$ $\quad$ reject H_0

16–9. H_0: $\mu = 8.5$ $\quad n = 20$
H_1: $\mu \neq 8.5$

$\alpha = .05$

CR: $R \leq R^*_{0.05} = 52$

$R^+ = 19 + 10.5 + \cdots + 1 = 88.5$
$R^- = 12 + 20 + \cdots + 3.5 = 121.5$
$R = 88.5 > R^*_{0.05}$
$\therefore$ do not reject H_0
conclude titanium content is 8.5%

16–10. H_0: $\mu = 8.5$ $\quad \mu_R = \dfrac{20(21)}{4} = 10.5$
H_1: $\mu \neq 8.5$ $\quad \sigma^2_R = \dfrac{10(21)(41)}{24} = 717.5$
$Z_0 = \dfrac{85 - 105}{\sqrt{717.5}} = -0.747$

16–12. $R^+ = 24.5$ $\quad R = \min(24.5, 11.5) = 11.5$
$R^- = 11.5$ $\quad R^*_{0.05} = 3$

Do not reject H_0, $\mu_d = 0$.

16–13. H_0: $\mu_1 = \mu_2$ $\quad n_1 = 8 \quad n_2 = 9$
H_1: $\mu_1 > \mu_2$

CR: $R_1 < R^*_\alpha = 51$ Table IX

$\alpha = 0.05$

$R_2 = 75$, $R_1 = 78$
Since R_2 is not less than or equal to 51, do not reject H_0. Conclude the two circuits are equivalent.

16–14. $n_1 = n_2$
$R_1 = 40$
$R_2 = 6(13) - 40 = 38$
$R^*_{0.05} = 26$

Do not reject H_0

Airline	Minutes	Rank
D	0	1
D	1	2.5
A	1	2.5
A	2	4
A	3	5
A	4	6.5
D	4	6.5
A	8	8
D	9	9
D	10	10
D	13	11
A	15	12

16–15. $n_1 = n_2 = 10$, $n > 8$ large-sample approximation

H_0: $\mu_1 = \mu_2$
H_1: $\mu_1 \neq \mu_2$

$\alpha = 0.05$

CR: $|Z_0| > Z_{0.025} = 1.96$

$$\mu_{R_1} = \frac{(20)(21)}{4} = 105, \qquad \sigma^2_{R_1} = \frac{10(10)(21)}{12} = 175$$

$$R_1 = 1 + 2 + 3 + \cdots + 19.5 = 77$$

$$Z_0 = \frac{77 - 105}{\sqrt{175}} = -2.117; |Z_0| = 2.117 > 1.96$$

Reject H_0; conclude unit 2 is superior.

16–16. H_0: techniques do not differ
H_1: techniques differ

	Ranks				R_i
1	14	11.5	6	7	38.5
2	16	11.5	9	15	51.5
3	5	8	10	13	36
4	15	3	1.5	4	10

$$S^2 = \frac{1}{15}\left[1495 - \frac{16(17)^2}{4}\right] = 22.6$$

$$K = \frac{1}{22.6}\left[\frac{38.5^2}{4} + \frac{51.5^2}{4} + \frac{36^2}{4} + \frac{10^2}{4} - \frac{16(17)^2}{4}\right] = 10.03$$

$\chi^2_{0.05,3} = 7.81$, reject H_0; conclude the techniques differ

16–17. H_0: methods do not differ
H_1: methods differ
$\alpha = 0.10$
CR: $K > \chi^2_{0.10,2} = 4.61$

Method		Ranks				r_i
1	10.5	9	12	7	5	43.5
2	10.5	15	14	8	6	53.5
3	1	4	3	2	13	83

$$K = \frac{12}{15(16)}\left(\frac{43.5^2}{5} + \frac{53.5^2}{5} + \frac{53.5^2}{5} + \frac{23^2}{5}\right) - 3(16) = 4.835$$

at 0.1 significance, reject H_0.

16–18. H_0: manufacturers do not differ
H_1: they differ

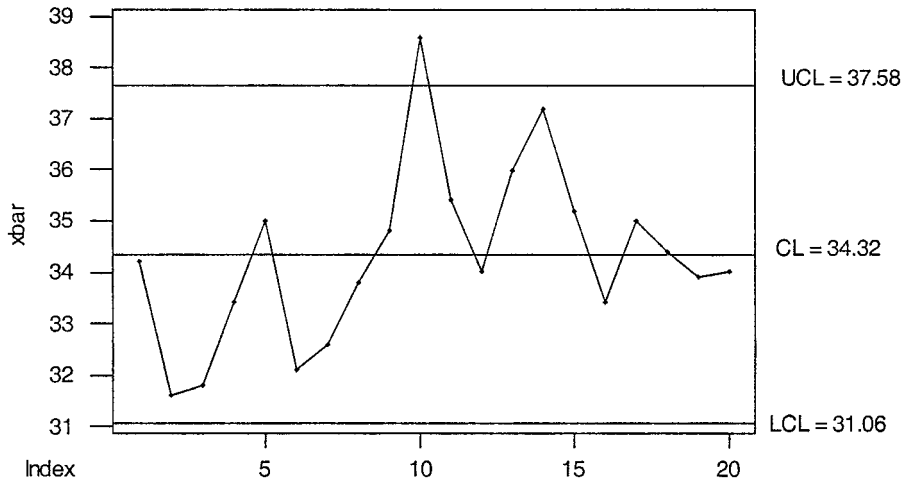
	Ranks					R_i
A	12	1	3	9	13	38
B	4	17	14	7	20	62
C	19	15	16	11	18	79
D	5	10	2	8	6	31

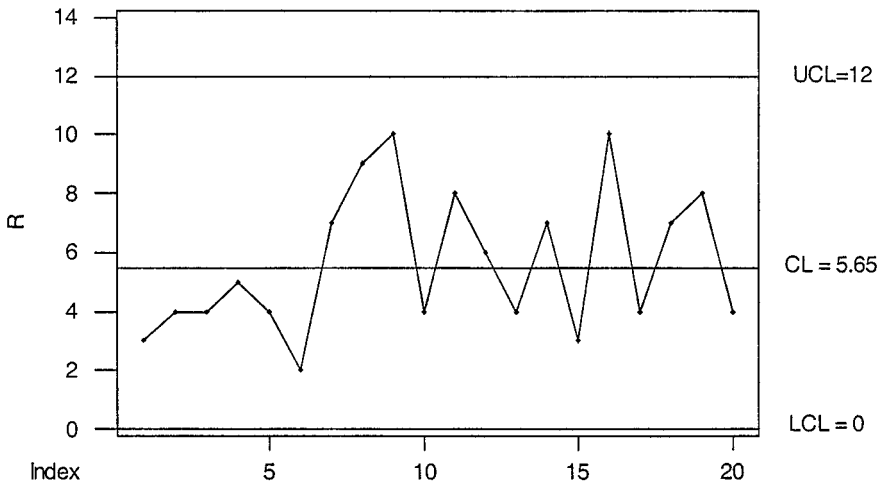
$$K = \frac{12}{20(21)}\left[\frac{38^2 + 62^2 + 79^2 + 31^2}{5}\right] - 3(21) = 8.37$$

$\chi^2_{0.05,3} = 7.81$. Reject H_0; conclude the manufacturers differ.

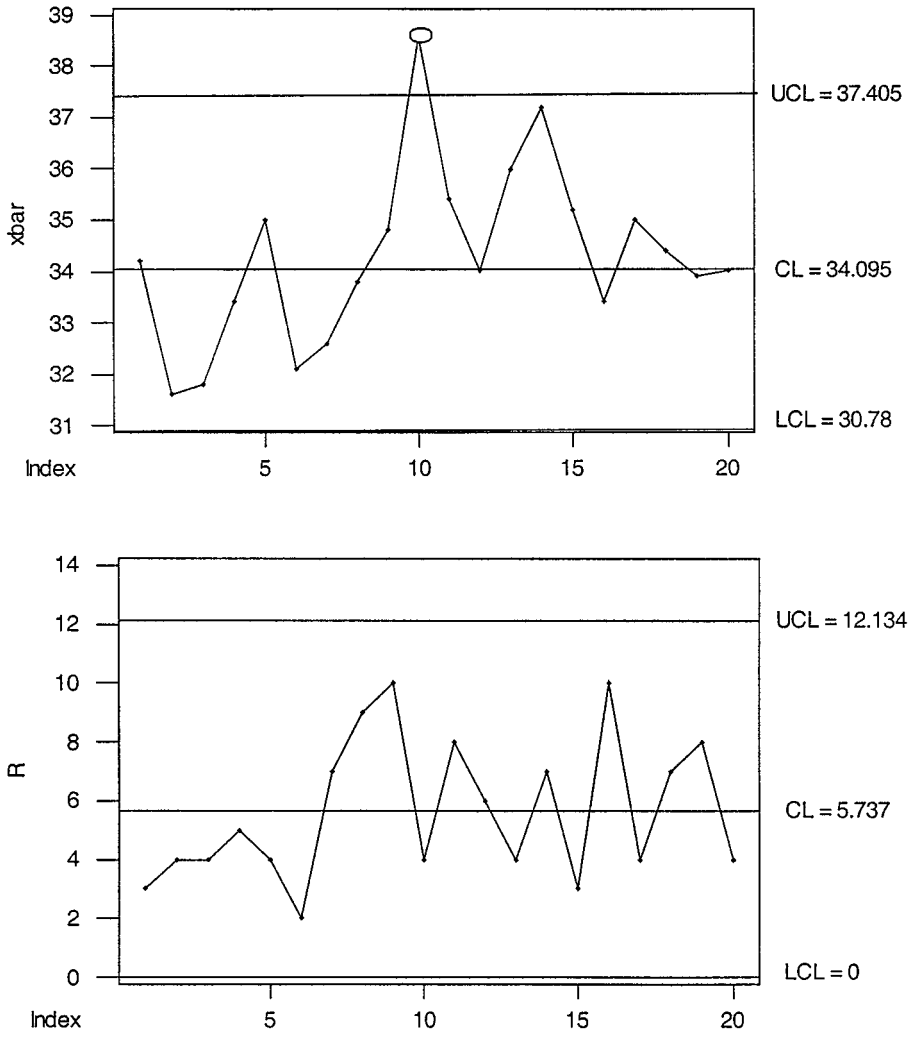
Chapter 17

17–1. (a) X-bar Chart: $UCL = 34.32 + 0.577(5.65) = 37.58$
$LCL = 34.32 - 0.577(5.65) = 31.06$
R chart: $UCL = 2.115(5.65) = 12$
$LCL = 0$





There is one observation beyond the upper control limit. Removal of this point results in the following control charts:



The process now appears to be in control.

(b) $\hat{\sigma} = \dfrac{\overline{R}}{d_2} = \dfrac{5.74}{2.326} = 2.468, PCR = \dfrac{USL - LSL}{6\sigma} = \dfrac{20}{6(2.468)} = 1.35$

$$PCR_K = \min\left[\frac{45 - 34.09}{3(2.468)}, \frac{34.09 - 25}{3(2.468)}\right] = \min[1.474, 1.2277] = 1.2277$$

(c) 0.205%

17–2. $P(\overline{X} < \mu + 3\sigma/\sqrt{n} | \mu_{\overline{X}} = \mu + 1.5\sigma)$

$= P(Z < \dfrac{\mu + 3\sigma/\sqrt{n} - (\mu + 1.5\sigma)}{\sigma/\sqrt{n}}$

$= P(Z < 3 - 1.5\sqrt{n})$
= probability of failing to detect shift on 1st sample following the shift.

$[P(Z < 3 - 1.5\sqrt{n})]^3 =$ prob of failing to detect shift for 3 consecutive samples following the shift.

For $n = 4$, $[P(Z < 0)]^3 = (0.5)^3 = 0.125$

For $n = 4$ with 2-sigma limits,

$$[P(Z < 2 - 3)]^3 = [P(Z < -1)]^3 = (0.1587)^3 = 0.003997$$

17–3. (a) $ARL = 1/\alpha$

(b) $ARL = 1/(1 - \beta)$

(c) If k changes from 3 to 2, the in-control ARL will get much shorter (from about 370 to 20). This is not desirable.

(d) For a 1-sigma shift, $\beta \simeq 0.8$, so the ARL is approximately $ARL = 1/(1 - 0.8) = 5$.

17–4. $\overline{\overline{X}} = \dfrac{362.75}{25} = 14.51$, $\overline{R} = \dfrac{8.60}{25} = 0.34$

(a) $\overline{X}$ Chart: $UCL = 14.706$, $CL = 14.31$, $LCL = 14.314$

$\overline{R}$ Chart: $UCL = 0.719$, $CL = 0.34$, $LCL = 0$

(b) $\hat{\sigma} = \overline{R}/d_2 = 0.34/2.326 = 0.14617$, $\hat{\mu} = 14.51$

6σ natural tolerance limits $= 14.51 \pm 3(0.14617) = 14.51 \pm 0.4385$

$P(X > 14.90) = 0.00379$, $P(X < 14.10) = 0.00252$

Fraction defective $= 0.00631$

(c) $PCR = \dfrac{15 - 14}{6(0.146)} = 1.141$ $\qquad PCR_K = \min[1.119, 1.164] = 1.119$

17–5. $D/2$

17–6. (a) $\overline{\overline{X}} = \dfrac{214.25}{20} = 10.7125 \qquad \overline{R} = \dfrac{133}{20} = 6.65$

$\overline{X}$ Chart: $UCL = 10.7125 + 0.729(6.65) = 15.56$

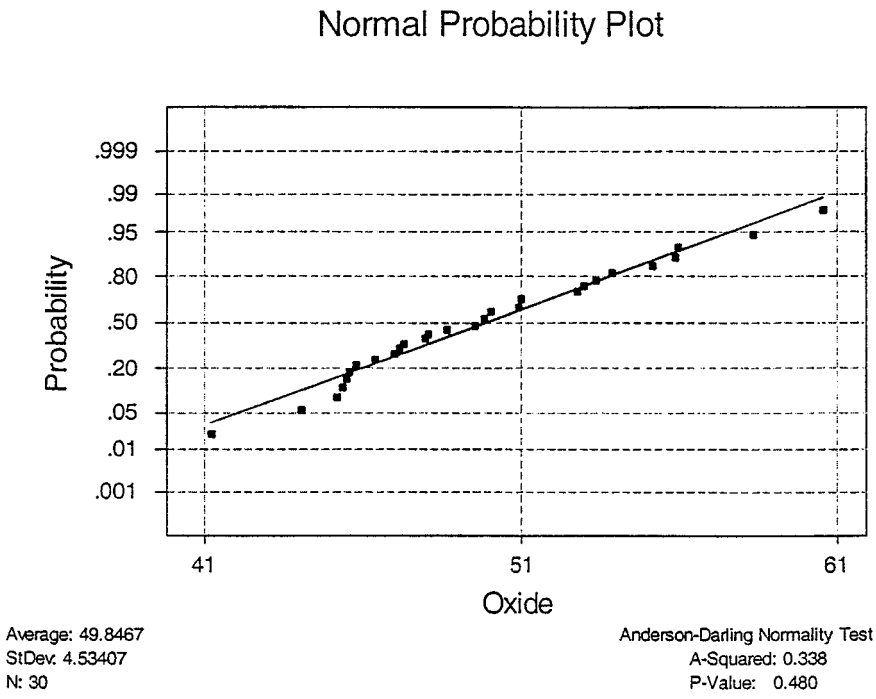
$LCL = 5.86$

R Chart: $UCL = 2.282(6.65) = 15.175$, $LCL = 0$; process in control

(b) $\hat{\sigma} = \overline{R}d_2 = 6.65/2.059 = 3.23$

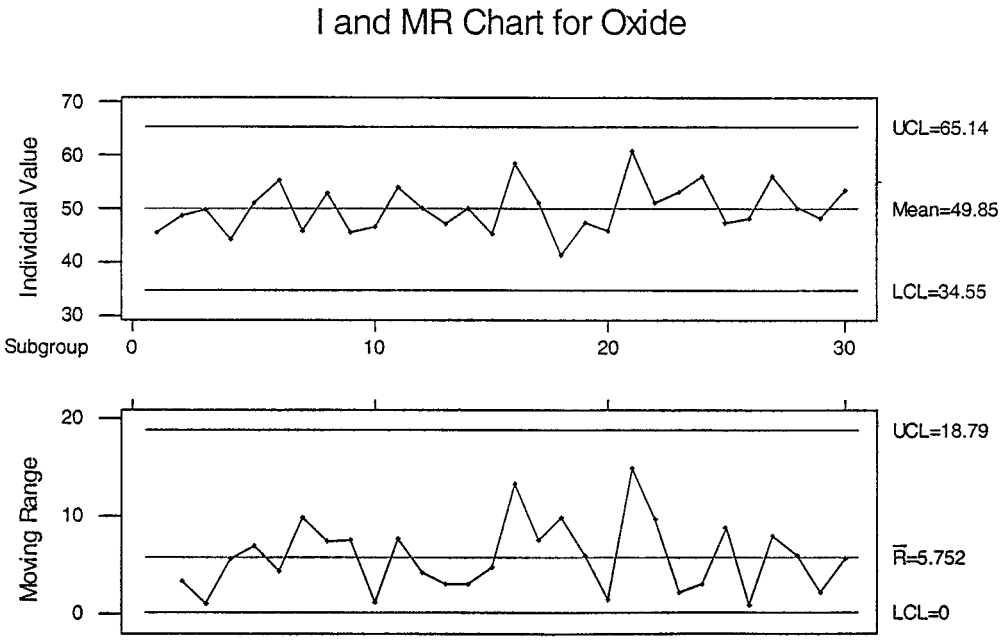
$$PCR = \frac{15 - 5}{6(3.23)} = 0.516$$

17–7. (a) Normal probability plot



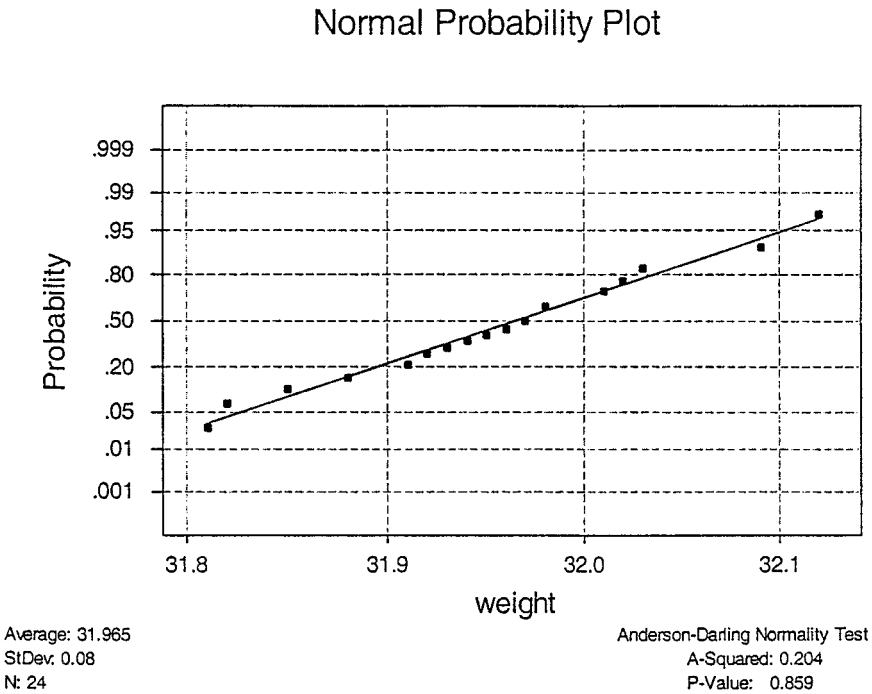
The normality assumption appears to be satisfied.

(b)



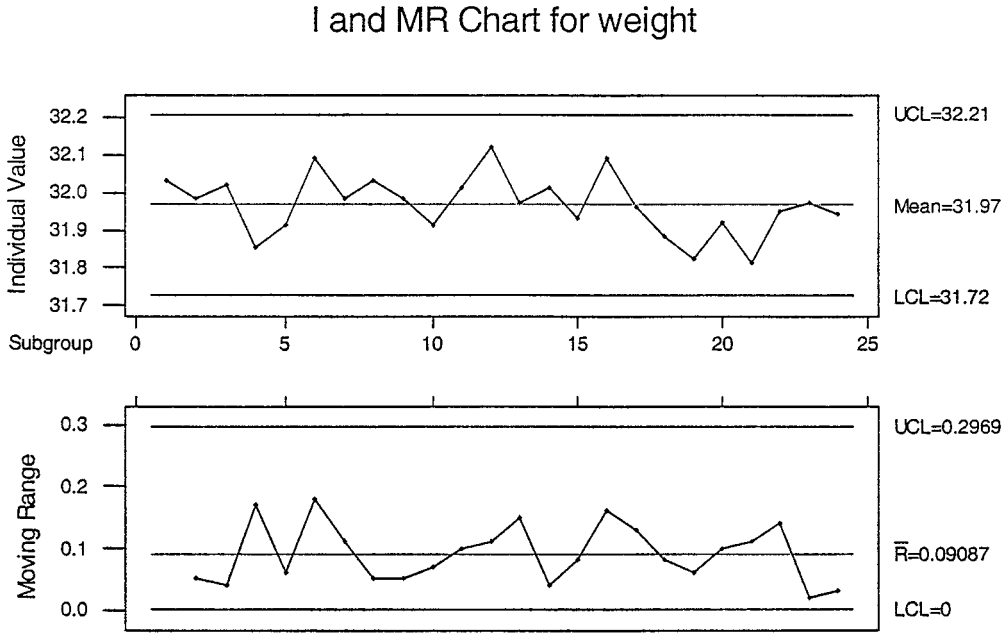
The process appears to be in control.

17–8. (a) Normal probability plot



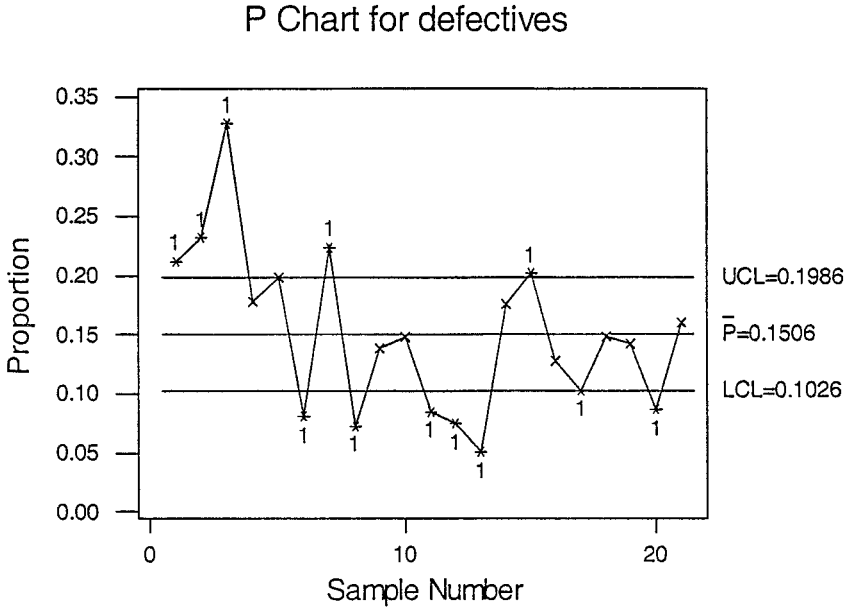
The normality assumption appears to be satisfied.

(b)



The process appears to be in control.

17–9.



The process is out of control.

17–10. $\overline{p} = 0.05 \qquad UCL = 0.05 + 3(0.0218) = 0.115$

$$P(\overline{X} < 0.115 | \mu = 0.08) = 1 - P\left(Z < \frac{0.115 - 0.08}{0.0654}\right) = 1 - P(Z < 0.535)$$

$= 1 - 0.7036 = 0.2964$ Probability of detecting shift on first sample following shift

$P(\text{detecting before 3rd sample}) = 1 - (0.7036)^2 = 0.5049$

17–11. For the detection probability to equal 0.5, the magnitude of the shift must bring the fraction nonconforming exactly to the upper control limit. That is, $\delta = k\sqrt{p(1-p)/n}$, where δ is the magnitude of the shift. Solving for n gives $n = (k/\delta)^2 p(1-p)$. For example, if $k = 3$, $p = 0.01$ (the in-control fraction nonconforming), and $\delta = 0.04$, then $n = (3/0.04)^2(0.01)(0.99) = 56$.

17–12. (a) $PCR = 1.5$

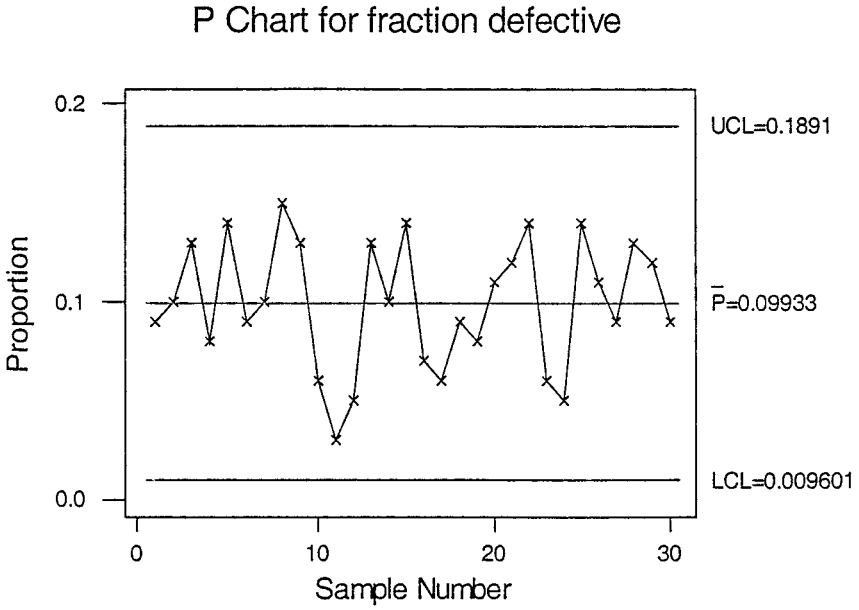
(b) About 7 defective parts per million.

(c) $PCR_k = 1$ PCR unchanged.

(d) About 0.135 percent defective.

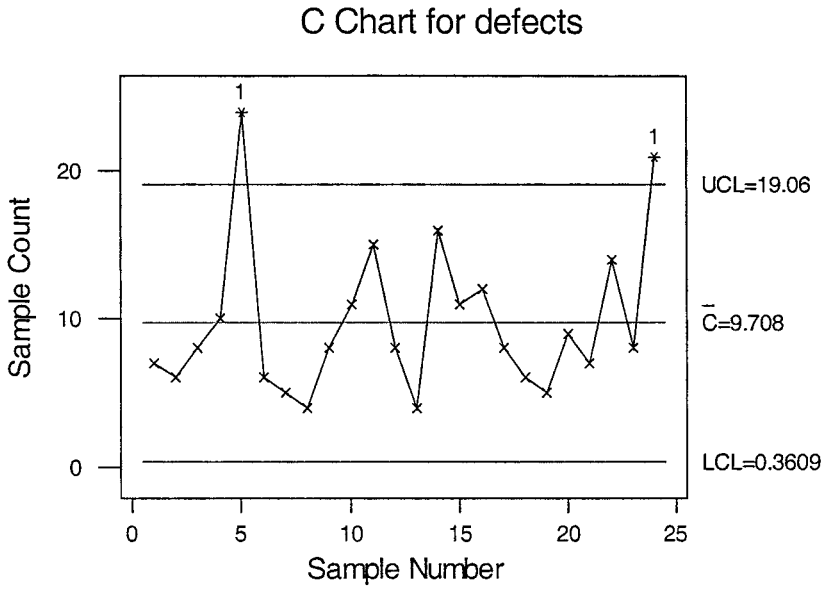
17–13. Center the process at $\mu = 100$. The probability that a shift to $\mu = 105$ will be detected on the first sample following the shift is about 0.15. A p-chart with $n = 7$ would perform about as well.

17–14.

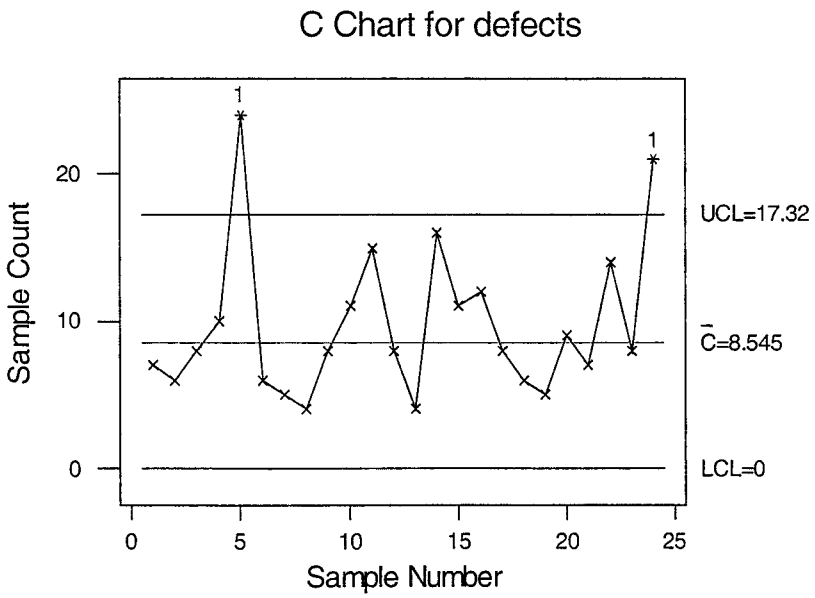


The process appears to be in control.

17–15.

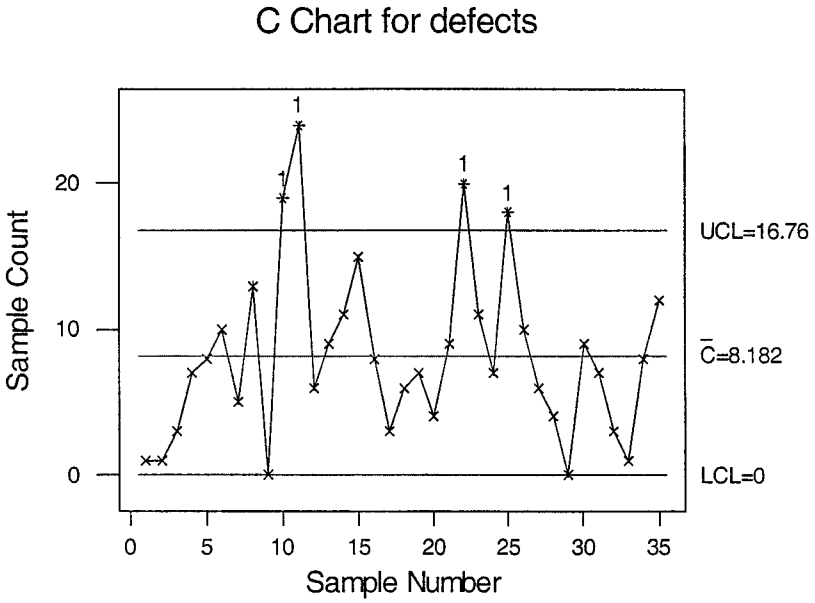


The process is out of control. Removing two out-of-control points and revising the limits results in:



The process is now in control.

17–16.

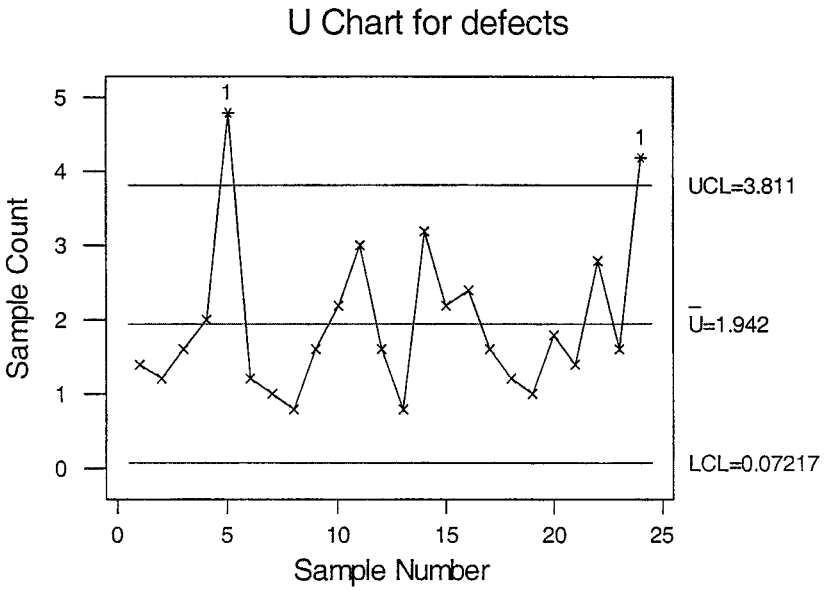


The process appears to be out of control.

17–17. $UCL = 16.485$; detection probability $= 0.434$

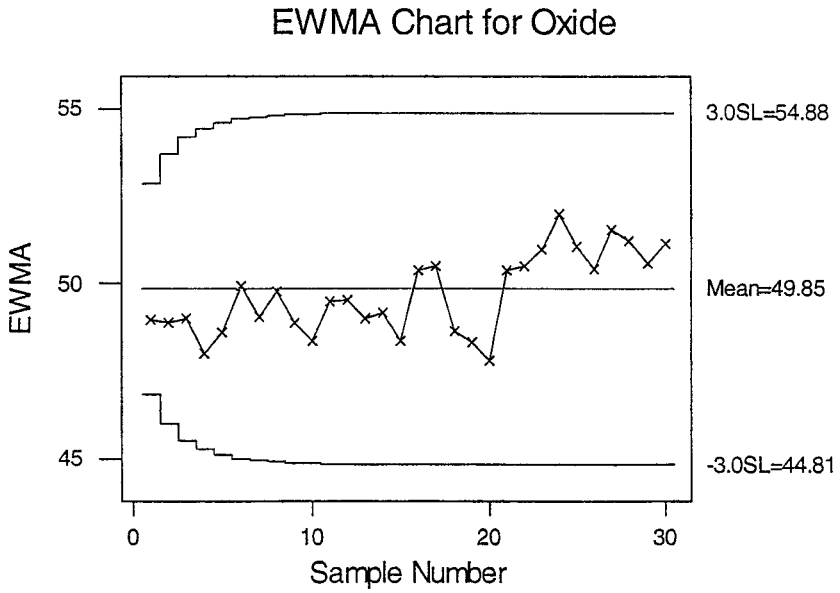
17–18. $UCL = 19.487$, $CL = 10$, $LCL = 0.513$

17–19.

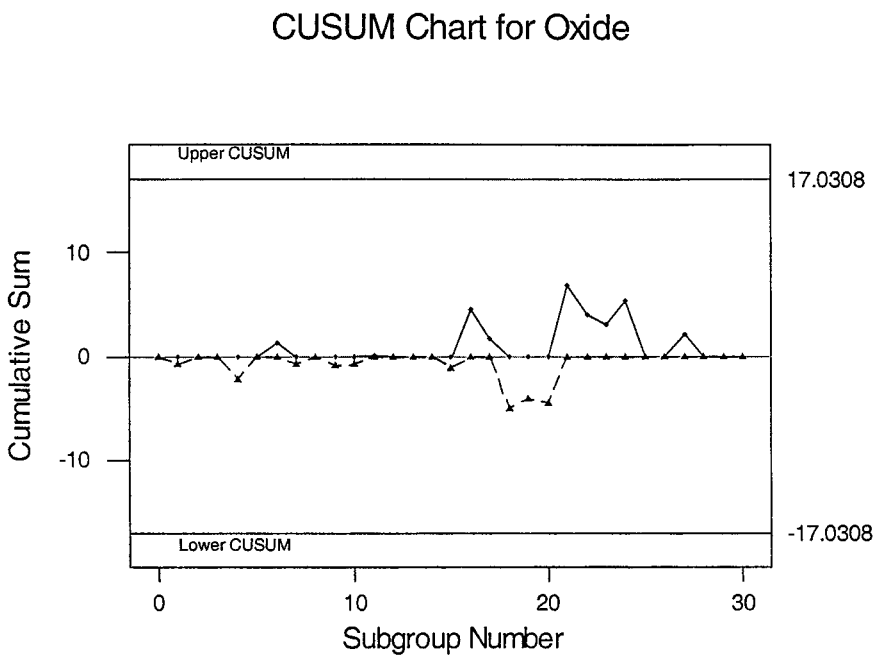


Since the sample sizes are equal, the c and u charts are equivalent.

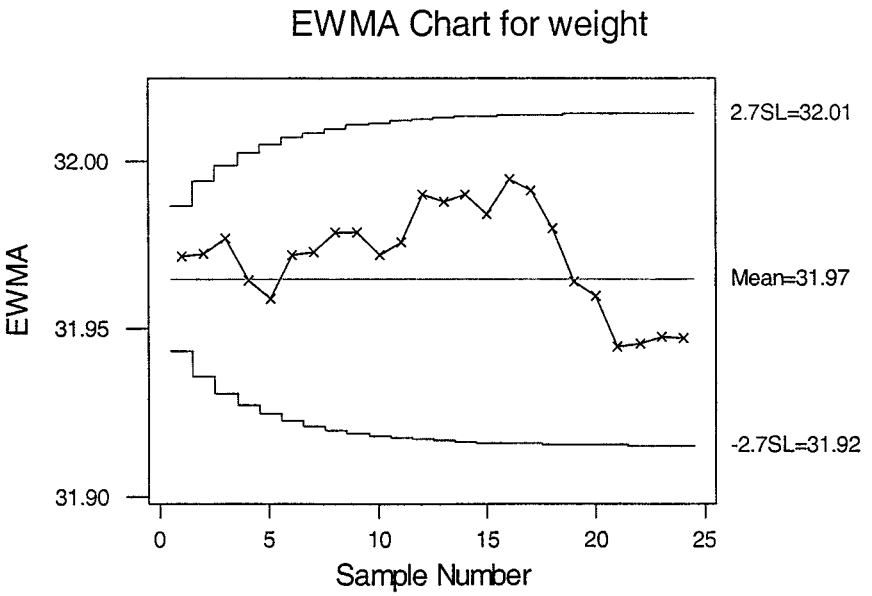
17–20.



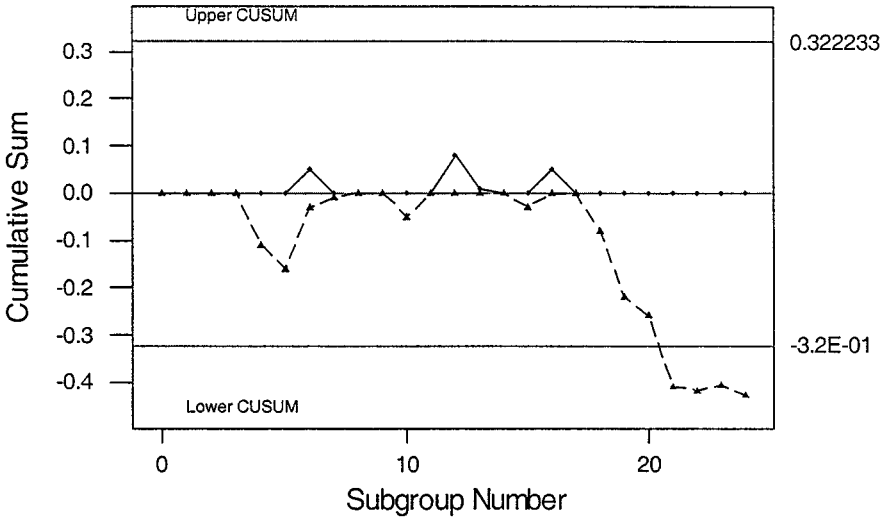
17–21.


CUSUM Chart for Oxide
Upper CUSUM
17.0308
10
0
-10
Cumulative Sum
Lower CUSUM
-17.0308
0
10
20
30
Subgroup Number

17–22.


EWMA Chart for weight
2.7SL=32.01
32.00
Mean=31.97
31.95
EWMA
-2.7SL=31.92
31.90
0
5
10
15
20
25
Sample Number

17–23.



17–24. (a)

$$R(t) \;=\; \int_t^\infty f(x)\,dx \;=\; \begin{cases} 1 & \text{if } t < \alpha \\ \frac{\beta - t}{\beta - \alpha} & \text{if } \alpha \le t \le \beta \\ 0 & \text{if } t > \beta \end{cases}$$

(b) $\displaystyle\int_0^\infty R(t)\,dt \;=\; \frac{\alpha+\beta}{2}$

(c) $\displaystyle h(t) \;=\; \frac{f(t)}{R(t)} \;=\; \frac{1}{\beta - t}, \quad \alpha \le t \le \beta.$

(d) $\displaystyle H(t) \;=\; \int_0^t h(t)\,dt \;=\; -\ell\mathrm{n}\left(\frac{\beta - t}{\beta - \alpha}\right)$

$$e^{-H(t)} \;=\; \frac{\beta - t}{\beta - \alpha} \;=\; R(t).$$

17–25. $R_S(t) = e^{-\lambda_s t},\ \lambda_s = \lambda_1 + \lambda_2 + \lambda_3 = 7.6 \times 10^{-2}$

(a) $R_S(60) = e^{-7.6\times10^{-2}\times 60} = 0.0105$

(b) $MTTF = 1/\lambda_s = \frac{1}{7.6\times10^{-2}} = 13.16$ hours

17–26. $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = \lambda_5 = 0.002$

(a) $R(1000) = \sum_{k=2}^{5} \binom{5}{k} (0.367)^k (0.633)^{5-k} = 0.6056$

(b) $R(1000) = 1 - (0.633)^5 = 0.8984$

17–27. $R(1000) = \sum_{k=0}^{3} \frac{e^{-1}(1)^k}{k!} = 0.98104$

17–28. $\lambda = 1/160 = 6.25 \times 10^{-3}$

17–29. 0.84, 0.85

17–30. If $\hat{\theta}$ is the maximum likelihood estimator of θ and $\phi = g(\theta)$ is a single-valued function of θ, then $\hat{\phi} = g(\hat{\theta})$ is the MLE of ϕ. To prove this, note that $L(\theta)$, the likelihood function, has a maximum at $\theta = \hat{\theta}$. Furthermore, $\theta = g^{-1}(\phi)$, so the likelihood function is $L[g^{-1}(\phi)]$, which has a maximum at $\hat{\theta} = g^{-1}(\phi)$ or at $\phi = g(\hat{\theta})$. In the problem stated, R is of the form $g(\theta) = e^{-t/\theta}$, so the problem is solved.

17–31. (a) 3842

(b) $[913.63, \infty)$

17–32. (a) $\hat{R}(300) = e^{-300/\hat{\theta}} = e^{-300/3842} = 0.9249$

$\hat{R}_L(300) = e^{-300/\hat{\theta}_L} = e^{-300/913.63} = 0.72$

(b) $\hat{L}_{0.9} = 3842\, \ell\mathrm{n}(1/0.9) = 404.795$

Chapter 18

18–1. (a) $\lambda = 2$ arrivals/hr, $\mu = 3$ services/hr, $\rho = 2/3$.

$$P(n > 5) = 1 - P(n \leq 5) = 1 - \sum_{j=0}^{5} \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^j = 0.088$$

(b) $L = \dfrac{\rho}{1-\rho} = 2$, $L_q = \dfrac{\lambda^2}{\mu(\mu-\lambda)} = 4/3$

(c) $W = \dfrac{1}{\mu - \lambda} = 1$ hour

18–2. (a) State 0 = rain; State 1 = clear.

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.1 & 0.9 \end{bmatrix}$$

(b)

$$\begin{aligned} p_0 &= 0.7p_0 + 0.1p_1 \\ 1 &= p_0 + p_1 \end{aligned}$$

This implies $p_0 = 1/4$, $p_1 = 3/4$.

(c) To find $p_{11}^{(3)}$, note that

$$P^{(3)} = P^3 = \begin{bmatrix} 0.412 & 0.588 \\ 0.196 & 0.804 \end{bmatrix}$$

Thus, $p_{11}^{(3)} = 0.804$.

(d) $f_{10}^{(2)} = p_{10}^{(2)} - f_{10}^{(1)} p_{10} = 0.16 - (0.1)(0.1) = 0.15$

(e) $\mu_0 = 1/p_0 = 4$ days

18–3.

$$P = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$

$$P^{\infty} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

18–4. (a)

$$P = \begin{bmatrix} 1-2\lambda\Delta t & 2\lambda\Delta t & 0 \\ \mu\Delta t & 1-(\mu+\frac{3}{2}\lambda)\Delta t & \frac{3}{2}\lambda\Delta t \\ 0 & \mu\Delta t & 1-\mu\Delta t \end{bmatrix}$$

(b)

$$\begin{aligned} -2\lambda p_0 + \mu p_1 &= 0 \\ 2\lambda p_0 - (\mu + \frac{3}{2}\lambda)p_1 + \mu p_2 &= 0 \\ \frac{3}{2}\lambda p_1 - \mu p_2 &= 0 \\ p_0 + p_1 + p_2 &= 1 \end{aligned}$$

Solving yields

$$\begin{aligned} p_0 &= \frac{\mu^2}{\mu^2 + 2\lambda\mu + 3\lambda^2} \\ p_1 &= \frac{2\lambda\mu}{\mu^2 + 2\lambda\mu + 3\lambda^2} \\ p_2 &= \frac{3\lambda^2}{\mu^2 + 2\lambda\mu + 3\lambda^2} \end{aligned}$$

18–5. (a) $p_{ii} = 1$ implies that State 1 is an absorbing state, so States 0 and 3 are absorbing.

(b) $p_0 = 1$. The system never leaves State 0.

(c) $b_{10} = \frac{2}{3} \cdot 1 + \frac{1}{6}b_{10} + \frac{1}{6}b_{20}$

$b_{20} = \frac{2}{3}b_{10} + \frac{1}{6}b_{20}$

$b_{10} = \frac{20}{21},\ b_{20} = \frac{16}{21}$

$b_{13} = \frac{1}{6}b_{13} + \frac{1}{6}b_{23}$

$b_{23} = \frac{2}{3}b_{13} + \frac{1}{6}b_{23} + \frac{1}{6} \cdot 1$

$b_{13} = \frac{1}{21},\ b_{23} = \frac{5}{21}$

$$p_0 = \frac{1}{2}\cdot\frac{20}{21} + \frac{1}{2}\cdot\frac{16}{21} = \frac{18}{21}$$

(d) $$p_0 = \frac{1}{4}\cdot 1 + \frac{1}{4}\cdot\frac{20}{21} + \frac{1}{4}\cdot\frac{16}{21} + \frac{1}{4}\cdot 1 = \frac{39}{42}$$

18–6. (a)

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ q & 0 & p & 0 & \cdots & 0 & 0 & 0 \\ 0 & q & 0 & p & \cdots & 0 & 0 & 0 \\ & & \vdots & & & & & \\ 0 & 0 & 0 & 0 & \cdots & q & 0 & p \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix}$$

(b)

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.7 & 0 & 0.3 & 0 & 0 \\ 0 & 0.7 & 0 & 0.3 & 0 \\ 0 & 0 & 0.7 & 0 & 0.3 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} b_{10} &= 0.7(1) + 0.3b_{20} \\ b_{20} &= 0.7b_{10} + 0.3b_{20} \\ b_{30} &= 0.7b_{20} \end{aligned}$$

$$\begin{aligned} b_{10} &= 0.953 \\ b_{20} &= 0.845 \\ b_{30} &= 0.591 \end{aligned}$$

$$\begin{aligned} b_{34} &= 0.7b_{24} + 0.3(1) \\ b_{24} &= 0.7b_{14} + 0.3b_{34} \\ b_{14} &= 0.3b_{24} \end{aligned}$$

$$\begin{aligned} b_{14} &= 0.0465 \\ b_{24} &= 0.155 \\ b_{34} &= 0.408 \end{aligned}$$

18–7. (a)

$$P = \begin{bmatrix} 0 & p & 0 & 1-p \\ 1-p & 0 & p & 0 \\ 0 & 1-p & 0 & p \\ p & 0 & 1-p & 0 \end{bmatrix}$$

$$A = [1 \;\; 0 \;\; 0 \;\; 0]$$

(b,c) The transition matrix P is said to be *doubly stochastic* since each row and each column add to one. For such matrices, it is easy to show that the steady-state probabilities are all equal. Thus, for $p = q = 1/2$ or $p = 4/5$, $q = 1/5$, we have $p_1 = p_2 = p_3 = p_4 = 1/4$.

18–9. $\lambda = 1/10$, $\mu = 1/3$, $\rho = 3/10$.

(a) $P(\text{wait}) = 1 - p_0 = \rho = 3/10$.

(b) $L_q = \dfrac{\lambda^2}{\mu(\mu - \lambda)} = 9/70$.

(c) $W_q = 3 = \dfrac{\lambda}{\mu(\mu - \lambda)} \;\Rightarrow\; \lambda = 1/6 =$ criteria for adding service.

(d) $P(W_q > 0) = \displaystyle\int_{10}^{\infty} \lambda(1-\rho)\, e^{-(\mu-\lambda)w_q}\, dw_q = \frac{3}{10}\, e^{-7/3} \doteq 0.03$.

(e) $P(W > 10) = e^{-\frac{1}{3}(\frac{7}{10})(0.10)} \doteq 0.10$.

(f) $\rho = 1 - p_0 = 3/10$.

18–10. $\lambda = 15$, $\mu = 27$, $N = 3$, $\rho = \dfrac{15}{27} = \dfrac{5}{9}$.

(a) $L = \dfrac{5}{9}\left[\dfrac{1 - 4(\frac{5}{9})^3 + 3(\frac{5}{9})^4}{\frac{4}{5}[1 - (\frac{5}{9})]^4}\right] \doteq 0.83$.

(b) $\left(\dfrac{5}{9}\right)^3 \left[\dfrac{\frac{4}{9}}{1 - (\frac{5}{9})^4}\right] \doteq 0.084$.

(c) Since both the number in the system and the number in the queue are the same when the system is empty.

18–11. The general steady-state equations for the $M/M/s$ queueing system are

$$\begin{aligned}
\rho &= \lambda/(s\mu) \\
p_0 &= \left\{ \left[\sum_{n=0}^{s-1} \frac{(s\rho)^n}{n!} \right] + \left[\frac{(s\rho)^s}{(s!)(1-\rho)} \right] \right\}^{-1} \\
L &= s\rho + \frac{(s\rho)^{s+1} p_0}{s(s!)(1-\rho)^2} \\
L_q &= \frac{(s\rho)^{s+1} p_0}{s(s!)(1-\rho)^2} \\
W &= L/\lambda \\
W_q &= W - \frac{1}{\mu}
\end{aligned}$$

In this problem, we have $\lambda = 0.0416$, $\mu = 0.025$.

(a) $\rho = 0.555$ for $s = 3$.

(b) Plugging into the appropriate equation above, we find that $p_0 = 0.1732$.

Then $L_q = 0.3725$.

And finally, $W = W_q + \dfrac{1}{\mu} = \dfrac{L_q}{0.0416} + 40 = 48.96$ min

(c) $\rho = 0.832$ for $s = 2$.

Plugging into the appropriate equation above, we find that $p_0 = 0.0917$.

Then $L_q = 3.742$.

And finally, $W = \dfrac{L_q}{0.0416} + 40 = 129.96$ min.

18–12. $\lambda = 18$, $\mu = 30$, $\phi = 0.6$, $\rho = 0.6/s$.

$$p_0 = \left[\sum_{j=0}^{s-1} \frac{\phi^j}{j!} + \frac{\phi^s}{s!(1-\rho)} \right]^{-1}.$$

$s = 1 \;\Rightarrow\; p_0 = 0.4$.

$s = 2 \Rightarrow p_0 = 0.5384$

and $P(\text{wait}) = 1 - p_0 - p_1 = 1 - 0.5384 - \dfrac{(0.6)^1}{1!}(0.5384) = 0.14.$

18–13. $\lambda = 50$, $\mu = 20$, $\phi = 2.5$, $\rho = 2.5/s$.

(a)

$$C_j = \frac{\lambda_{j-1}\lambda_{j-2}\cdots\lambda_0}{\mu_j\mu_{j-1}\cdots\mu_1},$$

where

$$\lambda_j = \begin{cases} \lambda, & j \le s \\ 0, & j > s \end{cases}$$

and

$$\mu_j = \begin{cases} j\mu, & j \le s \\ s\mu, & j > s \end{cases}$$

Therefore,

$$C_j = \frac{(\lambda/\mu)^j}{j!}, \quad j = 0, 1, \ldots, s.$$

$$p_j = \begin{cases} \frac{(\lambda/\mu)^j}{j!}p_0 & j = 0, 1, \ldots, s \\ 0 & \text{otherwise} \end{cases},$$

where

$$p_0 = \left[\sum_{j=0}^{s} \frac{(\lambda/\mu)^j}{j!}\right]^{-1}$$

(b) $p_s = \dfrac{(\lambda/\mu)^s}{s!}\left[\sum_{j=0}^{s} \dfrac{(\lambda/\mu)^j}{j!}\right]^{-1} \le 0.05$

Try $s = 5 \Rightarrow p_s = 0.065$

$s = 6 \Rightarrow \rho = 0.416$, $p_s = 0.0354$ (which works).

(c) $p_6 = 0.0354$

(d) Utilization would change from 41.6% to $100\rho = 100(0.5/6) = 8.33\%$.

(e) $\phi = 50/12 = 4.166$. $p_6 = 0.377$ fraction getting busy tone.

Chapter 19

19–1. Let n denote the number of coin flips. Then the number of heads observed is $X \sim \text{Bin}(n, 0.5)$. Therefore, we can expect to see about $n/2$ heads over the long term.

19–2. If $\hat{\pi}_n$ denotes the estimator for π after n darts have been thrown, then it is easy to see that $\hat{\pi}_n \sim (4/n)\text{Bin}(n, \pi/4)$. Then $E(\hat{\pi}_n) = \pi$, and we can expect to see the estimator converge towards π as n becomes large.

19–3. By the Law of the Unconscious Statistician,

$$\begin{aligned} E(\hat{I}_n) &= \frac{b-a}{n} E\left(\sum_{i=1}^{n} f(a + (b-a)U_i)\right) \\ &= (b-a)E[f(a + (b-a)U_i)] \\ &= (b-a)\int_0^1 f(a + (b-a)u) \cdot 1 \, du \\ &= I \end{aligned}$$

19–4. (a) The exact answer is $\Phi(2) - \Phi(0) = 0.4772$. The $n = 1000$ result will tend to be closer than the $n = 10$.

(b) We can instead integrate over $\int_0^4$, say, since $\int_4^{10} \approx 0$. This strategy will prevent the "waste" of observations on the trivial tail region.

(c) The exact answer is 0.

19–5. (a)

customer	arrival time	begin service	service time	depart time	wait
1	3	3.0	6.0	9.0	0.0
2	4	9.0	5.5	14.5	5.0
3	6	14.5	4.0	18.5	8.5
4	7	18.5	1.0	19.5	11.5
5	13	19.5	2.5	22.0	6.5
6	14	22.0	2.0	24.0	8.0
7	20	24.0	2.0	26.0	4.0
8	25	26.0	2.5	28.5	1.0
9	28	28.5	4.0	32.5	0.5
10	30	32.5	2.5	35.0	2.5

Time	Event	Customers in System
3	Cust 1 arrival	1
4	Cust 2 arrival	1 2
6	Cust 3 arrival	1 2 3
7	Cust 4 arrival	1 2 3 4
9	Cust 1 depart	2 3 4
13	Cust 5 arrival	2 3 4 5
14	Cust 6 arrival	2 3 4 5 6
14.5	Cust 2 depart	3 4 5 6
18.5	Cust 3 depart	4 5 6
19.5	Cust 4 depart	5 6
20	Cust 7 arrival	5 6 7
22	Cust 5 depart	6 7
24	Cust 6 depart	7
25	Cust 8 arrival	7 8
26	Cust 7 depart	8
28	Cust 9 arrival	8 9
28.5	Cust 8 depart	9
30	Cust 10 arrival	9 10
32.5	Cust 9 depart	10
35	Cust 10 depart	

Thus, the last customer leaves at time 35.

(b) The average waiting time for the 10 customers is 4.75.

(c) The maximum number of customers in the system is 5 (between times 14 and 14.5).

(d) The average number of customers over the first 30 minutes is calculated by adding up all of the customer minutes from the second table — one customer from times 3 to 4, two customers from times 4 to 6, etc.

$$\frac{1}{30}\int_0^{30} L(t)\,dt \;=\; \frac{79.5}{30} \;=\; 2.65$$

19–6. The following table gives a history for this (S, s) inventory system.

day	initial stock	customer order	end stock	reorder?	lost orders
1	20	10	10	no	0
2	10	6	4	yes	0
3	20	11	9	no	0
4	9	3	6	yes	0
5	20	20	0	yes	0
6	20	6	14	no	0
7	14	8	6	no	0

By using $s = 6$, we had no lost orders.

19–7. (a) Here is the complete table for the generator.

i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
X_i	0	1	16	15	12	13	2	11	8	9	14	7	4	5	10	3	0

Thus, $U_1 = 1/16$, $U_2 = 6/16$.

(b) Yes (see the table).

(c) Since the generator cycles, we have $X_0 = X_{16} = \cdots = X_{144} = 0$. Then $X_{145} = 1$, $X_{146} = 6$, …, $X_{150} = 2$.

19–8. (a) Using the algorithm given in Example 19–8 of the text, we find that $X_1 = 1422014746$ and $X_2 = 456328559$. Since $U_i = X_i/(2^{31} - 1)$, we have $U_1 = 0.6622$, $U_2 = 0.2125$.

19–9. (a) $X = -(1/2)\ell\text{n}(1 - U)$.

(b) $X = -(1/2)\ell\text{n}(0.025) = 0.693$.

19–10. (a) $Z = \Phi^{-1}(0.25) = -0.6745$.

(b) $X = \mu + \sigma Z = 1 + 3Z = -1.0235$.

19–11. $f(x) = |x/4|,\ -2 < x < 2$.

(a) If $-2 < x < 0$, then $F(x) = \displaystyle\int_{-2}^{x} -\frac{t}{4}\,dt = \frac{1}{2} - \frac{x^2}{8}$.

If $0 < x < 2$, then $F(x) = \displaystyle\frac{1}{2} + \int_{0}^{x} \frac{t}{4}\,dt = \frac{1}{2} + \frac{x^2}{8}$.

Thus, for $0 < U < 1/2$, we set $F(X) = \dfrac{1}{2} - \dfrac{X^2}{8} = U$.

Solving, we get $X = -\sqrt{4 - 8U}$.

For $1/2 < U < 1$, we set $F(X) = \dfrac{1}{2} + \dfrac{X^2}{8} = U$.

Solving this time, we get $X = \sqrt{8U - 4}$.

Recap:

$$X = \begin{cases} -\sqrt{4 - 8U}, & 0 < U < 1/2 \\ \sqrt{8U - 4} & 1/2 < U < 1 \end{cases}$$

(b) $X = \sqrt{8(0.6) - 4} = 0.894$.

19–12. (a)

x	$p(x)$	$F(x)$	U
-2.5	0.35	0.35	[0,0.35)
1.0	0.25	0.60	[0.35,0.60)
10.5	0.40	1.0	[0.60,1.0)

(b) $U = 0.86$ yields $X = 10.5$.

19–13. (a) $F(X) = 1 - e^{-(X/\alpha)^\beta} = U$.

Solving for X, we obtain $X = \alpha[-\ell\text{n}(1 - U)]^{1/\beta}$.

(b) $X = (1.5)[-\ell\text{n}(0.34)]^{1/2} = 1.558$.

19–14. We have

$$Z_1 = \sqrt{-2\ell\text{n}(0.45)}\,\cos(2\pi(0.12)) = 0.921$$

and

$$Z_2 = \sqrt{-2\ell\text{n}(0.45)}\,\sin(2\pi(0.12)) = 0.865.$$

19–15. $\displaystyle\sum_{i=1}^{12} U_i - 6 = 1.07.$

19–16. Since the X_i's are IID exponential(λ) random variables, we know that their m.g.f. is

$$M_{X_i}(t) \;=\; \frac{\lambda}{\lambda - t}, \quad t < \lambda, \quad i = 1, 2 \ldots, n.$$

Then the m.g.f. of $Y = \sum_{i=1}^{n} X_i$ is

$$M_Y(t) \;=\; \prod_{i=1}^{n} M_{X_i}(t) \;=\; \left(\frac{\lambda}{\lambda - t}\right)^n.$$

We will be done as soon as we can show that this m.g.f. matches that corresponding to the p.d.f. from Equation (19–4), namely,

$$\begin{aligned}
M_Y(t) &= \int_0^\infty e^{ty} \lambda^n e^{-\lambda y} y^{n-1}/(n-1)!\, dy \\
&= \frac{\lambda^n}{(n-1)!} \int_0^\infty e^{-(\lambda - t)y}\, y^{n-1}\, dy \\
&= \frac{\lambda^n}{(n-1)!} \int_0^\infty e^{-u} \left(\frac{u}{\lambda - t}\right)^{n-1} \frac{du}{\lambda - t} \\
&= \frac{\lambda^n}{(\lambda - t)^n (n-1)!} \int_0^\infty e^{-u}\, u^{n-1}\, du \\
&= \frac{\lambda^n}{(\lambda - t)^n (n-1)!}\, \Gamma(n) \\
&= \left(\frac{\lambda}{\lambda - t}\right)^n.
\end{aligned}$$

Since both versions of $M_Y(t)$ match, that means that the two versions of Y must come from the same distribution — and we are done.

19–17. $X = -\dfrac{1}{\lambda}\, \ell\mathrm{n}\left(\displaystyle\prod_{i=1}^{n} U_i\right) = -\frac{1}{3}\, \ell\mathrm{n}((0.73)(0.11)) = 0.841.$

19–18. (a) To get a Bernoulli(p) random variable X_i, simply set

$$X_i \;=\; \begin{cases} 1, & \text{if } U_i \le p \\ 0, & \text{if } U_i > p \end{cases}$$

(Note that there are other allocations of the uniforms that will do the trick just as well.)

(b) Suppose $X_1, X_2, \ldots, X_n$ are IID Bernoulli's, generated according to (a). To get a Binomial(n, p), let $Y = \sum_{i=1}^{n} X_i$.

19–19. Suppose success (S) on trial i corresponds to $U_i \leq 0.25$, and failure (F) corresponds to $U_i > 0.25$. Then, from the sequence of uniforms in Problem 19–15, we have $FFFFS$, i.e., we require $X = 5$ trials before observing the first success.

19–20. The grand sample mean is

$$\bar{Z}_b \;=\; \frac{1}{b}\sum_{i=1}^{b} Z_i \;=\; 4,$$

while

$$\hat{V}_B \;=\; \frac{1}{b-1}\sum_{i=1}^{b}(Z_i - \bar{Z}_b)^2 \;=\; 1.$$

So the 90% batch means CI for μ is

$$\begin{aligned}
\mu \;&\in\; \bar{Z}_b \pm t_{\alpha/2,b-1}\sqrt{\hat{V}_B/b} \\
&=\; 4 \pm t_{0.05,2}\sqrt{1/3} \\
&=\; 4 \pm 2.92\sqrt{1/3} \\
&=\; 4 \pm 1.686
\end{aligned}$$

19–21. The 90% confidence interval is of the form

$$\begin{aligned}
[-2.5,\, 3.5] \;&=\; \bar{X} \pm t_{\alpha/2,b-1}\, y \\
&=\; 0.5 \pm t_{0.05,4}\, y \\
&=\; 0.5 \pm 2.132\, y
\end{aligned}$$

Since the half-length of the CI is 3, we must have that $y = 3/2.132 = 1.407$.

The 95% CI will therefore be of the form

$$\bar{X} \pm t_{0.025,4}\, y \;=\; 0.5 \pm (2.776)(1.407) \;=\; 0.5 \pm 3.91 \;=\; [-3.41,\, 4.41].$$

19–23. The 95% batch means CI for μ is

$$\begin{aligned}\mu &\in \bar{Z}_b \pm t_{\alpha/2,b-1}\sqrt{\hat{V}_B/b} \\ &= 100 \pm t_{0.025,4}\sqrt{250/5} \\ &= 100 \pm 2.776\sqrt{50} \\ &= 100 \pm 19.63\end{aligned}$$

19–25. (a) (i) Both exponential(1).

(ii) $Cov(U_i, 1-U_i) = -V(U_i) = -1/12$.

(iii) Yes.

(b) After a little algebra,

$$\begin{aligned}V((\bar{X}_n + \bar{Y}_n)/2) &= \frac{1}{4}\left[V(\bar{X}_n) + V(\bar{Y}_n) + 2Cov(\bar{X}_n, \bar{Y}_n)\right] \\ &= \frac{1}{2}\left[V(\bar{X}_n) + Cov(\bar{X}_n, \bar{Y}_n)\right] \\ &= \frac{1}{2n}\left[V(X_i) + Cov(X_i, Y_i)\right] \\ &\leq \frac{1}{2n}V(X_i) \\ &= V(\bar{X}_{2n}).\end{aligned}$$

So the variance decreases compared to $V(\bar{X}_{2n})$.

(c) You get zero, which is the correct answer.

19–26. (a) $E(C) = E(\bar{X}) - E[k(Y - E(Y))] = \mu - [k(E(Y) - E(Y))] = \mu$.

(b) $V(C) = V(\bar{X}) + k^2V(Y) - 2k\,Cov(\bar{X}, Y)$. Comment: It would be nice, in terms of minimizing variance, if $k\,Cov(\bar{X}, Y) > 0$.

(c)

$$\frac{d}{dk}V(C) = 2kV(Y) - 2\,Cov(\bar{X}, Y) = 0.$$

This implies that the critical (minimizing) point is

$$k = \frac{Cov(\bar{X}, Y)}{V(Y)}.$$

Thus, the optimal variance is

$$\begin{aligned} V(C) &= V(\bar{X}) + \left(\frac{Cov(\bar{X},Y)}{V(Y)}\right)^2 V(Y) - 2\left(\frac{Cov(\bar{X},Y)}{V(Y)}\right) Cov(\bar{X},Y) \\ &= V(\bar{X}) - \frac{Cov^2(\bar{X},Y)}{V(Y)}. \end{aligned}$$

19–27. They are exponential(1).

19–28. They should look normal.

19–29. You should have a bivariate normal distribution (with correlation 0), centered at zero with symmetric tails in all directions.